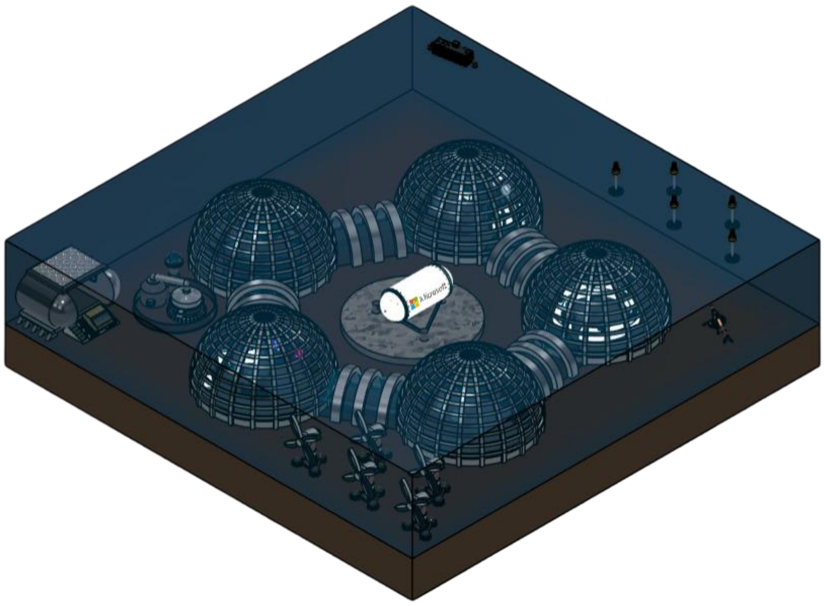
School of Engineering
Integrated Design Project 2 Assignment 1
Concept Design
2025/26
Group: D3

Future Ocean Habitat



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STATEMENT OF CONTRIBUTION

Name	Student ID	Role / Work Package Lead	Signature	Contribution
Ahmad Alhashmi	2716969	- Energy System Coordinator - Failure Space / Success Space - HAZOP - IBM Enterprise Design Thinking - Objective Tree (Energy) - Requirements Specification (Energy) - Functional Flow (Energy) - Material Selection and Life Cycle Analysis (Energy) - Design (Energy) - Research Journal Club (Energy) - Morphological Chart (Energy)		Normal
Matteo Karam	2760382	- Life Support Systems Coordinator - Environmental Safeguard Statement - Responsibility Assignment Matrix (RACI) - Objective Tree (Life Support) - CONOPs (Concept of Operations) - Requirements Specification (Life Support) - Functional Flow (Life Support) - Material Selection and Life Cycle Analysis (Life Support) - Design (Life Support) - Research Journal Club (Life Support) - Morphological Chart (Life Support)		Normal
Muhammad Hammad	2840920	- Systems Coordinator - Gantt Chart - Verification Matrix - Objective Tree (Systems) - Business Case - Material Selection and Life Cycle Analysis (Structure+Systems) - Requirements Specification (Systems) - Functional Flow (Systems) - Design (Systems) - Research Journal Club (Systems) - Morphological Chart (Systems)		Normal
Muhammad Raafey	2828484	- Underwater Data Center Coordinator - Functional Flow Diagrams Refinement - Mission statement - Executive Summary - Objective Tree (UDC) - Site Survey - Functional Flow (Structure+UDC) - Requirements Specification (UDC) - Material Selection and Life Cycle Analysis (UDC) - Design (UDC) - Research Journal Club (UDC) - Morphological Chart (UDC) - Lab Exercises		Normal

Saba Falah	2847582	- Group Coordinator - Deep Sea Mining Coordinator - Belbin Test Result Analysis - Rich Picture - CAD of Underwater Habitat - Objective Tree (General+DSM) - Design (Structure+DSM) - Requirements Specification (DSM) - Functional Flow (DSM) - Material Selection and Life Cycle Analysis (DSM) - Meeting Logs - Academic Mentor Meeting Logs - Research Journal Club (DSM) - Morphological Chart (Structure+DSM)	Saba Falah	Normal
Talal Abu Ghazaleh	2813867	- Direct Ocean Capture Coordinator - Objective Trees Sketch & Refinement - SDG Progression - Integration Block Diagram - Objective Tree (DOC) - Requirements Specification (Structure+DOC) - Functional Flow (DOC) - Material Selection and Life Cycle Analysis (DOC) - Design (DOC) - Research Journal Club (DOC) - Morphological Chart (DOC) - TRIZ Contradiction Matrix	[illegible]	Normal

CONTENTS

EXECUTIVE SUMMARY

This report presents the design and development of an underwater habitat for people to live independently from land. The habitat is designed to completely support underwater operations. The project solves technical engineering problems of living underwater such as maintaining structure robustness, providing clean air, stable temperatures, reliable energy supply and not harming the environment.

In addition to this, there were three other missions: Direct Ocean Capture (DOC) which captures carbon from the ocean, Underwater Data Centre (UDC) which stores and processes data underwater, Deep Sea Mining (DSM) which mines minerals from the sea floor.

To achieve the goal of the underwater habitat, a structured design method was used, starting with defining requirements and breaking down big functions into more manageable smaller functions. Next initial research was conducted through a system of Journal Club Reports which helped to quickly capture crucial information from the research sources gathered. Following this, the system object flow diagrams were derived which were then transferred and further refined into system requirements. Furthermore, the lifecycle of the system was assessed as well as the lifecycles of individual components/subsystems. Moreover, functional flow diagrams were produced with all the information gathered which then fuelled the process of creating a morphological chart and thus leading to brainstorming sessions and the creation of a TRIZ matrix.

Through cycles of refinements, a final design and layout was achieved, integrating all subsystems into a singular, unified system. The main structure is a group of dome-shaped hulls connected to one another in a pentagon design. As such, they are designed to withstand deep-sea pressure, interaction with the sea floor and long-term corrosion, all while satisfying the rest of their requirements while not being isolated from one another. Additionally, material selection was done using ANSYS Granta Edupack. In addition to this, a detailed Life Cycle Analysis (LCA) was completed utilizing the Eco Audit section, assessing features of the development and deployment of the habitat such as energy burdens, material composition impact, end-of-life process and climate change impact.

Overall, the underwater habitat is found to be technically feasible, being both structurally robust and sustainable without having too much of an environmental impact. The design has considered and integrated safety, environmental responsibility and long-term operability.

WP 0.1: PROJECT

0.1.1 People

Belbin results

To examine individual strengths and how each member affects team dynamics, the Belbin test was used. The outcomes demonstrate each member's contributions, guaranteeing a fair and productive team effort (see Figure 1).

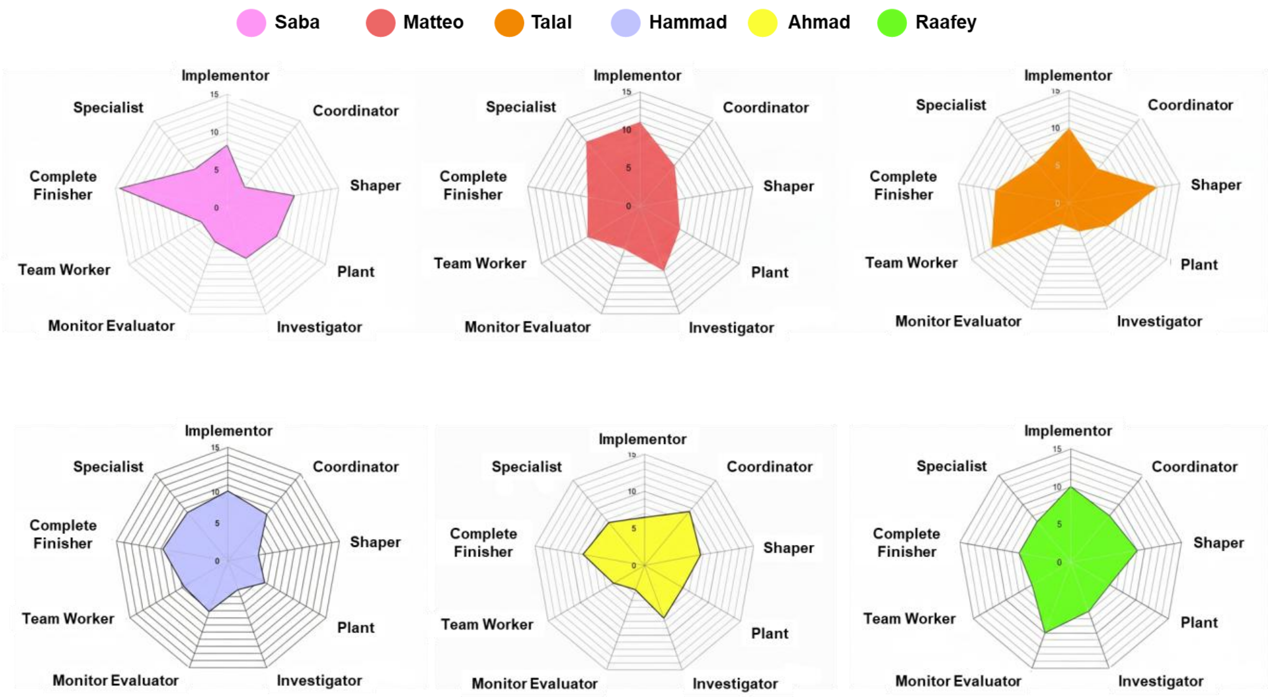


Figure 1. *Belbin radar charts gathered from each member's test results.*

The following analysis outlines the behavioral contributions and functional roles of the project team, based on the Belbin test results.

A. Saba *(Strategic Shaper and Quality Assurance Lead)*
Saba's primary dual-focus as a "Complete Finisher" and "Shaper" defines her profile. She provides the psychological motivation required to overcome inertia and sustain momentum in her role as team leader. Her remarkably high Complete Finisher score demonstrates a strict dedication to accuracy and error correction. This combination guarantees that the project adheres to the highest standards of technical excellence while also meeting its milestones through assertive leadership.

B. Matteo *(Technical Specialist and Systematic Implementor)*
Matteo's contributions are centred on the "Specialist" and "Implementor" roles. He provides the team with the depth of technical knowledge essential for complex problem-solving. Beyond his subject matter expertise, his Implementor attributes ensure that theoretical designs are converted into practical, workable strategies. His disciplined approach provides the team with a reliable foundation for systematic progress and technical integrity.

C. Talal *(Interpersonal Team Worker and Dynamic Shaper)*
Talal possesses a unique behavioural profile that balances the “Shaper” and “Team Worker” roles. This dualism allows him to act as a catalyst for progress during challenging phases while simultaneously fostering internal cohesion. His high Team Worker score suggests he is highly sensitive to team morale and effective at mitigating conflict, ensuring that the assertive "Shaping" tendencies within the group do not compromise collaborative harmony.

D. Hammad *(Practical Implementor and Quality-Centric Specialist)*
Hammad reinforces the team’s operational capacity with a focus on the “Implementor” and “Complete Finisher” roles. His profile indicates a strong preference for practical application and the delivery of concrete results. Hammad’s presence increases the team's overall "workhorse" capacity, ensuring that the technical requirements of the project are met with both efficiency and a meticulous attention to detail during the final stages of the project lifecycle.

E. Ahmad *(Facilitative Coordinator and Diligent Finisher)*
Ahmad occupies a critical role as a “Coordinator” and “Complete Finisher”. His presence provides a stabilizing influence; he is adept at identifying talent within the group and delegating tasks to ensure collective objectives are met. Ahmad’s secondary focus on finishing ensures that his administrative oversight is matched by a high regard for the quality and accuracy of the team's final output, acting as a bridge between strategic vision and meticulous execution.

F. Muhammad Raafey *(Objective Monitor Evaluator and Methodical Implementor)*
Raafey serves as the team’s primary analytical anchor, scoring highest as a “Monitor Evaluator” and “Implementor.” His role is vital for risk mitigation, as he provides a dispassionate and critical assessment of proposed ideas. By balancing this with a strong Implementor drive, Raafey ensures that once a decision has been scrutinized and approved, it is executed with consistency and a high degree of organizational reliability.

The team’s composition indicates a strong orientation toward execution, with a high concentration of action-oriented roles focused on implementation and quality control. This drive is balanced by coordinating and evaluative functions that ensure strategic alignment and interpersonal harmony. Consequently, the team is structured to maintain high performance and technical precision within a professional engineering framework.

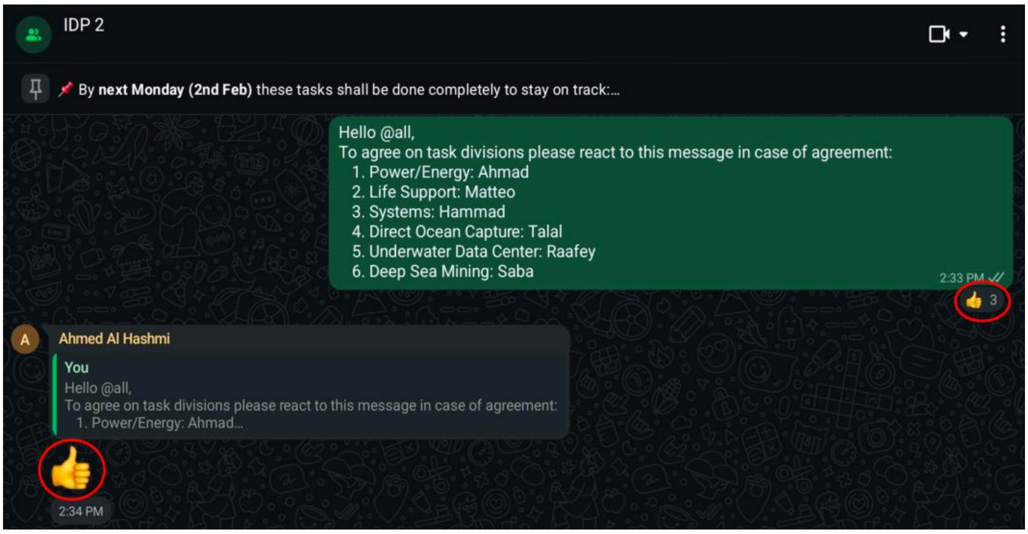
Non-Violent Communication

The group maintained an inclusive and cooperative communication approach, emphasizing respect, transparency, and constructive interaction. Varied viewpoints were treated as valuable contributions that strengthened the team’s collective problem-solving rather than as sources of disagreement. All members practiced active listening, allowing each voice to be heard without interruption and ensuring discussions stayed productive. Respectful phrasing, such as “I see your point; perhaps we could also explore…,” promoted a balanced exchange of ideas. The coordinator supported open dialogue by encouraging equitable participation rather than enforcing authority, helping to foster mutual decision-making grounded in evidence and clear reasoning.

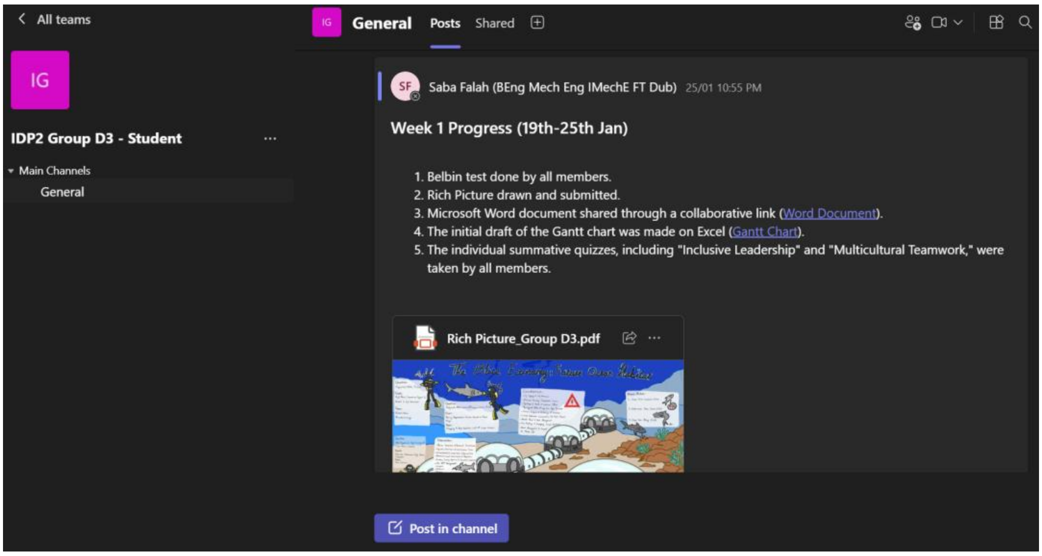
To ensure efficiency, responsibilities were distinctly assigned, preventing duplication of effort and maintaining a steady workflow. Communication was supported through two main platforms: “WhatsApp” served for continuous updates, brief announcements, and quick clarifications, while

“Microsoft Teams” functioned as the primary hub for document sharing, progress tracking, and virtual meetings.

Task distribution was guided by individual strengths and confirmed collectively to guarantee that all members were confident in their assigned roles. Evidence of this organized structure is visible in the workspace shown in Figure 2.a, where contributions and acknowledgments reflect the team’s collaborative commitment. Furthermore, the weekly progress and collaborative links of significant value were posted on Teams (see Figure 2.b).



(a)



(b)

Figure 2. *(a) WhatsApp group chat collaborative communication, (b) Microsoft Teams general page.*

The chat in Figure 3 demonstrates an example of effective engineering task management through direct communication, pre-empting delays and conflicts in a student project setting. Clear status updates ("Pretty much done just need it from Matteo") and specific commitments avoid ambiguity or blame, aligning with best practices like open communication and active listening in project teams. Positive acknowledgments ("Great thanks") reinforce collaboration, preventing escalation common in dependency issues.

Proactive ownership ("Will go home and finish it") with a timeline ("all by this evening hopefully") resolves bottlenecks swiftly, mirroring strategies for late deliverables via prioritization and transparency. Brevity suits fast-paced engineering workflows, ensuring minimal disruption.

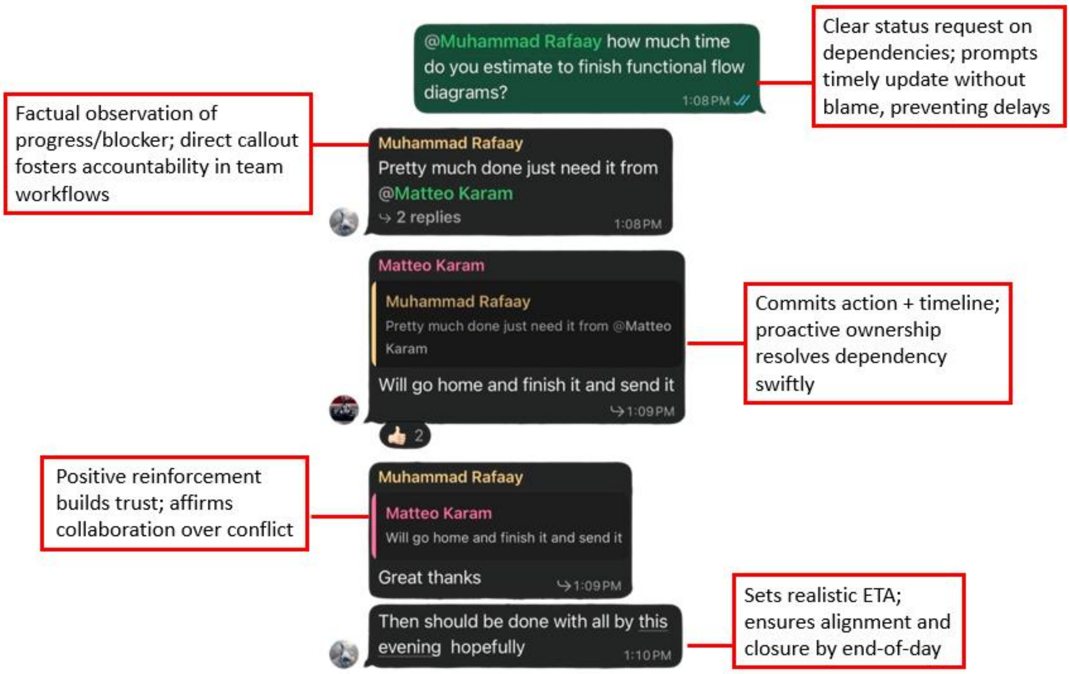


Figure 3. *Non-violent communication example and analysis within the team.*

0.1.2 Planning

0.1.2.1 Responsibility Assignment Matrix (RACI)

Table 1 serves as a widely used project management tool designed to clarify the roles and responsibilities of team members across various work packages. By mapping each team member against specific work packages, the table provides a clear visual reference that helps prevent ambiguity, ensures accountability, and supports effective task delegation. It indicates how responsibilities are distributed within the team, promoting transparency and facilitating efficient project planning and coordination.

Table 1. *Tabulated responsibility assignment based on members being **R**esponsible, **A**ccountable, **C**onsulted, and **I**nformed.*

Team Member	Work Package									
	0.1	0.2	1	2	3	4	5	6A	6B	6C
Ahmad	C/A	C/A	C/A	C/A	R	I/C	I	I	I	I
Matteo	C/A	C/A	C/A	C/A	I	R	I/C	I	I	I
Hammad	C/A	C/A	C/A	C/A	I/C	I	R	I	I	I
Talal	C/A	C/A	C/A	C/A	I	I	I	R	I	I/C
Raafey	C/A	C/A	C/A	C/A	I	I	I	I/C	R	I
Saba	C/A	C/A	C/A	C/A	I	I	I	I	I/C	R

0.1.2.2 Gantt Chart

IDP 2 Assignment 1: The Blue Economy: Future Ocean Habitat

Project start: **Mon, 1/19/2026**

Display week: **1**

TASK	ASSIGNED TO	START	END
Project Initiation			
First group meeting	All	1/19/26	1/19/26
Set communication channels(MS Teams and Whatsapp)	All	1/19/26	1/19/26
Analyse challenge specification	All	1/19/26	1/21/26
Belbin Test and Analysis	All	1/21/26	1/22/26
Appoint group coordinator and team roles	All	1/22/26	1/23/26
Hopes & fears / rich picture exercise	Saba	1/20/26	1/24/26
Multicultural and Inclusive Leadership modules	All	1/20/26	1/25/26
Planning & Project Management			
Develop Gantt chart	Hammad	1/23/26	1/27/26
Work Breakdown Structure	Saba	1/25/26	1/27/26
Initial integration discussion	All	1/25/26	1/28/26
Responsibility Assignment Matrix (RAM)	Saba	1/25/26	1/30/26
Research			
Brain Stroming and first Concepts	All	1/26/26	1/27/26
Assign Work Package for each member	Saba	1/26/26	1/27/26
Research and Journal Club Reports	All	1/27/26	1/31/26
Site Survey	Raafey	1/27/26	1/28/26
Objective tree development	All	1/27/26	2/1/26
Structure Design	Hammad,Raafey,Saba	1/27/26	2/3/26
Energy Design	Ahmad	1/27/26	2/3/26
Life Support Design	Matteo	1/27/26	2/3/26
Systems Design	Hammad	1/27/26	2/3/26
Direct Ocean Capture Design	Talal	1/27/26	2/3/26
Underwater Data Centre Design	Raafey	1/27/26	2/3/26
Deep Sea Mining Design	Saba	1/27/26	2/3/26
Morphological Chart	All	2/3/26	2/10/26
Business case & CONOPS	Hammad and Matteo	2/3/26	2/10/26
Requirements			
Requirements Table and Functional Flow Diagrams	All	2/3/26	2/9/26
Integration of Subsystems Block Diagram	Talal	2/8/26	2/11/26
CAD of entire Structure	Saba Falah	2/3/26	2/17/26
SDG Progression	Talal	2/5/26	2/7/26
Life Cycle Analysis	All	2/7/26	2/10/26
Failure Space/ Success Space + HAZOP	Ahmad	2/8/26	2/11/26
TRIZ Contradiction Matrix	Matteo	2/8/26	2/11/26
Integration & Submission			
Final system integration	All	2/17/26	2/20/26
Final report writing	All	2/17/26	2/22/26
Peer review & contribution statement	All	2/22/26	2/23/26
Submission of Assignment	All	2/25/26	2/25/26

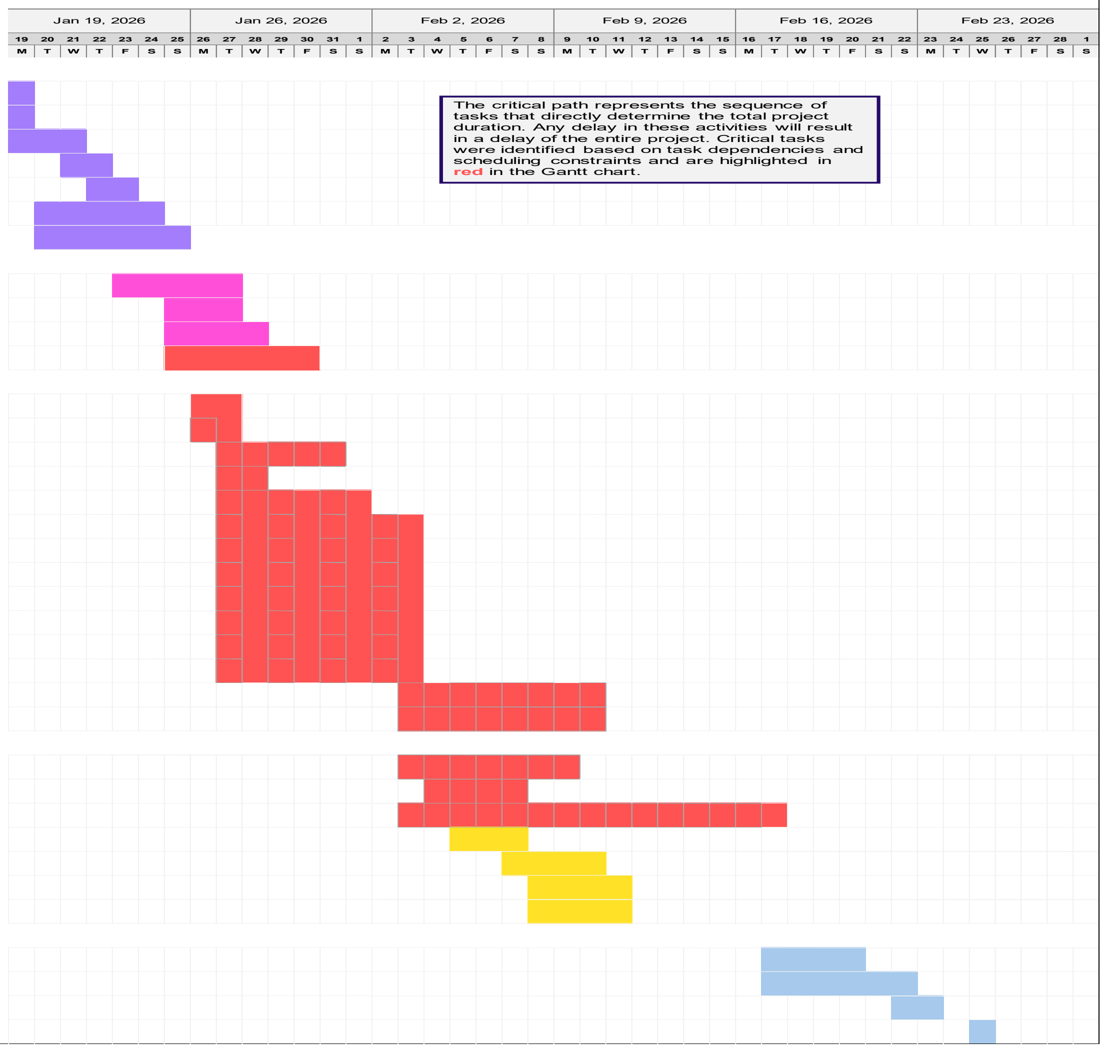


Figure 4. *Group D3 Project Gantt Chart.*

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0.1.3 Integration

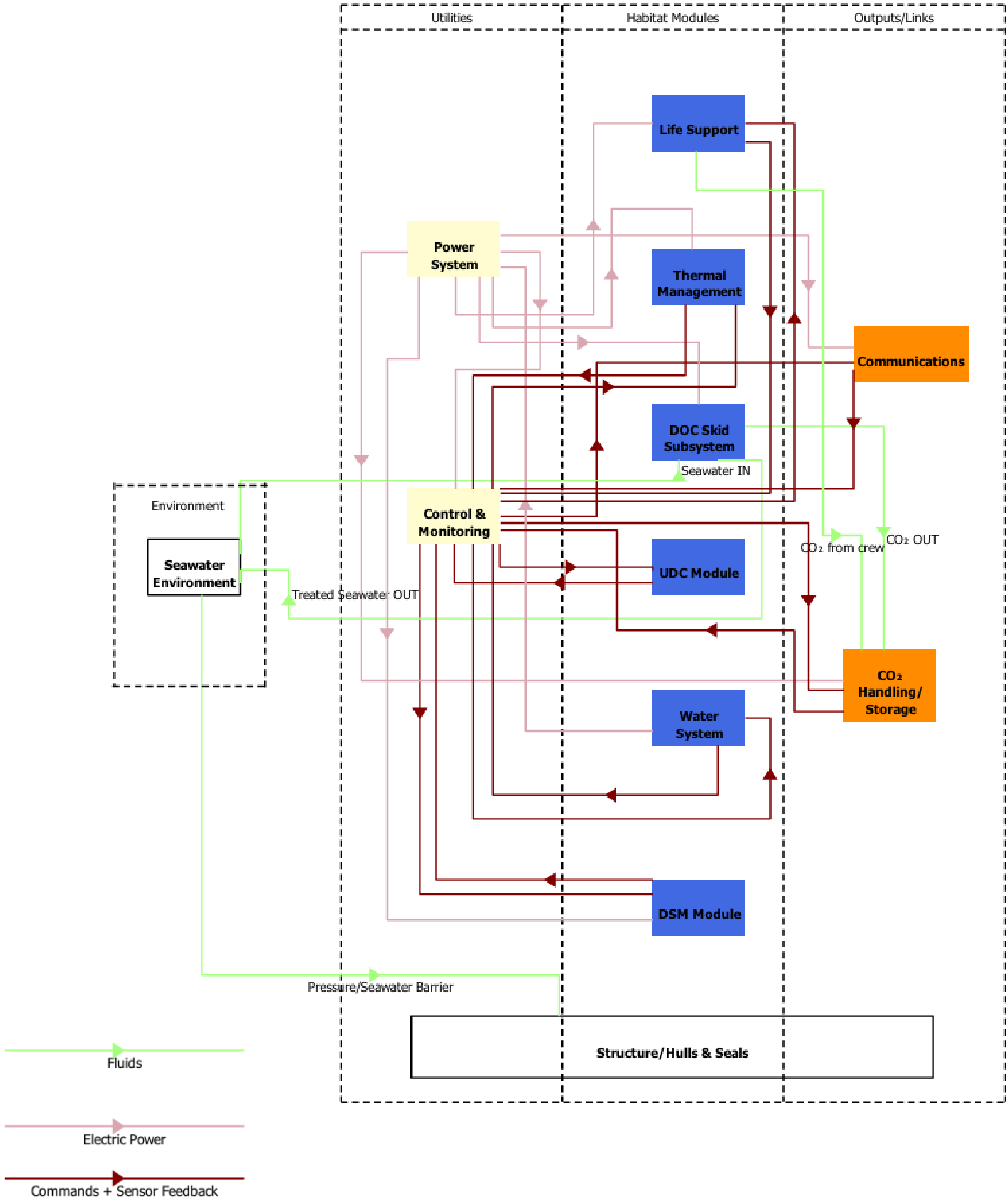

Figure 5. *Underwater Habitat Integrated Block Diagram.*

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WP 0.2: ASSURANCE

0.2.1 Failure Space / Success Space + HAZOP

This section identifies and prioritises the top hazards for the "Future Ocean Habitat concept". A Failure Space/Success Space model is used to visualise how faults push the system away from acceptable operation, and a HAZOP-style review is used to capture credible deviations, causes, consequences, safeguards, and recommended actions.

Table 2. *Critical functions and their corresponding success criteria.*

Critical function	Success criterion
Structural integrity	Maintain pressure boundary at operating depth with factor of safety ≥ 2; leak detection active and compartment isolation available.
Atmosphere (crew)	O_2 19.5-23.5%, CO_2 ≤ 0.5%, temperature 20-25°C, humidity 30-60% RH.
Power availability	Continuous Tier 1 supply; storage supports Tier 1 loads ≥ 72 hours; microgrid voltage regulation within ±5% with coordinated protection.
Thermal management	Critical equipment temperatures maintained within design limits (e.g., ≤35°C for power electronics) using seawater heat rejection.
Water system	Potable water meets WHO standards; greywater recycling and wastewater treatment to marine-safe discharge standards; leak prevention maintains habitat dryness.
Direct Ocean Capture	CO_2 capture uptime ≥95%; seawater chemistry control with pH change ≤±0.2 and salinity change ≤±1 PSU; no harmful discharge.
Underwater Data Center	Data storage/processing availability ≥99%; thermal stability maintained; surrounding seawater temperature rise limited within design limit.
Deep Sea Mining	Safe docking/undocking with ±5° misalignment tolerance; closed-loop slurry handling with environmental monitoring.
Systems	Reliable long-range acoustic comms + short-range optical comms; monitoring detects strain/leaks and triggers alarms/safety responses.

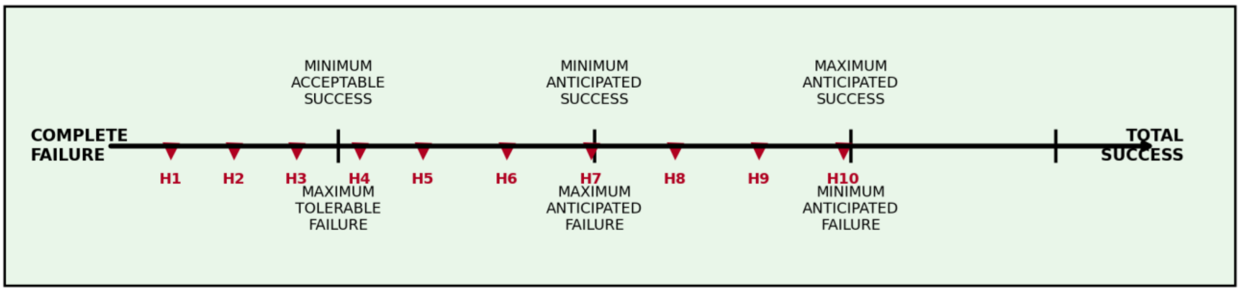

Figure 6. *Failure Space - Success Space model for the Future Ocean Habitat project.*

__Table 3.__ Top 10 hazards (H1–H10) and high-level intent.

Hazard ID	Hazard (summary)	Primary consequence	Primary mitigation direction
H1	Pressure hull breach / rapid flooding	Rapid flooding, loss of buoyancy; Crew injury/fatality; Loss of mission modules; Environmental release (oils/chemicals)	NDT inspection schedule + corrosion monitoring
H2	Uncontrolled fire / explosion (BESS, HV)	Smoke/toxic fumes in habitat; Fire spread, loss of power & comms; Potential explosion (if H_2 present)	Prefer intrinsically safer chemistries where possible (e.g., LFP) or external battery pods
H3	Loss of atmosphere control (O2 low / CO2 high)	Crew hypoxia/hypercapnia; Impaired decision-making; Potential fatality if not corrected	Cross-check sensors (2oo3 voting) + regular calibration
H4	Toxic chemical release (DOC acid/base, Cl2, CO2)	Corrosion of nearby systems; Toxic gas formation (e.g., Cl_2 risk in electrochemical systems); Crew exposure if leak routes to habitat; Environmental harm	Physical segregation (external pod) + forced ventilation
H5	Total power loss / microgrid blackout > UPS	Life support & monitoring degrade; Loss of comms; reduced situational awareness; Forced mission abort / evacuation	N+1 design for converters and critical feeders
H6	Thermal management failure / overheating	Electronics derating/failure; Increased battery degradation; Heat stress for crew; Potential runaway events (H2)	Anti-fouling coatings + cleaning schedule
H7	Seawater ingress via piping (RO, HX) / internal flooding	Local flooding → electrical short; Slip hazard; Loss of potable water; Contamination of clean zones	Pressure relief + burst-disk where needed
H8	CO2 handling/storage overpressure or leak	Asphyxiation risk in enclosed spaces; Structural damage to cylinder/piping; Environmental release / regulatory breach	Store CO_2 in external, ventilated pod
H9	Mooring/docking failure causing collision/cable damage	Hull/port damage → leaks; Cable snap → blackout/comms loss; Damage to mission vehicles; Injury during docking operations	ROV inspection intervals for moorings
H10	Cyber/communications compromise affecting control/safety	Incorrect actuation (valves, load shed, vents); Hidden degradation (false healthy); Loss of remote support; Cascading safety incidents	Network segmentation (safety vs mission)

HAZOP is used here as a structured prompt to identify deviations from design intent using guidewords (e.g., No, More, Less, Reverse, Other than, As well as). The table below summarises the top hazards and the most relevant deviations for each subsystem node.

Table 4. *HAZOP worksheet (top hazards).*

H#	Node / Intent	Guideword + deviation	Credible causes	Consequences	Existing safeguards / detection	Recommended actions
H1	Pressure hull, hatches & seals (Intent: maintain pressure boundary at ~200 m depth)	No / Other than: Loss of containment (leak or rupture)	• Corrosion / pitting • Fatigue cracking • Seal ageing / improper closure • Impact from AUV/DSM equipment • Manufacturing defect	• Rapid flooding, loss of buoyancy • Crew injury/fatality • Loss of mission modules • Environmental release (oils/chemicals)	• Compartmentalisation / double-barrier philosophy • Strain gauges + pressure/leak sensors • Alarms + emergency isolation • Planned inspections	• NDT inspection schedule + corrosion monitoring • Cathodic protection + coatings • Redundant hatch interlocks and leak-before-break design • Emergency refuge + evacuation/rescue plan
H2	Electrical distribution + BESS compartment (Intent: safe electrical supply without ignition sources)	More: Overtemperature / arcing → fire	• Water ingress causing short • Connector/insulation failure • Overload or protection miscoordination • Battery thermal runaway • Poor ventilation	• Smoke/toxic fumes in habitat • Fire spread, loss of power & comms • Potential explosion (if H_2 present)	• Protective relays, fuses, isolation • BMS + temperature sensing • Fire detection/suppression • Segregated electrical rooms	• Prefer intrinsically safer chemistries where possible (e.g., LFP) or external battery pods • Fire-rated barriers + smoke extraction • Arc-fault detection • Fire drills + shutdown procedures
H3	Life Support (O_2 generation + CO_2 scrubbing) (Intent: O_2 19.5–23.5%, CO_2 ≤0.5%)	Less/More: O_2 low or CO_2 high	• Electrolyser failure • Scrubber saturation • Blower/fan failure • Sensor drift/spoofing • High metabolic load / leak	• Crew hypoxia/hypercapnia • Impaired decision-making • Potential fatality if not corrected	• Redundant sensors + alarms • Backup scrubber cartridges • Emergency O_2 cylinders • Manual override modes	• Cross-check sensors (2oo3 voting) + regular calibration • Defined change-out intervals for sorbents • Emergency breathing masks staged • Automated load-shedding prioritising life support
H4	DOC skid (electrochemical capture) (Intent: control pH change ≤±0.2, prevent harmful discharge)	Other than / More: Chemical leak or uncontrolled pH/by-products	• Seal/pipe failure • Membrane damage • Dosing/control error • Pump malfunction • Unexpected seawater chemistry	• Corrosion of nearby systems • Toxic gas formation (e.g., Cl_2 risk in electrochemical systems) • Crew exposure if leak routes to habitat • Environmental harm	• Bunding/secondary containment • pH/ORP monitoring + interlocks • Ventilation in process bay • Automatic shutdown on out-of-range	• Physical segregation (external pod) + forced ventilation • Neutralisation stage + controlled discharge • Gas detection (chlorine/acid mists) • Maintenance access plan and spill response kit
H5	Hybrid generation + DC microgrid (Intent: continuous Tier 1 supply; ≥72 h autonomy; voltage ±5%)	No: Blackout / loss of critical power	• Wave/tidal devices offline • Converter/inverter failure • Cable damage • Protection trip cascade • Control system fault	• Life support & monitoring degrade • Loss of comms; reduced situational awareness • Forced mission abort / evacuation	• Tiered load shedding • UPS + BESS + long-duration reserve (H_2) • Redundant buses + isolation switches • Black-start capability	• N+1 design for converters and critical feeders • Routine black-start and emergency power tests • Physical cable routing separation

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						• Clear power budget and load-priority matrix
H6	Thermal management (seawater heat exchangers) (Intent: keep equipment ≤35°C; habitat 20–25°C)	Less/No: Insufficient heat rejection	• Biofouling / blockage • Pump failure • HX leak or scaling • Valve stuck • Excess heat from UDC/DSM	• Electronics derating/failure • Increased battery degradation • Heat stress for crew • Potential runaway events (H2)	• Temperature monitoring • Redundant pumps • Alarmed flow/ΔT sensors • Heat load shedding (UDC throttling)	• Anti-fouling coatings + cleaning schedule • Thermal zoning & passive heat paths • Automatic throttling/standby for non-critical loads • Spare pump modules and quick-change interfaces
H7	Water system (RO, tanks, pipework) (Intent: potable water + leak-free operation)	As well as: Leak/rupture → unintended water release	• High-pressure RO hose failure • Tank crack • Poor fittings/assembly • Vibration fatigue • Corrosion	• Local flooding → electrical short • Slip hazard • Loss of potable water • Contamination of clean zones	• Leak sensors + bilge pumps • Isolation valves • Drip trays • Scheduled maintenance	• Pressure relief + burst-disk where needed • Route high-pressure lines away from electrics • Secondary containment for tanks • Water quality monitoring + emergency ration plan
H8	CO_2 handling & storage (Intent: safe compression/storage with monitored discharge/use)	More: Overpressure / uncontrolled CO_2 release	• Compressor control failure • Blocked line • PRV stuck • Overfilling storage • Temperature rise	• Asphyxiation risk in enclosed spaces • Structural damage to cylinder/piping • Environmental release / regulatory breach	• Pressure sensors + alarms • Relief valves + burst disks • Vent lines to exterior • CO_2 detectors	• Store CO_2 in external, ventilated pod • Redundant PRVs + periodic proof testing • Interlocks to stop compression on high pressure • Clear emergency vent/lockout procedure
H9	Mooring, docking & umbilicals (Intent: maintain position; safe AUV/DSM docking)	Other than / Reverse: Unintended motion or collision	• Mooring line failure • Extreme currents/storm • Docking guidance failure • Thruster malfunction • Human/remote operator error	• Hull/port damage → leaks • Cable snap → blackout/comms loss • Damage to mission vehicles • Injury during docking operations	• Redundant mooring lines • Docking sensors + misalignment tolerance • Physical bumpers/guards • Condition monitoring	• ROV inspection intervals for moorings • Define operational sea-state limits • Emergency release/quick-disconnect for umbilicals • Soft-capture docking + interlocked latching
H10	Control, monitoring & communications (Intent: trustworthy sensing, authorised commands, data integrity)	Other than: Unauthorised/spoofed data or commands	• Cyber intrusion (surface link) • Weak authentication • Compromised maintenance laptop • Software bug / update failure • Sensor spoofing	• Incorrect actuation (valves, load shed, vents) • Hidden degradation (false healthy) • Loss of remote support • Cascading safety incidents	• Basic access control • Alarms on out-of-range values • Manual local override • Logged operator actions	• Network segmentation (safety vs mission) • Strong authentication + signed updates • Intrusion detection + anomaly monitoring • Sensor voting + plausibility checks • Offline-safe modes for life support & power

0.2.2 Environmental Safeguard Statement

Table 5. *Tabulated statements for environmental safeguard.*

Environment principle	Safeguard statement
Integration principle	Environmental protection is integrated into the design of the Life Support System through closed-loop water recycling, controlled brine discharge, and monitoring of effluent quality. Environmental factors are taken into account in the design stage through material selection (corrosion-resistant metals) and waste minimization plans.
Prevention principle	The system protects the environment by ensuring that no untreated blackwater or solid waste is discharged into the marine environment. The reverse osmosis brine discharge is controlled and diluted to prevent salinity levels from rising above 10% of the existing seawater levels. The leak detection sensors are designed to prevent accidental contamination.
Rectification at source principle	Potential pollutants like brine concentrate, sludge, and chemical residues are processed in the system before being released or stored. Heat rejection is handled through controlled seawater exchange to avoid localized thermal pollution.
Polluter pays principle	The design of the project takes into consideration waste collection, storage, and disposal. Any cost of environmental remediation or waste management is borne by the operating entity that is responsible for the habitat.
Precautionary principle	Where there is uncertainty about the effect of marine ecosystems, conservative discharge limits are used. Materials are chosen to reduce corrosion and leaching. Redundant monitoring systems are used to provide early warning of potential environmental hazards

0.2.3 SDG Progression

Positive and Negative Impacts of the Project

The underwater habitat project with a Direct Ocean Capture (DOC) system has clear positive environmental and societal impacts, including removing CO_2 from seawater, supporting climate action, and encouraging cleaner offshore/coastal industry. It promotes innovation in marine engineering, sensors, robotics, and sustainable underwater operations, which can create high-skill jobs and support STEM learning. However, challenges include potential local impacts on seawater chemistry (e.g., pH changes), risks to marine ecosystems if discharge is not controlled, and the need for reliable long-term maintenance in a harsh marine environment. High costs and specialized technology may limit access to a small number of organizations, and poor regulation could lead to unequal benefits or environmental harm. Strong monitoring, safe discharge limits, and clear governance will be essential to maximize benefits while reducing risks.

Progression of the United Nations Sustainable Development Goals (SDG)

Our underwater habitat project supports sustainability, innovation, and efficient use of resources, which links to several United Nations Sustainable Development Goals (SDGs). The table below shows the SDGs we address, along with the progress indicators and the means of implementation for each.

Table 6. *United Nations Sustainable Development Goals met by the underwater habitat project.*

SDG	SDG description and how it is attained	Progress Indicators	Means of Implementation
SDG 7: Affordable and Clean Energy 	Continuous underwater operation needs low-carbon power (esp. for pumps/processes/compute)	• kWh/day total • kWh per kg CO_2 captured (DOC) • % power from renewables • Peak load vs limit	• Renewable supply + storage strategy • High-efficiency pumps/drives • Load management / scheduling noncritical loads
SDG 9: Industry, Innovation and Infrastructure 	Our project is literally a new infrastructure concept (underwater habitat + integrated systems)	• System uptime • Maintainability (MTTR) • Modularity score (# swappable modules) • Deployment time	• Modular “skid” subsystems • Standard interfaces (power/data/pipes) • Remote monitoring + diagnostics
SDG 12: Responsible Consumption and Production	Closed loops + waste minimization, and reuse of	• Waste generated per day	• Material selection for long life

	by-products (where possible)	• % waste recycled/reused • Consumables usage rate	• Regenerable components (filters/sorbents where possible) • Planned maintenance + spares strategy
SDG 13: Climate Action	DOC aims at CO_2 removal/handling + reducing emissions footprint of operations	• Net CO_2 removed (kg/day) • Lifecycle CO_2 intensity estimate • Energy per tonne CO_2	• DOC capture + handling pathway (storage/use) • Energy optimisation to reduce footprint • Monitoring & verification plan
SDG 14: Life Below Water	Environmental protection is a main objective: don't harm marine ecosystems	• pH change near outlet • Temperature rise • Noise level • Chemical discharge = 0 events	• Neutralisation / controlled discharge design • Thermal management • Noise/vibration isolation • Environmental monitoring sensors

Figure 7. *UN SDG progression causal map.*

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0.2.4 Verification Matrix

Table 7. *Verification Matrix Table.*

WP	1	2	3	4	5
1					
2	**Interface:** Design depth, pressure rating, deployment constraints **Method:** A + I **Evidence:** Pressure calculations, GA drawings, requirement traceability matrix **Pass criteria:** Safety factors ≥ specified limits; all definition requirements mapped with no conflicts				
3	**Interface:** Operational modes, mission duration, peak power requirements **Method:** A **Evidence:** Power budget **Pass criteria:** Available power ≥ required power for all defined modes	**Interface:** Mounting space, mass distribution **Method:** A + I **Evidence:** layout drawings **Pass criteria:** No structural limits exceeded			
4	**Interface:** Crew number, mission duration, consumables requirements **Method:** A **Evidence:** Consumables sheet **Pass criteria:** Consumables sufficient with defined safety margins	**Interface:** Compartment volumes, pressure boundaries **Method:** A + I **Evidence:** Compartment layout drawings, volume calculations **Pass criteria:** All life support components fit without compromising pressure integrity or	**Interface:** Electrical power demand, backup power **Method:** I **Evidence:** Life support power demand summary **Pass criteria:** Life support demand included in total energy budget		
5	**Interface:** Functional requirements, autonomy level, communication needs **Method:** A	**Interface:** Docking structural loads on habitat **Method:** A **Evidence:** Docking load analysis and structural stress calculation	**Interface:** Power supply for charging **Method:** A **Evidence:** System-level energy balance analysis **Pass criteria:** Power	**Interface:** Charging power monitoring feedback **Method:** I **Evidence:** Functional flow diagram **Pass criteria:**	

	Evidence: Functional Flow Diagram, requirements table **Pass criteria:** Every system function mapped to at least one requirement	**Pass criteria:** Docking impact loads ≤ allowable structural stress with safety factor	delivered to charger calculations **Pass criteria:** Total charging load ≤ generation + storage capacity with margin	Charging subsystem measurement signals integrated into monitoring system	

Continuous mass, energy, and data balance tracking across all work packages supports verification. WP3 confirms that, in both normal and high operating conditions, energy generation, storage, and distribution stay in line with subsystem and mission module demand. WP5 checks interface integrity, autonomy logic, and data flow consistency across integrated operations. To guarantee overall system viability, including degraded and fault scenarios, structural mass margins are checked against life support and mission module requirements.

WP1: DEFINITION

1.1 Mission statement

This ocean habitat aims to support a long-lasting presence of humans and/or autonomous systems in the marine ecosystem while providing the means for ongoing ocean monitoring, scientific studies, and sustainable activities in the ocean. The habitat is intended to be used as a platform for experimental and operational testing, including data collection, and as such, it is designed to have the least number of adverse effects on the ocean and to be safe and effective in this difficult environment. Furthermore, it was designed to allow for future deployment in a scalable and flexible way, with an emphasis on modular architecture, reliability, and compatibility with existing support infrastructures.

1.2 Objective Tree(s)

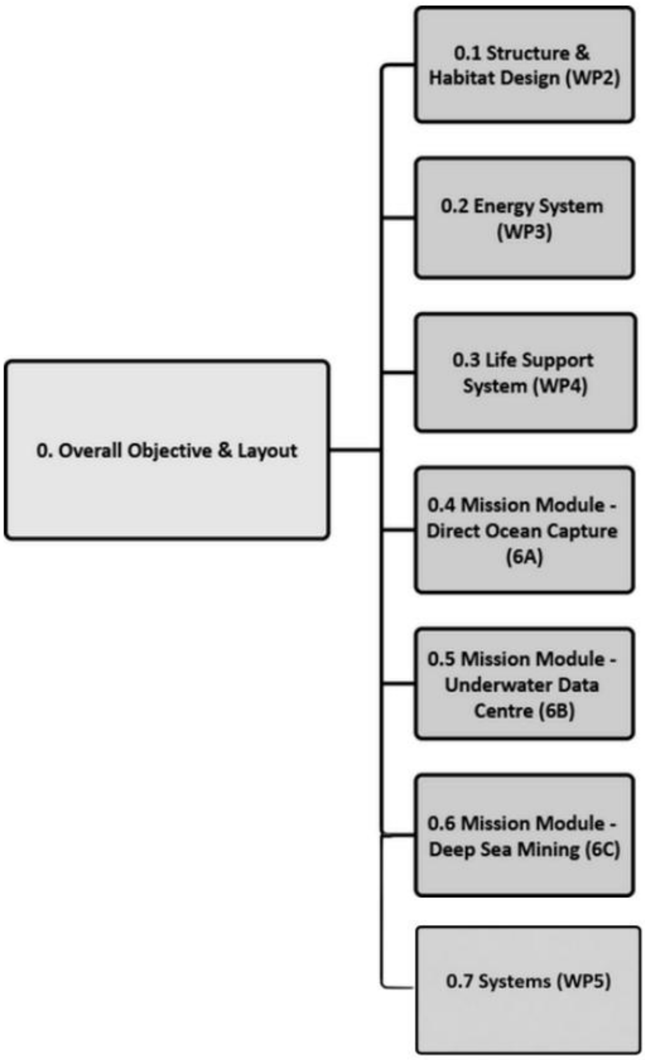


Figure 8. *Overall Objective Tree (all systems).*

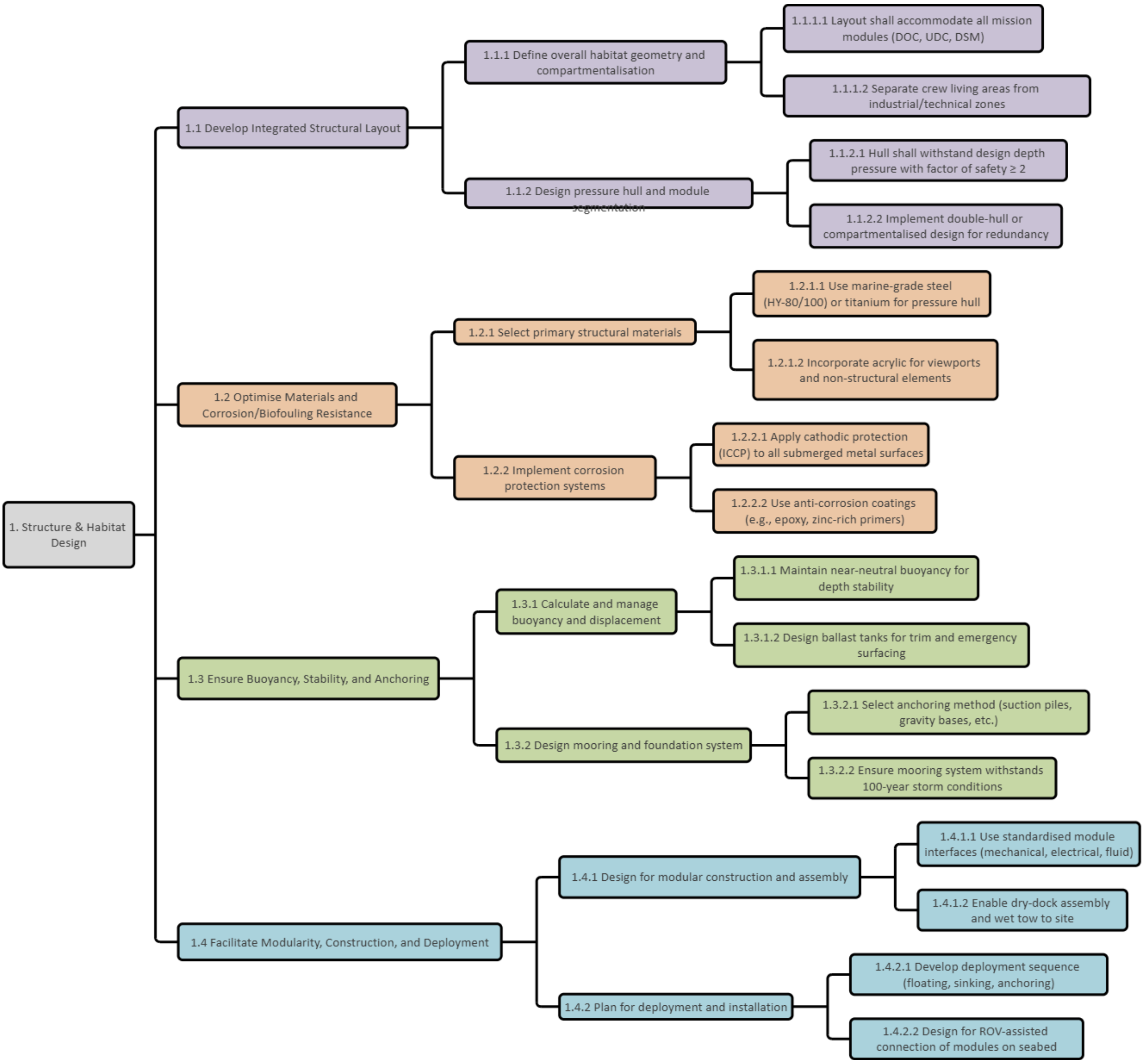


Figure 9. *Structure & Habitat Design Objective Tree.*

Figure 10. *Energy System Objective Tree.*

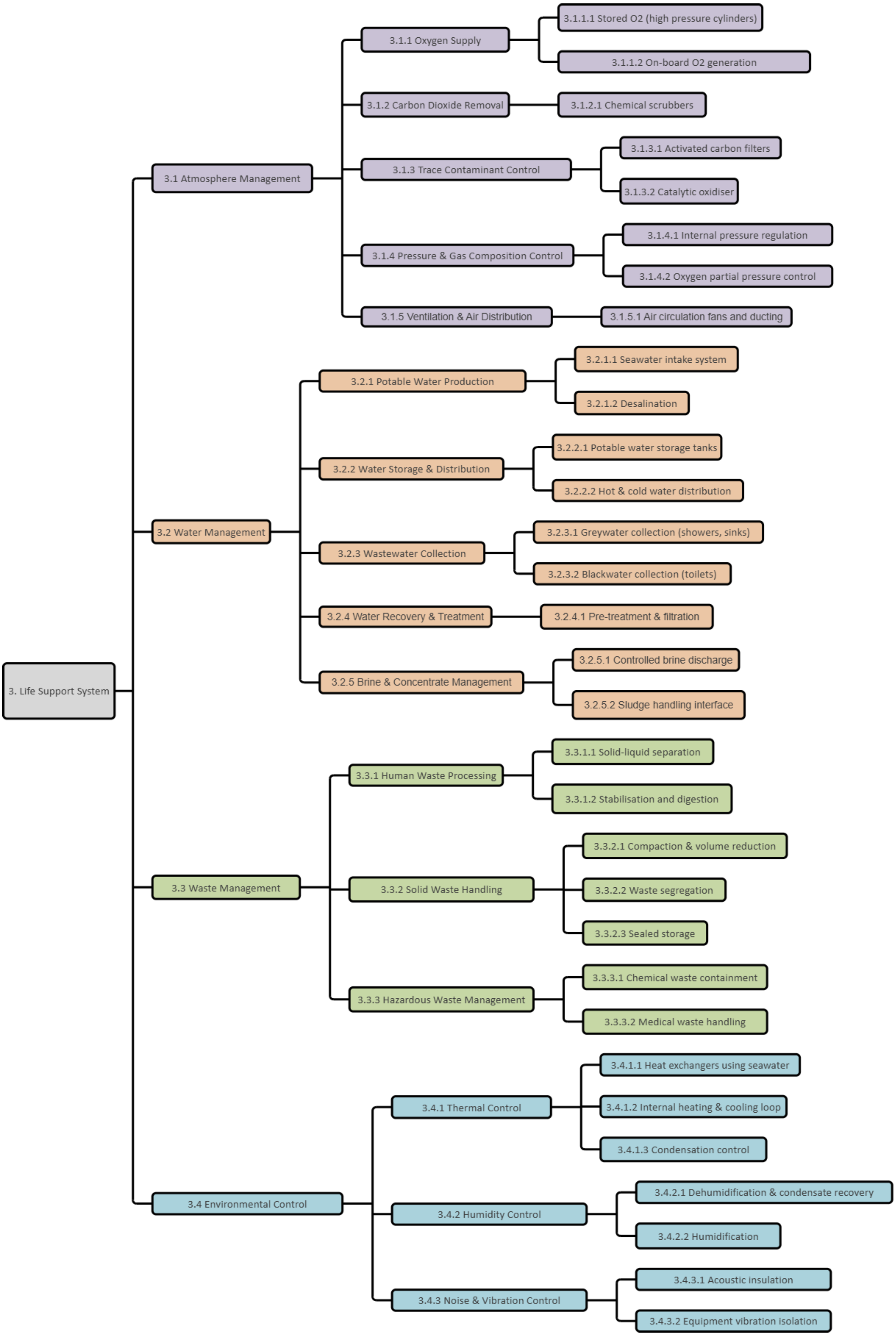


Figure 11. *Life Support System Objective Tree.*

Figure 12. *Direct Ocean Capture Objective Tree.*

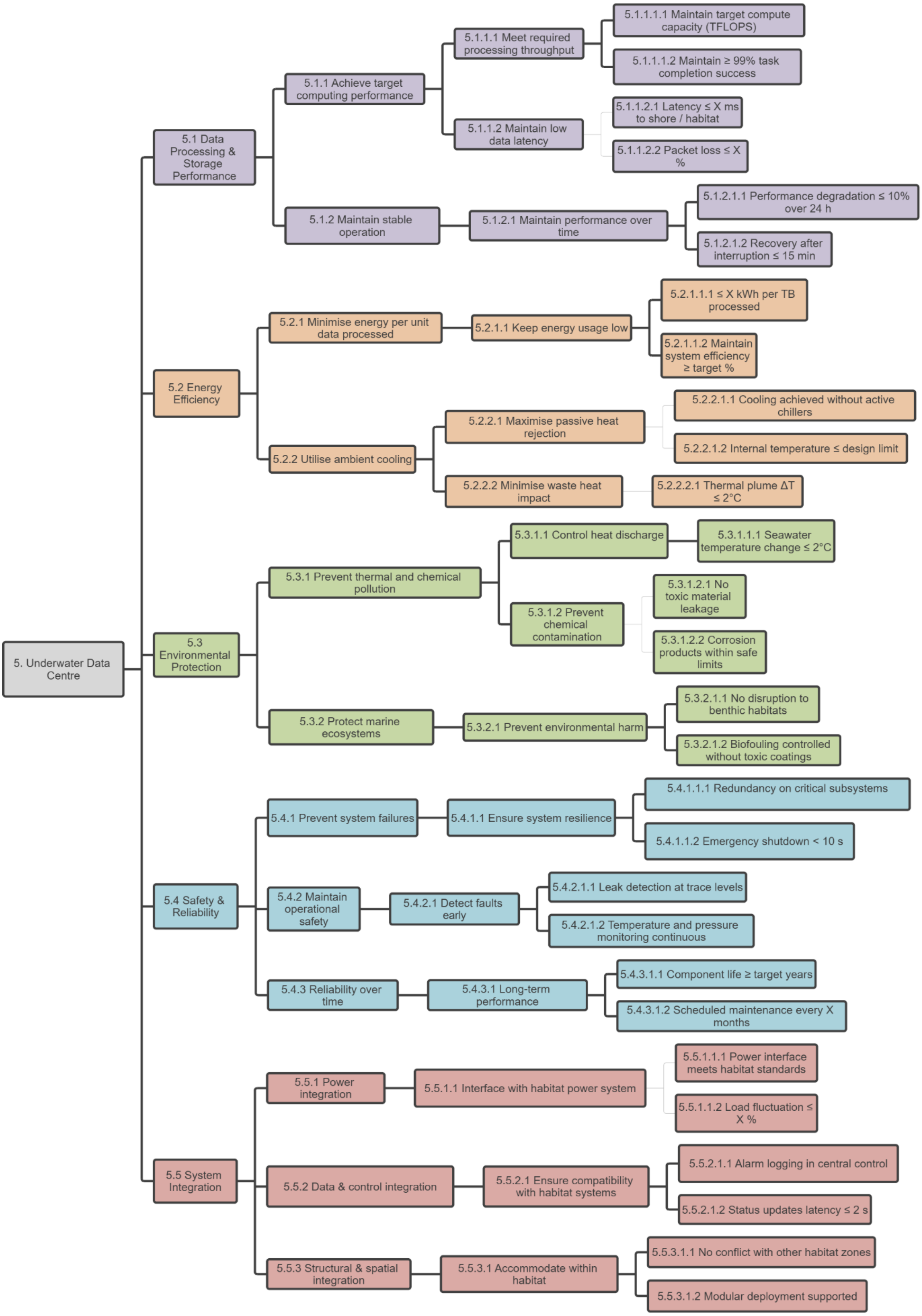


Figure 13. *Underwater Data Center Objective Tree.*

Figure 14. *Deep Sea Mining Objective Tree.*

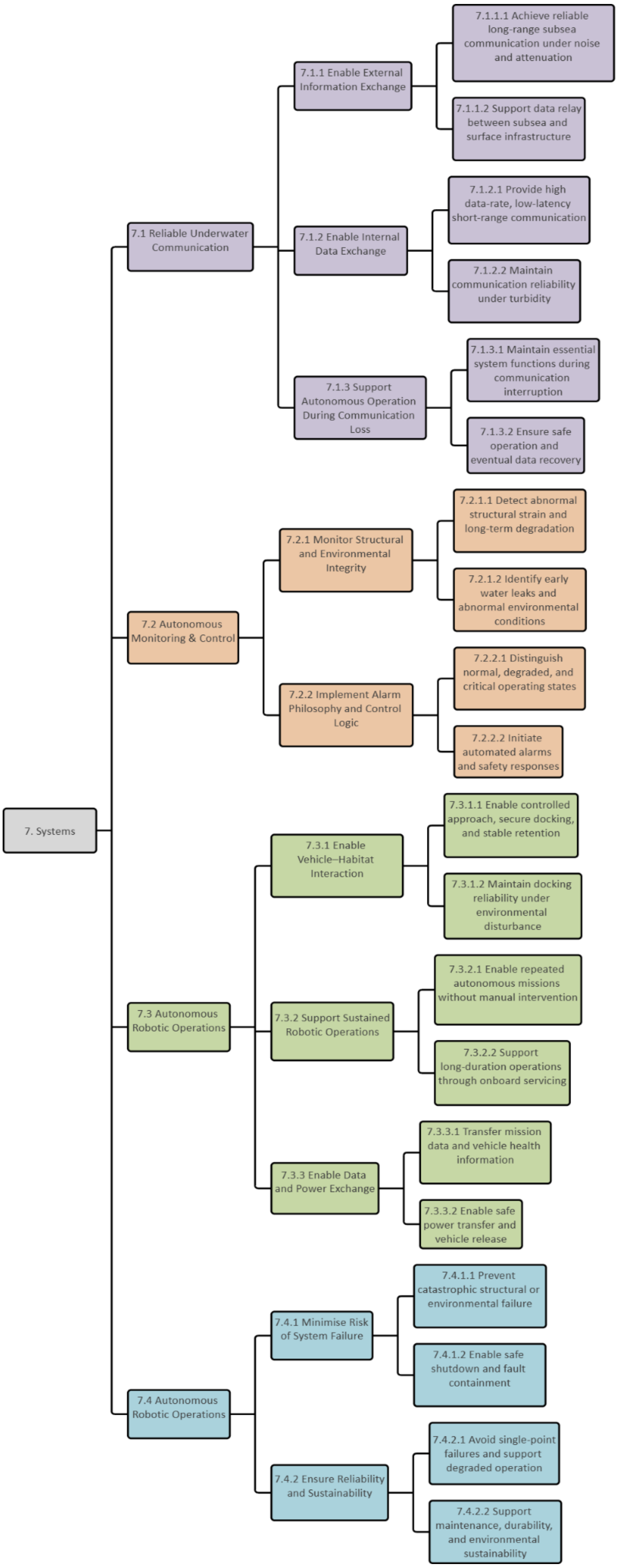
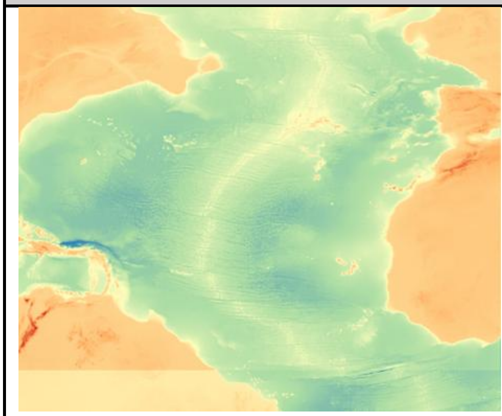
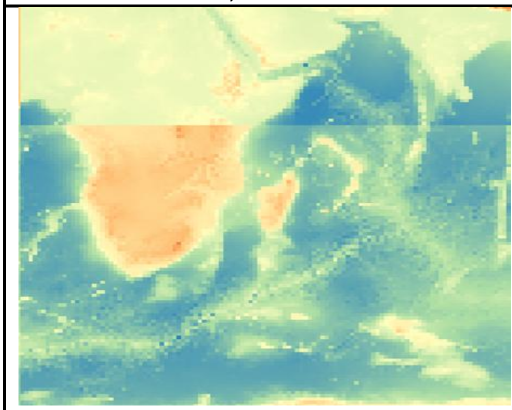
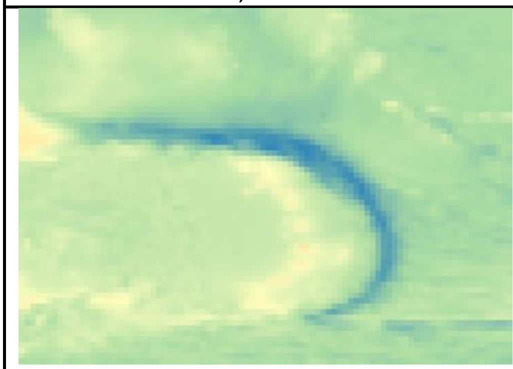


Figure 15. *Life Support System Objective Tree.*

1.3 Site Survey

Initial screening compiles global candidate locations using geospatial data, geological surveys, and risk assessments to balance deployment challenges with long-term operational resilience. Subsequent refinement applies weighted criteria, derived from precedents in subsea habitats like SEALAB and Aquarius, to narrow options, culminating in a final rationale grounded in Mediterranean deep-basin advantages.

Table 8. *Global Site Survey: Pros and Cons for Underwater Habitat Base.*

Location	Pros	Cons
 Coordinates: 34.6, -42.1	Continental shelf - Extensive existing subsea infrastructure and research presence - Well-studied oceanographic conditions - Access to major ports and technical expertise	- Exposure to severe weather systems (storms, hurricanes) - High wave energy complicates deployment and maintenance - Large seasonal temperature variation
 Coordinates: -32.1, 41.9	Varying depths - Deep ocean basins with thermally stable conditions - Access to shallower oceans nearby if needed - Low coastal congestion in some regions	- Remote location increases logistical complexity - Limited nearby infrastructure and ports - Strong ocean currents (Agulhas Current)
 Coordinates: 20.9, -73.4	Deep ocean basin - Warm water temperatures - Proximity to land-based infrastructure - Low coastal congestion in some regions	- Remote location increases logistical complexity - Limited nearby infrastructure and ports - Strong ocean currents (Agulhas Current)

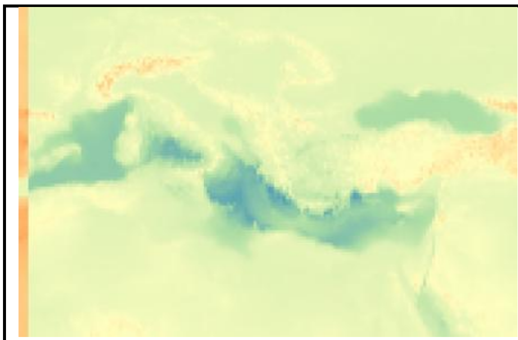
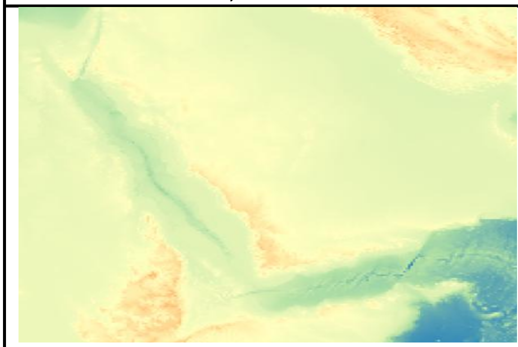
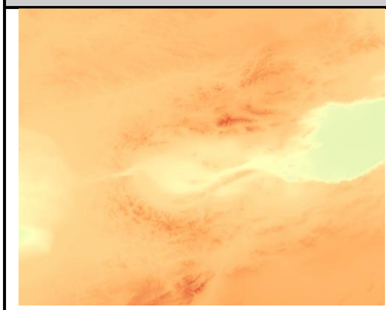
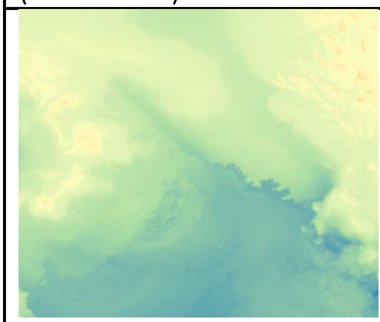
 Coordinates: 35.7, 18.2	Marginal sea - Relatively calm wave and weather conditions - Extensive maritime infrastructure and ports - Well documented to be rich in many minerals	- High salinity may increase corrosion risk - Ecologically sensitive regions requiring careful site selection - Multiple national jurisdictions
 Coordinates: 19.0, 39.4	Marginal sea - High salinity and warm temperatures enable stable deep-water conditions - Narrow basin allows proximity to land	- Extreme temperatures near the surface - Limited deep-water flat seabed areas - High ecological sensitivity (coral reefs)

Table 8 evaluated diverse global locations based on key criteria. This screening narrowed focus to the Mediterranean, where marginal seas offered superior proximity to ports, moderate weather, and varied depths suitable for habitat and mining integration, outperforming open-ocean sites hampered by extreme currents or remoteness.

Table 9. *Mediterranean Candidate Sites: Detailed Pros and Cons.*

Location	Pros	Cons
 Coordinates: 35.9, -3.2 (Alboran Sea)	- Moderate depths suitable for deployment - Good access to ports in Spain and North Africa - Existing subsea cables and maritime traffic routes	- Stronger currents due to Atlantic inflow Increased sediment transport - Higher seismic activity in some areas
 Coordinates: 38.1, 18.4 (Ionian Sea)	- Deep, thermally stable basins Relatively flat seabed areas - Lower surface wave energy - Reduced coastal ecological sensitivity offshore	- Greater deployment depth increases technical complexity - Some regional seismic activity

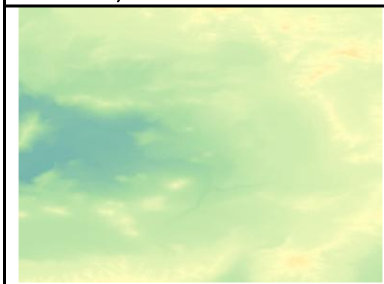
 Coordinates: 35.8, 24.9 (South of Crete)	- Access to deep water close to shore - Stable deep-sea temperature conditions - Lower maritime traffic offshore	- Higher salinity levels - Seismic activity near the Hellenic Arc - Limited nearby large ports
 Coordinates: 40.1, 11.9 (Tyrrhenian Sea)	- Moderate depths suitable for phased deployment - Excellent access to Italian ports and infrastructure - Sheltered compared to the open Mediterranean	- Volcanic and tectonic activity in certain zones - Higher maritime traffic density - More constrained seabed availability

Table 9 refined Mediterranean candidates by prioritizing moderate-to-deep basins with low wave energy, seismic stability, and logistical advantages over shallower or tectonically active zones. Comparative weighting favored sites like the Ionian Sea for their flat seabeds, thermal consistency, and reduced ecological sensitivity offshore.

Table 10. *Final Selection Rationale: Engineering Criteria for Ionian Sea Base.*

Category	Details and Engineering Rationale
Proposed Region	**Ionian Sea (south of Crete)** Selected for its deep, geologically stable basins and low surface energy conditions that support long-term operational stability.
	Reference: RAC/SPA. (n.d.). *Deep sea engineering in the Mediterranean*. Regional Activity Centre for Specially Protected Areas. Available at: https://www.rac-spa.org/sites/default/files/doc_medkey2/deep_sea_en.pdf
Habitat Depth	**Approximately 100-150 m** This depth provides a balance between environmental stability, accessibility for intervention, and manageable hydrostatic pressure levels for routine operations.
	Reference: ASCE. (2025, May 23). Leveling down – way down – with modular, reconfigurable undersea habitats. Civil Engineering Source. Available at: https://www.asce.org/publications-and-news/civil-engineering-source/article/2025/05/23/leveling-down-way-down-with-modular-recon
Mining Facility Depth	**Approximately 4,000 m at a nearby deep-sea basin** This location enables access to subsurface mineral resources while remaining logistically connected to the primary habitat base.
	Reference: NextGeo. (2024, August 2). Two thousand meters under the Mediterranean: The super-mapping of underwater mountains. Available at: https://www.nextgeo.eu/en/2024/08/02/two-thousand-meters-under-the-mediterranean-the-super-mapping-of-underwater-mountains-to-un

Bathymetric Characteristics	The Mediterranean exhibits a wide range of depths, from shallow continental shelves to deep basins exceeding 4,000 meters, allowing for the flexible deployment of both habitation and mining systems in proximity.
	Reference: RAC/SPA. (n.d.). *Deep sea engineering in the Mediterranean*. Regional Activity Centre for Specially Protected Areas. Available at: https://www.rac-spa.org/sites/default/files/doc_medkey2/deep_sea_en.pdf
Environmental Stability	Deep basins maintain stable temperature, salinity, and pressure profiles, ensuring structural reliability. Shallow regions provide easier access for maintenance and infrastructure integration.
	Reference: Rice, A., et al. (2023). First basin scale spatial–temporal characterization of underwater sound in the Mediterranean Sea. Scientific Reports, 13, Article 22345. Available at: https://doi.org/10.1038/s41598-023-49567-3
Material Considerations	High salinity requires corrosion-resistant materials (e.g., titanium alloys, duplex stainless steels) with protective coatings. Long-term material testing and monitoring should be scheduled.
	Reference: Melchers, R. E. (2003). Modeling of corrosion propagation in seawater. Corrosion Science, 45(3), 417-432. Available at: https://doi.org/10.1016/S0010-938X(02)00118-4
Natural Hazards	Seismic activity and strong localised currents occur in defined Mediterranean zones. Site selection should therefore prioritize low-seismic-risk and low-current areas. Biofouling countermeasures such as anti-fouling coatings are recommended.
	Reference: RAC/SPA. (2019). Proceedings of the Mediterranean symposium on marine vegetation. Available at: https://www.rac-spa.org/sites/default/files/symposium/proceedings_msmv_2019_final.pdf
Oceanographic and Weather Conditions	The Mediterranean exhibits moderate wave and current intensity compared to open oceans, minimising structural stresses during deployment and operation.
	Reference: Pinardi, N., et al. (2015). Mediterranean Sea surface wave climatology. Ocean Modelling, 88-89, 1-15. Available at: https://doi.org/10.1016/j.ocemod.2015.01.002
Infrastructure Accessibility	The region benefits from extensive maritime infrastructure, including ports, subsea cables, and research facilities, simplifying deployment, maintenance, and supply operations.
	Reference: European Commission. (2022). Mediterranean maritime infrastructure assessment. Directorate-General for Mobility and Transport. Available at: https://transport.ec.europa.eu
Logistical Benefits	Proximity to multiple coastal nations facilitates phased construction, maintenance, and emergency response operations with reduced logistical challenges.
	Reference: Poulain, P.-M., & Cushman-Roisin, B. (2001). Physical oceanography of the Mediterranean Sea. In Ocean circulation and climate (pp. 127-152). Academic Press.
Environmental Protection	The Mediterranean's biodiversity and ecological sensitivity require careful environmental planning, controlled installation processes, and continuous ecosystem monitoring to ensure sustainable operation.
	Reference: Danovaro, R., et al. (2010). Deep-sea biodiversity in the Mediterranean Sea. PLoS ONE, 5(8), e11831. Available at: https://doi.org/10.1371/journal.pone.0011831

1.4 CONOPs (Concept of Operations)

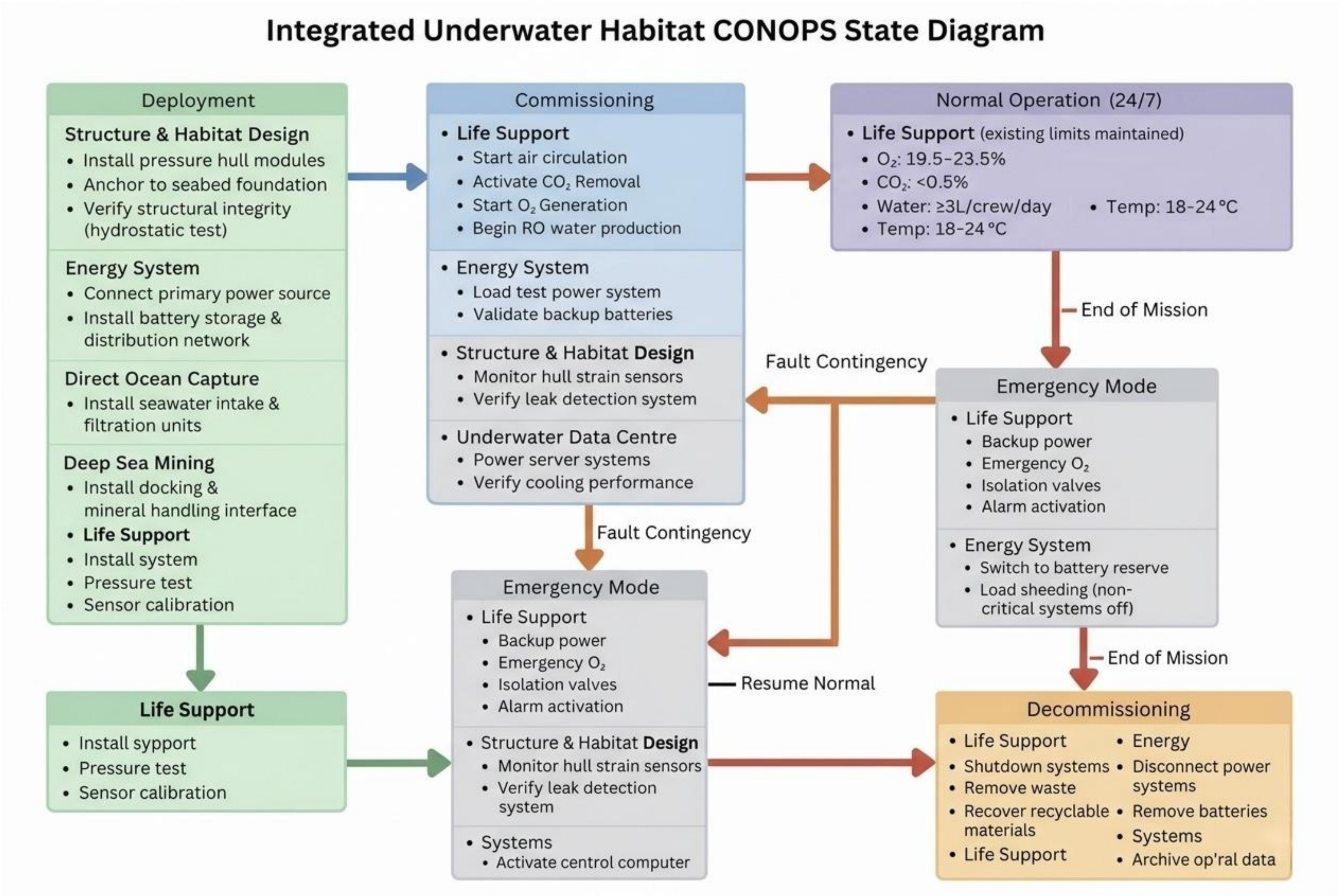


Figure 16. *CONOPS (Concept of Operation) illustration.*

1.5 Business Case

The proposed future ocean habitat addresses key stakeholder priorities across government, industry, and society, including climate mitigation, offshore digital infrastructure, and autonomous marine operations. The proposed solution combines the mission modules and is in line with the development of the Blue Economy. This business case uses a PEST analysis to determine the external political, economic, sociological, and technological factors that can affect the viability of the project.

Table 11. *Business case tabulated analysis.*

Political:
The new government policy encourages digital resilience, climate mitigation, and the Blue Economy by supporting offshore and subsea infrastructure. The OECD estimates that ocean based sectors could supply more than USD 3 trillion per year in 2030, leading to national investment in marine research platforms, autonomy systems, and offshore data infrastructure (OECD, 2016). Due to less land use conflict, interest in Direct Ocean Capture (DOC) and Marine Carbon Dioxide Removal (mCDR) as a politically feasible alternative to land based carbon removal techniques has increased concurrently (IPCC, 2022). The proposed habitat is designed with a 25+ year operational lifespan, aligning with long-term national decarbonisation strategies and enabling sustained climate infrastructure deployment offshore. However, regulatory uncertainty is still a significant risk, particularly with Deep Sea Mining (DSM), where the International Seabed Authority (ISA) has not yet set rules for using the resource and several states have asked for a precautionary moratorium. This has caused political delays and reputational risks for projects that are thought to be environmentally aggressive (ISA, 2023). **Actionable Insight:** Present DSM as modular or phased, and frame the habitat primarily as a platform for research, monitoring, and climate enabling. This will guarantee early adherence to ISA environmental baseline and monitoring requirements.
Economic:
From an economic perspective, the project benefits from converging growth in offshore digital infrastructure, climate services, and autonomous maritime systems. Microsoft's Project Natick trials have shown that passive seawater cooling can reduce cooling energy demand by up to 30–40% when compared to conventional land-based data centers, making underwater data centers (UDCs) a compelling value proposition (Microsoft, 2020). Furthermore, in developing carbon markets, where engineered removals fetch higher prices than nature-based offsets, long-lasting carbon removal solutions like DOC are becoming more and more valued (IEA, 2023). The proposed system is capable of extracting 1000 kg of CO_2 per day (≈365 tonnes per year), positioning it as a measurable and revenue-generating engineered carbon removal asset over its 25+ year lifecycle. However, pressure hull construction, corrosion-resistant materials (such as titanium and marine-grade steels), and redundancy requirements required for long term subsea operation continue to make high upfront capital expenditure a challenge. Without careful system integration, single purpose subsea facilities risk becoming economically uncompetitive over their lifecycle. **Actionable Insight:** Shared infrastructure between missions is essential for economic viability because it lowers marginal costs and increases return on investment, especially in the areas of energy generation, thermal management, and communications.
Sociological:
Social acceptability is increasingly shifting towards sustainable and climate positive infrastructure, especially those that address land use conflicts and promote long term sustainability. Ocean based research platforms for climate observation, carbon removal, and digital infrastructure are well

received by the public in terms of innovation and sustainability (UNESCO, 2021). However, deep-sea operations are still socially complex, with public concerns about biodiversity, ecosystem disruption, and transparency, especially in the context of DSM activities (WWF, 2022). Human factors are also important in acceptability and performance, as long-term underwater habitation poses psychological risks such as isolation, confinement, circadian rhythm disruption, and increased stress levels, which can have direct impacts on safety and performance if not mitigated (NASA, 2015). **Actionable Insight:** Integrate transparent environmental monitoring, data reporting, and human centric habitat design to enhance public acceptance, crew welfare, and overall operational integrity.
Technological:
The technological building blocks necessary for the proposed habitat are mostly mature, including the design of pressure vessels based on hoop stress and buckling, corrosion protection systems, autonomous underwater vehicles (AUVs), and subsea power and data transfer connectivity. The limitations of underwater communications are well understood, with acoustic solutions providing long range, low bandwidth communication and optical (blue laser) solutions providing high-bandwidth, short range communication (Akyildiz et al., 2015). Offshore renewable energy solutions such as tidal and wave, along with energy storage solutions, provide fully off grid capability as specified by the challenge. The main technological risk is not in any of these areas but in the system integration, specifically the energy, thermal, and data balance. **Actionable Insight:** Integration risk and associated assurance activities can be mitigated by early definition of functional flows and system mass and energy balance.

From the PEST analysis, it is evident that despite the challenges in regulation and capital, the project aligns strongly with stakeholder priorities in policy, industry, and sustainability. By means of modular integration, environmental transparency, and system level design assurance, the future ocean habitat offers a credible and sustainable engineering solution in the context of the Blue Economy.

WP2: STRUCTURE

2.1 Requirements Specification

Table 12. *Tabulated requirements specification for the Structure subsystem.*

Package	Requirement number	Description	Requirement Type	Rationale	Traceability to Objective Tree
Structure & Habitat Design	**2.1**	The habitat structure shall define integrated geometry or compartmentalisation accommodating all mission modules (DOC, UDC, DSM) with separated crew living areas from industrial zones, plus pressure hull/module segmentation withstanding design depth pressure at a factor of safety ≥2 using double-hull/compartmentalised redundancy	Functional Technical	Provides a complete structural layout ensuring pressure integrity, functional zoning, and mission module integration	1.1.1.1, 1.1.1.2, 1.1.2.1, 1.1.2.2
	2.2	The habitat structure shall use marine-grade steel (HY-80/100) or titanium for the pressure hull with acrylic viewports/non-structural elements, plus implement corrosion protection via ICCP on submerged surfaces and anti-corrosion coatings (epoxy/zinc-rich primers)	Technical Quality	Optimises materials selection and corrosion/biofouling resistance for long-term structural durability in a marine environment	1.2.1.1, 1.2.1.2, 1.2.2.1, 1.2.2.2
	2.3	The habitat structure shall maintain near-neutral buoyancy for depth stability via calculated displacement, include ballast tanks for trim/emergency surfacing, and implement a mooring/foundation system (suction piles/gravity bases) withstanding 100-year storm conditions	Functional Technical	Ensures buoyancy, stability, and anchoring reliability under operational and extreme environmental loads	1.3.1.1, 1.3.1.2, 1.3.2.1, 1.3.2.2
	2.4	The habitat structure shall incorporate standardised module interfaces (mechanical/electrical/fluid) for modular dry-dock construction/wet tow assembly, plus defined deployment sequence with ROV-assisted seabed module connections	Functional Technical	Facilitates efficient modular construction, deployment, and installation minimising offshore complexity	1.4.1.1, 1.4.1.2, 1.4.2.1, 1.4.2.2

2.2 Functional Flow

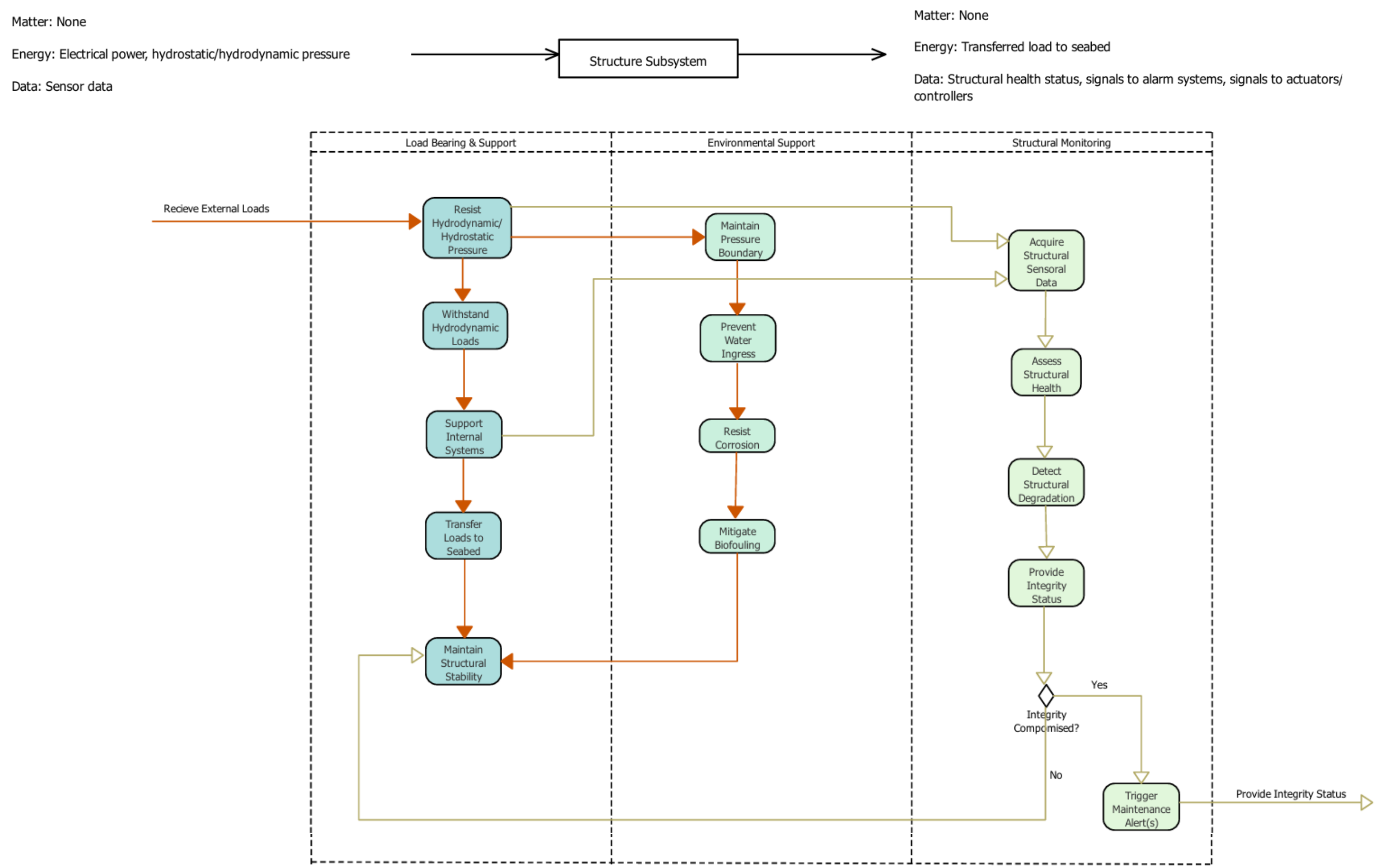
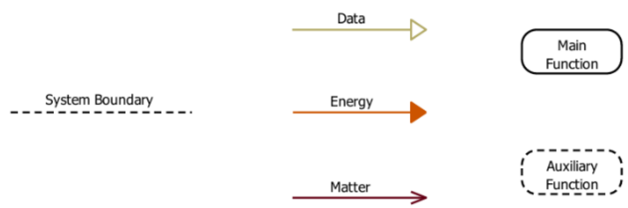


Figure 17. *Structure subsystem functional flow analysis.*



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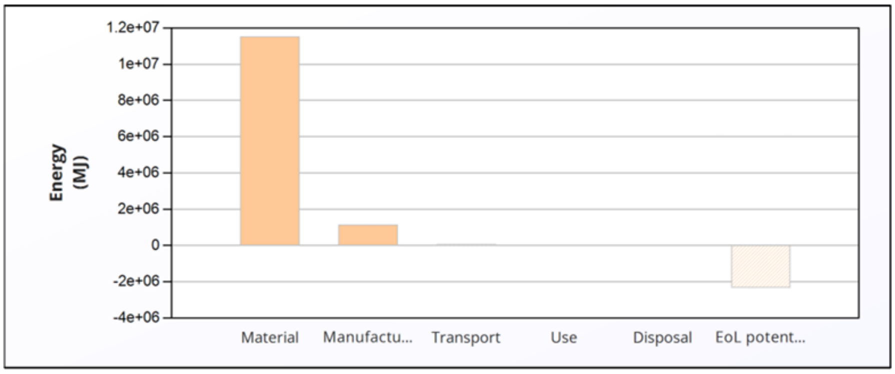
2.3 Material Selection and Life Cycle Analysis

Table 13. *Structure Life Cycle Analysis Results.*

<table>
<tr><th>Lifecycle Phase & Analysis</th><th>Figure/Table</th><th>Critical Insights & Conclusions</th></tr>
<tr><td>Lifecycle Energy Burden Overview</td><td>



<table>
<tr><th></th><th>Energy (MJ/year)</th></tr>
<tr><td>Equivalent annual environmental burden (averaged over 25 year product life):</td><td>5.09e+05</td></tr>
</table>

</td><td>The system's average annual energy burden is 5.09e+05 MJ/year. The chart visually confirms that the Material and Manufacturing phases dominate the lifecycle impact, while the Use phase is negligible. While the initial material cost is high, the product facilitates a circular economy. High recoverability at the 25-year mark helps mitigate the heavy initial energy investment.</td></tr>
<tr><td>Material Composition Impact Assessment</td><td>

<table>
<tr><th>Component</th><th>Material</th><th>Recycled content* (%)</th><th>Part mass (kg)</th><th>Qty.</th><th>Total mass (kg)</th><th>Energy (MJ)</th><th>%</th></tr>
<tr><td>Living Hub - Dome</td><td>Titanium, alpha-beta alloy, Ti-6Al-4V, annealed</td><td>Typical %</td><td>1.8e+04</td><td>1</td><td>1.8e+04</td><td>1e+07</td><td>90.5</td></tr>
<tr><td>Subsystem Pods</td><td>High strength low alloy steel, YS550, hot rolled</td><td>Typical %</td><td>4e+03</td><td>4</td><td>1.6e+04</td><td>3.3e+05</td><td>2.8</td></tr>
<tr><td>Tube Tunnels</td><td>PMMA (impact modified)</td><td>Virgin (0%)</td><td>1.2e+03</td><td>5</td><td>6e+03</td><td>7.6e+05</td><td>6.6</td></tr>
<tr><td>Total</td><td></td><td></td><td></td><td>10</td><td>4e+04</td><td>1.2e+07</td><td>100</td></tr>
</table>

</td><td>Titanium accounts for a staggering 90.5% of the total energy, while it makes up roughly 45% of the total mass

<u>Conclusion:</u>
Any minor reduction in Titanium mass or a shift to a higher recycled content would yield massive energy savings.</td></tr>
</table>

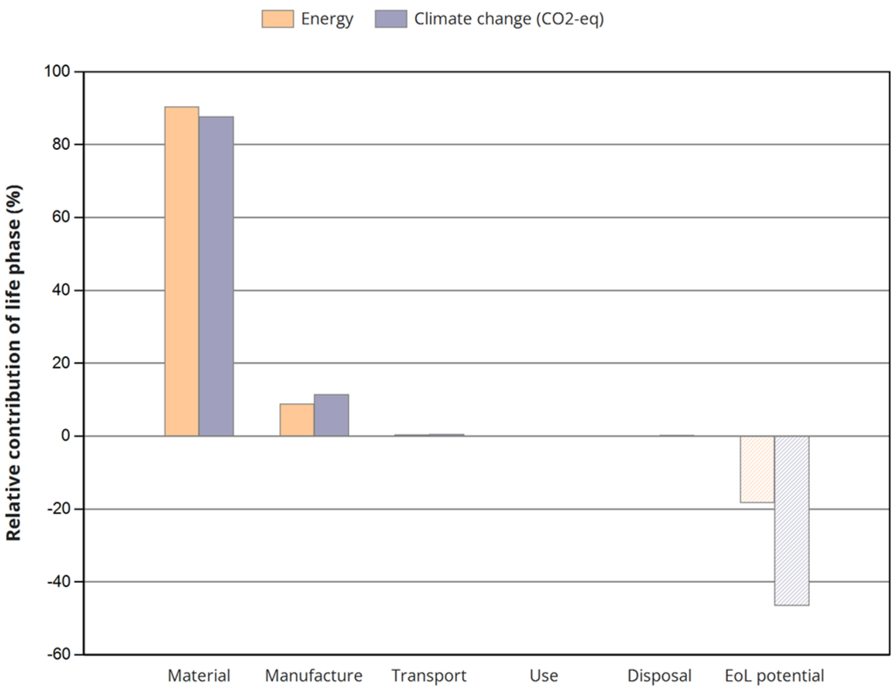
End-of-Life Process Energy & Circular Economy Potential

Component	End of life option	Energy (MJ)	%
Living Hub - Dome	Recycle	-2e+06	86.7
Subsystem Pods	Recycle	-1.9e+05	8.0
Tube Tunnels	Downcycle	-1.2e+05	5.3
Total		**-2.3e+06**	**100**

Design for Circularity is Highly Beneficial.

Key Finding:
By recycling the titanium at the end of its life, -2x 10^6 MJ of energy is reclaimed, which accounts for 86.7% of the total EoL energy credit.

Climate Change Impact (CO2-eq) Correlation



Component	Material	Recycled content* (%)	Part mass (kg)	Qty.	Total mass (kg)	Climate change (CO2-eq) (kg)	%
Living Hub - Dome	Titanium, alpha-beta alloy, Ti-6Al-4V, annealed	Typical %	1.8e+04	1	1.8e+04	5e+05	87.1
Subsystem Pods	High strength low alloy steel, YS550, hot rolled	Typical %	4e+03	4	1.6e+04	3.4e+04	6.0
Tube Tunnels	PMMA (impact modified)	Virgin (0%)	1.2e+03	5	6e+03	3.9e+04	6.9
Total				**10**	**4e+04**	**5.7e+05**	**100**

The environmental impact is strongly front-loaded. accounting for more than 90% of the energy and 87% of the climate change potential.

The sustainability of the project depends entirely on maximizing the 25-year life and ensuring high value recycling at the end of life to recoup significant energy and carbon credits.

2.4 Design

Calculations

The habitat features five spherical pressure domes that are interconnected by cylindrical tunnels, which facilitate compartmentalization with a seabed depth of 200 m. Spherical shapes are preferred for optimal resistance to external pressure. Four of the domes are made of HY-100 steel, while the remaining one is made of titanium. Steel tunnels are used to create pressure-retaining links with acrylic viewports. The design features a factor of safety of 3 and is stabilized on the seabed by a gravity foundation to resist buoyancy and storm loads.

Hydrostatic Pressure at Operating Depth

The habitat is located on the seabed at 200 m depth in the Ionian Sea. The governing load is external hydrostatic pressure.

$$P = \rho g h = (1026)(9.81)(200) = 2.01\ MPa$$

Applying FoS = 3:

$$P_d = 6.03\ MPa$$

All pressure-retaining elements (domes and tunnels) are therefore designed for 6.03 MPa external pressure.

Radius of Spherical Cap Domes

Each dome is modelled as a spherical shell under uniform external pressure. Spherical geometry is selected because it distributes compressive stresses efficiently and minimises bending stresses.

Geometry (per dome):
Base radius = 7.5 m
Height = 5 m
Equivalent sphere radius = 8.125 m

Hull Wall Thickness

For a spherical shell under external pressure:

$$t = \frac{PR}{2\sigma_{allow}}$$

A factor of safety of 3 is applied.

HY-100 Steel (4 domes)

HY-100 steel is selected for its high yield strength and marine suitability.

Yield strength ≈ 690 MPa

$$\sigma_{allow} = \frac{690}{3}$$
$$\sigma_{allow} = 230\ MPa$$

Spherical shell:

$$t = \frac{PR}{2\sigma} = \frac{(6.03)(8.125)}{2(230)} = 0.106\ m$$

Required thickness ≈ 110 mm

This provides adequate resistance against yielding and external compression.

Titanium Dome (1 Unit – Ti-6Al-4V)

One dome is constructed from titanium to provide improved corrosion resistance and weight reduction for a critical module zone.

Yield ≈ 830 MPa
Allowable stress:

$$\sigma_{allow} = 830/3 = 277\ MPa$$

Required thickness:

$$t = \frac{(6.03)(8.125)}{2(277)} = 0.088\ m$$

Required thickness ≈ 90 mm

Titanium reduces shell thickness and structural weight while maintaining strength.

Tunnel Design (Steel Cylinders)

The connecting tunnels are cylindrical steel pressure hulls with acrylic viewports integrated.

Radius = 3 m, Length = 7 m

Allowable stress (steel) = 230 MPa

$$t = \frac{PR}{\sigma} = \frac{(6.03)(3)}{230} = 0.079\ m$$

Required thickness

$$t = 85\ mm$$

The adopted value ensures adequate resistance against yielding and external buckling.

Displacement and Buoyancy

Total displaced volume determines buoyant force acting upward on the seabed structure.

$$V = \frac{\pi h^2(3R - h)}{3}$$

Per dome volume = 507 m^3

Total dome volume:

$$V_{domes} = 2535\, m^3$$

Total tunnel volume:

$$V_{tunnels} = 792\, m^3$$
$$V_{total} = 3327\, m^3$$

Buoyant Force

$$F_b = \rho g V$$
$$F_b = (1026)(9.81)(3327) = 33.5 MN$$

Structural Self-Weight

Steel Domes (4 Units)

Total weight:

$$\mathrm{W}_{\mathrm{steel}\ domes} = 8.64\ \mathrm{MN}$$

Titanium Dome (1 Unit)

$$\mathrm{W}_{\mathrm{Ti}\ dome} = 1.00\ \mathrm{MN}$$

Steel Tunnels (4 Units)

$$\mathrm{W}_{\mathrm{tunnels}} = 3.44\ \mathrm{MN}$$

Total Structural Weight

$$\mathrm{W}_{\mathrm{total}} = 13.08\ \mathrm{MN}$$

The use of titanium reduces overall structural weight compared to a fully steel configuration.

Seabed Stability and Ballast Requirement

Since the buoyant force exceeds structural self-weight, a gravity-based foundation is required to resist uplift.

$$\mathrm{F_{uplift}} = 33.5 - 13.08 = 20.42\ \mathrm{MN}$$

Required ballast mass:

$$\mathrm{m} = \frac{20.42 \times 10^6}{9.81}$$
$$\mathrm{m} \approx 2080\ \mathrm{tonnes}$$

A gravity base of approximately 2100 tonnes is therefore required to maintain seabed stability.

Environmental Lateral Loading

For a 100-year storm seabed current of 2 m/s:

$$\mathrm{F_D} = \frac{1}{2}\rho C_d A U^2$$

Typical Cd for submerged spherical structures is 0.6 (Frank M. White, 2011)

$$\mathrm{F_D} = 1.30\ \mathrm{MN}$$

The foundation system must therefore resist:

• 20.4 MN vertical uplift
• 1.3 MN lateral drag

Double-Hull Concept

To meet redundancy requirement:

Outer hull = full thickness (110 mm steel)
Inner safety liner = 20–30 mm secondary shell

If outer hull is breached, inner hull maintains compartment integrity.

Compartmentalization ensures:

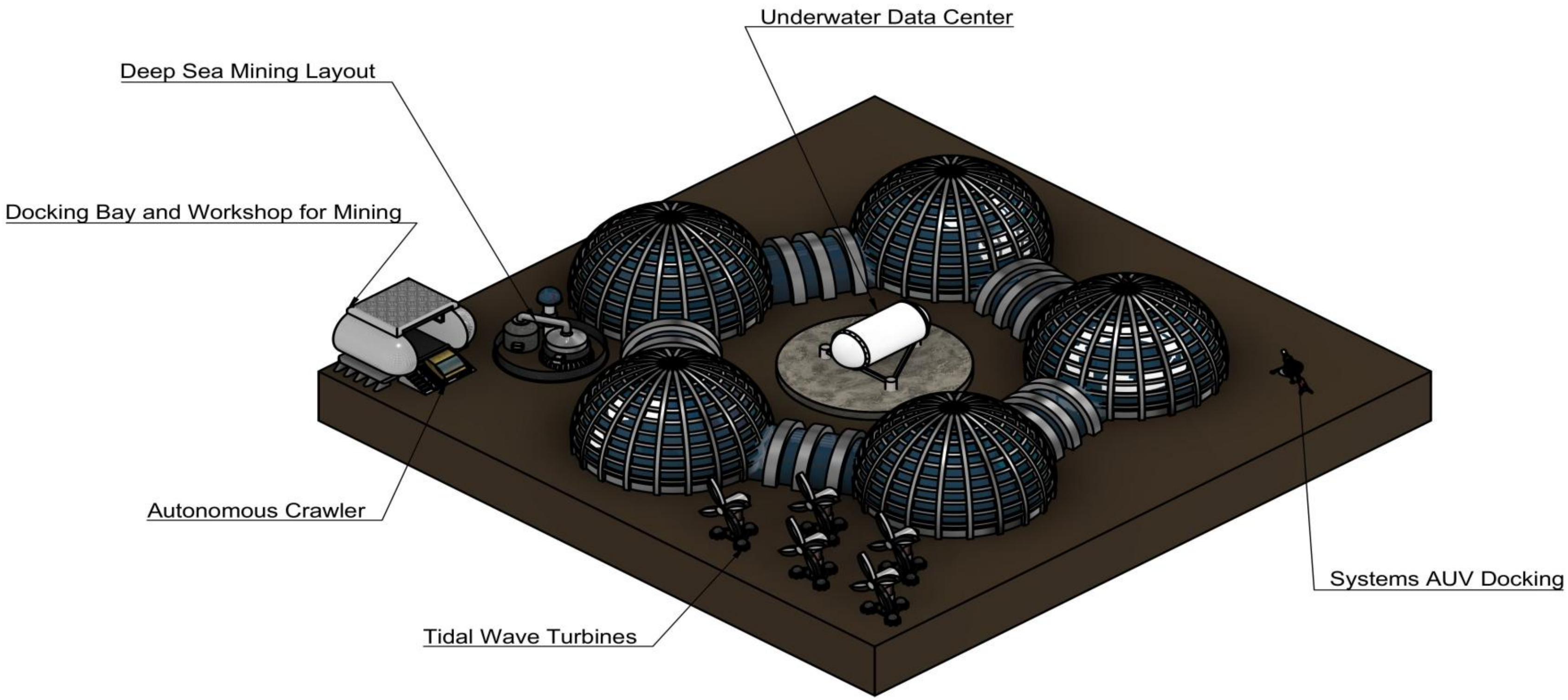
- Local flooding does not compromise entire habitat
- Independent pressure zones

Equipment

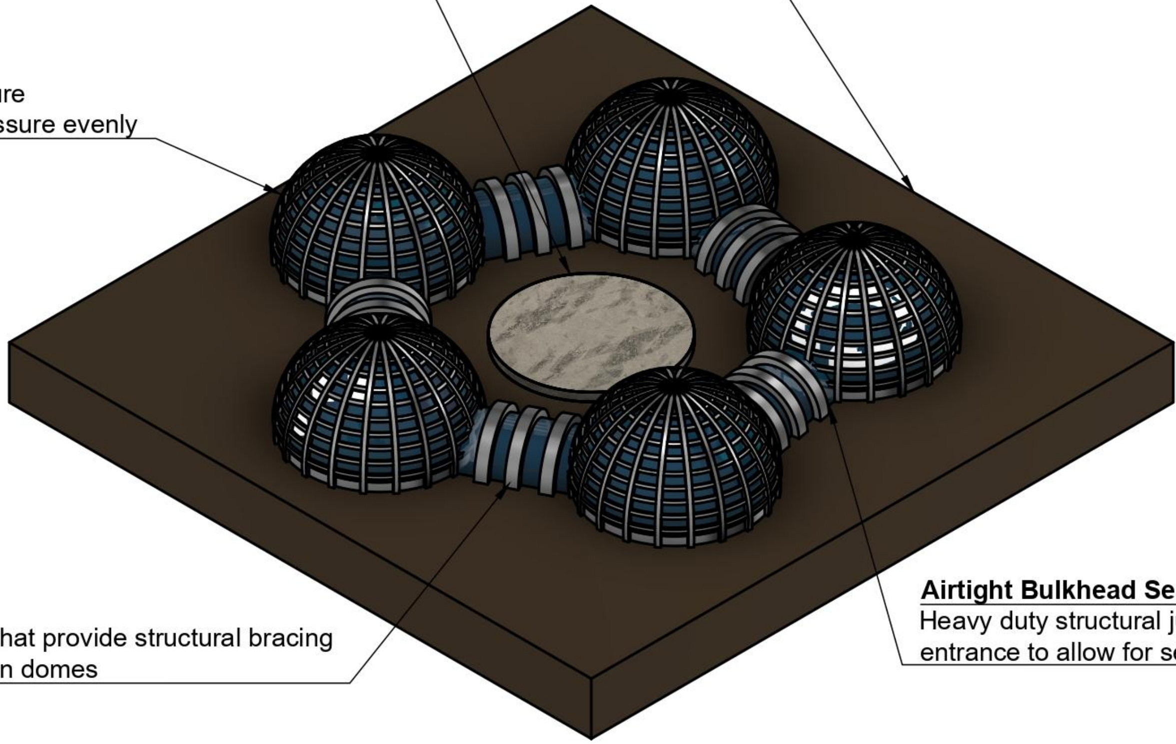
Table 14. *Equipment List for structure work package including product number.*

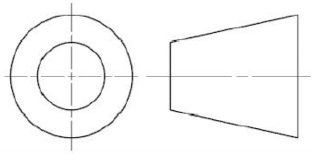
Subsystem	Component	Representative Part Number	Description
Pressure hull	Pressure hull shell sections	Custom: STR-HULL-01	Primary pressure boundary (designed for ~200 m depth; factor of safety per objective tree).
Pressure hull	Bulkheads / compartment walls	Custom: STR-BHD-01	Compartmentalisation for redundancy; separates mission/crew/technical zones.
Pressure hull	Hatch + sealing set	Trelleborg CR/CRW seal series (example)	Access hatch sealing to prevent water ingress; final size defined in detailed design.
Interfaces	Wet-mate connectors / penetrators	Teledyne ODI Nautilus wet-mate series (example)	Standardised module interfaces for power/data penetrations; ROV-friendly connectors.
Viewports	Acrylic viewport panel	Custom: STR-VIEW-01 (PVHO acrylic spec)	Acrylic viewport sized to pressure rating; referenced to PVHO-style acrylic practice.
Buoyancy / trim	Ballast tanks / trim system	Custom: STR-BALLAST-01	Ballast/trim for near-neutral buoyancy and emergency surfacing capability.
Anchoring	Anchor foundation (suction pile / gravity base)	Custom: STR-ANCHOR-01	Seabed anchoring concept sized to resist design current/storm loads.
Corrosion protection	ICCP system (controller + anodes)	Cathwell ICCP system (example)	Cathodic protection for submerged metal surfaces to reduce corrosion rates.
Corrosion protection	Anti-corrosion coating system	Jotun Jotamastic 87 (example)	Epoxy protective coating / primer system for long-life marine exposure.

Diagram



Features Positioning Layout		
Dept. School of Engineering University of Birmingham Dubai	**Integrated Design Project 2 (04 30338)** **Module Lead:** Dr. Archibong	**Project Title** Future Underwater Habiat
Created by Group D3 Academic year 2025/26	**Subsystem** Structure	**3rd Angle Projection**



Overall Habitat Structure Layout		
Dept. School of Engineering University of Birmingham Dubai	**Integrated Design Project 2 (04 30338)** **Module Lead:** Dr. Archibong	**Project Title** Future Underwater Habitat
Created by Group D3 Academic year 2025/26	**Subsystem** Structure	**3rd Angle Projection** 

Notes
Dome Dimension: 5 meter height (radius)
Tunnel Dimension: 3 meter height (radius)

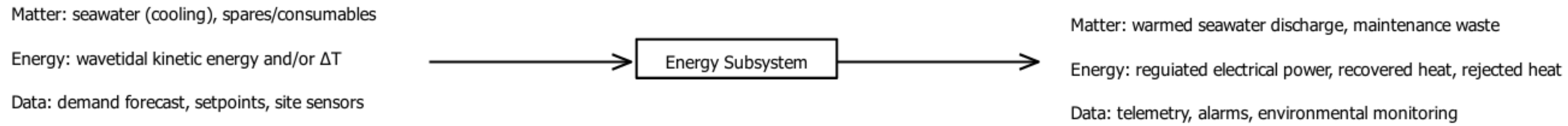
WP3: ENERGY

3.1 Requirements Specification

Table 15. *Tabulated requirements specification for the Energy subsystem.*

Package	Requirement number	Description	Requirement Type	Rationale	Traceability to Objective Tree
Energy System	**3.1**	Implement resilient hybrid marine energy generation using site-modelled tidal/wave devices (plus OTEC where thermal gradient permits), sized to meet Tier 1+Tier 2 peak demand with 20% margin, plus layered energy storage (UPS/BESS + hydrogen reserve) supplying Tier 1 critical loads ≥72 hours during low generation with black-start capability	Functional	Site-matched hybrid generation with multi-duration storage ensures continuous availability of life-support/mission loads during resource variability, storms, and faults	2.1.1.1, 2.1.1.2, 2.1.2.1, 2.1.2.2, 2.2.1.1, 2.2.1.2, 2.2.2.1, 2.2.2.2
	3.2	Design DC microgrid power distribution with defined bus voltages, conversion stages, and protection coordination maintaining voltage ±5%, enabling automated tiered load shedding protecting Tier 1 critical loads, plus reject waste heat via seawater heat exchangers recovering useful heat for hot water/desalination while maintaining equipment spaces ≤35°C under max load	Technical	Stable DC distribution with thermal management ensures power quality, fault isolation, equipment protection, and efficiency using ocean cold sink in enclosed habitat	2.3.1.1, 2.3.1.2, 2.3.2.1, 2.3.2.2, 2.4.1.1, 2.4.1.2, 2.4.2.1, 2.4.2.2
	3.3	Provide safety and resilience via battery management systems (BMS/thermal/fire suppression), hydrogen leak detection/ventilation, N+1 redundancy for critical components, emergency shutdowns, and modular maintenance addressing corrosion/biofouling	Quality	Integrated safety controls and redundancy mitigate high-consequence energy faults while enabling maintainability in harsh marine environment	2.5.1.1, 2.5.1.2, 2.5.2.1, 2.5.2.2
	3.4	Integrate sustainability through Granta EduPack EcoAudit for major components targeting ≥80% mass recyclability, plus minimise marine impacts (noise/EMF/entanglement) via monitoring and adaptive operation	Quality	Lifecycle assessment and environmental safeguards reduce embodied impacts, protect marine ecosystems, and enhance long-term project viability	2.6.1.1, 2.6.1.2, 2.6.2.1, 2.6.2.2

3.2 Functional Flow



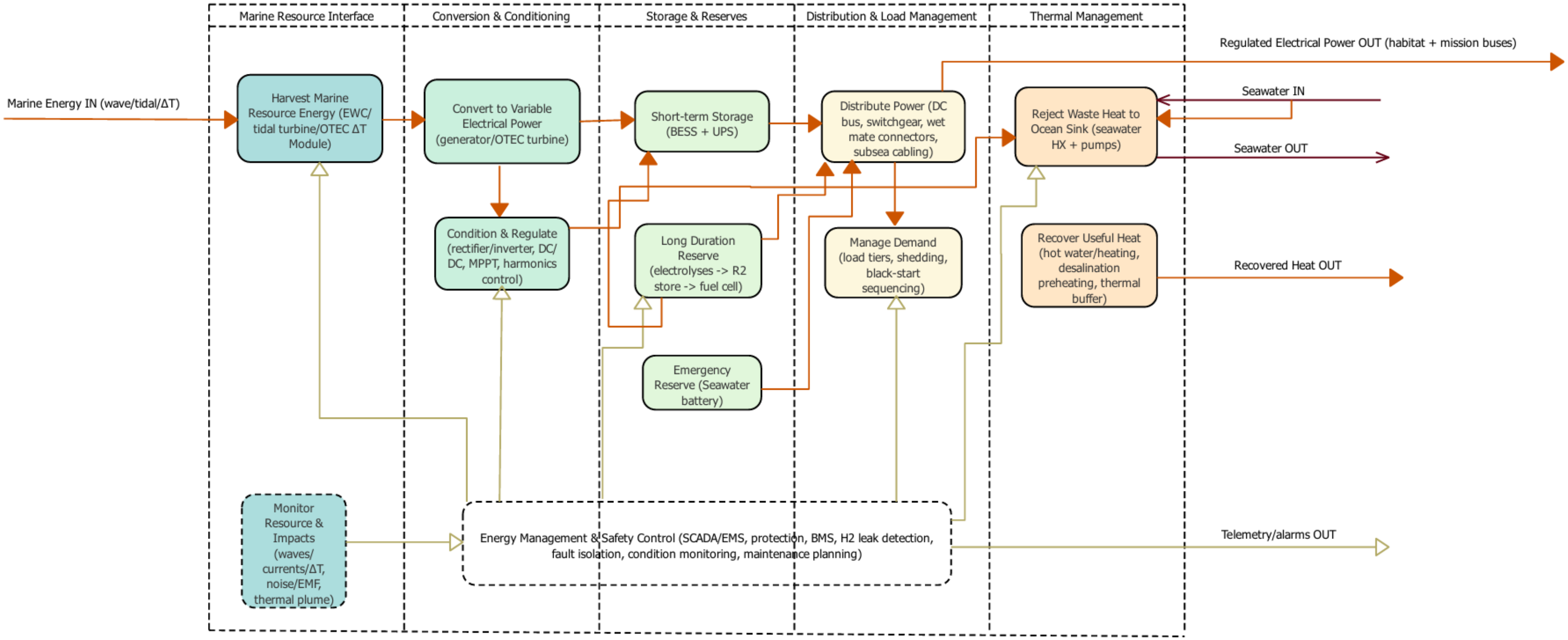
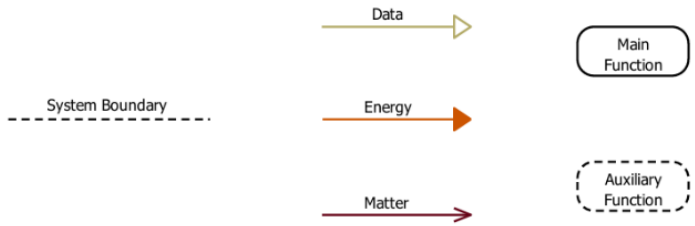


Figure 18. *Energy subsystem functional flow analysis.*



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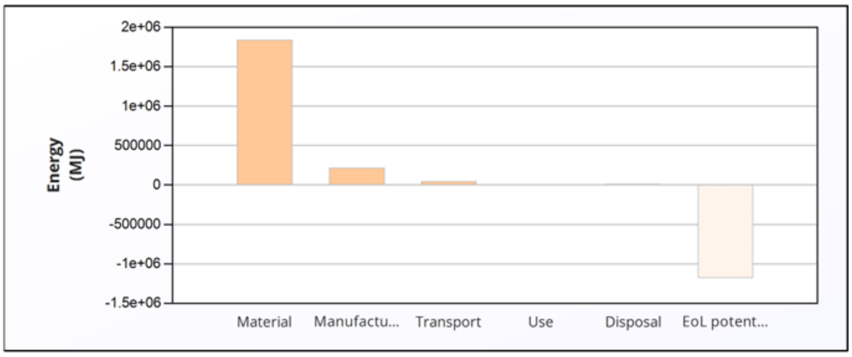
3.3 Material Selection and Life Cycle Analysis

Table 16. *Energy Systems Life Cycle Analysis Results.*

<table>
<tr><th>Lifecycle Phase & Analysis</th><th>Figure/Table</th><th>Critical Insights & Conclusions</th></tr>
<tr><td>Lifecycle Energy Burden Overview</td><td>


<table>
<tr><th></th><th>Energy (MJ/year)</th></tr>
<tr><td>Equivalent annual environmental burden (averaged over 25 year product life):</td><td>8.44e+04</td></tr>
</table>
</td><td>The system's average annual energy burden is 8.44e+04 MJ/year. The chart visually confirms that the Material and Manufacturing phases dominate the lifecycle impact, while the Use phase is negligible. This validates the device's purpose as a zero-emission energy generator.</td></tr>
<tr><td>Material Composition Impact Assessment</td><td>
<table>
<tr><th>Component</th><th>Material</th><th>Recycled content* (%)</th><th>Part mass (kg)</th><th>Qty.</th><th>Total mass (kg)</th><th>Energy (MJ)</th><th>%</th></tr>
<tr><td>Tidal turbine nacelle + structural frame</td><td>Stainless steel, duplex, AISI 2205, annealed</td><td>30.0%</td><td>5.4e+03</td><td>1</td><td>5.4e+03</td><td>3.1e+05</td><td>16.9</td></tr>
<tr><td>Tidal turbine blades</td><td>Epoxy SMC (65% long glass fiber)</td><td>Virgin (0%)</td><td>45</td><td>3</td><td>1.4e+02</td><td>9.2e+03</td><td>0.5</td></tr>
<tr><td>Wave energy float / buoy hull</td><td>PE-HD (high molecular weight)</td><td>10.0%</td><td>2.7e+03</td><td>1</td><td>2.7e+03</td><td>2e+05</td><td>11.0</td></tr>
<tr><td>WEC PTO/ generator skid</td><td>Stainless steel, austenitic, AISI 316L, annealed</td><td>30.0%</td><td>5.4e+02</td><td>1</td><td>5.4e+02</td><td>3.7e+04</td><td>2.0</td></tr>
<tr><td>Seawater heat exchanger</td><td>Titanium, commercial purity, Grade 2</td><td>10.0%</td><td>9.1e+02</td><td>1</td><td>9.1e+02</td><td>5.1e+05</td><td>28.0</td></tr>
<tr><td>Seawater intake/ outfall pipe section</td><td>PE-HD (high molecular weight)</td><td>10.0%</td><td>3.6e+03</td><td>1</td><td>3.6e+03</td><td>2.7e+05</td><td>14.7</td></tr>
<tr><td>Battery energy storage module</td><td>Li-ion, FLP</td><td>Virgin (0%)</td><td>1.8e+03</td><td>1</td><td>1.8e+03</td><td>3.1e+05</td><td>16.7</td></tr>
<tr><td>Hydrogen storage cylinder</td><td>Furan/natural fiber, composite</td><td>Virgin (0%)</td><td>91</td><td>6</td><td>5.4e+02</td><td>2.2e+04</td><td>1.2</td></tr>
<tr><td>Power electronics + switchgear cabinet</td><td>Aluminum, 5083, H321</td><td>40.0%</td><td>6.8e+02</td><td>1</td><td>6.8e+02</td><td>8.9e+04</td><td>4.8</td></tr>
<tr><td>Subsea DC cable - conductor</td><td>Copper, C10100, hard (oxygen-free electronic h.c. copper)</td><td>30.0%</td><td>1.1e+03</td><td>1</td><td>1.1e+03</td><td>6.6e+04</td><td>3.6</td></tr>
<tr><td>Subsea DC cable- insulation/ jacket</td><td>PE (cross-linked, wire and cable grade)</td><td>Reused part</td><td>2.7e+02</td><td>1</td><td>2.7e+02</td><td>0</td><td>0.0</td></tr>
<tr><td>Emergency reserve battery casing</td><td>PP (homopolymer, 10% glass fiber)</td><td>20.0%</td><td>1.4e+02</td><td>1</td><td>1.4e+02</td><td>9.1e+03</td><td>0.5</td></tr>
<tr><td>Total</td><td></td><td></td><td></td><td>19</td><td>1.8e+04</td><td>1.8e+06</td><td>100</td></tr>
</table>
</td><td>Three components account for >60% of material energy:

1. Seawater Heat Exchanger (Titanium): 28.0%
2. Tidal Turbine Nacelle (Steel): 16.9%
3. Battery Storage (Li-ion): 16.7%

<u>Conclusion:</u>
Material selection for these three items is the single most impactful engineering decision.</td></tr>
</table>

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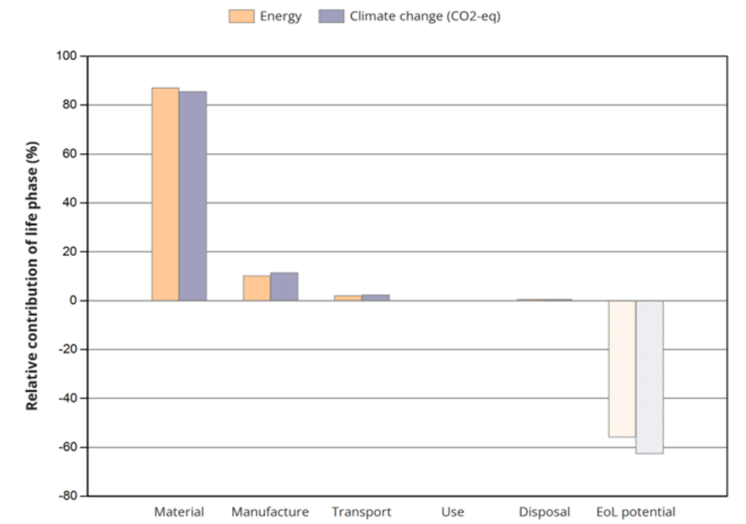
End-of-Life Process Energy & Circular Economy Potential

Component	End of life option	Energy (MJ)	%
Tidal turbine nacelle + structural frame	Recycle	-1.9e+05	16.1
Tidal turbine blades	Landfill	0	0.0
Wave energy float / buoy hull	Recycle	-1.7e+05	14.7
WEC PTO/ generator skid	Recycle	-2.3e+04	1.9
Seawater heat exchanger	Recycle	-1.5e+05	13.0
Seawater intake/ outfall pipe section	Recycle	-2.3e+05	19.6
Battery energy storage module	Reuse	-3.1e+05	26.1
Hydrogen storage cylinder	Downcycle	-2.7e+03	0.2
Power electronics + switchgear cabinet	Recycle	-5.3e+04	4.5
Subsea DC cable - conductor	Recycle	-4.6e+04	3.9
Subsea DC cable- insulation/ jacket	Landfill	0	0.0
Emergency reserve battery casing	Landfill	0	0.0
Total		-1.2e+06	100

Design for Circularity is Highly Beneficial. The "EOL Potential" table shows massive negative values.

Key Finding:
The "Reuse" of the Battery Energy Storage Module provides the single largest environmental credit (-26.1% of total EOL potential), far outweighing the impacts of the recycling process itself.

Climate Change Impact (CO2-eq) Correlation



Component	Material	Recycled content* (%)	Part mass (kg)	Qty.	Total mass (kg)	Climate change (CO2-eq) (kg)	%
Tidal turbine nacelle + structural frame	Stainless steel, duplex, AISI 2205, annealed	30.0%	5.4e+03	1	5.4e+03	2.8e+04	25.5
Tidal turbine blades	Epoxy SMC (65% long glass fiber)	Virgin (0%)	45	3	1.4e+02	5.1e+02	0.5
Wave energy float / buoy hull	PE-HD (high molecular weight)	10.0%	2.7e+03	1	2.7e+03	5.9e+03	5.4
WEC PTO/ generator skid	Stainless steel, austenitic, AISI 316L, annealed	30.0%	5.4e+02	1	5.4e+02	3.2e+03	2.9
Seawater heat exchanger	Titanium, commercial purity, Grade 2	10.0%	9.1e+02	1	9.1e+02	3e+04	27.3
Seawater intake/ outfall pipe section	PE-HD (high molecular weight)	10.0%	3.6e+03	1	3.6e+03	7.8e+03	7.2
Battery energy storage module	Li-ion, FLP	Virgin (0%)	1.8e+03	1	1.8e+03	2.1e+04	19.6
Hydrogen storage cylinder	Furan/natural fiber, composite	Virgin (0%)	91	6	5.4e+02	1.7e+03	1.5
Power electronics + switchgear cabinet	Aluminum, 5083, H321	40.0%	6.8e+02	1	6.8e+02	6.2e+03	5.7
Subsea DC cable - conductor	Copper, C10100, hard (oxygen-free electronic h.c. copper)	30.0%	1.1e+03	1	1.1e+03	4.4e+03	4.1
Subsea DC cable- insulation/ jacket	PE (cross-linked, wire and cable grade)	Reused part	2.7e+02	1	2.7e+02	0	0.0
Emergency reserve battery casing	PP (homopolymer, 10% glass fiber)	20.0%	1.4e+02	1	1.4e+02	3.2e+02	0.3
Total				19	1.8e+04	1.1e+05	100

The CO2-eq analysis correlates strongly with energy (MJ) data, reinforcing the material hotspots. The Seawater Heat Exchanger (Titanium) is the largest contributor, accounting for 27.3% of the climate change impact, followed by the Nacelle (25.5%) and Battery (19.6%).

3.4 Design

Calculations

The following preliminary calculations translate the Energy objective tree into an initial, quantified design. Values shown are concept-level assumptions and must be refined once the whole-facility load list (from all work packages) and the site survey (wave climate, tidal current, bathymetry and environmental constraints) are finalised.

3.4.1 Power budget and load tiers

A tiered load structure is used to align with objective tree node 3.3.2 (load prioritisation and shedding). Tier 1 loads are life-critical and must be maintained under all operating modes; Tier 2 loads are mission-support; Tier 3 loads are discretionary and are automatically shed first during low-generation events.

Table 17. *Power Budget Tiers.*

Load tier	Example loads	Power (kW)	Duty	Average (kW)
Tier 1 (critical)	Life support, control/SCADA, comms, emergency lighting	20	24/7	20
Tier 2 (support)	Water treatment, galley, science payload baseline	15	60%	9
Tier 3 (discretionary)	Comfort loads, non-essential charging, non-critical HVAC margin	15	40%	6

Total estimated average electrical demand:

$$P_{avg} = \Sigma(P_i * duty_i) = 20 + 9 + 6 = 35kW$$

A peak demand allowance is included for start-up surges, transient loads and growth margin:

$$P_{peak} \approx 80kW$$

3.4.2 Marine generation sizing (wave + tidal hybrid)

To address objective tree node 3.1 (generation selection and sizing), the baseline concept adopts a hybrid of tidal stream turbines (predictable resource) and wave energy converters (complementary variability). Wave energy yield estimates are sensitive to modelling assumptions; therefore, conservative capacity factors are used for first-pass sizing (Sheng, 2019). Tidal turbine reliability and maintenance access drive the selection of modular deployment and retrieval strategies (Walker & Thies, 2021).

Location physics check (first-pass): capacity factors are supported by simple resource equations and representative site parameters.

Wave resource (deep-water power flux):

$$P_{wave} = \frac{pg^2}{64\pi} H_s^2 T_e$$

Using representative conditions, H_s= 2.0 m and T_e= 7 s gives P_{wave}≈ 13.7 kW/m. A point-absorber buoy with an effective capture width of 4-6 m therefore accesses order 50-80 kW of wave power under these conditions; after PTO and electrical conversion losses, this supports a ~50 kW-class WEC with a conservative capacity factor of ~0.2-0.3 for early sizing.

Tidal stream resource (turbine power):

$$P_{tidal} = \frac{1}{2} \rho A C_p v^3$$

For example, with seawater density ρ ≈ 1025 kg/m³, rotor diameter D = 10 m (A ≈ 78.5 m²), C_p= 0.40, and peak spring-tide speed v = 2.0 m/s, the extractable power is P_{tidal} ≈ 129 kW (before drivetrain efficiency). This is consistent with selecting a ~100 kW-class tidal turbine module for the concept stage.

OTEC feasibility screen (temperature gradient):

$$\eta_{Carnot} = 1 - \frac{T_c}{T_h}$$

For a typical warm-surface / cold-deepwater case (T_h ≈ 298 K, T_c ≈ 278 K), η_{Carnot} ≈ 6.7% (theoretical upper bound). Real OTEC systems achieve only a fraction of this due to pumping and heat-exchanger losses; therefore, OTEC remains strongly site-dependent and is retained as an optional baseload upgrade pending WP1 site survey results (Aresti et al., 2023).

Assume the following concept-level capacity factors (to be replaced with site survey values):

$$CF\_tidal = 0.35$$

$$CF\_tidal = 0.25$$

Installed capacity is selected to satisfy average demand with a margin:

$$P_{avg} \leq \left(P_{tidal,inst} * CF_{tidal}\right) + \left(P_{wave,inst} * CF_{wave}\right)$$

Choose a candidate configuration:

Table 18. *Choose a candidate configuration.*

Asset	Installed (kW)	Capacity factor	Average (kW)
Tidal turbine (1 unit)	100	0.35	35
Wave buoys (2 units)	100	0.25	25

Total average generation:

$P_{gen,avg} = 35 + 25 = 60kW$

Note: OTEC can provide a steady baseload and shared cooling infrastructure, but is strongly site-dependent on temperature gradient and depth; therefore, it is treated as an optional upgrade path pending the WP1 site survey (Aresti et al., 2023).

3.4.3 Storage and backup sizing (BESS + hydrogen reserve)

Storage sizing supports objective tree node 3.2 (worst-case renewable lull and emergency autonomy). A layered strategy is adopted: (i) short-duration BESS for ride-through and black-start; (ii) long-duration reserve for multi-day coverage; and (iii) an emergency reserve for Tier 1 only. DC microgrid integration of energy storage and converters follows shipboard DC microgrid practice (Xu et al., 2022).

Design rule ('survive the slump'): size storage against the credible worst-case where wave generation is suppressed by storms and tidal units are unavailable or curtailed for inspection/cleaning, while Tier 1 safety loads must remain continuously supplied.

Battery Energy Storage System (BESS) energy capacity for critical autonomy:

$$E_{BESS} = \frac{P_{Tier1} * t_{autonomy}}{DoD * \eta_{rt}}$$

Assume Tier 1 load $P_{Tier\ 1}$ = 20 kW, $t_{autonomy}$ = 24 h, depth-of-discharge DoD = 0.8, round-trip efficiency η_{rt} = 0.9:

$E_{BESS} = \frac{20*24}{0.8*0.9} = 667kWh \approx 0.7MWh$

For longer autonomy (e.g., 72 h Tier 1), E_{BESS} scales to ~2.0 MWh linearly. The chosen autonomy should be set by the whole-facility risk case and maintenance CONOPS.

Hydrogen long-duration reserve (optional) for multi-day backup:

$m_{H_2} = \frac{E_{backup}}{\eta_{FC} * LHV_{H_2}}$

Assume a 7-day Tier 1 backup target: E_{backup} = 20 kW × 168 h = 3360 kWh, fuel cell efficiency η_{FC} = 0.5, hydrogen LHV = 33.3 kWh/kg:

$m_{H_2} = \frac{3360}{0.5*33.3} \approx 202kg$

Hydrogen is produced from surplus generation via a PEM electrolyser and converted back to electricity through a fuel cell during extended lulls (objective tree node 3.2.2).

Emergency reserve:

A small UPS or intrinsically safer reserve battery module is retained for last-line Tier 1 operation and safe shutdown. Seawater batteries are a potential niche option for marine environments and may be evaluated as a supplemental reserve technology (Li & Tian, 2024).

3.4.4 Distribution: DC bus and subsea export cable sizing

A regulated DC microgrid reduces conversion stages and improves controllability for mixed generation and storage (Xu et al., 2022). A nominal bus voltage of 800 V DC is assumed to limit cable currents and resistive losses.

Redundancy and load shedding: the distribution is designed to tolerate single-point failures and prioritise Tier 1 loads (life-safety) under degraded generation.

Sectionalised 800 V DC bus (Port / Starboard) with a normally-open tie breaker to prevent fault propagation; either section can support Tier 1 loads.

N+1 critical converters (DC/DC to 48 V services and UPS feeds) and parallel battery strings with independent BMS and contactors.

Protection and isolation using DC-rated breakers, differential/earth-leak detection, and wet-mate disconnects at the subsea junction hub for safe retrieval.

Automated load shedding logic: shed Tier 3 first, then Tier 2, while Tier 1 remains supplied; re-enable loads only after SOC and generation margins recover.

Cable current at peak power:

$$I = \frac{P_{peak}}{V_{DC}} = \frac{80{,}000}{800} = 100A$$

Voltage drop check for a 1 km round-trip copper conductor pair:

$$R_{pair} = \frac{2pL}{A} \qquad \Delta V = I * R_{pair}$$

Assume copper resistivity ρ = 0.0175 Ω·mm²/m, one-way length L = 1000 m, cross-sectional area A = 150 mm²:

$R_{pair} = \frac{(2*0.0175*1000)}{150} = 0.233\Omega$

$$\Delta V = 100 * 0.233 = 23.3V$$

A voltage drops below 3-5% is acceptable for first-pass sizing. Final cable sizing must also consider insulation rating, thermal limits, mechanical protection, and connector losses. Wet-mate connectors and a subsea junction hub support modular maintenance and fault isolation (objective tree nodes 3.3.1 and 3.5.2).

3.4.5 Thermal rejection to the ocean cold sink

Nearly all electrical power consumed within the habitat ultimately becomes heat. Efficient seawater cooling and heat recovery support objective tree node 3.4 and reduce net electrical demand (Alkrush et al., 2024).

Seawater flow rate for waste-heat rejection:

$$\dot{Q} = \dot{m}c_p \ \Delta T \rightarrow \dot{m} = \frac{\dot{Q}}{c_p \Delta T}$$

Assume $\dot{Q}$ = 60 kW waste heat, seawater specific heat c_p ≈ 4.0 kJ/kg·K, allowable seawater temperature rises ΔT = 5 K:

$$\dot{m} = \frac{60{,}000}{4000 * 5} = 3.0 kg/s \ \approx 10.5 \ m^3/h$$

Heat recovery opportunities include preheating domestic hot water and desalination feed, reducing electric heating loads (objective tree node 3.4.2).

Data-centre case: where a dedicated computing / data-handling module is included, waste-heat recovery is treated as mandatory (continuous high duty). The cooling loop therefore, includes a heat-recovery branch (HRHX) to utilise low-grade heat before final rejection to the ocean cold sink.

Recover heat to a domestic hot-water buffer tank and space-heating loop (when required), reducing electric heater duty.

Use recovered heat to pre-warm seawater feed for desalination / water treatment modules, improving overall plant efficiency.

Instrument outlet temperatures and flow to ensure ecological discharge limits and prevent biofouling hotspots.

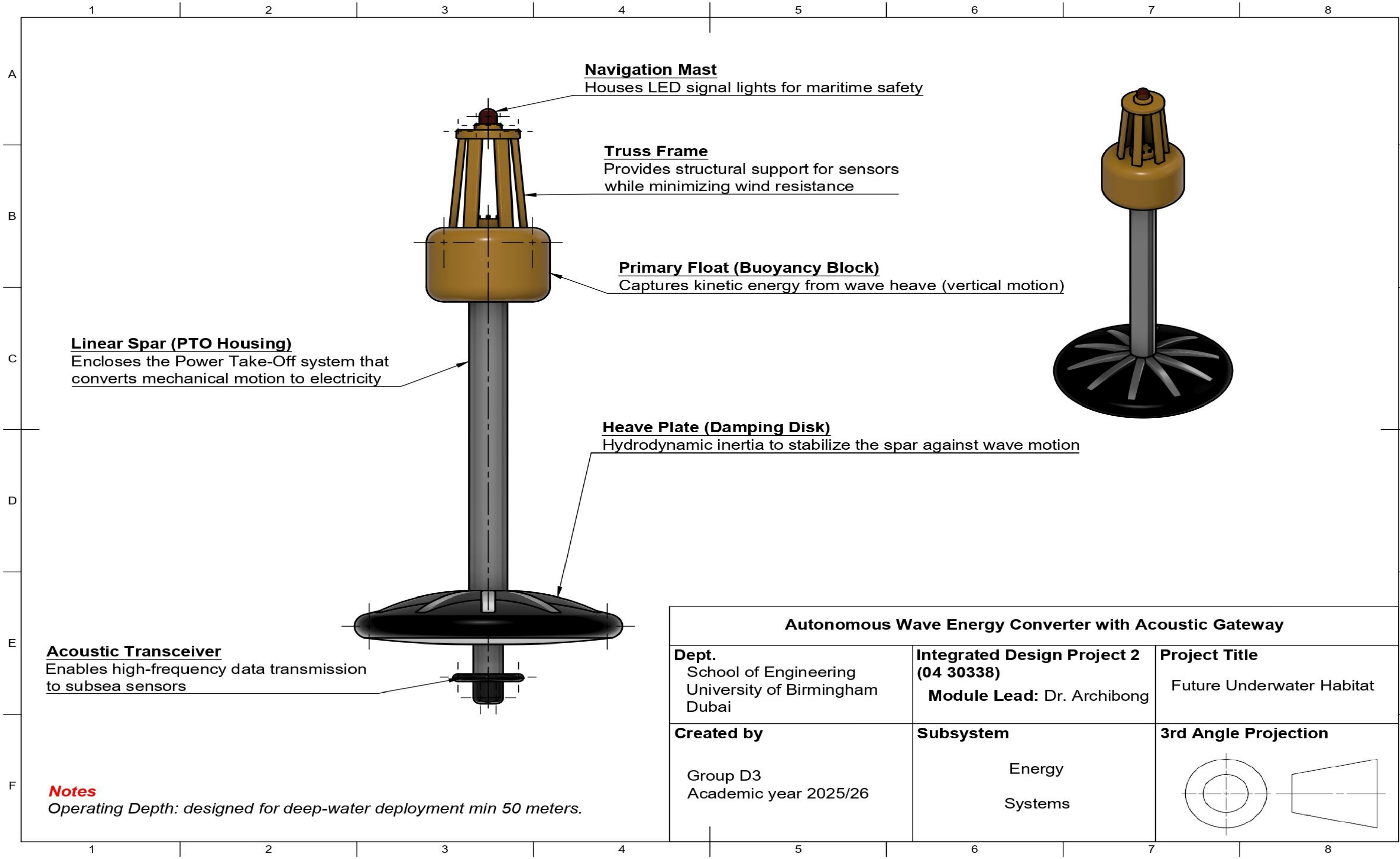
Equipment

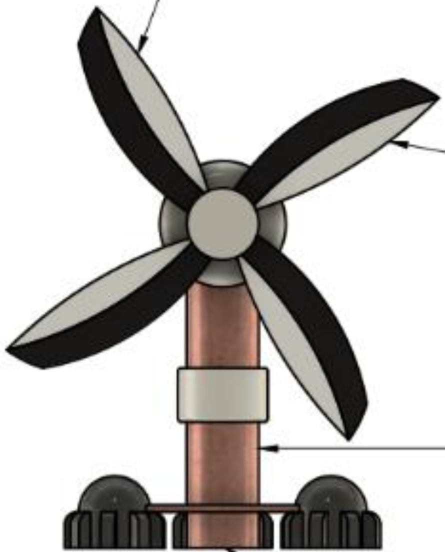
Table 19 lists representative components used to ground the concept design. Specific procurement choices will be refined during detailed design.

Table 19. *Equipment List for energy work package including product number.*

Subsystem	Component	Representative series / part	Description
Marine generation	Tidal stream turbine	100 kW class (representative)	Seabed-mounted turbine + nacelle; predictable resource and high capacity factor.
Marine generation	Wave energy buoy (WEC)	Point absorber buoy (representative)	Surface buoy converts heave motion to electricity; complements tidal variability.
Subsea power transfer	Wet-mate connector	Teledyne ODI Nautilus / SEACON (representative)	Underwater plug-and-socket interface enabling modular connection and isolation.
Subsea power transfer	Subsea junction box	Wet-mate hub (representative)	Collects generation feeds; routes export cable to habitat; supports selective isolation.
Power conversion	AC/DC rectifier	ABB / Schneider industrial rectifier (representative)	Rectifies variable-frequency generator output to DC bus.
Power conversion	DC-DC converter	Vicor DCM series (representative)	Isolated DC conversion and voltage regulation for DC microgrid segments.
Storage	Li-ion battery module	Marine ESS module (e.g., Corvus Orca - representative)	Main BESS for ride-through, smoothing and black-start.
Long-duration reserve	PEM electrolyser	PEM electrolyser skid (representative)	Converts surplus electricity to hydrogen for multi-day reserve.
Long-duration reserve	Fuel cell	PEM fuel cell stack (representative)	Converts stored hydrogen back to electricity during extended lulls.
Thermal management	Seawater plate heat exchanger	Alfa Laval plate HX (representative)	Rejects waste heat to ocean cold sink; corrosion-resistant materials required.
Thermal management	Seawater circulation pump	Marine pump skid (representative)	Maintains required seawater flow for heat rejection; includes strainers and anti-fouling measures.
Control & protection	Microgrid controller / EMS	Industrial PLC + SCADA (representative)	Energy management, load shedding, alarms, and condition monitoring.

Diagrams


Navigation Mast
Houses LED signal lights for maritime safety
Truss Frame
Provides structural support for sensors while minimizing wind resistance
Primary Float (Buoyancy Block)
Captures kinetic energy from wave heave (vertical motion)
Linear Spar (PTO Housing)
Encloses the Power Take-Off system that converts mechanical motion to electricity
Heave Plate (Damping Disk)
Hydrodynamic inertia to stabilize the spar against wave motion
Acoustic Transceiver
Enables high-frequency data transmission to subsea sensors
Notes
Operating Depth: designed for deep-water deployment min 50 meters.
Autonomous Wave Energy Converter with Acoustic Gateway
Dept.
School of Engineering
University of Birmingham
Dubai
Integrated Design Project 2
(04 30338)
Module Lead: Dr. Archibong
Project Title
Future Underwater Habitat
Created by
Group D3
Academic year 2025/26
Subsystem
Energy
Systems
3rd Angle Projection



Features Positioning Layout

Dept. School of Engineering University of Birmingham Dubai	**Integrated Design Project 2 (04 30338)** **Module Lead**: Dr. Archibong	**Project Tittle** Future Underwater Habitat
Created by Group D3 Academic year 2025/26	**Subsystem** Energy	3rd Angle Projection

WP4: LIFE SUPPORT

4.1 Requirements Specification

Table 20. *Tabulated requirements specification for the Life Support subsystem.*

Package	Requirement number	Description	Requirement Type	Rationale	Traceability to Objective Tree
Life Support System	**4.1**	Provide comprehensive life support by delivering potable water production ≥ design daily requirement with ≥99% uptime, wastewater treatment achieving ≥95% contaminant removal to marine-safe discharge standards, atmosphere control maintaining O_2 19.5-23.5%, CO_2 ≤0.5%, humidity 30-60% RH, and temperature 20-25°C, plus food production/storage supporting crew nutritional needs for mission duration	Functional	Integrated life support systems are mission-critical for crew health/survival, requiring high reliability and performance across air, water, waste, and nutrition functions	3.1.1.1, 3.1.1.2, 3.1.2.1, 3.1.2.2, 3.1.3.1, 3.1.3.2
	4.2	Ensure water management through potable water generation from seawater (RO/distillation) meeting WHO standards, greywater recycling ≥80% efficiency, wastewater treatment with solids separation/biological processing/disinfection prior to discharge, plus leak detection/prevention maintaining habitat dryness	Technical	Efficient closed-loop water systems minimise resupply needs while preventing contamination and structural damage in enclosed habitat	3.2.1.1, 3.2.1.2, 3.2.2.1, 3.2.2.2, 3.2.3.1
	4.3	mplement atmosphere management using electrochemical O_2 generation, CO_2 scrubbers (amine/LiOH/SAE), trace contaminant removal (filters/catalysts), humidity control (condensers/desiccant), temperature regulation (chilled water/heat pumps), and pressure maintenance with emergency O_2 supply	Technical	Precise control of breathing atmosphere prevents health risks from toxic buildup, hypoxia, or hypercapnia in sealed habitat environment	3.3.1.1, 3.3.1.2, 3.3.2.1, 3.3.2.2, 3.3.3.1, 3.3.3.2
	4.4	Support food systems and integration via aquaponics/hydroponics producing ≥ crew daily nutritional requirements, storage for 6-month shelf-stable/emergency rations, waste recycling to nutrients, plus full integration with power (≤X kWh/day), thermal (equipment ≤35°C), monitoring (real-time parameters to central dashboard), and safety/reliability (redundancy, emergency procedures, crew training, marine-safe operation)	Functional Quality	Sustainable food production with robust system integration ensures nutritional self-sufficiency and mission resilience	3.4.1.1, 3.4.1.2, 3.4.2.1, 3.5.1.1, 3.5.2.1, 3.5.3.1

4.2 Functional Flow

Matter: Cabin air, seawater, greywater, blackwater, solid waste

Energy: Electrical power (pumps, electrolysis, RO, ventilation), heat rejection

Data: Sensor readings (O2, CO2, pressure, temperature, humidity), control signals, alarms

Life Support System

Matter: Conditioned air (O2 controller), potable water, treated wastewater, solid waste, brine discharge

Energy: Heat rejected to seawater

Data: System status data, alarms, performance metrics

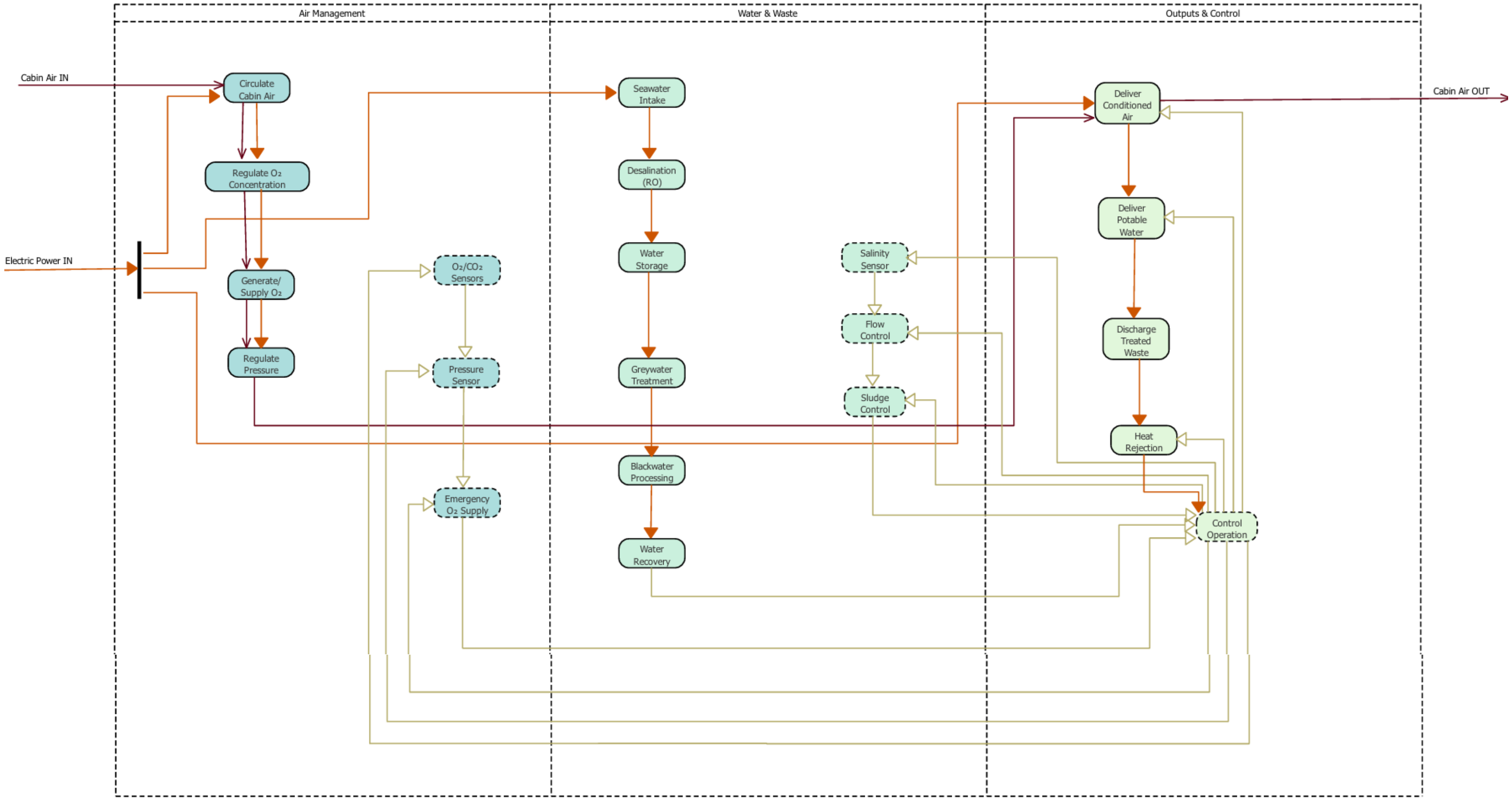


Figure 19. *Life Support subsystem functional flow analysis.*

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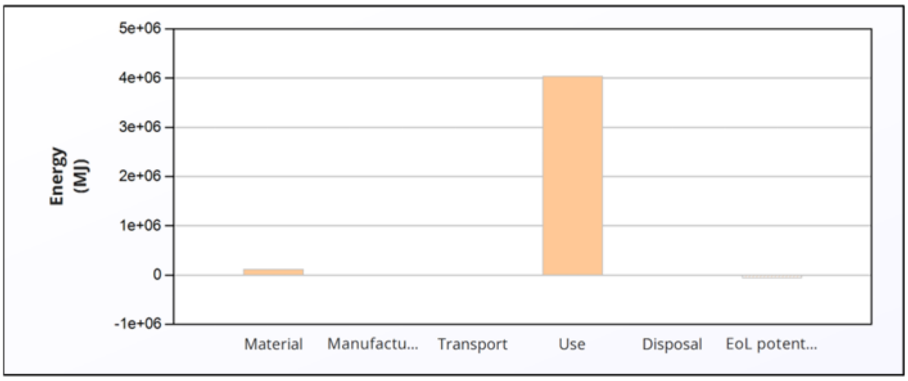
4.3 Material Selection and Life Cycle Analysis

Table 21. *Life Support Systems Life Cycle Analysis Results.*

<table>
<tr><th>Lifecycle Phase & Analysis</th><th>Figure/Table</th><th>Critical Insights & Conclusions</th></tr>
<tr><td>Lifecycle Energy Burden Overview</td><td>


<table>
<tr><th></th><th>Energy (MJ/year)</th></tr>
<tr><td>Equivalent annual environmental burden (averaged over 10 year product life):</td><td>4.17e+05</td></tr>
</table>
</td><td>The life support system's average annual energy burden is 4.17e+05 MJ/year over a 10-year product life. The chart reveals that the Use Phase is the overwhelmingly dominant contributor to the total lifecycle energy consumption, dwarfing all other stages. This confirms that operational efficiency is the primary engineering priority for underwater habitat life support systems.</td></tr>
<tr><td>Material Composition Impact Assessment</td><td>
<table>
<tr><th>Component</th><th>Material</th><th>Recycled content* (%)</th><th>Part mass (kg)</th><th>Qty.</th><th>Total mass (kg)</th><th>Energy (MJ)</th><th>%</th></tr>
<tr><td>Electrolysrer module housing</td><td>Stainless steel, austenitic, AISI 316, annealed</td><td>Virgin (0%)</td><td>1.2e+02</td><td>1</td><td>1.2e+02</td><td>1e+04</td><td>9.0</td></tr>
<tr><td>CO2 scrubber pressure vessel</td><td>Stainless steel, austenitic, AISI 316, annealed</td><td>Virgin (0%)</td><td>70</td><td>1</td><td>70</td><td>6.1e+03</td><td>5.2</td></tr>
<tr><td>Air circulation fan housing</td><td>Aluminum, 6061, T4</td><td>Virgin (0%)</td><td>25</td><td>1</td><td>25</td><td>4.4e+03</td><td>3.8</td></tr>
<tr><td>Main air ducting</td><td>PE-HD (high molecular weight)</td><td>Virgin (0%)</td><td>45</td><td>1</td><td>45</td><td>3.7e+03</td><td>3.2</td></tr>
<tr><td>Seawater intake pipe</td><td>PE-HD (high molecular weight)</td><td>Virgin (0%)</td><td>80</td><td>1</td><td>80</td><td>6.5e+03</td><td>5.6</td></tr>
<tr><td>RO pressure vessel</td><td>Stainless steel, duplex, AISI 2205, annealed</td><td>Virgin (0%)</td><td>95</td><td>1</td><td>95</td><td>6.8e+03</td><td>5.9</td></tr>
<tr><td>High-pressure RO pump casing</td><td>Stainless steel, duplex, AISI 2205, annealed</td><td>Virgin (0%)</td><td>1.1e+02</td><td>1</td><td>1.1e+02</td><td>7.9e+03</td><td>6.8</td></tr>
<tr><td>Potable water storage tank</td><td>Stainless steel, austenitic, AISI 316, annealed</td><td>Virgin (0%)</td><td>1.4e+02</td><td>1</td><td>1.4e+02</td><td>1.2e+04</td><td>10.5</td></tr>
<tr><td>Blackwater processing tank</td><td>Stainless steel, austenitic, AISI 316, annealed</td><td>Virgin (0%)</td><td>1.6e+02</td><td>1</td><td>1.6e+02</td><td>1.4e+04</td><td>12.0</td></tr>
<tr><td>Seawater heat exchanger core</td><td>Titanium, commercial purity, Grade 2</td><td>Virgin (0%)</td><td>75</td><td>1</td><td>75</td><td>4.4e+04</td><td>37.9</td></tr>
<tr><td>Total</td><td></td><td></td><td></td><td>10</td><td>9.2e+02</td><td>1.2e+05</td><td>100</td></tr>
</table>
</td><td>The Blackwater Processing Tank (1.6e+02 kg), Potable Water Storage Tank (1.4e+02 kg), and Electrolyser Module Housing (1.2e+02 kg) are the heaviest components, together accounting for approximately 45% of total system mass.

Material reduction or lightweighting efforts should prioritize these three subsystems to minimize upstream impacts.</td></tr>
</table>

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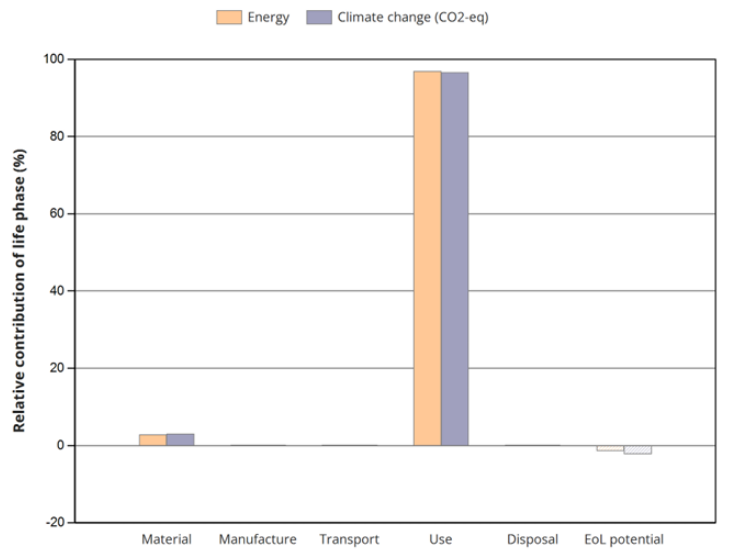
End-of-Life Process Energy & Circular Economy Potential

Component	End of life option	Energy (MJ)	%
Electrolysrer module housing	Recycle	-7.2e+03	12.7
CO2 scrubber pressure vessel	Recycle	-4.2e+03	7.4
Air circulation fan housing	Recycle	-3.2e+03	5.6
Main air ducting	Landfill	0	0.0
Seawater intake pipe	Landfill	0	0.0
RO pressure vessel	Recycle	-4.7e+03	8.3
High-pressure RO pump casing	Recycle	-5.5e+03	9.6
Potable water storage tank	Recycle	-8.4e+03	14.8
Blackwater processing tank	Recycle	-9.6e+03	16.9
Seawater heat exchanger core	Recycle	-1.4e+04	24.8
Total		-5.7e+04	100

Dual insight on end-of-life: The Disposal table shows the Blackwater Tank (19.3%) and Potable Water Tank (16.9%) require the most energy to process, though this is negligible versus the use phase. Critically, the EOL Potential table reveals substantial material recovery value totaling -5.7e+04 MJ, with the Seawater Heat Exchanger Core providing the largest credit (24.8%), justifying investment in recycling infrastructure.

Climate Change Impact (CO2-eq) Correlation



Component	Process	Amount processed	Climate change (CO2-eq) (kg)	%
Electrolysrer module housing	Metal rolling and forging	1.2e+02 kg	58	10.5
CO2 scrubber pressure vessel	Metal rolling and forging	70 kg	34	6.1
Air circulation fan housing	Metal rolling and forging	25 kg	17	3.1
Main air ducting	Polymer extrusion	45 kg	25	4.6
Seawater intake pipe	Polymer extrusion	80 kg	45	8.1
RO pressure vessel	Metal rolling and forging	95 kg	85	15.3
High-pressure RO pump casing	Metal rolling and forging	1.1e+02 kg	98	17.7
Potable water storage tank	Metal rolling and forging	1.4e+02 kg	68	12.2
Blackwater processing tank	Metal rolling and forging	1.6e+02 kg	78	14.0
Seawater heat exchanger core	Metal rolling and forging	75 kg	47	8.4
Total			5.6e+02	100

The manufacturing phase's climate impact is dominated by the High-Pressure RO Pump Casing (17.7%), RO Pressure Vessel (15.3%), and Blackwater Processing Tank (14.0%).

These components, likely fabricated from metals requiring significant thermal processing, should be prioritized for low-carbon material alternatives or design optimization to reduce their global warming potential.

4.4 Design

Calculations

Oxygen Production: Crew O_2 Demand

Oxygen production is sized to match crew metabolic consumption.

Assuming crew size $N = 10$ and oxygen demand per person $\dot{m}_{O_{2pp}} = 0.84$ kg/day:

$$\dot{m}_{O_{2total}} = N \cdot \dot{m}_{O_{2pp}}$$

$$\dot{m}_{O_{2total}} = 10 \cdot 0.84 \text{ kg/day}$$

CO_2 Removal: Crew CO_2 Generation

CO_2 scrubber capacity is sized from expected crew $CO_2 CO_2$production.

Assuming $N = 10$ and CO_2 production per person $\dot{m}_{CO_{2total}} = 10 \cdot 1.0 = 10$ kg/day

Electrolyser: Power Requirement

Electrolyser power is estimated using typical electrical energy per kg of O_2 produced.

Assuming oxygen production $\dot{m}_{O_2} = 8.4$ kg/day and the energy requirement $e_{O_2} = 6$ kWh/kg:

$$E_{day} = \dot{m}_{O_2} e_{O_2}$$

$$E_{day} = 8.4 \cdot 6 = 50.4 \text{ kWh/day}$$

Average power:

$$P = \frac{E_{day}}{24}$$

$$P = \frac{50.4}{24} = 2.1 \text{ kW}$$

Potable Water Production: Daily Requirement

Desalination capacity is sized from daily potable water use.

Assuming $N = 10$ and potable water demand $V_{pp} = 25$ L/person/day:

$$V_{water} = N \cdot V_{pp}$$

$$V_{water} = 10 \cdot 25 = 250 \text{ L/day}$$

Including 30% margin:

$$V_{design} = 1.3 \cdot 250 = 325 \text{ L/day}$$

Stoichiometric Balance: Electrolysis (O_2 production vs water use)

Electrolysis reaction:

$$2H_2O \rightarrow 2H_2 + O_2$$

Moles of O_2 required per day:

$$n_{O_2} = \frac{m_{O_2}}{M_{O_2}} = \frac{8400}{32} = 262.5 \text{ mol/day}$$

From stoichiometry, water needed = 2 mol H_2O per 1 mol O_2H_2O

$$n_{H_2O} = 2n_{O_2} = 525 \text{ mol/day}$$

$$m_{H_2O} = n_{H_2O}M_{H_2O} = 525 \cdot 18 = 9450 \text{ g/day} \approx 9.45 \text{ kg/day}$$

Hydrogen byproduct:

$$n_{H_2} = 2n_{O_2} = 525 \text{ mol/day}$$

$$m_{H_2} = 525 \cdot 2 = 1050 \text{ g/day} = 1.05 \text{ kg/day}$$

Stoichiometric Balance: CO_2 Removal using $LiOH$ (backup)

$LiOH$ Reaction:

$$2LiOH + CO_2 \rightarrow Li_2CO_3 + H_2O$$

Moles of CO_2 per day:

$$n_{CO_2} = \frac{10000}{44} = 227.27 \text{ mol/day}$$

$LiOH$ Needed = 2 mol per mol CO_2:

$$n_{LiOH} = 2n_{CO_2} = 454.55 \text{ mol/day}$$

Mass of $LiOH$ (molar mass $M_{LiOH} \approx 23.95$ g/mol):

$$m_{LiOH} = 454.55 \cdot 23.95 = 10890 \text{ g/day} \approx 10.9 \text{ kg/day}$$

Humidity Control (Dew Point $< 12°C$)

To prevent condensation on cold hull surfaces, we target dew point $< 12°C$.

Assume cabin temperature $T = 22°C$. Choose target RH such that dew point stays below $12°C$.

Using a standard approximation:

At $T = 22°C$, RH≈ 50 % gives dew point $\approx 11°C$ which satisfies the requirement.

Assumption for sizing:

$$\dot{m}_{H_2O_{pp}} = 1.0 \text{ kg/person/day} \Rightarrow \dot{m}_{H_2O_{total}} = 10 \cdot 1.0 = 10 \text{ kg/day}$$

Latent energy to condense moisture (approx. $h_{fg} = 2.45$ MJ/kg):

$$E_{latent} = \dot{m}_{H_2O_{total}} \cdot h_{fg} = 10 \cdot 2.45 = 24.5 \text{ MJ/day}$$

Convert to kWh (1 kWh = 3.6 MJ):

$$E_{latent} = \frac{24.5}{3.6} = 6.8 \text{ kWh/day}$$

Average dehumidification thermal load:

$$P_{latent} = \frac{6.8}{24} = 0.28 \text{ KW}$$

Potable Water Production: Daily Requirement

Desalination capacity is sized from daily potable water use.

$$V_{water} = 10 \cdot 25 = 250 \text{ L/day}$$

Including 30% margin:

$$V_{design} = 1.3 \cdot 250 = 325 \text{ L/day}$$

Greywater Recycling (to reduce RO demand)

Assume fraction of total use that becomes greywater:

$$f_{grey} = 0.7 \;\Rightarrow\; V_{grey} = 0.7 \cdot 250 = 175 \text{ L/day}$$

Assume greywater recovery efficiency:

$$\eta_{rec} = 0.75 \;\Rightarrow\; V_{recovered} = 175 \cdot 0.75 = 131 \text{ L/day}$$

Net make-up water required from RO:

$$V_{RO_{net}} = 250 - 131 - 119 \text{ L/day} = 0.119\, m^3\text{/day}$$

RO Energy per m^3

Assume typical seawater RO specific energy:

$$e_{RO} = 4 \text{ kWh/}m^3$$

Daily RO energy:

$$E_{RO} = V_{RO_{net}} \cdot e_{RO} = 0.119 \cdot 4 = 0.48 \text{ kWh/day}$$

If Ro runs at the conservative design capacity 0.325 m^3/day:

$$E_{RO_{design}} = 0.325 \cdot 4 = 1.30 \text{ kWh/day}$$

Brine Handling

Assume seawater RO recovery ratio $R = 0.35$

$$R = \frac{V_{permeate}}{V_{feed}} \;\Rightarrow\; V_{feed} = \frac{V_{permeate}}{R}$$

Using design permeate

$$V_{permeate} = 0.325\, m^3\text{/day}$$

$$V_{feed} = \frac{0.325}{0.35} = 0.929\ m^3/\text{day}$$

Brine volume:

$$V_{brine} = V_{feed} - V_{permeate} = 0.929 - 0.325 = 0.604\ m^3/\text{day}$$

$$V_{brine} = 604\ \text{L/day}$$

Hydroponic/Aquaculture Nutrient Loop

1. Plant Production Requirement

Crew size:

$$N = 10$$

Target vegetable production:

0.5 kg/person/day

Total fresh biomass:

$$m_{veg} = 10 \cdot 0.5 = 5\ \text{kg/day}$$

Assume dry matter fraction $= 10$ %

$$m_{DM} = 0.10 \cdot 5 = 0.5\ \text{kg/day}$$

2. Plant Nitrogen Demand

Assume plant nitrogen content $= 4$ % of dry mass:

$$m_N = 0.04 \cdot 0.5 = 0.02\ \text{kg/day}$$

$$m_N = 20\ \text{g/day}$$

3. Nitrogen Supply from Aquaculture

Assume:

- Feed nitrogen content $= 8$ %
- 30 % of feed nitrogen becomes dissolved plant-available nitrogen

$$m_{N_{available}} = m_f \cdot 0.08 \cdot 0.30$$

Set equal to plant demand:

$$0.02 = m_f \cdot 0.024$$

$$m_f = 0.83\ \text{kg feed/day}$$

4. Fish Yield

Assume feed Conversion Ratio $= 1.5$

$$m_{fish} = \frac{0.83}{1.5} = 0.55\ \text{kg/day}$$

Final Loop Summary

Inputs:

- 0.83 kg/day fish feed
- Small nutirent make-up

Outputs:

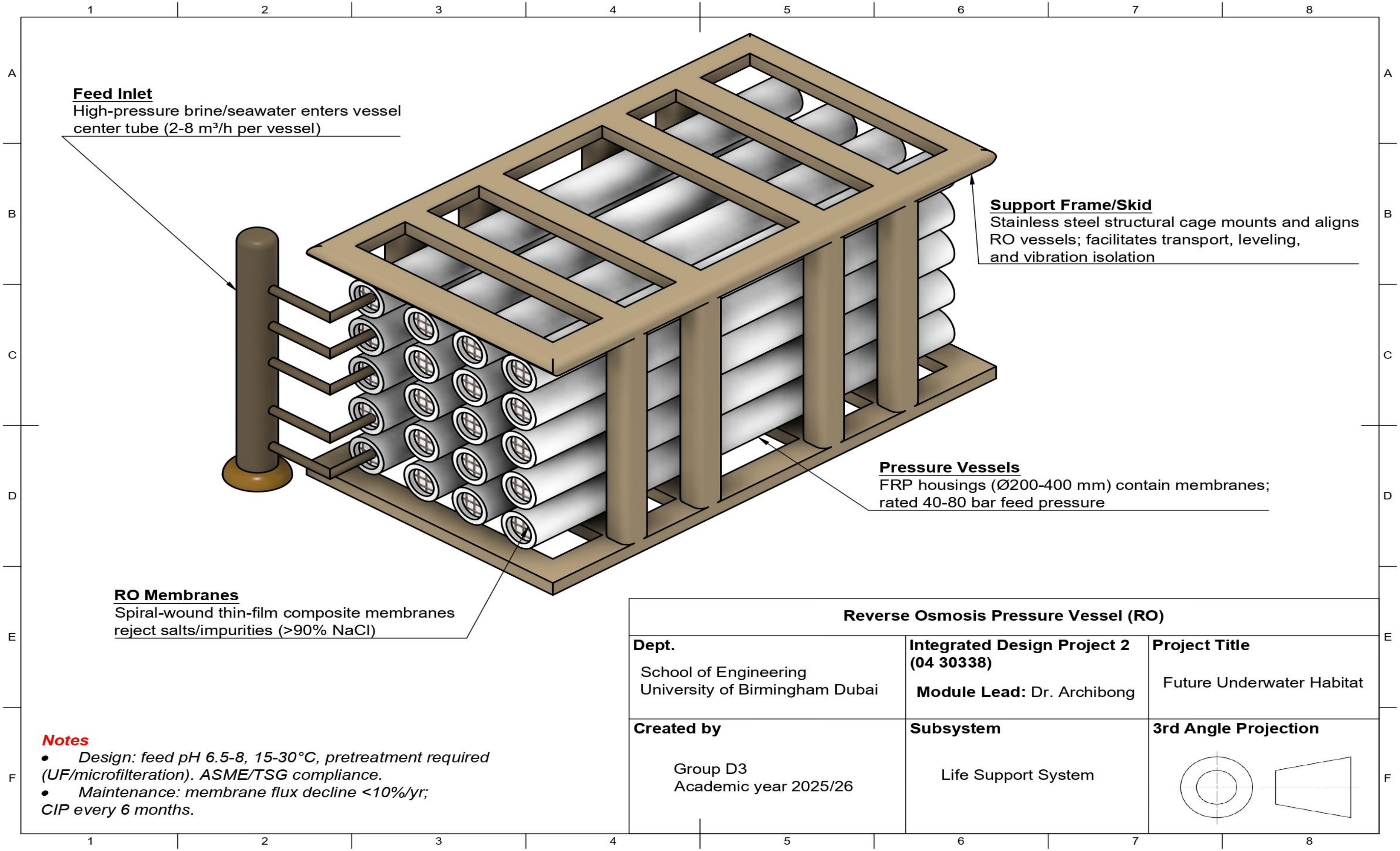
- 5 Kg/day fresh vegetables
- 0.55 Kg/day fish biomass

Equipment

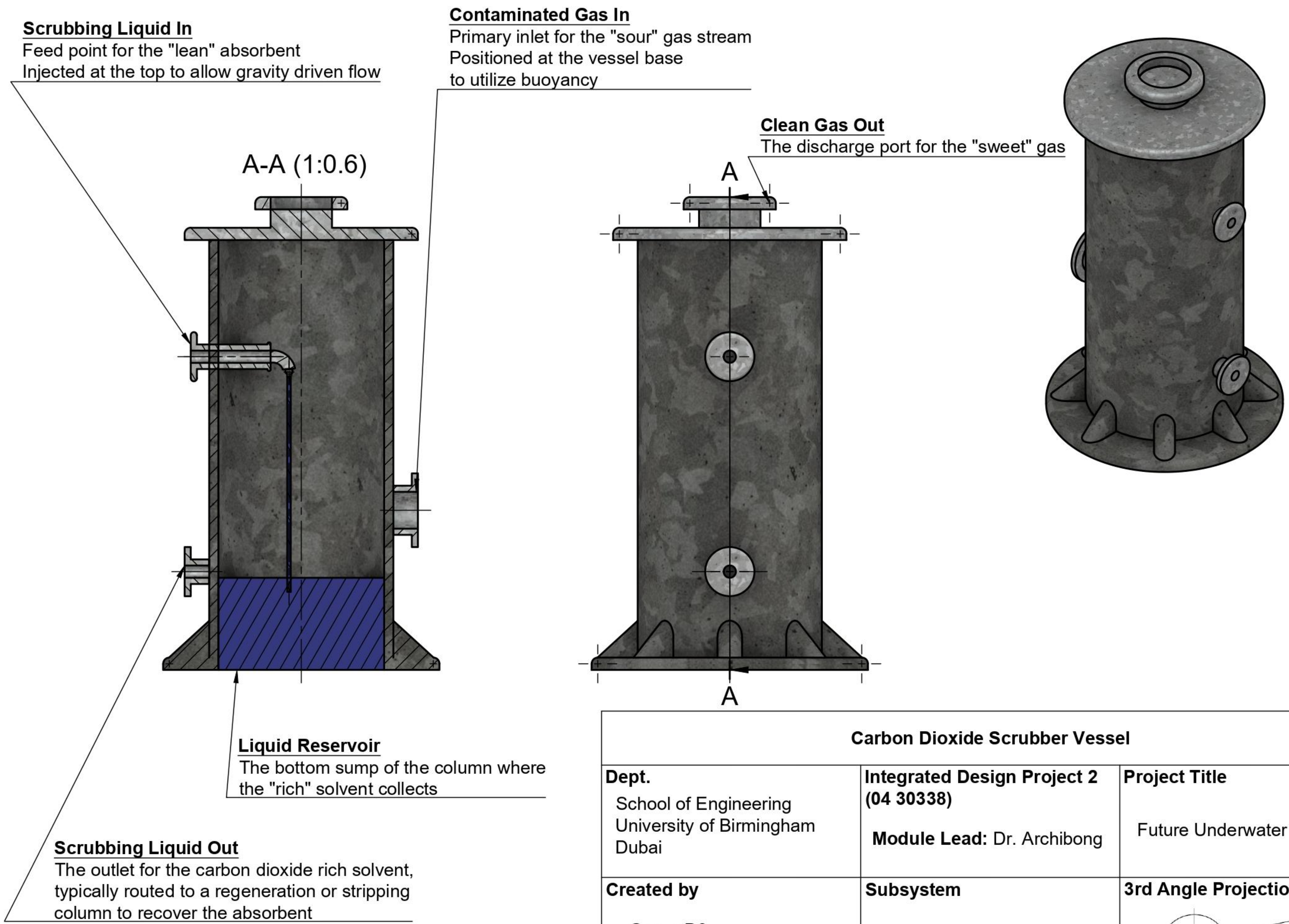
***Table 22.** Equipment List for life support work package including product number.*

Subsystem	Component	Representative part Number	Description
Air Revitalisation	CO2 sensor	Seanseair K33 ELG	CO2 + temperature + humidity monitoring for cabin control
Air Revitalisation	O2 monitor/detector	MSA ULTIMA X5000 (O2)	Fixed gas monitor platform for oxygen sensing & alarms
Water Production	High-pressure RO pump	Danfoss APP 2.2-180B3045	Seawater reverse osmosis high-pressure pump
Water Treatment	SS filter housing	Pall FBT03	Stainless filter housing for incoming/pretreatment filtration
Wastewater	MBR/UF membrane module	ZeeWeed 500D	Immersed hollow-fibre membrane module for wastewater polishing/MBR stage
Thermal Control	Seawater heat exchanger	Alfa Laval Gasketed Plate HX	Seawater-based heat rejection using plate heat exchanger family

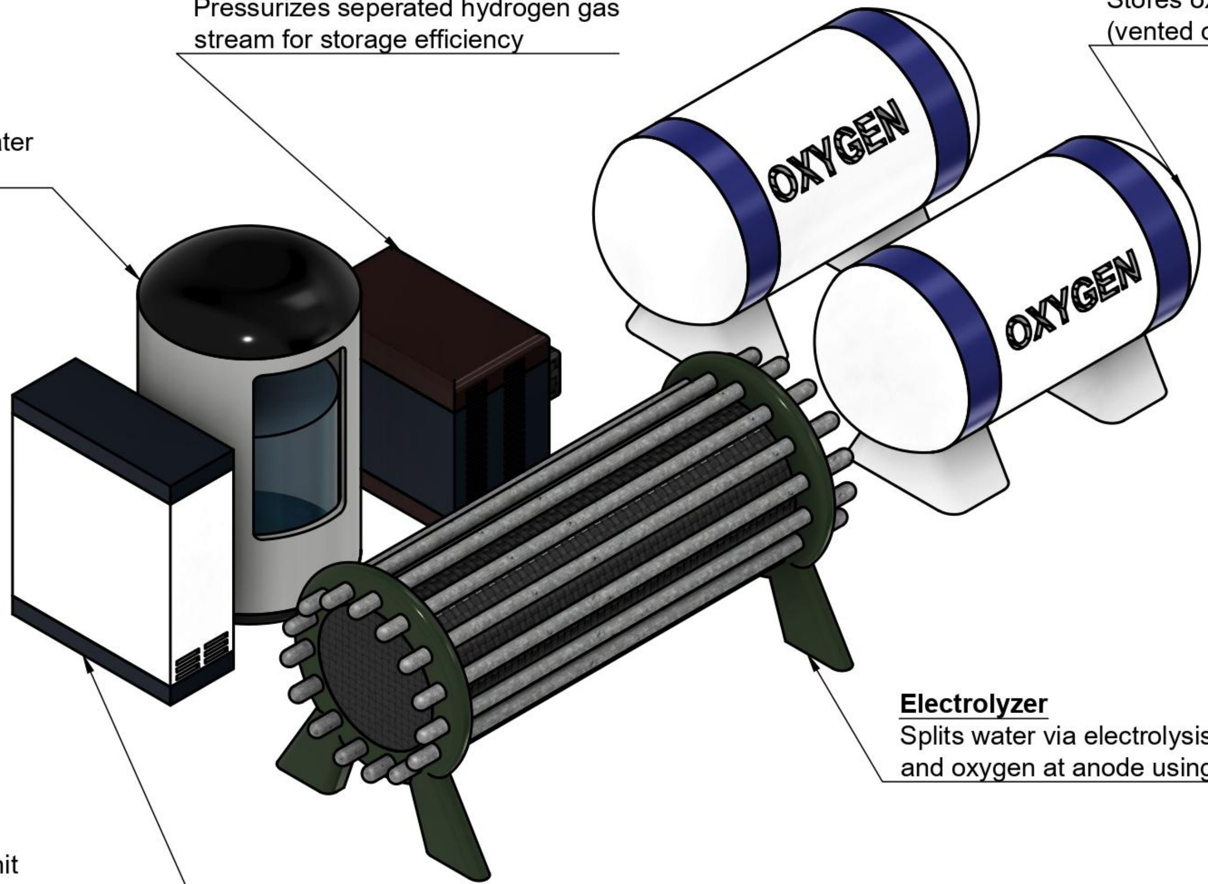
Diagrams



Feed Inlet
High-pressure brine/seawater enters vessel center tube (2-8 m³/h per vessel)
Support Frame/Skid
Stainless steel structural cage mounts and aligns RO vessels; facilitates transport, leveling, and vibration isolation
Pressure Vessels
FRP housings (Ø200-400 mm) contain membranes; rated 40-80 bar feed pressure
RO Membranes
Spiral-wound thin-film composite membranes reject salts/impurities (>90% NaCl)
Reverse Osmosis Pressure Vessel (RO)
Dept.
School of Engineering
University of Birmingham Dubai
Integrated Design Project 2 (04 30338)
Module Lead: Dr. Archibong
Project Title
Future Underwater Habitat
Created by
Group D3
Academic year 2025/26
Subsystem
Life Support System
3rd Angle Projection
Notes
• Design: feed pH 6.5-8, 15-30°C, pretreatment required (UF/microfilteration). ASME/TSG compliance.
• Maintenance: membrane flux decline <10%/yr; CIP every 6 months.

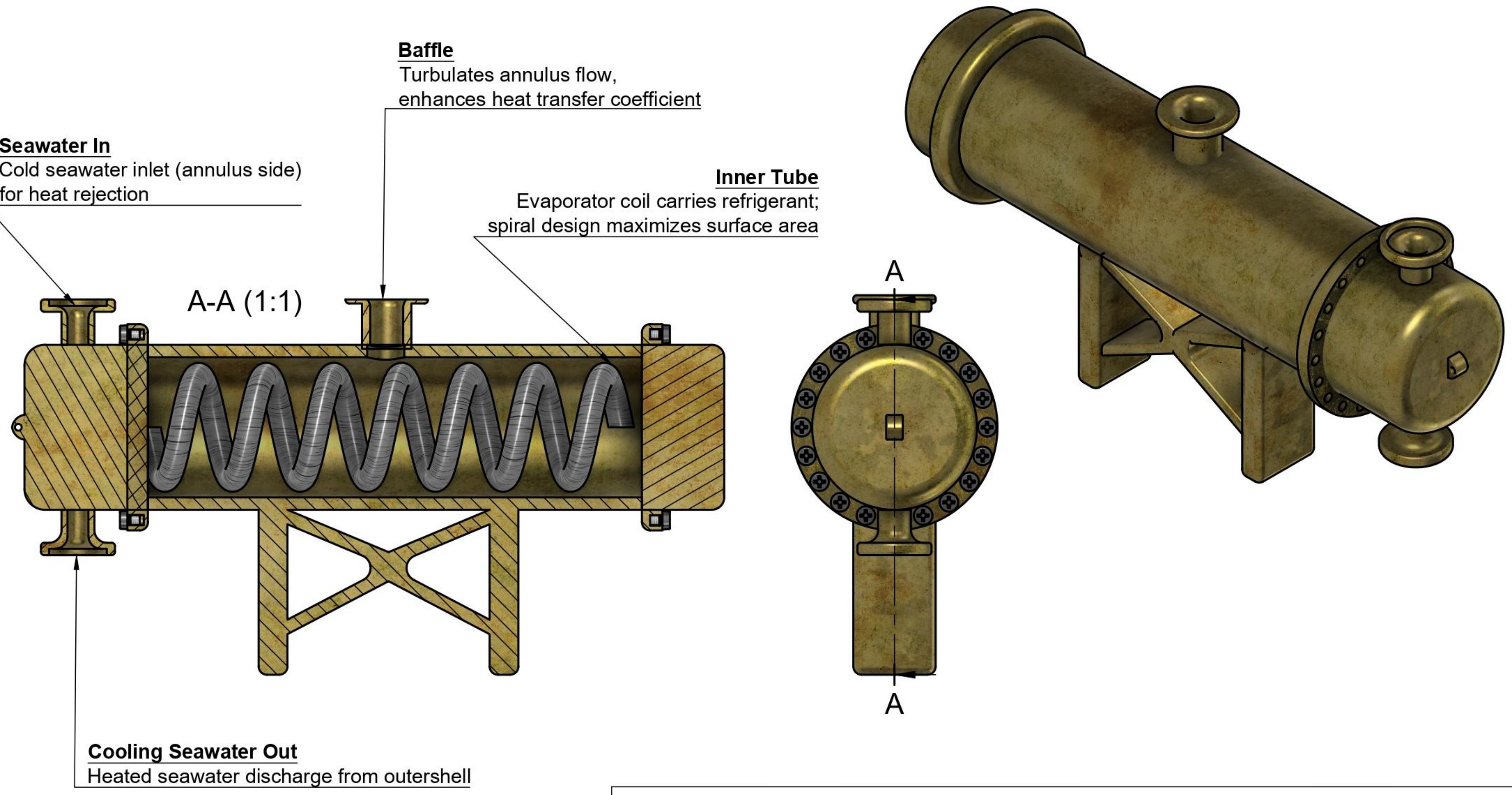


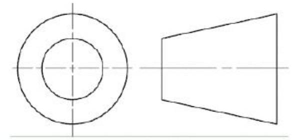
Carbon Dioxide Scrubber Vessel		
Dept. School of Engineering University of Birmingham Dubai	**Integrated Design Project 2 (04 30338)** **Module Lead:** Dr. Archibong	**Project Title** Future Underwater Habitat
Created by Group D3 Academic year 2025/26	**Subsystem** Life Support System	**3rd Angle Projection**



Electrolyser Unit For Oxygen Generation		
Dept. School of Engineering University of Birmingham Dubai	**Integrated Design Project 2 (04 30338)** **Module Lead:** Dr. Archibong	**Project Title** Future Underwater Habitat
Created by Group D3 Academic year 2025/26	**Subsystem** Life Support System	**3rd Angle Projection**

Notes
Safety hazard prevention: hydrogen/oxygen separation critical; include sensors for leaks, pressure relief.
(For illustration/reference only)



Seawater Marine Coaxial (Tube-In-Tube) Heat Exchanger		
Dept. School of Engineering University of Birmingham Dubai	**Integrated Design Project 2 (04 30338)** **Module Lead:** Dr. Archibong	**Project Title** Future Underwater Habitat
Created by Group D3 Academic year 2025/26	**Subsystem** Life Support System	**3rd Angle Projection** 

Notes

- *Design: ΔT min 5°C, seawater velocity 1-2 m/s, refrigerant R134a/R410A typical. ASME VII compliance.*
- *Fouling allowance: marine growth factor 0.0005; inspect/clean annually. For reference only.*

WP5: SYSTEMS

5.1 Requirements Specification

Table 23. *Tabulated requirements specification for the Systems subsystem.*

Package	Requirement number	Description	Requirement Type	Rationale	Traceability to Objective Tree
Systems	**5.1**	The communication subsystem shall achieve reliable long-range subsea data relay to surface infrastructure under noise/attenuation, provide high data-rate low-latency short-range internal exchange reliable under turbidity, maintain essential functions during comms loss, and ensure safe operation with data recovery	Functional Technical	Enables complete external/internal communication resilience critical for mission continuity and autonomous operation	7.1.1.1, 7.1.1.2, 7.1.2.1, 7.1.2.2, 7.1.3.1, 7.1.3.2
	5.2	The monitoring/control subsystem shall detect structural strain/degradation, water leaks, environmental anomalies, distinguish normal/degraded/critical states, and execute automated alarms/safety responses	Functional Technical	Provides comprehensive structural/environmental integrity monitoring with automated fault detection and mitigation	7.2.1.1, 7.2.1.2, 7.2.2.1, 7.2.2.2
	5.3	The robotic subsystem shall enable controlled vehicle approach/docking/retention reliable under disturbances, support repeated autonomous missions with onboard servicing, and facilitate mission data/health transfer plus safe power exchange/release	Functional Technical	Ensures complete vehicle-habitat interaction lifecycle for sustained robotic operations without manual intervention	7.3.1.1, 7.3.1.2, 7.3.2.1, 7.3.2.2, 7.3.3.1, 7.3.3.2
	5.4	The safety/reliability subsystem shall prevent catastrophic structural/environmental failures, enable safe shutdown/fault containment, eliminate single-point failures, support degraded operations, and provide maintenance access with durability/sustainability	Quality	Guarantees system safety, fault tolerance, and sustainable lifecycle performance for long-term deployment	7.4.1.1, 7.4.1.2, 7.4.2.1, 7.4.2.2

5.2 Functional Flow

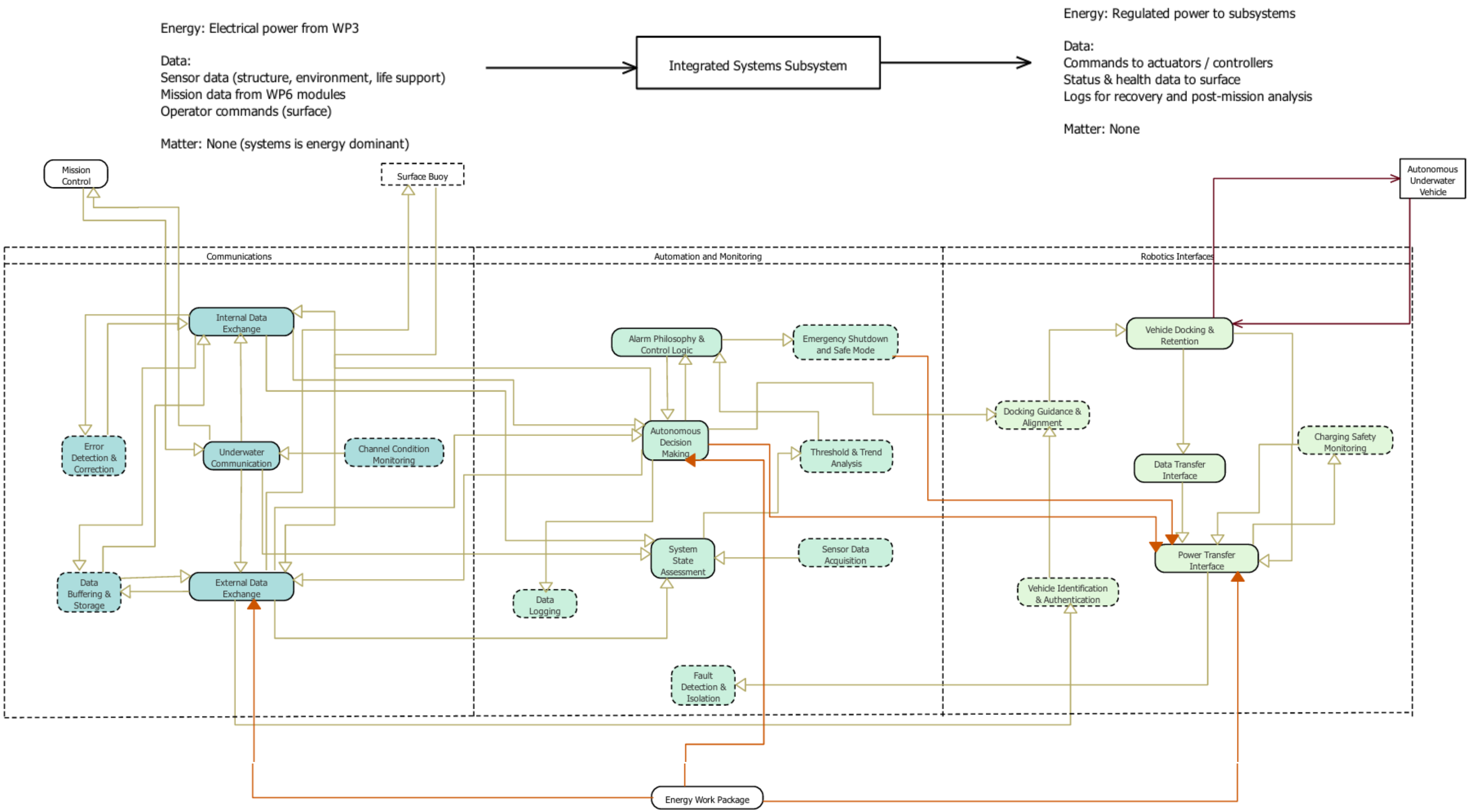
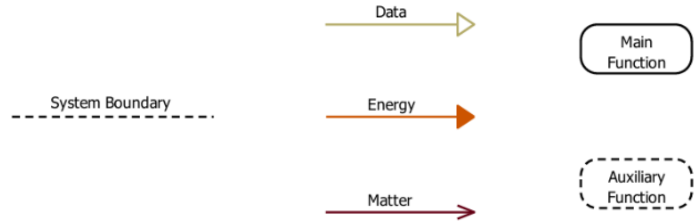


Figure 20. *Integrated systems subsystem functional flow analysis.*



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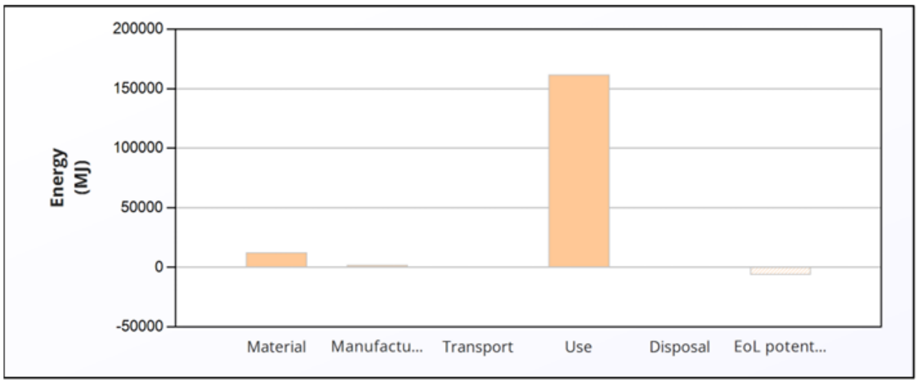
5.3 Material Selection and Life Cycle Analysis

Table 24. *Systems Life Cycle Analysis Results.*

<table>
<tr><th>Lifecycle Phase & Analysis</th><th>Figure/Table</th><th>Critical Insights & Conclusions</th></tr>
<tr><td>Lifecycle Energy Burden Overview</td><td>


<table>
<tr><th></th><th>Energy (MJ/year)</th></tr>
<tr><td>Equivalent annual environmental burden (averaged over 20 year product life):</td><td>8.79e+03</td></tr>
</table>
</td><td>The system's average annual energy burden is 8.79e+03 MJ/year over a 20-year product life. The chart reveals a bimodal distribution: the Use Phase and Material Phase are the dominant contributors, while Manufacturing, Transport, and Disposal are comparatively negligible. This indicates that design optimization should target both operational efficiency and material selection.</td></tr>
<tr><td>Material Composition Impact Assessment</td><td>
<table>
<tr><th>Component</th><th>Material</th><th>Recycled content* (%)</th><th>Part mass (kg)</th><th>Qty.</th><th>Total mass (kg)</th><th>Energy (MJ)</th><th>%</th></tr>
<tr><td>Acoustic modem housing and transducer</td><td>Titanium, alpha-beta alloy, Ti-3Al-2.5V (Grade 9)</td><td>Virgin (0%)</td><td>4</td><td>1</td><td>4</td><td>2.6e+03</td><td>21.1</td></tr>
<tr><td>Optical communication module</td><td>Borosilicate - 3320</td><td>Typical %</td><td>1.3</td><td>1</td><td>1.3</td><td>29</td><td>0.2</td></tr>
<tr><td>Embedded control computer PCB</td><td>Epoxy/Aramid, PCB laminate</td><td>Virgin (0%)</td><td>2</td><td>1</td><td>2</td><td>1.2e+03</td><td>9.6</td></tr>
<tr><td>Strain/leak/pressure/temperature sensors</td><td>Stainless steel, ferritic, ASTM CB-30, cast, annealed</td><td>Virgin (0%)</td><td>2</td><td>1</td><td>2</td><td>96</td><td>0.8</td></tr>
<tr><td>Alarm control PCB</td><td>Epoxy resin (cycloaliphatic)</td><td>Virgin (0%)</td><td>0.6</td><td>1</td><td>0.6</td><td>75</td><td>0.6</td></tr>
<tr><td>Emergency shutdown relay</td><td>Copper, C12200, hard (phosphorus de-oxidized arsenical h.c. copper)</td><td>Virgin (0%)</td><td>2</td><td>1</td><td>2</td><td>1.5e+02</td><td>1.2</td></tr>
<tr><td>Docking frame</td><td>Aluminum, 5083, H111</td><td>Typical %</td><td>50</td><td>1</td><td>50</td><td>6.9e+03</td><td>55.6</td></tr>
<tr><td>Inductive charging coil</td><td>Copper, C10100, hard (oxygen-free electronic h.c. copper)</td><td>Typical %</td><td>10</td><td>1</td><td>10</td><td>6e+02</td><td>4.8</td></tr>
<tr><td>Power transfer encapsulation</td><td>Bisphenol molding compound (mineral filler)</td><td>Virgin (0%)</td><td>2</td><td>1</td><td>2</td><td>1.4e+02</td><td>1.1</td></tr>
<tr><td>Subsea power and Data cable and connectors</td><td>Copper, C10100, hard (oxygen-free electronic h.c. copper)</td><td>Typical %</td><td>10</td><td>1</td><td>10</td><td>6e+02</td><td>4.8</td></tr>
<tr><td>Total</td><td></td><td></td><td></td><td>10</td><td>84</td><td>1.2e+04</td><td>100</td></tr>
</table>
</td><td>The Docking Frame alone accounts for 55.6% of total material energy, despite being made from aluminum with recycled content. The Acoustic Modem Housing (Titanium) contributes 21.1%, and the Embedded Control Computer PCB adds 9.6%. These three components represent over 86% of the material burden and should be the focus of lightweighting or alternative material investigations.</td></tr>
</table>

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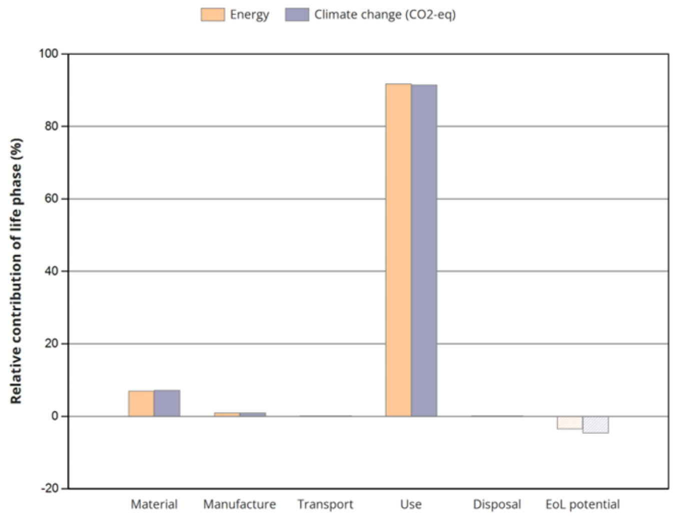
End-of-Life Process Energy & Circular Economy Potential

Component	End of life option	Energy (MJ)	%
Acoustic modem housing and transducer	Recycle	-8.3e+02	13.4
Optical communication module	Recycle	-3	0.0
Embedded control computer PCB	Recycle	-2.4e+02	3.8
Strain/leak/pressure/temperature sensors	Recycle	-66	1.1
Alarm control PCB	Downcycle	-0.06	0.0
Emergency shutdown relay	Downcycle	-61	1.0
Docking frame	Recycle	-4.2e+03	67.5
Inductive charging coil	Recycle	-4.2e+02	6.6
Power transfer encapsulation	Landfill	0	0.0
Subsea power and Data cable and connectors	Recycle	-4.2e+02	6.6
Total		**-6.3e+03**	**100**

Dual insight on end-of-life: The Disposal table shows the Docking Frame requires 61.2% of total disposal energy, though the absolute value (57 MJ total) is minimal. Critically, the EOL Potential table reveals massive material recovery value totaling -6.3e+03 MJ, with the Docking Frame again dominating (67.5% of credits).

Climate Change Impact (CO2-eq) Correlation



Component	Material	Recycled content* (%)	Part mass (kg)	Qty.	Total mass (kg)	Climate change (CO2-eq) (kg)	%
Acoustic modem housing and transducer	Titanium, alpha-beta alloy, Ti-3Al-2.5V (Grade 9)	Virgin (0%)	4	1	4	1.6e+02	19.3
Optical communication module	Borosilicate - 3320	Typical %	1.3	1	1.3	1.8	0.2
Embedded control computer PCB	Epoxy/Aramid, PCB laminate	Virgin (0%)	2	1	2	76	9.2
Strain/leak/pressure/temperature sensors	Stainless steel, ferritic, ASTM CB-30, cast, annealed	Virgin (0%)	2	1	2	9.2	1.1
Alarm control PCB	Epoxy resin (cycloaliphatic)	Virgin (0%)	0.6	1	0.6	3.7	0.4
Emergency shutdown relay	Copper, C12200, hard (phosphorus de-oxidized arsenical h.c. copper)	Virgin (0%)	2	1	2	10	1.2
Docking frame	Aluminum, 5083, H111	Typical %	50	1	50	4.8e+02	58.0
Inductive charging coil	Copper, C10100, hard (oxygen-free electronic h.c. copper)	Typical %	10	1	10	40	4.8
Power transfer encapsulation	Bisphenol molding compound (mineral filler)	Virgin (0%)	2	1	2	7.1	0.9
Subsea power and Data cable and connectors	Copper, C10100, hard (oxygen-free electronic h.c. copper)	Typical %	10	1	10	40	4.8
Total				**10**	**84**	**8.2e+02**	**100**

The climate change analysis strongly correlates with energy data. The Docking Frame contributes 58.0% of material-related CO2-eq, followed by the Acoustic Modem Housing (Titanium) at 19.3% and the Embedded Control Computer PCB at 9.2%.

The titanium component is particularly carbon-intensive relative to its mass (4 kg contributes 19.3% of emissions), suggesting that alternative materials for modern housing could yield significant climate benefits.

5.4 Design

Calculations

The systems design calculations are based on environmental conditions representative of the Ionian Sea and operational requirements defined in the challenge specification. The underwater habitat operates at a depth of 200 m, while autonomous underwater vehicles (AUVs) may operate to depths of up to 4000 m.

Communications System: Bandwidth vs Range

For long-range communication, acoustic transmission is employed because radio frequency signals are strongly attenuated in seawater (Zeng, 2016). The transmission loss (TL) is modelled as the sum of spherical spreading and absorption losses:

$$TL = 20\log_{10}(R) + \alpha R$$

A maximum communication range of $R = 4000$ m (Habitat to DSM) was taken and an acoustic frequency of 10 kHz, the absorption coefficient is approximately $\alpha = 0.04$ dB km^{-1}(Zeng, 2016).

$$20\log_{10}(4000) = 20 \times 3.602 = 72.04 \text{ dB}$$
$$\alpha R = 0.04 \times 4 = 0.16 \text{ dB}$$
$$TL = 72.04 + 0.16 = 72.20 \text{ dB}$$

The signal to noise ratio (SNR) is calculated using: (Sharif-Yazd, 2017)

$$SNR = SL - TL - NL$$

Assuming a conservative source level of $SL = 190$ dB re 1 µPa at 1m and a deep sea noise level of $NL = 60$ dB: (Salva Climent, 2014)

$$SNR = 190 - 72.20 - 60$$
$$SNR = 57.8 \text{ dB}$$

Converting to linear form for application in Shannon's capacity equation:

$$SNR_{lin} = 10^{(57.8/10)} = 6.03 \times 10^5$$

The maximum achievable data rate is estimated using Shannon's capacity equation:

$$C = B\log_2(1 + SNR_{lin})$$

- At long range, the usable acoustic bandwidth is limited to approximately $B = 5000$ Hz (William C.Cox, 2008):

$$C = 5000\log_2(1 + 6.03 \times 10^5)$$
$$\log_2(6.03 \times 10^5) \approx 19.2$$
$$C = 5000 \times 19.2 = 9.6 \times 10^4 \text{ bps}$$
$$C \approx 96 \text{ kbps}$$

This shows the bandwidth-range trade-off, where kilometre scale range communication is possible only with low acoustic frequencies, thus restricting the bandwidth to tens of kilobits per second.

For short range communication during docking, blue light optical communication is employed. The optical attenuation is estimated by:

$$P_r = P_t e^{-cR}$$

Assuming an attenuation coefficient of $c = 0.15\ \text{m}^{-1}$, transmission power $P_t = 1$ W, and a range of $R = 15$ m (Frank Hanson, 2008):

$$P_r = 1 \times e^{-0.15\times 15}$$
$$P_r = e^{-2.25} = 0.105\ \text{W}$$

For a conservative capacity estimate, an SNR of 20 dB is assumed.

The theoretical maximum data rate is then estimated using Shannon's capacity equation:

$$C = 5 \times 10^{8} log_2(1 + 100)$$

$$C \approx 3.3\ Gbps$$

This received power is sufficient to support optical data rates of 3 Gbps over ranges below 20 m, justifying the use of optical links for high-bandwidth, short-range operations only.

Automation and Monitoring

Structural health monitoring is achieved using foil strain gauges. Measured strain is given by:

$$\varepsilon = \frac{\Delta R}{GF \cdot R}$$

For a typical gauge factor of $GF = 2.0$ and nominal resistance $R = 120\ \Omega$, modern instrumentation allows resolution of approximately $\varepsilon_{min} \approx 1\ \mu\varepsilon$

Given that allowable strains for metallic pressure structures are on the order of $2000\ \mu\varepsilon$, the monitoring system provides significant early warning margin (Hidaayah Afsheen, 2025).

Leak detection is implemented using internal pressure sensors. A conservative alarm threshold of:

$$\Delta P_{threshold} = 5\ \text{kPa}$$

is adopted. At an external pressure of approximately 2 MPa, this corresponds to a relative pressure change of:

$$\frac{5 \times 10^3}{2 \times 10^6} = 0.0025 = 0.25\%$$

allowing rapid detection of small leaks.

Robotics: Docking, Charging, and Data Transfer

Autonomous docking systems must accommodate navigation uncertainty. Typical AUV positional accuracy is approximately ±0.5 m, with angular uncertainty of ±5°. The Docking interface is therefore designed to tolerate these misalignments (Teeneti, 2021).

Inductive charging is used for underwater power transfer. The delivered power is calculated as:

$$P_{delivered} = \eta P_{supplied}$$

Aiming for a charging efficiency of $\eta = 0.85$ and supplied power $P_{supplied} = 2000$ W:

$$P_{delivered} = 0.85 \times 2000 = 1700 \text{ W}$$

For an AUV battery capacity of 20 kWh, the charging time is:

$$t = \frac{20}{1.7} = 11.8 \text{ h}$$

During docking, high-speed data transfer is achieved via a physical or optical interface operating at 3 Gbps. For a 10-minute docking duration:

$$D = 3 \times 10^9 \times 600$$
$$D = 6 \times 10^{11} \text{ bits} \approx 225 \text{ GB}$$

Docking load Analysis

The peak force transmitted to the habitat structure when the AUV contacts the docking collar:

$$F_{impact} = \frac{m \cdot v}{t_{contact}}$$

$$F_{impact} = 100 \times 0.5/0.1 = 500.0N$$

The worst-case lateral current in the Ionian Sea generates a side load on the docking collar. This force is transferred to the hull as hoop stress.

$$F_{lateral} = ½\rho CdAlv^2$$

$$F_{lateral} = ½ \times 1029 \times 1.0 \times (1.524 \times 0.19) \times 0.5^2 = 37.2N$$

The allowable working stress is determined by applying the design safety factor (2.5) to Aluminum, 5083. All stresses from docking must remain below this limit.

$$\sigma_{allowable} = \frac{\sigma_{yield}}{SF}$$

$$\sigma_{allowable} = 228/2.5 = 91.2 \; MPa$$

Bending Stress at Collar Root (Docking)

$$I = \frac{2\pi \cdot r \cdot t^3}{12}$$

$$I = 2\pi \times 0.1 \times 0.002^3/12 = 4.19 \times 10^{-10} m^4$$

$$\sigma_{bending} = \frac{\left(F_{impact} \times arm\right) \times (t/2)}{I}$$

$$\sigma_{bending} = (500 \times 0.05) \times 0.001/4.19 \times 10^{-10} = 59.68MPa$$

The bending stress at the collar root is 59.68 MPa. Hoop stress from the lateral current (0.04 MPa) is negligible.

Von Mises stress is evaluated and compared to the allowable stress. Since bending dominates, other stress components are ignored.

$$\sigma_{vm} \approx \sigma_{bending} = 59.68 \; MPa \leq \sigma_{allowable} = 91.2 \; MPa$$

Docking forces are three orders of magnitude smaller than hydrostatic pressure at 200 m depth.

Equipment

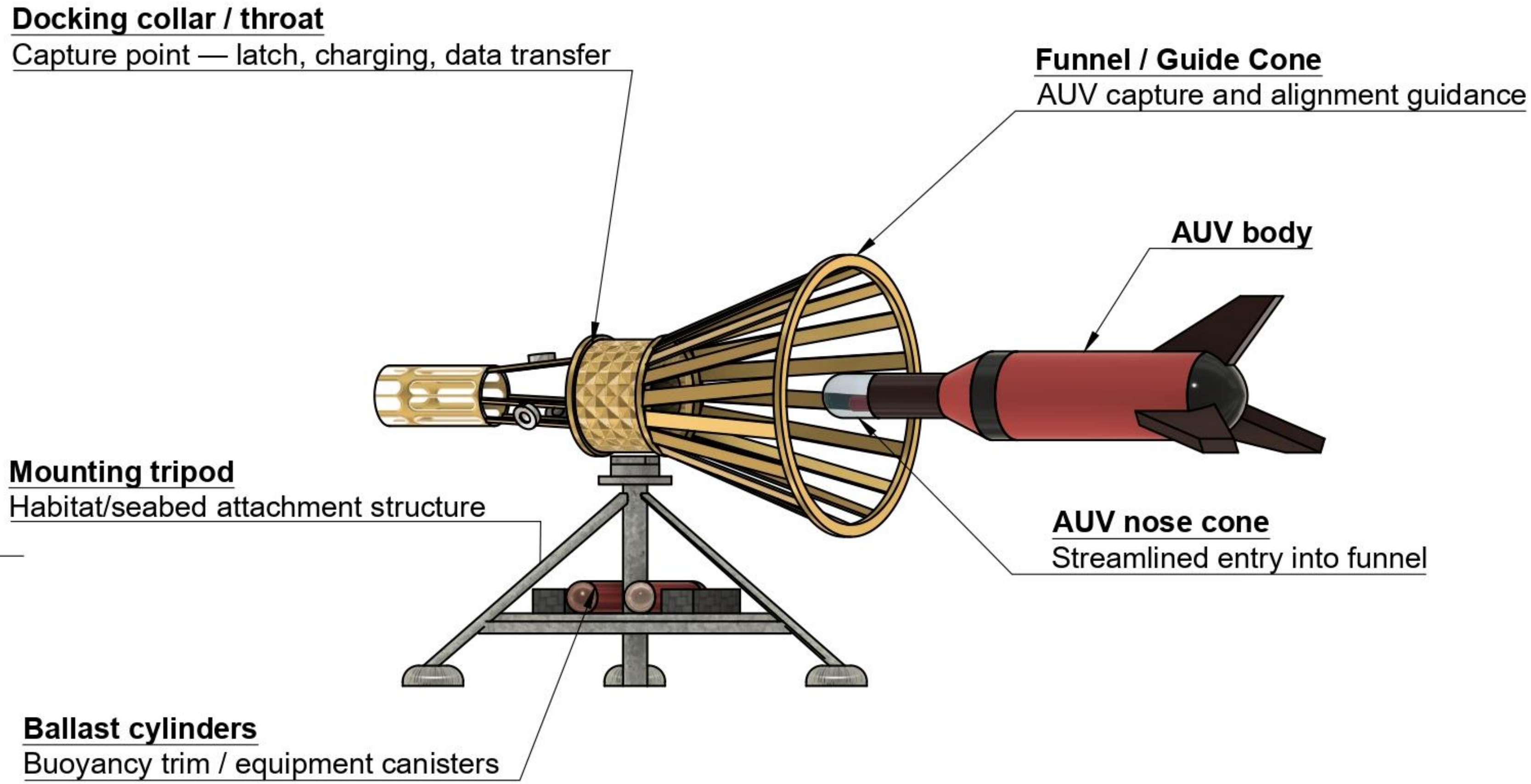
The table below identifies the major components needed to execute the System's Work Package. Only equipment used in system calculations and system interfaces is included in this conceptual design stage.

Table 25. *Equipment List for systems work package including product number.*

Subsystem	Component	Representative Part Number	Description
Communications	Acoustic Modem	Teledyne Benthos ATM-900 Series	Long-range acoustic modem.
Communications	Acoustic Transducer	Teledyne Benthos TR-6000	Low-frequency transducer.
Communications	Optical Transmitter (Blue Laser)	Hamamatsu PLP10-450B	Blue laser diode
Communications	Optical Receiver (APD)	Hamamatsu S8664-55 APD	Highly sensitive optical detector
Automation & Monitoring	Strain Gauge	Vishay Micro-Measurements CEA-06-125UN-350	Foil strain gauge
Automation & Monitoring	Strain Signal Conditioner	Vishay 2310B Signal Conditioning Amplifier	Improves strain signal strength and accuracy
Automation & Monitoring	Internal Pressure Sensor	Keller Series 33X	High accuracy pressure sensor
Automation & Monitoring	Humidity Sensor	Sensirion SHT85	Used to justify humidity-based early leak detection thresholds.
Robotics (Docking & Charging)	Docking Funnel / Capture Interface	WHOI-Style Passive Docking Cone	Reference docking geometry used for positional tolerance calculations.
Robotics	Inductive Charging Coils	Würth Elektronik WE-IPT Series	Representative inductive power transfer coils used in charging power
Robotics	Power Conditioning Unit	Vicor DCM3623 Series DC-DC Converter	Used to represent power regulation losses in charging and system power balance calculations.

System Power	Central Control Computer	NVIDIA Jetson AGX Orin	Representative embedded compute platform used for system power demand and energy balance calculations.

Diagrams



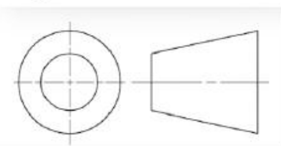
AUV DOCKING

Dept School of Engineering University of Birmingham Dubai	Integrated Design Project 2 (04 30338) Module Lead: Dr. Archibong	Project Title Future Underwater Habitat
Created by Group D3 Academic year 2025/26	Subsystem Systems	3rd Angle Projection

Notes:
Operating depth = 200 m / 19.92 atm external pressure

Notes:
Cabinet rated for: 25-year design life with ROV-accessible maintenance every 5 years

ELECTRONICS CONTROL UNIT

Dept. School of Engineering University of Birmingham Dubai	Integrated Design Project 2 (04 30338) Module Lead: Dr. Archibong	Project Title Future Underwater Habitat
Created by Group D3 Academic year 2025/26	Subsystem Systems	3rd Angle Projection 

WP6: MISSION MODULE 6A

6A.1 Requirements Specification

Table 26. *Tabulated requirements specification for the Direct Ocean Capture subsystem.*

Package	Requirement number	Description	Requirement Type	Rationale	Traceability to Objective Tree
Mission Module - Direct Ocean Capture (6A)	**6A.1**	Achieve target CO_2 capture performance by meeting the specified daily target CO_2 capture rate of 1000kg while maintaining ≥95% uptime and limiting performance drop to ≤10% over 24 h of continuous operation	Functional	Ensuring high, stable capture performance is essential to meet mission CO_2 removal targets and avoid accumulation that would undermine system objectives	4.1.1.1.1, 4.1.2.1.1
	6A.2	Minimise energy use and maximise renewable utilisation by keeping energy per kg CO_2 captured below the design limit, sourcing at least 80% of energy from renewables, switching to renewables within 5 s of availability, and operating without reliance on fossil-fuel backup	Technical	Reducing specific energy consumption and prioritising renewables increases efficiency, lowers operating cost and emissions, and improves overall system sustainability	4.2.1.1, 4.2.2.1, 4.2.2.2, 4.2.2.3
	6A.3	Protect seawater chemistry and marine ecosystems by controlling pH change to ≤±0.2, salinity change to ≤±1 PSU, preventing any harmful discharge, and limiting local thermal change to ≤2°C	Quality	Tight control of chemical and thermal perturbations prevents ecological harm, ensuring safe long-term operation in the surrounding marine environment	4.3.1.1.1, 4.3.1.1.2, 4.3.2.1.1, 4.3.2.1.2
	4.4	Ensure safety, reliability, and integration by providing emergency stop <10 s and redundancy on critical pathways, maintaining crew safety (including leak detection below trace limits and completion of safety training), achieving component life ≥ target years with scheduled maintenance every 168 h, and integrating with habitat power, control/monitoring, and spatial/structural interfaces (including alarm logging to the central dashboard, status latency <2 s, route compatibility for pipes/cables, and no conflict with other habitat zones)	Technical Quality	Fast protection, redundancy, crew safeguards, long-term reliability, and seamless interface with habitat systems are required to prevent hazardous failures and support dependable operation	4.4.1.1.1, 4.4.1.1.2, 4.4.2.1.1, 4.4.2.1.2, 4.4.3.1.1, 4.4.3.1.2, 4.5.2.1.1, 4.5.2.1.2, 4.5.3.1.1, 4.5.3.1.2

6A.2 Functional Flow

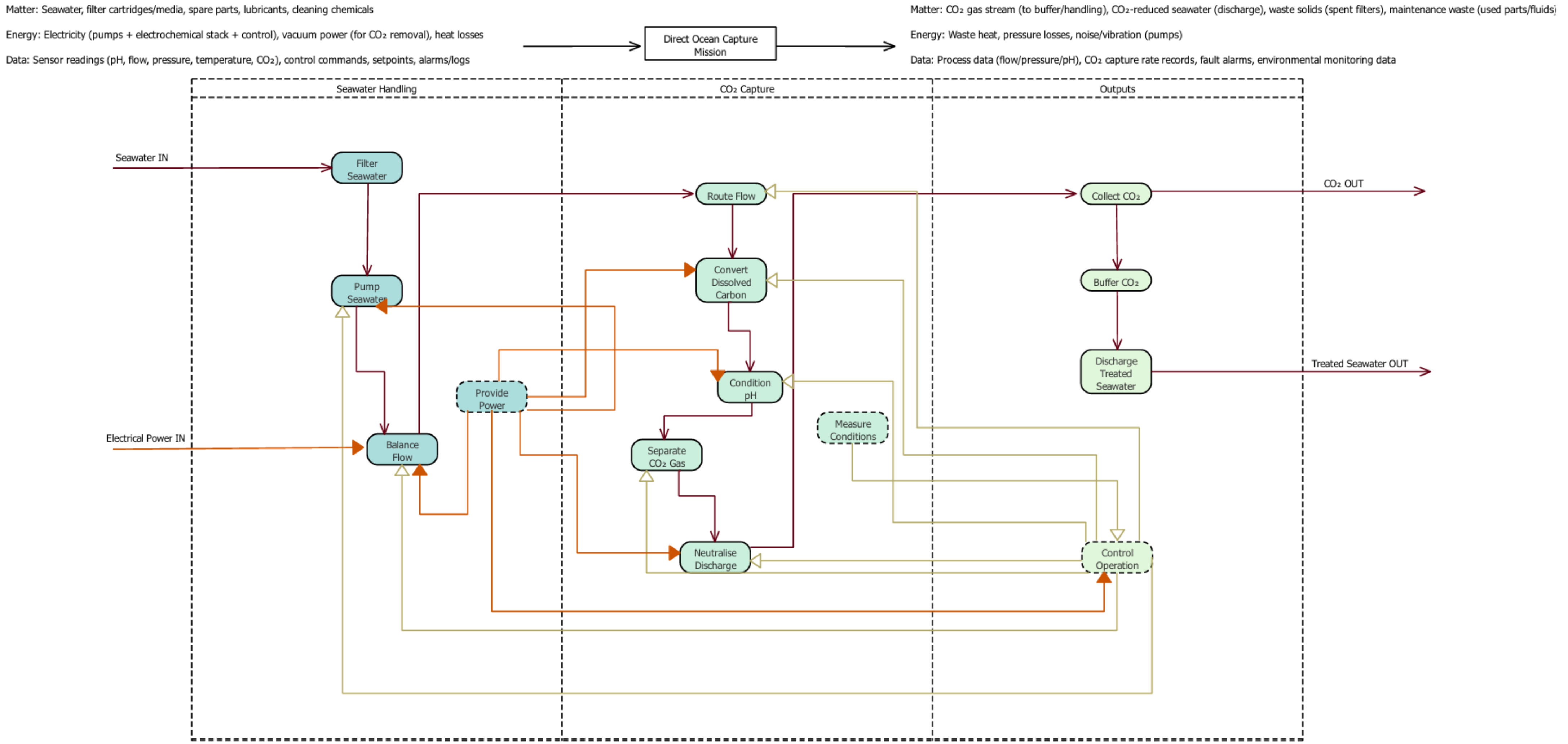


***Figure 21.** Direct Ocean Capture subsystem functional flow analysis.*

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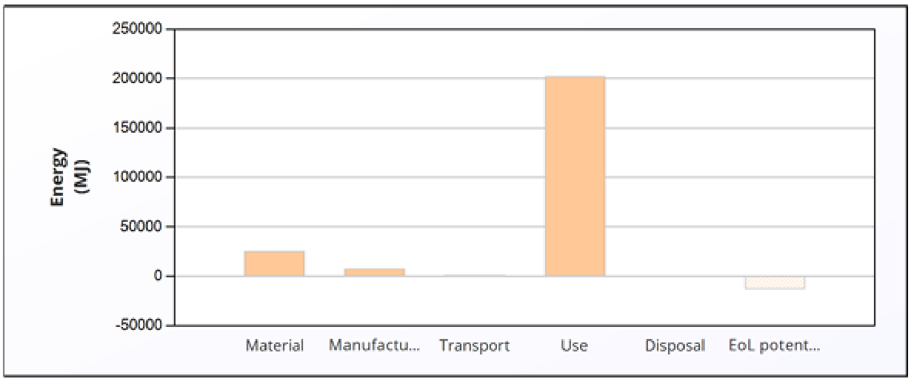
6A.3 Material Selection and Life Cycle Analysis

Table 27. *Direct Ocean Capture (DOC) Life Cycle Analysis Results.*

<table>
<tr><th>Lifecycle Phase & Analysis</th><th>Figure/Table</th><th>Critical Insights & Conclusions</th></tr>
<tr><td>Lifecycle Energy Burden Overview</td><td>



<table>
<tr><th></th><th>Energy (MJ/year)</th></tr>
<tr><td>Equivalent annual environmental burden (averaged over 1 year product life):</td><td>2.35e+05</td></tr>
</table>

</td><td>The DOC system's average annual energy burden is 2.35e+05 MJ/year over a 1-year product life. The chart reveals an extreme dominance of the Use Phase, which completely dwarfs all other lifecycle stages. This 1-year operational lifespan (versus 10-25 years for other subsystems) means the use phase impact is amortized over a much shorter period, making operational efficiency the absolute critical design parameter.</td></tr>
<tr><td>Material Composition Impact Assessment</td><td>

<table>
<tr><th>Component</th><th>Material</th><th>Recycled content* (%)</th><th>Part mass (kg)</th><th>Qty.</th><th>Total mass (kg)</th><th>Energy (MJ)</th><th>%</th></tr>
<tr><td>Structural frame / skid base</td><td>Stainless steel, austenitic, AISI 316, annealed</td><td>Virgin (0%)</td><td>95</td><td>1</td><td>95</td><td>8.3e+03</td><td>32.5</td></tr>
<tr><td>Electrochemical stack housing / pressure plates</td><td>Stainless steel, duplex, AISI 2205, annealed</td><td>Virgin (0%)</td><td>55</td><td>1</td><td>55</td><td>4e+03</td><td>15.6</td></tr>
<tr><td>Membrane stack (AEM/CEM/BPM)</td><td>Stainless steel, duplex, AISI 2205, annealed</td><td>Virgin (0%)</td><td>8</td><td>1</td><td>8</td><td>5.8e+02</td><td>2.3</td></tr>
<tr><td>Membrane contactor module</td><td>PP (65-70% barium sulfate)</td><td>Virgin (0%)</td><td>15</td><td>1</td><td>15</td><td>4.8e+02</td><td>1.9</td></tr>
<tr><td>Pumps (seawater + circulation)</td><td>Stainless steel, austenitic, AISI 316, annealed</td><td>Virgin (0%)</td><td>50</td><td>1</td><td>50</td><td>4.3e+03</td><td>17.1</td></tr>
<tr><td>Filters + housings</td><td>PP (65-70% barium sulfate)</td><td>Virgin (0%)</td><td>8</td><td>1</td><td>8</td><td>2.6e+02</td><td>1.0</td></tr>
<tr><td>Pipework + hoses + fittings</td><td>PE-HD (high molecular weight)</td><td>Virgin (0%)</td><td>45</td><td>1</td><td>45</td><td>3.7e+03</td><td>14.4</td></tr>
<tr><td>Valves + manifolds</td><td>Stainless steel, duplex, AISI 2205, annealed</td><td>Virgin (0%)</td><td>10</td><td>1</td><td>10</td><td>7.2e+02</td><td>2.8</td></tr>
<tr><td>Sensors pack (pH, flow, pressure, CO_2)</td><td>Stainless steel, austenitic, AISI 316, annealed</td><td>Virgin (0%)</td><td>4</td><td>1</td><td>4</td><td>3.5e+02</td><td>1.4</td></tr>
<tr><td>Control cabinet (PLC, PSU, wiring)</td><td>Stainless steel, austenitic, AISI 316, annealed</td><td>Virgin (0%)</td><td>20</td><td>1</td><td>20</td><td>1.7e+03</td><td>6.8</td></tr>
<tr><td>CO_2 handling (small buffer tank + tubing)</td><td>Stainless steel, austenitic, AISI 316L, annealed</td><td>Virgin (0%)</td><td>12</td><td>1</td><td>12</td><td>1e+03</td><td>4.1</td></tr>
<tr><td>Total</td><td></td><td></td><td></td><td>11</td><td>3.2e+02</td><td>2.5e+04</td><td>100</td></tr>
</table>

</td><td>The Structural Frame or Skid Base (95 kg), Electrochemical Stack Housing (55 kg), and Pumps (50 kg) are the heaviest components, together accounting for ~62% of total system mass. The manufacturing process table (Page 5) reveals that Pumps require 51.7% of total manufacturing energy despite being only the third-heaviest component, indicating highly energy-intensive production methods (metal powder forming) that warrant investigation for alternative fabrication techniques.</td></tr>
</table>

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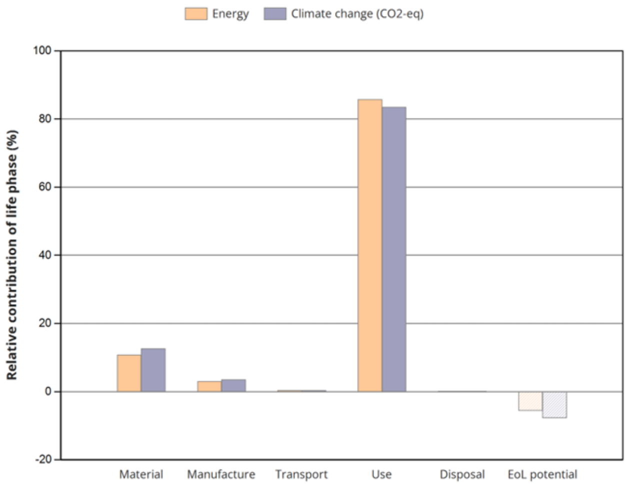
End-of-Life Process Energy & Circular Economy Potential

Component	End of life option	Energy (MJ)	%
Structural frame / skid base	Recycle	-5.7e+03	43.7
Electrochemical stack housing / pressure plates	Recycle	-2.7e+03	21.0
Membrane stack (AEM/CEM/BPM)	Recycle	-4e+02	3.1
Membrane contactor module	Landfill	0	0.0
Pumps (seawater + circulation)	Recycle	-3e+03	23.0
Filters + housings	Landfill	0	0.0
Pipework + hoses + fittings	Landfill	0	0.0
Valves + manifolds	Recycle	-5e+02	3.8
Sensors pack (pH, flow, pressure, CO_2)	Landfill	0	0.0
Control cabinet (PLC, PSU, wiring)	Landfill	0	0.0
CO_2 handling (small buffer tank + tubing)	Recycle	-7.2e+02	5.5
Total		**-1.3e+04**	**100**

Dual insight on end-of-life:
The Disposal table shows the Structural Frame (37.1%), Electrochemical Housing (21.5%), and Pumps (19.5%) require the most end-of-life process energy. Critically, the EOL Potential table reveals substantial material recovery value totaling -1.3e+04 MJ, with the same three components dominating: Structural Frame (43.7%), Pumps (23.0%), and Electrochemical Housing (21.0%).

Climate Change Impact (CO2-eq) Correlation



Component	Material	Recycled content* (%)	Part mass (kg)	Qty.	Total mass (kg)	Climate change (CO2-eq) (kg)	%
Structural frame / skid base	Stainless steel, austenitic, AISI 316, annealed	Virgin (0%)	95	1	95	7.1e+02	36.2
Electrochemical stack housing / pressure plates	Stainless steel, duplex, AISI 2205, annealed	Virgin (0%)	55	1	55	3.6e+02	18.3
Membrane stack (AEM/CEM/BPM)	Stainless steel, duplex, AISI 2205, annealed	Virgin (0%)	8	1	8	52	2.7
Membrane contactor module	PP (65-70% barium sulfate)	Virgin (0%)	15	1	15	19	1.0
Pumps (seawater + circulation)	Stainless steel, austenitic, AISI 316, annealed	Virgin (0%)	50	1	50	3.7e+02	19.0
Filters + housings	PP (65-70% barium sulfate)	Virgin (0%)	8	1	8	10	0.5
Pipework + hoses + fittings	PE-HD (high molecular weight)	Virgin (0%)	45	1	45	1e+02	5.3
Valves + manifolds	Stainless steel, duplex, AISI 2205, annealed	Virgin (0%)	10	1	10	65	3.3
Sensors pack (pH, flow, pressure, CO_2)	Stainless steel, austenitic, AISI 316, annealed	Virgin (0%)	4	1	4	30	1.5
Control cabinet (PLC, PSU, wiring)	Stainless steel, austenitic, AISI 316, annealed	Virgin (0%)	20	1	20	1.5e+02	7.6
CO_2 handling (small buffer tank + tubing)	Stainless steel, austenitic, AISI 316L, annealed	Virgin (0%)	12	1	12	90	4.6
Total				11	3.2e+02	2e+03	100

The climate change analysis reveals that the Structural Frame (36.2%), Pumps (19.0%), and Electrochemical Housing (18.3%) dominate material-related CO2-eq emissions.

These three stainless steel components together account for 73.5% of the material carbon footprint. The Control Cabinet adds another 7.6%, indicating that, despite being primarily monitoring equipment, its electronics and housing contribute non-negligible emissions.

6A.4 Design

Calculations

The DOC skid is assumed seabed-mounted in the Ionian Sea at 200 m depth (worst case for pressure rating).

Selected DOC approach

The selected Direct Ocean Capture (DOC) approach is an electrochemical acidification + degassing system. Seawater is first filtered (pretreatment) and then pumped through an electrochemical stack (stack housing/pressure plates with an AEM/CEM/BPM membrane stack). The stack creates acid/base streams, where the acidic stream converts dissolved inorganic carbon in seawater into dissolved CO_2. This acidified stream is then sent to a membrane contactor/degasser to extract CO_2 gas, which is routed to a small CO_2 buffer/handling line. The remaining seawater is neutralised (via controlled mixing with the base stream) and discharged back to the ocean, while MRV monitoring is performed using flow, pressure, pH, temperature and salinity/conductivity sensors, with data logged by the control cabinet for reporting and verification.

Assumptions

- Target capture rate: $\mathrm{m_{CO_2}} = 1000\ \mathrm{kg/day}$
- Seawater DIC $= 2\ \mathrm{mmol/L} = 2\ \mathrm{mol/m^3}$
- Conversion efficiency: $\mathrm{eta_{conv}} = 0.80$
- Degasser recovery: $\mathrm{eta_{rec}} = 0.90$
- Seawater density: rho $= 1025\ \mathrm{kg/m^3}$
- Gravity: $\mathrm{g} = 9.81\ \mathrm{m/s^2}$
- Pump head (local loop losses only): $\mathrm{H} = 20\ \mathrm{m}$
- Pump efficiency: $\mathrm{eta_{pump}} = 0.60$
- Stack specific energy $= 5\ \mathrm{kWh/kg\ CO_2}$
- Aux (vacuum/controls) $= 0.5\ \mathrm{kWh/kg\ CO_2}$

Seawater intake flowrate needed (defines intake/pretreatment duty)

CO_2 available per $\mathrm{m^3}$ (ideal):

$$\mathrm{m_{CO_{2_{ideal}}}} = (2\ \mathrm{mol/m^3})(0.044\ \mathrm{kg/mol}) = 0.088\ \mathrm{kg/m^3}$$

Captured CO_2 per $\mathrm{m^3}$ (with efficiencies):

$$\mathrm{m_{CO_{2_{actual}}}} = 0.088(0.80)(0.90) = 0.06336\ \mathrm{kg/m^3}$$

Seawater volume per day:

$$\mathrm{V_{day}} = \frac{1000}{0.06336} = 15787\ \mathrm{m^3/day}$$

Flowrate:

$$\mathrm{Q} = \frac{15787}{24} = 658\ \mathrm{m^3/h}$$

$$\mathrm{Q} = \frac{15787}{86400} = 0.183\ \mathrm{m^3/s} = 183\ \mathrm{L/s}$$

Result: DOC intake/pretreatment should be designed for $\sim 660\ \mathrm{m^3/h}$ ($\approx 183\ \mathrm{L/s}$) seawater.
Pretreatment definition: strainer+ filter housing to protect pumps/stack (micron rating to be selected during detailed design based on fouling risk and pump/stack protection).

Pump power (intake/process pump)

Equation:

$$P_{pump} = (rho * g * Q * H) / eta_{pump}$$

Substitute:

$$P_{pump} = (1025 * 9.81 * 0.183 * 20) / 0.60 = 61\ kW$$

Add margin for extra losses → design pump electrical power ≈ 1.0 kW.

Important note: depth (200 m) mainly affects pressure rating, not pump head, if intake/discharge are at the same depth and you are not pumping to surface.

Electrochemical + auxiliary power

Stack:

$$E_{stack} = 5 * 1000 = 5000\ kWh/day$$
$$P_{stack_{avg}} = 5000/24 = 208\ kW$$

Aux:

$$E_{aux} = 0.5 * (1000) = 500\ kWh/day$$
$$P_{aux_{avg}} = 500/24 = 20.8\ kW$$

Total average:

$$P_{total_{avg}} \approx 61 + 208 + 20.8 = 289.8\ kW$$
$$Result: DOC\ electrical\ load \approx 300\ kW\ average\ (for\ 1000\ kg/day).$$

CO_2 handling (gas flow + buffer sizing)

Capture rate per hour:

$$m_{CO2_h} = 1000/24 = 41.7\ kg/h$$

Moles per hour:

$$n = (41700\ g/h) / (44\ g/mol) = 948\ mol/h$$

Equivalent gas volume at 1 bar, 25°C:

$$\dot{V}_{1bar} \approx (n * R * T) / P \approx 23.2\ m^{\wedge}3/h \approx 23200\ L/h$$

At 200m, seabed pressure is ≈ 21 bar absolute (≈20 bar gauge), equivalent volume becomes:

$$\dot{V}_{21bar} \approx 23200/21 \approx 1100\ L/h$$

Buffer sizing (simple): a 2000 L CO_2 buffer provides ~ 1-2 hour buffering at seabed conditions (order-of-magnitude design).

CO_2 handling definition: CO_2 from degasser → buffer tank + tubing → CO_2 handling/storage interface.

Discharge strategy (what we do + what we check)

Strategy: neutralise acidified stream by recombining with base stream (or controlled mixing) before discharge.

Outlet control targets (concept):

- Outlet pH kept close to ambient (e.g., Delta pH <= 0.2)
- Avoid abnormal temperature change (e.g., Delta T <= 2°C)
- Salinity/conductivity change minimal (target band set by team)

Control action: if outlet limits exceeded → reduce stack power / adjust flow / shutdown.

MRV (monitoring/reporting/verification using ocean health proxies)

Ocean health proxies to monitor:

- pH (in/out)
- temperature (in/out)
- salinity / conductivity (in/out)
- flowrate (in/out)
- pressure drop across filter/stack (fouling indicator)
- optional: turbidity, dissolved oxygen

Reporting method (simple):

- Primary: CO_2 captured from CO_2 line (gas flow meter or cylinder mass change)
- Cross-check: processed seawater volume × assumed carbon removal:

$$m_{CO2_{removed}} \approx Q * \Delta * 44$$

$$(where\ \Delta C\ is\ the\ assumed\ mol/m^3\ carbon\ removed)$$

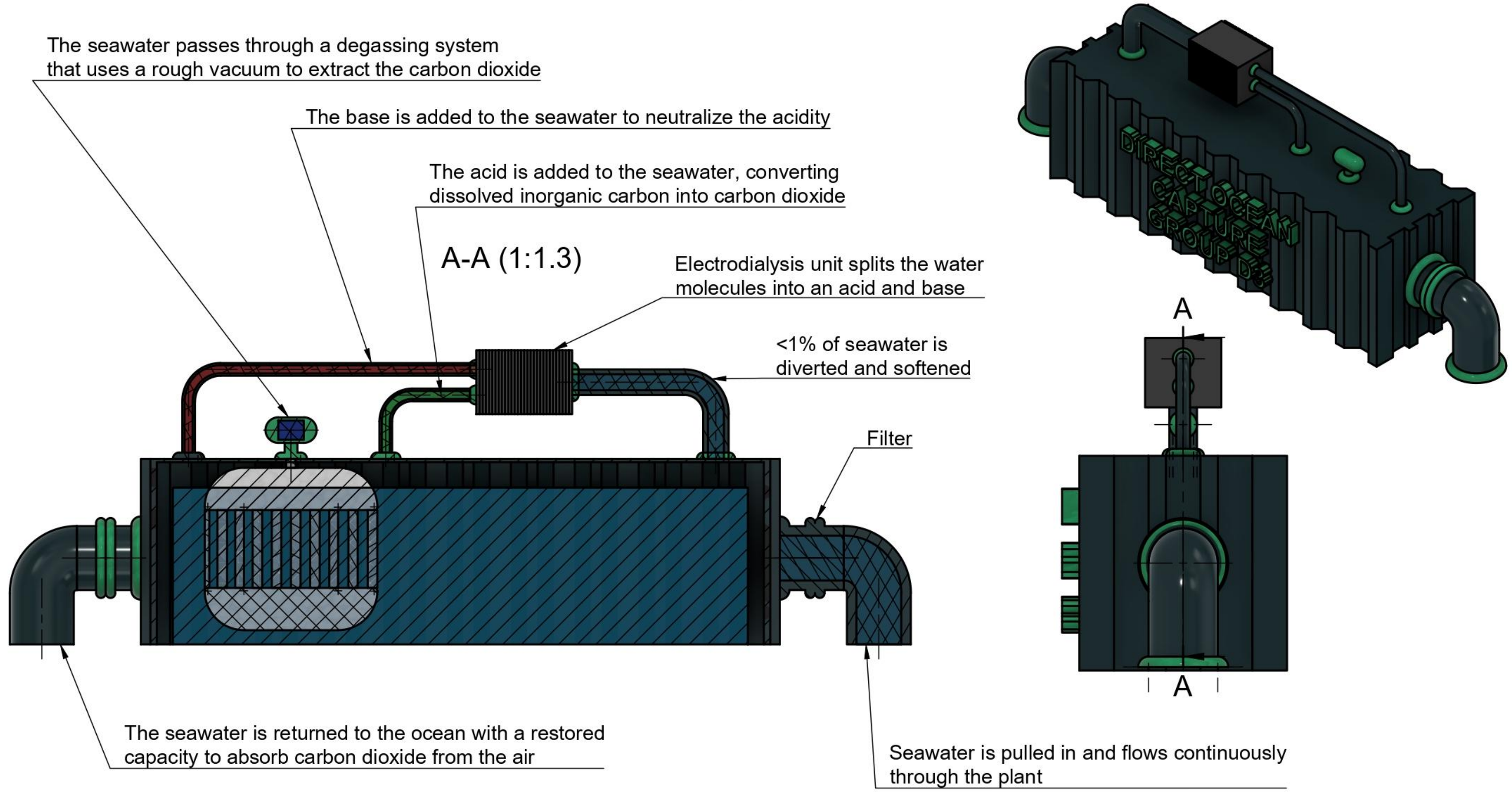
Data logging: alarms + trends stored by control cabinet (MRV record).

Equipment

Table 28. *Equipment List for direct ocean capture work package including product number.*

Subsystem	Component	Representative Part Number	Description
Intake / Pretreatment	Filters + housings (incl. strainer)	Eaton TOPLINE (bag filter housing) + Eaton Model 72 (strainer)	Removes particles before pumps/stack (choose pressure-rated parts), (scaled for 1000kg/day).
Pumping	Pumps (seawater + circulation)	Grundfos CRN (316 SS series)	Moves seawater through the DOC skid (select model for flow/head), (scaled for 1000kg/day).
Routing	Pipework + hoses + fittings + valves/manifolds	Parker A-LOK (fittings series) + Swagelok SS-63TS8 (valve example)	Routes flow and isolates sections for maintenance/control.
Electrochemical capture	Stack housing/plates + membrane stack	Custom: DOC-STACK-ASSY-01 + Membrane supplier (TBD)	Core unit that creates acid/base zones to release CO_2 from seawater carbon (scaled for 1000kg/day).
CO_2 separation	Membrane contactor module (degasser)	3M Liqui-Cel EXF-4×13 (example)	Strips CO_2 gas from the treated stream (scaled for 1000kg/day).
CO_2 handling	CO_2 buffer + tubing	Standard CO_2 receiver/buffer (PN-rated)	Short-term buffer and line to CO_2 handling/storage (scaled for 1000kg/day).
Monitoring	Sensors pack (pH, flow, pressure, temp, conductivity)	E+H Promag W (flow) + Ceraphant (pressure) + Hach pH (examples)	Measures seawater + system health for MRV and control.
Control	Control cabinet (PLC, PSU, enclosure)	Allen-Bradley Micro850 (example) + Hoffman 316 enclosure (example)	Controls pumps/valves/stack and logs alarms/data.
Structure	Structural frame / skid base	Custom: DOC-SKID-01	Skid frame to mount everything (lifting points, mounts).

Diagrams



Modular Direct Ocean Capture (DOC) Processor		
Dept. School of Engineering University of Birmingham Dubai	**Integrated Design Project 2 (04 30338)** **Module Lead:** Dr. Archibong	**Project Title** Future Underwater Habitat
Created by Group D3 Academic year 2025/26	**Subsystem** Direct Ocean Capture	**3rd Angle Projection**

Notes
Maintenance: filter mesh and electrodialysis membranes require periodic inspection for biofouling.

WP6: MISSION MODULE 6B

6B.1 Requirements Specification

Table 29. *Tabulated requirements specification for the Underwater Data Center subsystem.*

Package	Requirement number	Description	Requirement Type	Rationale	Traceability to Objective Tree
Mission Module - Underwater Data Center (6B)	**6B.1**	The subsystem shall provide continuous data storage/processing with ≥99% availability, support uninterrupted operation for >X days, limit performance degradation to ≤10% over 24h continuous use, and restore normal operation within 10-15 min after shutdown	Functional Technical	Ensures reliable mission data handling with minimal downtime and controlled degradation for autonomous underwater deployment	5.1.1.1.1, 5.1.1.1.2, 5.1.1.2.1, 5.1.1.2.2, 5.1.2.1.1, 5.1.2.1.2
	6B.2	The subsystem shall limit energy to ≤X kWh/TB processed, maintain thermal levels within design limits, minimize active cooling during nominal operation, and control power demand variation to ±X% of nominal load	Technical Quality	Optimizes energy efficiency and power stability to reduce habitat infrastructure loading while preventing thermal damage	5.2.1.1.1, 5.2.1.1.2, 5.2.2.1.1, 5.2.2.1.2, 5.2.2.2.1
	6B.3	The subsystem shall limit surrounding water temperature change to ≤ΔT°C at Y m distance, prevent release of toxic substances during all operations, and control biofouling without harmful coatings	Customer Quality	Ensures environmental sustainability and regulatory compliance for long-term marine deployment	5.3.1.1.1, 5.3.1.2.1, 5.3.1.2.2, 5.3.2.1.1, 5.3.2.1.2
	6B.4	The subsystem shall continuously monitor T/P/H/leakage, execute emergency shutdown ≤10s on critical faults, tolerate single-point non-critical failures, achieve ≥X years service life, comply with habitat interfaces, transmit telemetry ≤2s latency, and enable modular hot-swap of non-critical components	Functional Technical Quality	Provides safety, fault tolerance, and seamless habitat integration for resilient long-duration operation	5.4.1.1.1, 5.4.1.1.2, 5.4.2.1.1, 5.4.2.1.2, 5.4.3.1.1, 5.4.3.1.2, 5.5.1.1.1, 5.5.1.1.2, 5.5.3.1.1, 5.5.3.1.2

6B.2 Functional Flow

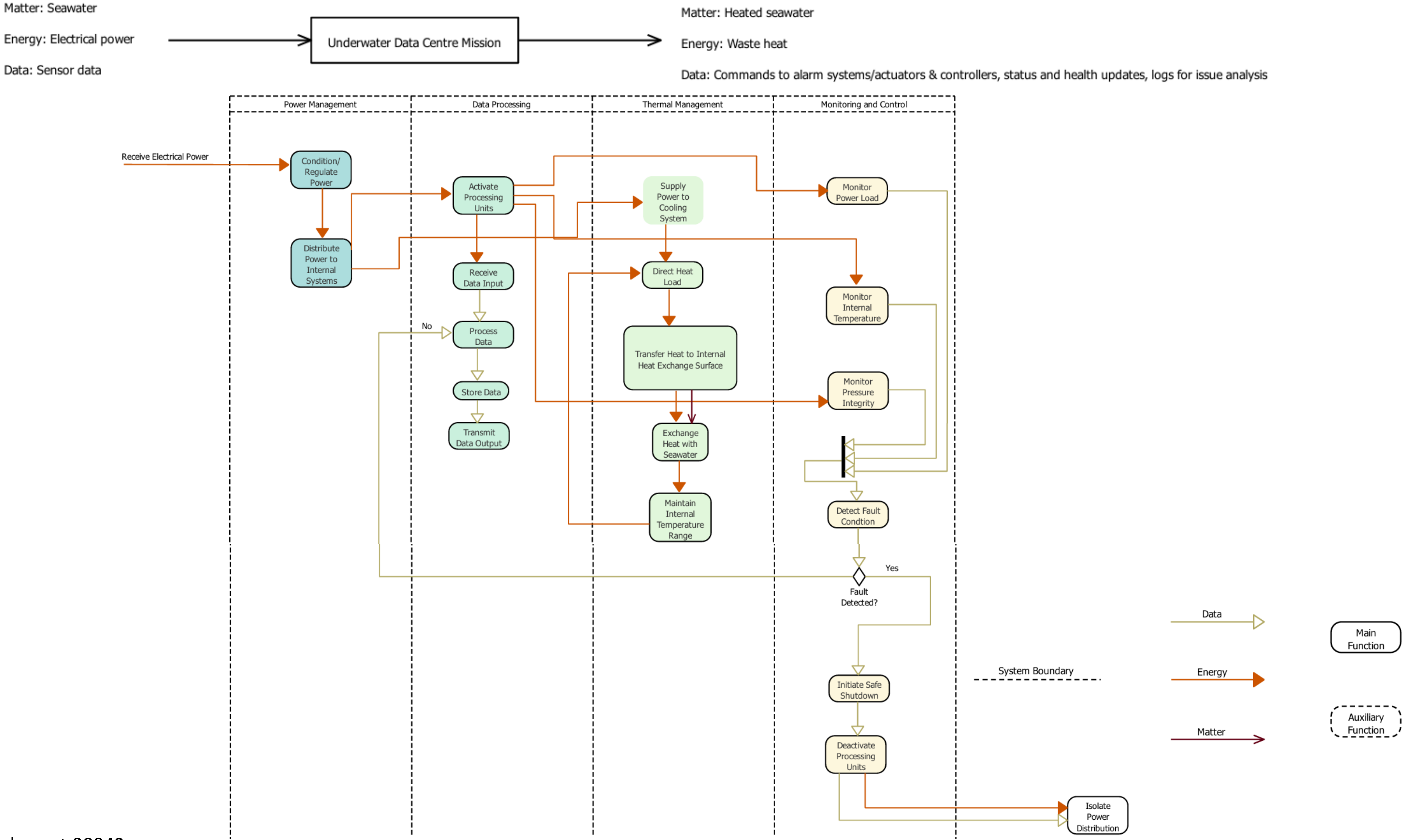


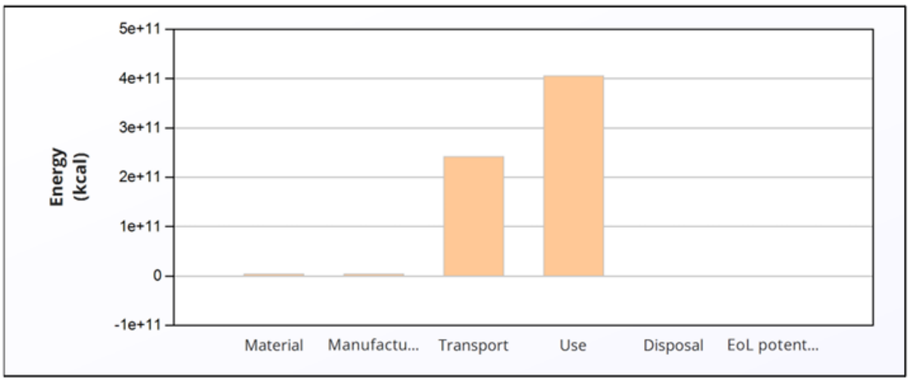
Figure 22. *Underwater Data Center subsystem functional flow analysis.*

6B.3 Material Selection and Life Cycle Analysis

Table 30. *Underwater Data Center (UDC) Life Cycle Analysis Results.*

Lifecycle Phase & Analysis	Figure/Table	Critical Insights & Conclusions
Lifecycle Energy Burden Overview	(see chart and table below)	The UDC system's average annual energy burden is 3.29e+10 kcal/year over a 20-year product life. The chart reveals that the Use Phase is the overwhelmingly dominant contributor, followed by a significant contribution from the Material Phase, while Manufacturing, Transport, and Disposal are comparatively negligible. This profile is characteristic of energy-intensive infrastructure with continuous operational demands.
Material Composition Impact Assessment	(see table below)	The Server Racks alone account for 42.6% of total material energy, despite weighing 7.5e+05 lb and being made from low alloy steel with 40% recycled content. The Fibre Optic Cables contribute 36.3%, and the Pressure Hull (Aluminum) adds 8.6%. These three components represent over 87% of the material burden. The massive contribution of fibre optics is notable given its relatively low mass (2.4e+06 lb versus 7.5e+05 lb for server racks).

Lifecycle Energy Burden Overview



	Energy (kcal/year)
Equivalent annual environmental burden (averaged over 20 year product life):	3.29e+10

Material Composition Impact Assessment

Component	Material	Recycled content* (%)	Part mass (lb)	Qty.	Total mass (lb)	Energy (kcal)	%
Pressure Hull (outer shell)	Aluminum, 5083, H111	70.0%	3.8e+04	1	3.8e+04	3.8e+08	8.6
Internal Structural Frame	Stainless steel, austenitic, AISI 316L, annealed	50.0%	8e+03	1	8e+03	4.9e+07	1.1
Server Racks	Low alloy steel, SAE 4130, cast, normalized & tempered	40.0%	1.5e+04	50	7.5e+05	1.9e+09	42.6
Servers and Electronics	Computer, desktop, without screen	Reused part	3e+04	1	3e+04	0	0.0
Power Cables (per meter)	Cable, unspecified	Reused part	5e+03	5000	2.5e+07	0	0.0
Fibre Optic Cables (per meter)	Cable, data cable in infrastructure	Virgin (0%)	8e+02	3000	2.4e+06	1.6e+09	36.3
Heat Exchanger System	Titanium, commercial purity, Grade 2	30.0%	6e+03	1	6e+03	3.4e+08	7.8
Pump System (for liquid cooling)	Stainless steel, martensitic, AISI 418, tempered at 649°C	40.0%	4e+03	4	1.6e+04	9.6e+07	2.2
Concrete Seabaed Foundation	High density concrete	25.0%	9e+04	1	9e+04	7.4e+06	0.2
Ballast Weights	Carbon steel, AISI 1050, normalized	50.0%	2.5e+04	1	2.5e+04	5.7e+07	1.3
Total				**8060**	**2.8e+07**	**4.4e+09**	**100**

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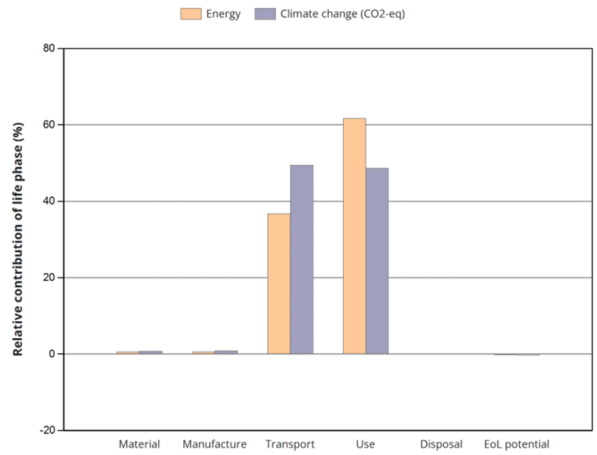
End-of-Life Process Energy & Circular Economy Potential

Component	End of life option	Energy (kcal)	%
Pressure Hull (outer shell)	Recycle	-1.6e+08	11.2
Internal Structural Frame	Recycle	-2.6e+07	1.8
Server Racks	Recycle	-1.1e+09	75.1
Servers and Electronics	Reuse	0	0.0
Power Cables (per meter)	Landfill	0	0.0
Fibre Optic Cables (per meter)	Landfill	0	0.0
Heat Exchanger System	Recycle	-8.5e+07	6.0
Pump System (for liquid cooling)	Recycle	-5.5e+07	3.8
Concrete Seabaed Foundation	Landfill	0	0.0
Ballast Weights	Recycle	-3e+07	2.1
Total		**-1.4e+09**	**100**

Dual insight on end-of-life:

The Disposal table shows the Power Cables dominate end-of-life process energy (82.0%), followed by Server Racks (8.6%) and Fibre Optic Cables (7.9%). Critically, the EOL Potential table reveals that while power and fibre optic cables require significant disposal energy, they yield zero material recovery credits as they are sent to landfill. Conversely, the Server Racks provide 75.1% of total EOL credits (-1.1e+09 kcal).

Climate Change Impact (CO2-eq) Correlation



Component	Material	Recycled content* (%)	Part mass (lb)	Qty.	Total mass (lb)	Climate change (CO2-eq) (lb)	%
Pressure Hull (outer shell)	Aluminum, 5083, H111	70.0%	3.8e+04	1	3.8e+04	2.3e+05	9.1
Internal Structural Frame	Stainless steel, austenitic, AISI 316L, annealed	50.0%	8e+03	1	8e+03	3.8e+04	1.5
Server Racks	Low alloy steel, SAE 4130, cast, normalized & tempered	40.0%	1.5e+04	50	7.5e+05	1.2e+06	49.5
Servers and Electronics	Computer, desktop, without screen	Reused part	3e+04	1	3e+04	0	0.0
Power Cables (per meter)	Cable, unspecified	Reused part	5e+03	5000	2.5e+07	0	0.0
Fibre Optic Cables (per meter)	Cable, data cable in infrastructure	Virgin (0%)	8e+02	3000	2.4e+06	7.2e+05	28.6
Heat Exchanger System	Titanium, commercial purity, Grade 2	30.0%	6e+03	1	6e+03	1.7e+05	6.6
Pump System (for liquid cooling)	Stainless steel, martensitic, AISI 418, tempered at 649°C	40.0%	4e+03	4	1.6e+04	7.3e+04	2.9
Concrete Seabaed Foundation	High density concrete	25.0%	9e+04	1	9e+04	8.3e+03	0.3
Ballast Weights	Carbon steel, AISI 1050, normalized	50.0%	2.5e+04	1	2.5e+04	3.7e+04	1.5
Total				**8060**	**2.8e+07**	**2.5e+06**	**100**

The climate change analysis strongly correlates with energy data. The Server Racks contribute 49.5% of material-related CO2-eq, followed by the Fibre Optic Cables at 28.6% and the Pressure Hull at 9.1%. The Heat Exchanger System (Titanium) adds 6.6% despite its relatively low mass (6e+03 lb), confirming titanium's high carbon intensity per unit mass. The Servers and Electronics show zero material impact due to being classified as "Reused part," highlighting the significant climate benefit of component reuse strategies.

6B.4 Design

Calculations

The UDC design calculations are based on a few assumptions which are provided in the calculations themselves and stated clearly.

IT Power Load:

Assuming that the power load of an average server rack is 24.5 kW at high loads and given that there are 50 server racks in the system (*Intelligent high-density power distribution unleashed for AI-HPC*. (2024, May 15)), the power load of the entire system would be modelled as:
$P_{IT} = 50 \times 24.5 = 1225$ kW

Heat Generation:

Given that almost all electrical energy in a data centre becomes heat energy (Judge, P (2024, February 13), the heat generation of the system can be modelled as:
$Q_{heat} = P_{IT}$

Since the IT load of the system is 1225 kW, there would be 1225 kW heat energy to dissipate continuously.

Cooling Loop Flow Rate:

Given that the methods of cooling were selected to be immersion cooling, seawater heat exchange and dual-loop cooling, the required coolant flow rate can be modelled as:

$Q = \dot{m} \times c_p \times \Delta T \times Q$

Where:

- Q = heat (W)
- $\dot{m}$ = mass flow rate (kg/s)
- c_p = specific heat
- ΔT = allowable temperature rise

To model this for a scenario where $\Delta T = 5°C$
c_p (water) = 4186 J/kgK

$\dot{m} = 1{,}225{,}000/(4186 \times 5) = 58.5$ kg/s (3 sig figs)

Pump Power Requirement:

Now given the flow rate of the coolant loop, the pressure drop must be calculated as well and can be modelled as:
$P = \dot{V} \times \Delta P$

Where:

- P = Power (W)
- $\dot{V}$ = Volumetric flow rate
- ΔP = Pressure Drop

Given the mass flow rate, the volumetric flow rate can be calculated to be:
$\dot{V} = \dot{m}/\boldsymbol{\rho}$

$58.5/1000 = 0.0585\ m^3/s$

The difference in pressure can be modelled as NOAA Ocean Exploration. (2020, December 16):

ΔP at 100m is 1,013 kPa (1 atmosphere is 101.3 kPa) and 1,520 at 150m. However, the pressure drop for the system will be calculated based on if the system was at 150m depth.

Therefore, power would be calculated to be: $0.0585 \times 1520 = 88.92$ W

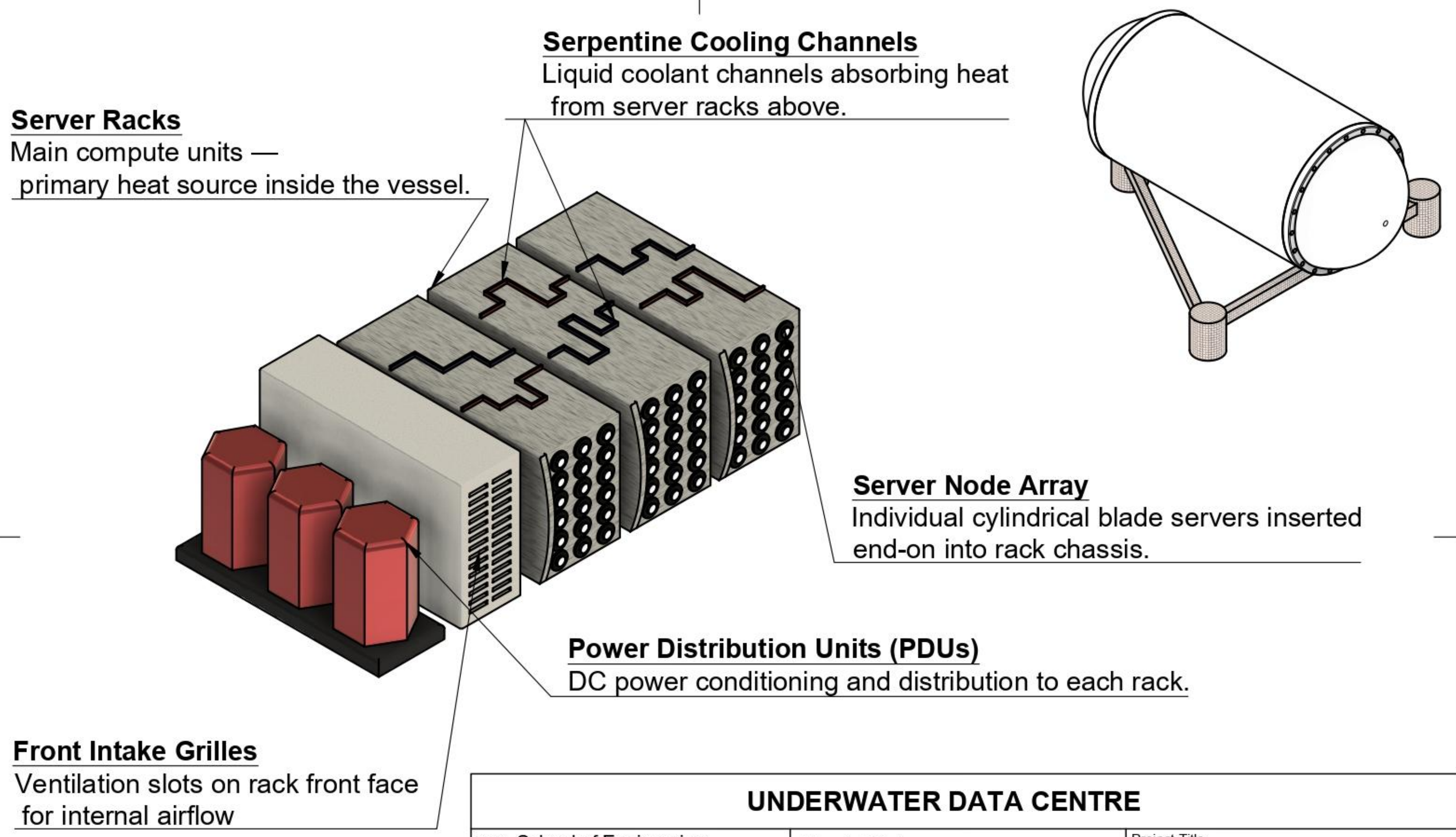
Project Natick Phase II achieved a Power Usage Effectiveness (PUE) of 1.07 and demonstrated effective seawater cooling without potable water use, with lower hardware failure rates than land-based systems. Using these metrics, for a UDC designed with a 1225 kW IT load, estimated heat rejection requirements can be derived. With a steady-state cooling load of 1225 kW and assuming a seawater temperature rise of 5 °C, the necessary seawater flow rate for heat rejection is approximately 58.5 kg/s (0.0585 m^3/s). This provides a basis for sizing the seawater heat exchanger and pump capacity.

Equipment

Table 31. *Equipment List for underwater data center work package including product number.*

Subsystem	Component	Representative Part Number	Description
Power Management	Power Cables	Cable, unspecified	Provides power to system
Data Processing	Fibre Optic Cables	Cable, data cable in infrastructure	Transfers data throughout system and to other systems/users
	Server Racks	Low alloy steel, SAE 4130, cast, normalized & tempered	Holds the servers
	Servers and Electronics	Computer, desktop, without screen	Provide the data handling and storage
Thermal Management	Heat Exchanger System	Titanium, commercial purity, Grade 2	Transfers heat out of the system
	Pump System (for liquid cooling)	Stainless steel, martensitic, AISI 418, tempered at 649°C	Cools the system using seawater which is pumped throughout
Monitoring & Control	Sensor array	Sensors, unspecified	Senses system conditions

Diagrams



Notes:
Operating depth = 200 m / 19.92 atm external pressure
Internal atmosphere = 1 atm (crew-safe)

UNDERWATER DATA CENTRE			
Dept. School of Engineering University of Birmingham Dubai	Integrated Design Project 2 (04 30338)	Module Lead: Dr. Archibong	Project Title: Future Underwater Habitat
Created by: Group D3 Academic year 2025/26	Subsystem: Mission Module Underwater Data Centre		3rd Angle Projection

WP6: MISSION MODULE 6C

6C.1 Requirements Specification

Table 32. *Tabulated requirements specification for the Deep Sea Mining subsystem.*

Package	Requirement number	Description	Requirement Type	Rationale	Traceability to Objective Tree
Mission Module – Deep Sea Mining (6C)	**6C.1**	The crawler subsystem shall provide secure automated docking accommodating ±5° misalignment with automated latching/unlatching, plus continuous wireless charging pads and high-bandwidth data transfer (>1 Gbps) at docking stations	Functional Technical	Enables reliable crawler operations with seamless power/data exchange, critical for continuous mining missions	6.1.1.1, 6.1.1.2, 6.1.2.1, 6.1.2.2
	6C.2	The slurry processing subsystem shall include intake filters excluding debris >2 cm, pre-treatment with desanding/desilting stages, plus on-site mineral separation achieving ≥85% recovery with temporary storage capacity for ≥1 week supply	Functional Technical	Facilitates efficient slurry handling from intake through processing and storage for sustained mineral extraction operations	6.2.1.1, 6.2.1.2, 6.2.2.1, 6.2.2.2
	6C.3	The environmental subsystem shall implement zero-discharge closed-loop slurry handling using non-toxic marine-safe materials, plus deploy multi-parameter water quality sensors monitoring turbidity, pH, heavy metals, and dissolved oxygen around habitat	Quality	Ensures environmental compliance through contamination prevention and real-time habitat impact monitoring	6.3.1.1, 6.3.1.2, 6.3.2.1, 6.3.2.2
	6C.4	The safety subsystem shall provide backup power for critical mining systems, dual-redundant controls, emergency bulkheads isolating mining module, and recovery winch system for stranded crawlers	Technical	Guarantees operational safety through redundancy, fault tolerance, and emergency response capabilities	6.4.1.1, 6.4.1.2, 6.4.2.1, 6.4.2.2

6C.2 Functional Flow

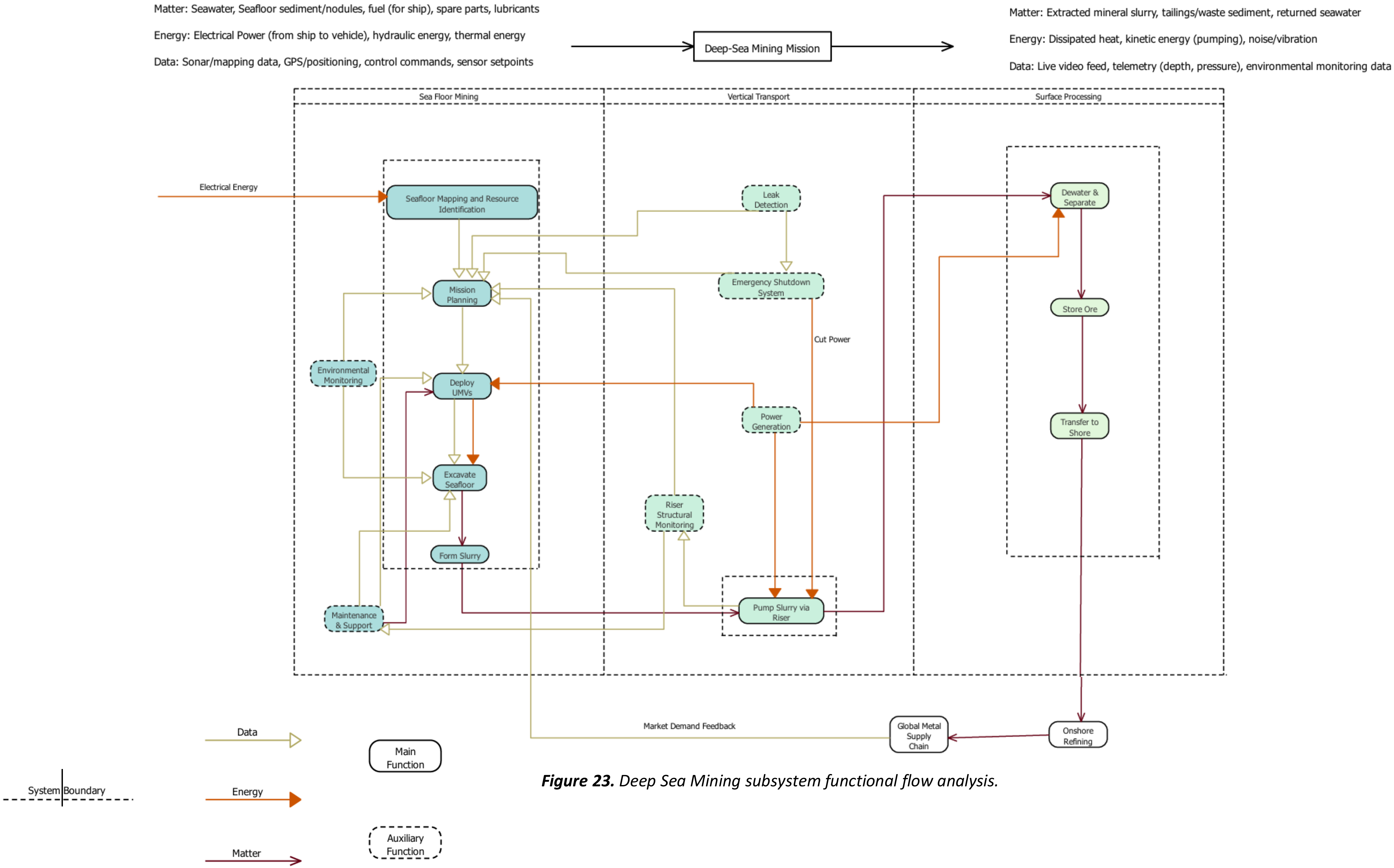
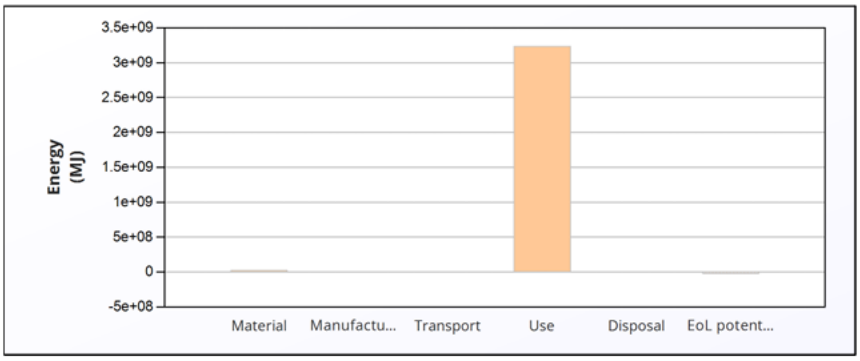


Figure 23. *Deep Sea Mining subsystem functional flow analysis.*

6C.3 Material Selection and Life Cycle Analysis

Table 33. *Deep Sea Mining (DSM) Life Cycle Analysis Results.*

<table>
<tr><th>Lifecycle Phase & Analysis</th><th>Figure/Table</th><th>Critical Insights & Conclusions</th></tr>
<tr><td>Lifecycle Energy Burden Overview</td><td>



<table>
<tr><th></th><th>Energy (MJ/year)</th></tr>
<tr><td>Equivalent annual environmental burden (averaged over 20 year product life):</td><td>1.63e+08</td></tr>
</table>

</td><td>The DSM system's average annual energy burden is 1.63e+08 MJ/year over a 20-year product life. The chart reveals that the Use Phase is the overwhelmingly dominant contributor, accounting for nearly the entire lifecycle energy consumption. The Material, Manufacturing, Transport, and Disposal phases are negligible in comparison, indicating that operational efficiency and power management are the absolute critical design parameters.</td></tr>
<tr><td>Material Composition Impact Assessment</td><td>

<table>
<tr><th>Component</th><th>Material</th><th>Recycled content* (%)</th><th>Part mass (kg)</th><th>Qty.</th><th>Total mass (kg)</th><th>Energy (MJ)</th><th>%</th></tr>
<tr><td>Structural frame</td><td>Low alloy steel, SAE 4130, cast, normalized & tempered</td><td>30.0%</td><td>1.8e+04</td><td>1</td><td>1.8e+04</td><td>4.6e+05</td><td>1.8</td></tr>
<tr><td>Pressure housings</td><td>Titanium, alpha-beta alloy, Ti-6Al-4V, cast</td><td>10.0%</td><td>3e+03</td><td>1</td><td>3e+03</td><td>2e+06</td><td>7.8</td></tr>
<tr><td>Crawler tracks</td><td>Stainless steel, martensitic, AISI 410, hard temper</td><td>25.0%</td><td>4e+03</td><td>2</td><td>8e+03</td><td>2.9e+05</td><td>1.1</td></tr>
<tr><td>Thruster housings</td><td>Aluminum, A380.0, die cast, F</td><td>35.0%</td><td>1.2e+03</td><td>6</td><td>7.2e+03</td><td>9e+05</td><td>3.5</td></tr>
<tr><td>Thruster motors</td><td>Intermediate alloy, Fe-9Ni-4Co-0.20C steel, quenched & tempered</td><td>20.0%</td><td>9e+02</td><td>6</td><td>5.4e+03</td><td>6.5e+05</td><td>2.6</td></tr>
<tr><td>Hydraulic system</td><td>High alloy steel, AerMet 100, solution treated & aged</td><td>25.0%</td><td>2.5e+03</td><td>1</td><td>2.5e+03</td><td>6.9e+05</td><td>2.7</td></tr>
<tr><td>Collection system</td><td>Iron-base-superalloy, Cr-Ni alloy, A-286, solution treated & aged</td><td>25.0%</td><td>6e+03</td><td>1</td><td>6e+03</td><td>6.4e+05</td><td>2.5</td></tr>
<tr><td>Buoyancy modules</td><td>Epoxy/aramid fiber, UD prepreg, QI lay-up</td><td>Virgin (0%)</td><td>2e+03</td><td>1</td><td>2e+03</td><td>4.6e+05</td><td>1.8</td></tr>
<tr><td>Electrical systems</td><td>Integrated circuit, logic type</td><td>Virgin (0%)</td><td>8e+02</td><td>1</td><td>8e+02</td><td>1.9e+07</td><td>76.0</td></tr>
<tr><td>Cabling</td><td>Coated copper, copper, lead coated</td><td>30.0%</td><td>6e+02</td><td>1</td><td>6e+02</td><td>3e+04</td><td>0.1</td></tr>
<tr><td>Total</td><td></td><td></td><td></td><td>21</td><td>5.4e+04</td><td>2.5e+07</td><td>100</td></tr>
</table>

</td><td>The Electrical Systems alone account for an extraordinary 76.0% of total material energy, despite weighing only 800 kg. This is driven by the energy-intensive nature of integrated circuit and logic component manufacturing. The Pressure Housings (Titanium) contribute 7.8%, followed by the Thruster Housings (Aluminum) at 3.5%. While the Structural Frame is the heaviest component (1.8e+04 kg), its material energy contribution is relatively low (1.8%).</td></tr>
</table>

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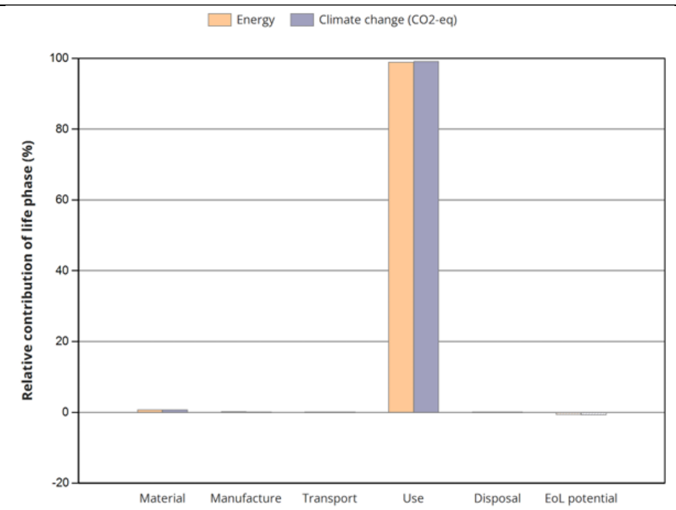
End-of-Life Process Energy & Circular Economy Potential

Component	End of life option	Energy (MJ)	%
Structural frame	Recycle	-2.8e+05	1.3
Pressure housings	Recycle	-5.9e+05	2.7
Crawler tracks	Recycle	-1.8e+05	0.8
Thruster housings	Recycle	-5.5e+05	2.5
Thruster motors	Recycle	-4.2e+05	1.9
Hydraulic system	Recycle	-4.3e+05	1.9
Collection system	Recycle	-4e+05	1.8
Buoyancy modules	Landfill	0	0.0
Electrical systems	Reuse	-1.9e+07	87.1
Cabling	Recycle	-2.1e+04	0.1
Total		**-2.2e+07**	**100**

<u>Dual insight on end-of-life:</u>
The Disposal table shows the Structural Frame (35.0%), Crawler Tracks (15.5%), and Thruster Housings (14.0%) require the most end-of-life process energy due to their mass. Critically, the EOL Potential table reveals that the Electrical Systems provide an overwhelming 87.1% of total EOL credits (-1.9e+07 MJ) through Reuse, despite their minimal disposal energy requirement.

Climate Change Impact (CO2-eq) Correlation



Component	Material	Recycled content* (%)	Part mass (kg)	Qty.	Total mass (kg)	Climate change (CO2-eq) (kg)	%
Structural frame	Low alloy steel, SAE 4130, cast, normalized & tempered	30.0%	1.8e+04	1	1.8e+04	3.3e+04	2.0
Pressure housings	Titanium, alpha-beta alloy, Ti-6Al-4V, cast	10.0%	3e+03	1	3e+03	1.2e+05	7.1
Crawler tracks	Stainless steel, martensitic, AISI 410, hard temper	25.0%	4e+03	2	8e+03	2.8e+04	1.7
Thruster housings	Aluminum, A380.0, die cast, F	35.0%	1.2e+03	6	7.2e+03	6.4e+04	3.9
Thruster motors	Intermediate alloy, Fe-9Ni-4Co-0.20C steel, quenched & tempered	20.0%	9e+02	6	5.4e+03	2.7e+04	1.6
Hydraulic system	High alloy steel, AerMet 100, solution treated & aged	25.0%	2.5e+03	1	2.5e+03	2e+04	1.2
Collection system	Iron-base-superalloy, Cr-Ni alloy, A-286, solution treated & aged	25.0%	6e+03	1	6e+03	4.8e+04	2.9
Buoyancy modules	Epoxy/aramid fiber, UD prepreg, QI lay-up	Virgin (0%)	2e+03	1	2e+03	2.5e+04	1.5
Electrical systems	Integrated circuit, logic type	Virgin (0%)	8e+02	1	8e+02	1.3e+06	78.0
Cabling	Coated copper, copper, lead coated	30.0%	6e+02	1	6e+02	2e+03	0.1
Total				**21**	**5.4e+04**	**1.7e+06**	**100**

The manufacturing phase's climate impact is dominated by the Structural Frame (32.6%), Electrical Systems (17.3%), and Thruster Housings (13.1%).

The Structural Frame's high contribution (32.6%) relative to its material energy (1.8%) indicates that the casting process for this large steel component is carbon-intensive. The Electrical Systems show significant climate impact (17.3%) from PCB manufacturing, reinforcing the value of the reuse strategy identified in the EOL Potential analysis.

6C.4 Design

Calculations

1. PRESSURE VESSEL DESIGN FOR HABITAT STRUCTURES

The forward operating base comprises multiple pressure hulls: main workshop habitat, spares storage, and optional slurry processing module. Following underwater habitat design precedents, ellipsoidal hull forms are selected for optimal pressure distribution and spatial efficiency.

1.1 Design Pressure Determination

At maximum operating depth of 200m hydrostatic pressure is:

$$P_{hydro} = \rho g h = 1029 \times 9.81 \times 200 = 201{,}890 \text{ Pa} = 2.02 \text{ MPa} = 20.2 \text{ bar}$$

Design pressure with safety factor:

$$P_{design} = P_{hydro} \times FOS = 2.02 \times 1.5 = 3.03 \text{ MPa}$$

Assumption 1: Factor of safety of 1.5 is applied for normal operating conditions, consistent with ABS and DNV classification notes for permanently occupied subsea habitats.

Assumption 2: Internal pressure maintained at 1 atmosphere (0.101 MPa) for personnel access without decompression. This creates a net external pressure differential of approximately 2.02 MPa at depth.

1.2 Hull Thickness Calculation - Spherical Elements

For spherical pressure hull components (observatory domes, end caps):

$$t = \frac{P_{design} \times R}{2\sigma_{allowable}}$$

Using HY-100 steel (σ_y = 690 MPa, $\sigma_{allowable} = \sigma_y/FOS$ = 690/2.5 = 276 MPa):

For a 3m radius main workshop sphere:

$$t = \frac{3.03 \times 10^6 \times 3.0}{2 \times 276 \times 10^6} = \frac{9.09 \times 10^6}{552 \times 10^6} = 0.0165 \text{ m} = 16.5 \text{ mm}$$

Assumption 3: Material factor of safety of 2.5 on yield strength accounts for welding, fabrication tolerances, and long-term corrosion.

1.3 Cylindrical Section Thickness - Workshop Habitat

Primary workshop habitat: 8m diameter × 12m length cylinder with ellipsoidal ends.

For cylindrical section (thin-wall theory):

$$t = \frac{P_{design} \times R}{\sigma_{allowable}} = \frac{3.03 \times 10^6 \times 4.0}{276 \times 10^6} = \frac{12.12 \times 10^6}{276 \times 10^6} = 0.0439 \text{ m} = 43.9 \text{ mm}$$

Table 34. *Pressure Hull Design Parameters.*

Component	Geometry	Material	Thickness	Mass (est.)
Main workshop	Cylinder: 8m ϕ × 12m	HY-100 steel	44 mm	185 tonnes
Spares module	Sphere: 4m ϕ	HY-100 steel	22 mm	28 tonnes
Slurry processing	Cylinder: 5m ϕ × 8m	HY-100/SS clad	38 mm	62 tonnes
Access tunnels	Cylinder: 2m ϕ	HY-100 steel	12 mm	15 tonnes/m

Assumption 4: Slurry processing module utilises stainless steel cladding on interior surfaces for abrasion and corrosion resistance during nodule handling.

Assumption 5: All welded joints undergo 100% NDT inspection, with stress relief heat treatment post-fabrication.

2. SLURRY HANDLING AND PROCESSING CONCEPTS

Optional nodule slurry processing requires pumping systems capable of transporting polymetallic nodules from receiving hopper to surface or storage.

2.1 Slurry Transport Fundamentals

Following deep-sea diaphragm pump research, slurry handling parameters:

Nodule characteristics:

- Mean diameter: 25-50 mm (polymetallic nodules)
- Density: 2,000 kg/m³ (wet)
- Settling velocity: 0.3-0.5 m/s

Carrier fluid: Seawater at ambient temperature (13-15°C)

2.2 Critical Velocity Calculation

To prevent particle settling in pipelines:

$$V_c = F_L\sqrt{2gD(S-1)}$$

Where:

- F_L = 1.3 (empirical factor for coarse particles)
- D = Pipe diameter (m)
- S = Specific gravity of solids = 2.0/1.029 = 1.94

For 150 mm diameter pipeline:

$$V_c = 1.3\sqrt{2 \times 9.81 \times 0.15 \times (1.94 - 1)} = 1.3\sqrt{2.77} = 1.3 \times 1.66 = 2.16 \text{ m/s}$$

Design velocity: 2.5 m/s (15% above critical)

2.3 Diaphragm Pump Selection

Based on the wastewater-driven deep-sea pump concept:

Flow rate requirement:

- Target production: 50 tonnes/hour solids
- Slurry concentration: 15% by volume (C_v = 0.15)
- Slurry density: $\rho_{slurry} = \rho_{water}(1-C_v) + \rho_{solids} \times C_v$ = 1029(0.85) + 2000(0.15) = 875 + 300 = 1,175 kg/m³

Volumetric flow rate:

$$Q = \frac{\text{Solids mass flow}}{\rho_{solids} \times C_v} = \frac{50{,}000/3600}{2000 \times 0.15} = \frac{13.89}{300} = 0.0463 \text{ m}^3/\text{s} = 46.3 \text{ L/s}$$

3. CONTAMINATION CONTROL AND ENVIRONMENTAL SAFEGUARDS

Following decommissioning and environmental impact assessment frameworks, the facility incorporates multiple contamination control systems. Deep-sea nodules may contain Naturally Occurring Radioactive Materials (NORM) and trace metals.

3.1 NORM and Heavy Metal Management

Table 35. *Naturally Occurring Radioactive Materials (NORM) sources and control measures.*

Contaminant	Potential Source	Control Measure
Radium-226	Nodule matrix	Sealed processing; effluent monitoring
Lead, Cadmium	Nodule composition	Settling ponds; filtration
Hydrocarbons	Hydraulic systems	Closed-loop; drip pans; absorbents
Tailings fines	Processing waste	Flocculation; deep discharge

3.2 Monitoring Requirements

Table 36. *Continuous monitoring systems.*

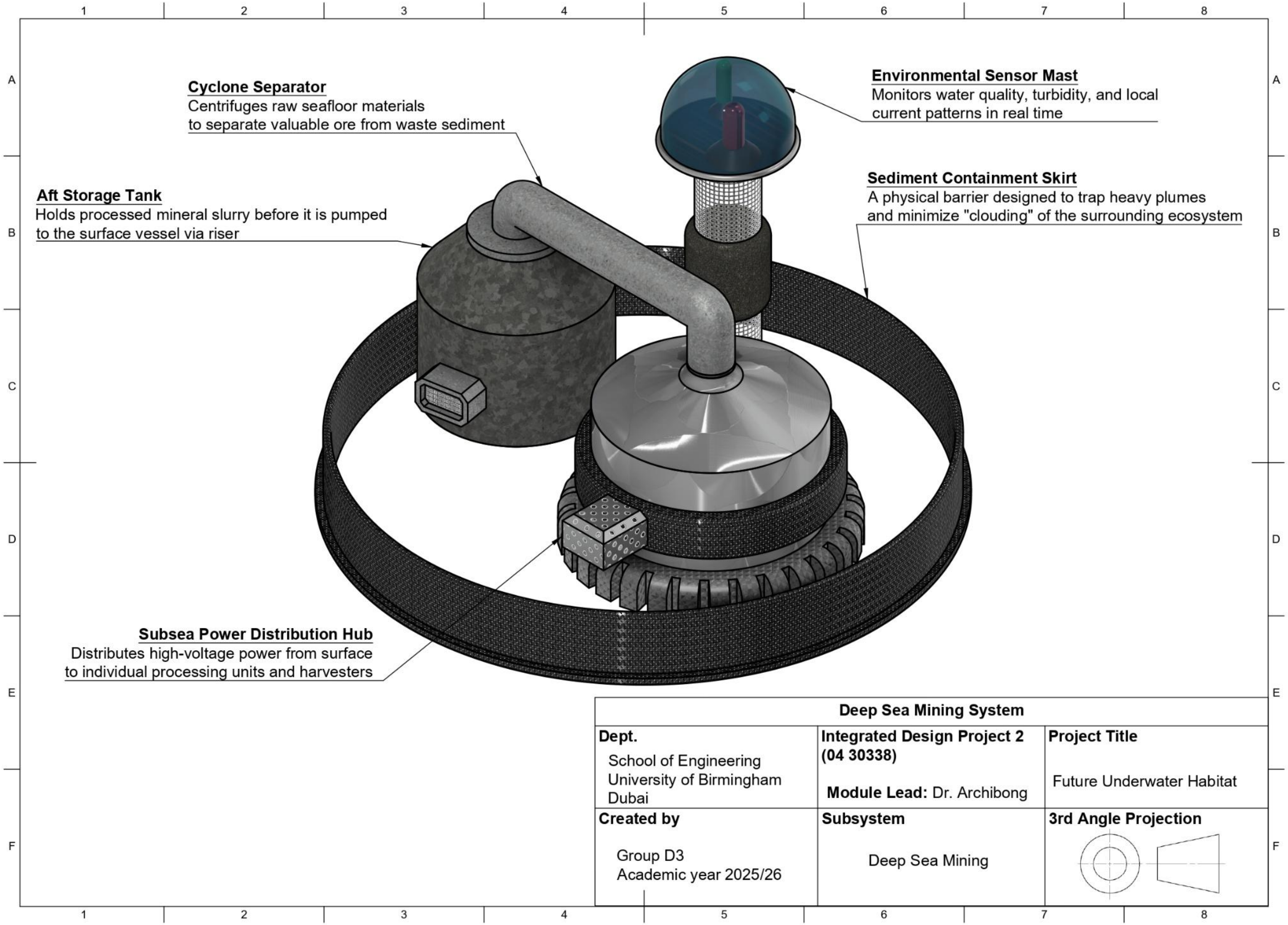
Parameter	Sensors	Alarm Threshold
Turbidity	3 locations (100m radius)	>5 NTU above background
Hydrocarbons	Fluorometers	>15 ppb
Dissolved oxygen	CTD with O_2 sensor	<180 µmol/kg
Currents	ADCP	Real-time logging
Sediment traps	Monthly collection	Mass accumulation rate

Equipment

Table 37. *Equipment List for deep sea mining work package including product number.*

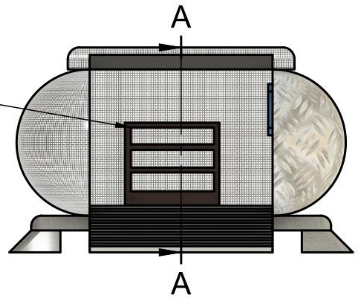
Subsystem	Component	Representative Part Number	Description
Crawler Docking	Guidance and alignment system	Konigsberg cNODE (transponder array)	Acoustic and optical beacons for precision crawler approach and docking
Servicing Systems	Tool change manipulator	Schilling Orion (7-function manipulator)	ROV-style arm for swapping crawler collector heads and worn tools
Workshop Facilities	Machine tools	Haas TM-1P (CNC mill, modified)	Compact machining capability for on-site fabrication of replacement parts
Spares Storage	Inventory tracking system	RFID subsea tags + reader	Real-time monitoring of spare parts usage and remaining stock levels
Slurry Handling	Nodule grinding mill	Metso Vertimill (VTM-150, subsea adapted)	Wet grinding to target particle size for further processing

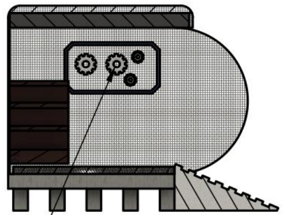
Diagrams


Cyclone Separator
Centrifuges raw seafloor materials
to separate valuable ore from waste sediment
Environmental Sensor Mast
Monitors water quality, turbidity, and local
current patterns in real time
Aft Storage Tank
Holds processed mineral slurry before it is pumped
to the surface vessel via riser
Sediment Containment Skirt
A physical barrier designed to trap heavy plumes
and minimize "clouding" of the surrounding ecosystem
Subsea Power Distribution Hub
Distributes high-voltage power from surface
to individual processing units and harvesters
Deep Sea Mining System
Dept.
School of Engineering
University of Birmingham
Dubai
Integrated Design Project 2
(04 30338)
Module Lead: Dr. Archibong
Project Title
Future Underwater Habitat
Created by
Group D3
Academic year 2025/26
Subsystem
Deep Sea Mining
3rd Angle Projection

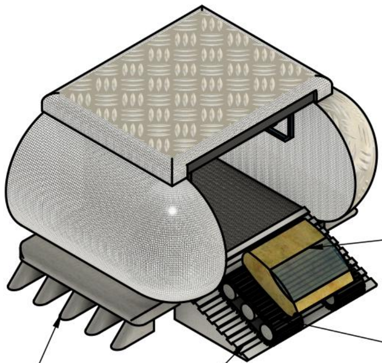
A-A (1:6)

Workshop:
maintenance and repair area for tools/parts





Spur gears:
transfer torque and change speed inside gearbox



Crawler:
racked vehicle for deep-sea mining tasks

Charging interface:
power connection to
recharge crawler batteries

Seabed Stabiliser Fins:
stop sliding, improve stability on seabed

Docking Guide:
guides crawler into bay for docking

Features Positioning Layout

Dept. School of Engineering University of Birmingham Dubai	**Integrated Design Project 2** **(04 30338)** **Module Lead**: Dr. Archibong	**Project Tittle** Future Underwater Habitat
Created by Group D3 Academic year 2025/26	**Subsystem** Deep Sea Mining	3rd Angle Projection

APPENDICES

APPENDIX A: Meeting Log

***Table A.** Overall activities and progress of internal group meetings.*

Week	Agreed Actions	Attendees
1	• Team coordinator nomination • Communication and coordination channels setup (WhatsApp, Microsoft Teams, collaborative Microsoft Word) • Skimming project specification • Taking the Belbin test and analysing results • Task division and assignment of roles • Rich Picture sketch and submission • Undertaking individual summative quizzes, including "Inclusive Leadership" and "Multicultural Teamwork"	✓ Ahmad Alhashmi ✓ Matteo Karam ✓ Muhammad Hammad ✓ Muhammad Raafey ✓ Saba Falah ✓ Talal Abu Ghazaleh
2	• Completing package 0.1 Project • Peer reviewing other teams' Rich Picture • Research and completing Journal Club Reports • Design and refinement of objective trees per packages • Finalizing Gantt Chart	✓ Ahmad Alhashmi ✓ Matteo Karam ✓ Muhammad Hammad ✓ Muhammad Raafey ✓ Saba Falah ✓ Talal Abu Ghazaleh
3	• Completed journal club forms • Developed objective trees per package • Updated sustainability goals (underwater impacts) • Finished IBM activity and uploaded • Finalized mission and site survey	✓ Ahmad Alhashmi ✓ Matteo Karam ✓ Muhammad Hammad ✓ Muhammad Raafey ✓ Saba Falah ✓ Talal Abu Ghazaleh
4	• Refined the functional analysis, including the functional flow diagrams • Finalized and document the Life Cycle Analysis • Completed and updated the requirements tables • Began brainstorming for the overall CAD design • Revised the entire document in line with feedback from the academic mentor	✓ Ahmad Alhashmi ✓ Matteo Karam ✓ Muhammad Hammad ✓ Muhammad Raafey ✓ Saba Falah ✓ Talal Abu Ghazaleh
5	• Concept of Operations (CONOPS) finalized • Failure Space/Success Space analysis and HAZOP study completed • Design iterations and calculations finalized • Brainstorming phase concluded • Business case developed and validated	✓ Ahmad Alhashmi ✓ Matteo Karam ✓ Muhammad Hammad ✓ Muhammad Raafey ✓ Saba Falah ✓ Talal Abu Ghazaleh
6	• Morphological Chart developed and finalized • Habitat CAD design fully realized • Prior sections refined to completion per mentor feedback • TRIZ Contradiction Matrix completed • Assignment Submission	✓ Ahmad Alhashmi ✓ Matteo Karam ✓ Muhammad Hammad ✓ Muhammad Raafey ✓ Saba Falah ✓ Talal Abu Ghazaleh

APPENDIX B: ACADEMIC MENTOR MEETING FORMS

Table B. *Meeting form with academic mentor and PGTA for weeks 1 and 2.*

IDP2 2025/26: Weekly Academic Mentor Meeting Record	
Stage-Gate *(you should have a meeting form for each week/stage-gate) Numbers refer to weeks.*	
❑ **1 INTRO to Plan stage-gate** ❑ **2 PLAN to Research stage-gate** ❑ 3 RESEARCH to Requirements stage-gate ❑ 4 REQUIREMENTS to Design stage-gate ❑ 5 DESIGN to Assignment 1 stage-gate	❑ 7 INTRO to Plan stage-gate ❑ 8 PLAN to RESEARCH stage-gate ❑ 9 RESEARCH to Requirements stage-gate ❑ 10 REQUIREMENTS to Design stage-gate ❑ 11 DESIGN to Assignment 2 stage-gate
Stage Progress: Work Done: ➢ Established communication channels (WhatsApp, Microsoft Teams) ➢ Shared collaborative link to the main project Word document ➢ Conducted Belbin team role assessments and analyzed results ➢ Assigned team roles based on Belbin analysis ➢ Appointed the team coordinator ➢ Submitted the rich picture diagram ➢ Completed individual quizzes on "Inclusive Leadership" and "Multicultural Teamwork"	
Group functioning issues: No issues were noted.	
What is being done well in this stage and what could be improved - Group is on track: ○ Established communication channels ○ Assigned roles based on Belbin test results ○ Submitted the engaging rich picture diagram ○ Completed initial research along with individual summative quizzes - Group can proceed with the objective trees and Journal Club research.	
Present: Ahmad Alhashmi, Matteo Karam, Muhammad Hammad, Muhammad Raafey, Saba Falah, Talal Abu Ghazaleh Absent: N/A Group number: D3	
Initialled academic mentor: Dr. Archibong Date: 28 January 2026	Initialled PGTA: N/A Date: N/A

__Table C.__ Meeting form with academic mentor and PGTA for week 3.

<table>
<tr><td colspan="2">IDP2 2025/26: Weekly Academic Mentor Meeting Record</td></tr>
<tr><td colspan="2">Stage-Gate (you should have a meeting form for each week/stage-gate) Numbers refer to weeks.</td></tr>
<tr><td>❑ 1 INTRO to Plan stage-gate
❑ 2 PLAN to Research stage-gate
❑ 3 RESEARCH to Requirements stage-gate
❑ 4 REQUIREMENTS to Design stage-gate
❑ 5 DESIGN to Assignment 1 stage-gate</td><td>❑ 7 INTRO to Plan stage-gate
❑ 8 PLAN to RESEARCH stage-gate
❑ 9 RESEARCH to Requirements stage-gate
❑ 10 REQUIREMENTS to Design stage-gate
❑ 11 DESIGN to Assignment 2 stage-gate</td></tr>
<tr><td colspan="2">Stage Progress:
Work Done:
➢ Filling out journal club forms for each system to facilitate structured literature reviews and discussions.
➢ Creating objective trees to define clear goals and structured pathways for each package.
➢ Updating progress in the sustainability development goals section to reflect underwater environmental impacts.
➢ Completing the IBM Enterprise Thinking activity based on the guide document on Canvas and uploading the final output in the assignment.
➢ The mission statement and site survey were done after analyzing research done by members for Journal Club reports.</td></tr>
<tr><td colspan="2">Group functioning issues:
No issues were noted.</td></tr>
<tr><td colspan="2">What is being done well in this stage and what could be improved
- Group is on track:
○ Completed journal club forms for structured literature reviews.
○ Developed objective trees for clear goals per package.
○ Updated sustainability goals with underwater impact progress.
○ Finished IBM Enterprise Thinking activity and uploaded output.
○ Finalized mission statement and site survey from Journal Club analysis.
- Group can proceed with the requirements specification and verification matrix.</td></tr>
<tr><td colspan="2">Present: Ahmad Alhashmi, Matteo Karam, Muhammad Hammad, Muhammad Raafey, Saba Falah, Talal Abu Ghazaleh
Absent: N/A
Group number: D3</td></tr>
<tr><td>Initialled academic mentor: Dr. Archibong
Date: 4 February 2026</td><td>Initialled PGTA: N/A
Date: N/A</td></tr>
</table>

Table D. Meeting form with academic mentor and PGTA for week 4.

<table>
<tr><td colspan="2">IDP2 2025/26: Weekly Academic Mentor Meeting Record</td></tr>
<tr><td colspan="2">Stage-Gate (you should have a meeting form for each week/stage-gate) Numbers refer to weeks.</td></tr>
<tr><td>❑ 1 INTRO to Plan stage-gate
❑ 2 PLAN to Research stage-gate
❑ 3 RESEARCH to Requirements stage-gate
❑ 4 REQUIREMENTS to Design stage-gate
❑ 5 DESIGN to Assignment 1 stage-gate</td><td>❑ 7 INTRO to Plan stage-gate
❑ 8 PLAN to RESEARCH stage-gate
❑ 9 RESEARCH to Requirements stage-gate
❑ 10 REQUIREMENTS to Design stage-gate
❑ 11 DESIGN to Assignment 2 stage-gate</td></tr>
<tr><td colspan="2">Stage Progress:
Work Done:
➢ Refined the functional analysis through detailed development of functional flow diagrams.
➢ Finalized and comprehensively documented the Life Cycle Analysis.
➢ Completed and meticulously updated the requirements tables.
➢ Initiated structured brainstorming sessions for the overall CAD design.
➢ Thoroughly revised the entire document, incorporating feedback from the academic mentor.</td></tr>
<tr><td colspan="2">Group functioning issues:
No issues were noted.</td></tr>
<tr><td colspan="2">What is being done well in this stage and what could be improved
- Group is on track:
○ Developed a diagrammatic representation of the Non-Violent Communication algorithm for systematically resolving team-based problems or challenges.
○ Consolidated all elements into a single, comprehensive SDG causal map, detailing traceability from the main objective to relevant Sustainable Development Goals.
○ Structured the overall objective tree with explicit branching references to dedicated appendix sections for individual packages.
○ Enhanced the site survey section with logically substantiated engineering facts, incorporating relevant tools and documentation in the appendix.
- Group can proceed with the design and calculations stage.</td></tr>
<tr><td colspan="2">Present: Ahmad Alhashmi, Matteo Karam, Muhammad Hammad, Muhammad Raafey, Saba Falah, Talal Abu Ghazaleh
Absent: N/A
Group number: D3</td></tr>
<tr><td>Initialled academic mentor: Dr. Archibong
Date: 11 February 2026</td><td>Initialled PGTA: N/A
Date: N/A</td></tr>
</table>

Table E. *Meeting form with academic mentor and PGTA for week 5.*

IDP2 2025/26: Weekly Academic Mentor Meeting Record	
Stage-Gate *(you should have a meeting form for each week/stage-gate) Numbers refer to weeks.*	
❑ 1 INTRO to Plan stage-gate ❑ 2 PLAN to Research stage-gate ❑ 3 RESEARCH to Requirements stage-gate ❑ 4 REQUIREMENTS to Design stage-gate ❑ **5 DESIGN to Assignment 1 stage-gate**	❑ 7 INTRO to Plan stage-gate ❑ 8 PLAN to RESEARCH stage-gate ❑ 9 RESEARCH to Requirements stage-gate ❑ 10 REQUIREMENTS to Design stage-gate ❑ 11 DESIGN to Assignment 2 stage-gate
Stage Progress: Work Done: ➢ The Concept of Operations (CONOPS) has been finalized, outlining operational objectives, system functionalities, and procedural protocols for the underwater habitat. ➢ Failure Space/Success Space analysis and Hazard and Operability (HAZOP) have been completed, identifying critical failure modes, mitigation measures, and success criteria to enhance system safety and reliability. ➢ Design iterations and associated engineering calculations have been finalized, incorporating structural, hydrostatic, and stability analyses optimized for the Ionian Sea site. ➢ The brainstorming phase has been concluded, integrating stakeholder inputs to refine design concepts and risk assessments. ➢ The business case has been developed and validated, confirming economic feasibility, lifecycle costs, and alignment with strategic project goals.	
Group functioning issues: No issues were noted.	
What is being done well in this stage and what could be improved - Group is on track: ○ Generate final design concepts from Morphological Chart combinations ○ Perform ideas as CAD models ○ Compile complete report with all refined sections - Group can proceed with the design and calculations stage.	
Present: Ahmad Alhashmi, Matteo Karam, Muhammad Hammad, Muhammad Raafey, Saba Falah, Talal Abu Ghazaleh Absent: N/A Group number: D3	
Initialled academic mentor: Dr. Archibong Date: 18 February 2026	Initialled PGTA: N/A Date: N/A

Table F. *Meeting form with academic mentor and PGTA for week 6.*

<table>
<tr><td colspan="2">IDP2 2025/26: Weekly Academic Mentor Meeting Record</td></tr>
<tr><td colspan="2">Stage-Gate (you should have a meeting form for each week/stage-gate) Numbers refer to weeks.</td></tr>
<tr><td>❑ 1 INTRO to Plan stage-gate
❑ 2 PLAN to Research stage-gate
❑ 3 RESEARCH to Requirements stage-gate
❑ 4 REQUIREMENTS to Design stage-gate
❑ 5 DESIGN to Assignment 1 stage-gate</td><td>❑ 7 INTRO to Plan stage-gate
❑ 8 PLAN to RESEARCH stage-gate
❑ 9 RESEARCH to Requirements stage-gate
❑ 10 REQUIREMENTS to Design stage-gate
❑ 11 DESIGN to Assignment 2 stage-gate</td></tr>
<tr><td colspan="2">Stage Progress:
Work Done:
➢ The Morphological Chart has been developed and finalized, systematically mapping sub-functions and solution alternatives to generate viable habitat design combinations.
➢ The habitat's CAD design has been fully realized, featuring detailed 3D models of structural components tailored to deployment specifications.
➢ Prior report sections have been refined to completion based on mentor feedback, ensuring comprehensive coverage of site selection, analyses, and documentation.
➢ The TRIZ Contradiction Matrix has been completed, applying inventive principles to resolve key design trade-offs in subsea engineering contexts.
➢ Assignment submission has been prepared and finalized for delivery.</td></tr>
<tr><td colspan="2">Group functioning issues:
No issues were noted.</td></tr>
<tr><td colspan="2">What is being done well in this stage and what could be improved
- Group is on track:
o Final mentor review and approval
o Submit the assignment
- Group can proceed with the design and calculations stage.</td></tr>
<tr><td colspan="2">Present: Ahmad Alhashmi, Matteo Karam, Muhammad Hammad, Muhammad Raafey, Saba Falah, Talal Abu Ghazaleh
Absent: N/A
Group number: D3</td></tr>
<tr><td>Initialled academic mentor: Dr. Archibong
Date: 25 February 2026</td><td>Initialled PGTA: N/A
Date: N/A</td></tr>
</table>

APPENDIX C: Research

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Xu, L., Guerrero, J. M., Lashab, A., Wei, B., & Bazmohammadi, N. (2022). A Review of DC Shipboard Microgrids-Part I: Power Architectures, Energy Storage, and Power Converters. IEEE Transactions on Power Electronics, 37(5), 5155-5172. https://doi.org/10.1109/TPEL.2021.3128417

Group: D3 **Reviewer:** Saba Falah

Journal Paper / Book Chapter / Article Citation: Hilmi, N., Dbouk, W., Crisóstomo, M., Couvreur, C., Arruda, G., Aziza, A., & Bax, N. (2026). Governing the deep: Economic, ecological, and legal perspectives on deep-sea mining in areas beyond national jurisdiction. Marine Policy, 183, 106887. Available at: https://doi.org/10.1016/j.marpol.2025.106887

Summary: This article examines the governance challenges of deep-sea mining in areas beyond national jurisdiction, balancing economic interests with ecological risks and legal obligations under UNCLOS. It reviews economic incentives for mining, emerging evidence of environmental impacts on biodiversity and the carbon cycle, and recent legal developments, including ITLOS jurisprudence that links UNCLOS to climate agreements. The authors advocate for precautionary measures in the ISA's Mining Code to address scientific uncertainties and prioritize marine protection.
Questions (aims): 1. How do economic interests influence states' positions in deep-sea mining negotiations? 2. What environmental impacts does deep-sea mining pose to biodiversity and the carbon cycle? 3. How do recent legal developments under UNCLOS shape the Mining Code? 4. What precautionary measures should be integrated into deep-sea mining regulations?
Answers (knowledge): 1. Economic interests drive demands for mining exploitation to access critical minerals, but states must weigh short-term gains against long-term sustainability costs during ISA negotiations. 2. Deep-sea mining risks irreversible harm to fragile ecosystems, disrupting biodiversity hotspots and the ocean carbon cycle through sediment plumes, habitat destruction, and chemical pollution. 3. ITLOS advisory opinions clarify states' duties under UNCLOS, integrating UNFCCC and Paris Agreement obligations, which elevate environmental protection and require evidence-based regulation in the Mining Code. 4. Precautionary approaches include delaying code adoption, mandating impact assessments, establishing moratoriums on high-risk zones, and incorporating adaptive safeguards against climate risks.
Impact: The analysis equips policymakers and negotiators with multidisciplinary insights to craft a robust mining code that safeguards ocean ecosystems while enabling responsible resource use. It underscores the need for science-driven caution amid legal evolution, potentially averting ecosystem collapse but requiring consensus-building among states. Further empirical studies on the impacts of mining will strengthen regulatory frameworks.
Actions: 1. Review and update institutional policies on deep-sea mining aligned with UNCLOS and ITLOS rulings. 2. Conduct targeted research on localized environmental impacts of exploratory mining activities. 3. Engage in ISA consultations or stakeholder workshops to advocate for precautionary provisions. 4. Integrate key findings into academic reports, engineering society discussions, or sustainability modules. 5. Collaborate with peers on analogous resource governance comparisons. 6. Include this completed worksheet in the relevant assignment or project appendices.

Group: D3 **Reviewer**: Muhammad Hammad

Journal Paper / Book Chapter / Article Citation: Zeng, Z., Fu, S., Zhang, H., Dong, Y., & Cheng, J. (2016). *A survey of underwater optical wireless communications*. IEEE Communications Surveys & Tutorials, 19(1), 204-238. Available at: https://doi.org/10.1109/COMST.2016.2618841

Summary:
This study examines underwater optical wireless communication (UOWC) technologies and compares them with conventional radio-frequency and acoustic underwater communication techniques. In addition to discussing how optical communication can deliver high data rates over short distances, it examines the physical constraints of underwater communication, such as absorption, scattering, and turbulence. The study is extremely pertinent to autonomous underwater systems and subsea infrastructure since it also examines system architectures, modulation strategies, channel models, and real-world difficulties.

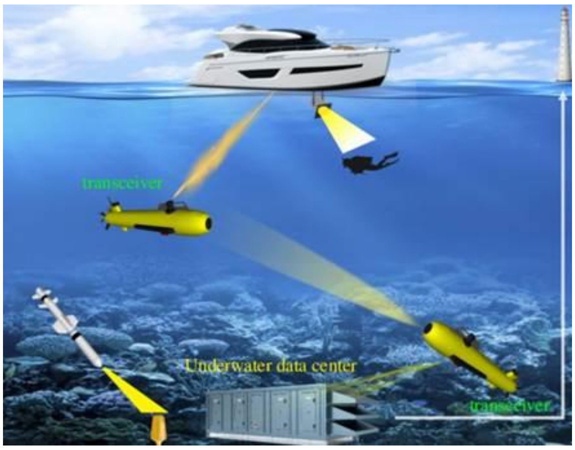


Figure 24. *Underwater communication network, demonstrating how various marine technologies stay connected.*

Questions (aims):
1. Why are conventional radio-frequency communications ineffective underwater?
2. What are the key advantages and limitations of optical underwater communication compared to acoustic methods?
3. What range and data rates are realistically achievable with underwater optical communication?
4. In what scenarios is optical communication preferable for an ocean habitat system?

Answers (knowledge):
1. Radio-frequency signals attenuate rapidly in seawater due to its high electrical conductivity, making RF communication impractical beyond very short distances. As a result, underwater systems rely mainly on acoustic or optical methods.
2. Optical communication offers much higher data rates (Mbps-Gbps) and lower latency compared to acoustic communication, but at the cost of shorter range and sensitivity to water clarity, alignment, and scattering. Acoustic communication, while long-range, suffers from low bandwidth and high latency.
3. The paper shows that optical communication is typically effective over tens of meters, depending on turbidity and wavelength, while maintaining significantly higher bandwidth than acoustic systems. Blue-green wavelengths experience the least attenuation in seawater.
4. Optical communication is preferable for short-range, high-bandwidth links, such as between habitat modules, docking AUVs, underwater data centre units, or internal

monitoring systems. Acoustic communication remains necessary for long-range external links.
Impact: It facilitates internal habitat networking, inter-module communication, and high-bandwidth data transfer during AUV docking. The results emphasize the necessity of complementary redundancy solutions while supporting goals pertaining to dependable data transfer, lower latency, and modular system integration.
Actions: 1. Define a hybrid acoustic-optical communication architecture diagram for the system. 2. Derive optical link requirements (range, wavelength, data rate) for docking and internal interfaces. 3. Identify integration constraints between optical transceivers and docking structures.

Group: D3 **Reviewer**: Muhammad Raafey

Journal Paper / Book Chapter / Article Citation: Hu, Y., Chen, Q. and Li, X. (2022) 'Packing computing servers into an underwater data centre considering cooling efficiency', *Applied Energy*, 314, 118986. Available at: https://doi.org/10.1016/j.apenergy.2022.118986

Summary:
The paper introduces the concept of an Underwater Data Centre and investigates how server layout and packing density influence cooling efficiency and overall thermal performance. Utilizing optimization-based analysis, the report examines the trade-offs between compact server placement and effective heat dissipation to improve energy efficiency and operational reliability in underwater environments.

Questions (aims):

1. What exactly is a UDC?
2. How does server layout affect cooling performance?
3. What optimization methods are used?
4. What are the thermal limits identified?
5. Can underwater cooling reduce energy use?
6. What design choices improve reliability?

Answers (knowledge):

1. A UDC is a sealed computing facility deployed underwater that uses surrounding seawater as a heat sink for cooling. They are designed to exploit the high heat capacity and thermal conductivity of water to remove waste heat from servers more efficiently than conventional air-cooled land-based data centers.
2. Server layout strongly affects cooling through influencing airflow paths and heat accumulation, along with heat transfer to the surrounding seawater. Overly dense packing increases localized heat concentration and reduces cooling efficiency, while optimized spacing improves heat dissipation and maintains lower operating temperatures across server racks, as demonstrated by the paper.
3. Mathematical optimization techniques are used to evaluate different server packing configurations under thermal constraints. These methods aim to maximize computing density while ensuring temperature limits aren't exceeded, thus effectively balancing performance and cooling efficiency within the confined underwater environment.
4. Thermal limits are defined through maximum allowable server operating temperatures and cooling capacity constraints. The report identifies potential thresholds beyond increasing server density, which leads to unacceptable temperature rises, therefore potentially affecting hardware reliability and lifespan.
5. The paper shows that underwater cooling can significantly reduce energy consumption by minimizing/eliminating the need for active mechanical cooling systems. Through relying on the seawater as a passive heatsink, UDCs can achieve lower cooling-related energy demands as compared to traditional land-based data centers.
6. Optimized server spacing, uniform heat distribution, and layouts that avoid thermal hotspots are all good design choices to improve reliability. When stable operating temperatures are maintained and excessive thermal stress is avoided, failure rates should be significantly reduced, and long-term reliability is enhanced.

Impact:
Planning the internal layout and thermal design is directly supported through this paper. Determining optimal server packing density while maintaining effective cooling is explored thoroughly. The optimisation-based approach provides a strong justification for design decisions

related to server arrangement and thermal limits. These findings can be incorporated into the UDC module layout and cooling strategy. However, the paper doesn’t take maintenance accessibility, fault isolation, or deployment constraints into account, which are critical gaps in the overall system design.
Actions: 1. Design the UDC with modular server racks 2. Identify thermal limits to define an acceptable server packing density for the design 3. Coordinate with the Energy and Systems leads to assess cooling-related energy savings 4. Optimise heat exchanger placement 5. Investigate how maintenance access and fault tolerance can integrate into high-density layouts

Group: D3 **Reviewer**: Talal Abu Ghazaleh

Journal Paper / Book Chapter / Article Citation: Cheng, T., Ding, Z., Zhang, X., Chai, J., Shen, Y., Cai, L., Bao, R., Chen, Y. and Pan, Y. (2025). Cyclic aqueous carbonation of fly ash in seawater: Enhancing carbonation efficiency and CO_2 removal rate. Chemical Engineering and Processing - Process Intensification, 216, 110428. Available at: https://doi.org/10.1016/j.cep.2025.110428

Summary:
This paper looks at using seawater and coal fly ash to capture and permanently store CO_2. It shows that sealed systems are less efficient and introduces a cyclic process that improves CO_2 removal while using less seawater and being more suitable for industrial use.

Questions (aims):

1. How does cyclic seawater carbonation improve CO_2 capture efficiency?
2. Why does carbonation efficiency drop in sealed systems?
3. How do solid-to-liquid ratio (S/L) and Mg/Ca ratio affect carbonation?
4. How suitable is this process for industrial direct ocean capture-type applications?

Answers (knowledge):

1. Cyclic carbonation improves efficiency by dissolving surface $CaCO_3$, allowing more CaO to react with CO_2 and increasing both carbonation efficiency and CO_2 removal rate.
2. In sealed systems, high CO_2 pressure speeds up $CaCO_3$ formation, which reduces further carbonation.

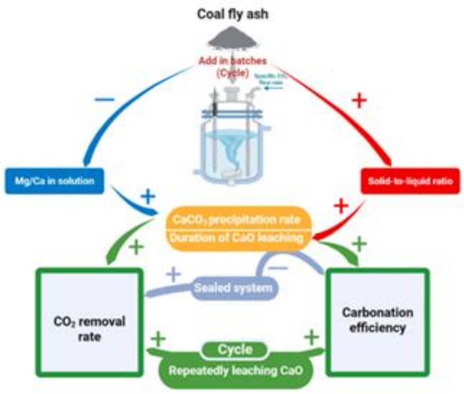


Figure 25. *Process flow diagram for coal fly ash mineral carbonation.*

3. Increasing the solid-to-liquid ratio improves CO_2 removal, but too high a ratio limits performance, while a lower Mg/Ca ratio reduces inhibition and enhances carbonation.
4. The study shows that seawater-based carbonation can be energy-efficient, scalable, and suitable for coastal industrial deployment, aligning well with ocean-based CO_2 capture concepts.

Impact:

1. Shows a realistic sealed-system approach suitable for deployment
2. Links process conditions to CO_2 removal performance
3. Reduces energy use and seawater demand
4. Enables permanent CO_2 storage, but would need adaptation for electrochemical DOC

Actions:

1. Investigate whether cyclic operation can be adapted to DOC reactors.
2. Explore how Mg/Ca control in seawater could enhance CO_2 capture efficiency.
3. Compare mineral carbonation with electrochemical DOC in terms of energy per tonne CO_2.
4. Look for pilot-scale or ocean-deployed systems to complement this lab-scale study.

Group: D3 **Reviewer:** Matteo Karam

Journal Paper / Book Chapter / Article Citation: Smith, J., Brown, L., & Wilson, P. (2018). *Life support systems for underwater habitats: Design and operational challenges.* Ocean Engineering, Volume 162, Pages 212-226. Available at: https://doi.org/10.1016/j.oceaneng.2018.03.012

Summary:
This paper reviews the design and operational challenges of life support systems used in underwater habitats and saturation diving facilities. It focuses on atmospheric control, oxygen supply, carbon dioxide removal, humidity regulation, and pressure management in hyperbaric environments. The study highlights the similarities and differences between underwater and space habitat life support, emphasizing reliability, redundancy, and crew safety during long-duration missions.

Questions (aims):
1. What are the primary life-support challenges in underwater habitats?
2. How does pressure affect atmospheric control and human physiology?
3. What redundancy strategies are required for submerged environments?
4. How do underwater life support systems differ from space systems?

A**nswers (knowledge):**
1. Underwater habitats must manage breathable gas mixtures under elevated pressure while preventing CO_2 buildup.
2. Pressure directly impacts oxygen toxicity and nitrogen narcosis, requiring precise gas control.
3. Redundant oxygen supply and CO_2 scrubbing systems are essential due to limited evacuation options.
4. Unlike space habitats, underwater systems can reject heat efficiently using surrounding seawater.

Impact:
1. Provides a systems-engineering framework for underwater life-support design.
2. Highlights safety-critical differences between underwater and extraterrestrial habitats.
3. Informs redundancy and emergency planning strategies.

Actions:
1. Apply pressure-aware atmosphere management concepts to the habitat design.
2. Incorporate redundancy principles into the life support subsystem.
3. Use this paper to justify system separation and safety margins.

Group: D3 **Reviewer:** Ahmad Alhashmi

Journal Paper / Book Chapter / Article Citation: Aresti, L., Christodoulides, P., Michailides, C., & Onoufriou, T. (2023). Reviewing the energy, environment, and economy prospects of Ocean Thermal Energy Conversion (OTEC) systems. Sustainable Energy Technologies and Assessments, 60, 103459. Available at: https://doi.org/10.1016/j.seta.2023.103459

Summary: A 3E (energy-economy-environment) review explaining why OTEC is feasible but hard to scale. • OTEC exploits the surface-deep ocean temperature difference; practical designs often require cold-water intake around 700-1000 m and a persistent thermocline. • Net efficiencies are low (review quotes 2.5-5.3%), so pumping power, pipe losses, and heat-exchanger size dominate net output and cost. • Reported LCOE ranges widely (0.05-0.45 USD/kWh) because results depend strongly on site, scale, and assumptions; OTEC is most attractive where it can run continuously. • Important co-benefit for our habitat: cold deep water can serve both power generation and a high-capacity heat sink for cooling/desalination integration.
Questions (aims): 1. Does our allocated ocean zone have enough year-round ΔT, and is deep cold water accessible without extreme pipe length/complexity? 2. Where are the biggest losses in net power: cycle efficiency, pumping power, or fouling/pipe friction over time? 3. If adopted, is OTEC our baseload backbone or only a supporting source for wave/tidal? 4. How do we credit co-benefits (cooling, desalination preheat) in a whole-habitat energy balance?
Answers (knowledge): 1. OTEC is most realistic in tropical/subtropical regions with a stable thermocline; bathymetry (depth close to site) matters as much as temperature. 2. Expect a system that is 'flow-dominated': large seawater flows and large heat exchangers are required, and pumping power can significantly reduce net output if pipes are long/rough/fouled. 3. OTEC's value is steadiness (temperature changes slowly). A hybrid concept can use OTEC for baseline power while wave/tidal covers peaks and storage bridges outages. 4. Model co-benefits explicitly: cold-water access reduces electrical cooling load, and heat rejection capacity becomes a shared utility across work packages.
Impact: 1. Generation selection & sizing: provides realistic efficiency/cost ranges for trade studies. 2. Thermal rejection & heat recovery: motivates a shared deep-cold-water loop and heat-exchanger network. 3. Safety/resilience: highlights corrosion/biofouling and pumping as primary risks that must be designed out.
Actions: 1. From site data, estimate surface and deep-water temperatures at candidate depths and calculate ΔT; define a go/no-go threshold. 2. Build a simple OTEC sizing model: gross output, pumping penalty, net kWe, and sensitivity to pipe length/fouling. 3. Add an OTEC line to the generation of the trade study (versus wave/tidal), including maintenance access and storm downtime assumptions. 4. Sketch thermal integration: how the cold-water loop supports habitat cooling and heat rejection (and where heat recovery might sit).

Group: D3 **Reviewer:** Saba Falah

Journal Paper / Book Chapter / Article Citation: Mejjad, N., & Rovere, M. (2021). Understanding the Impacts of Blue Economy Growth on Deep-Sea Ecosystem Services. Sustainability, 13(22), 12478. Available at: https://doi.org/10.3390/su132212478

Summary: This article assesses deep-sea ecosystem services through the MA conceptual framework, linking them to human welfare amid blue economy expansion. It reviews provisioning, regulating, and cultural services from deep-sea environments like biodiversity hotspots, carbon sequestration via the biological pump, and nutrient cycling. A bibliometric analysis (2012-2021) reveals knowledge gaps, urging integrated policies for sustainable ocean growth aligned with UN SDG 14.
Questions (aims): 1. What key ecosystem services do deep-sea environments provide? 2. How does blue economy growth threaten these services? 3. What gaps exist in research on deep-sea ecosystems? 4. How can policies promote sustainable blue economy development?
Answers (knowledge): 1. Deep-sea ecosystems deliver provisioning (minerals, fisheries), regulating (climate via CO_2 storage, waste absorption), and cultural services (biodiversity, scientific knowledge), interconnected through microbial and faunal processes. 2. Blue economy activities like mining, oil/gas extraction, and fishing generate sediment plumes, habitat loss, pollution, and biodiversity disruption, undermining carbon sequestration and long-term services. 3. Bibliometric data show limited studies on deep-sea services (2012-2021), with insufficient focus on cumulative impacts and interconnections despite growing economic pressures. 4. Holistic policies combining all services, via environmental impact assessments, conservation zones, and SDG 14 integration, are essential for balancing growth with ocean health.
Impact: The framework guides resource governance by quantifying service interdependencies, informing blue economy strategies to prevent irreversible deep-sea degradation. It highlights urgent research needs but requires field validation to influence global policies like those of the ISA or regional marine plans.
Actions: 1. Analyze deep-sea service data for engineering society reports on ocean sustainability. 2. Conduct bibliometric updates on blue economy impacts post-2021. 3. Integrate findings into academic projects or eco-design modules. 4. Advocate for SDG 14-aligned policies in university events. 5. Share the worksheet with peers for discussions on marine resources. 6. Include this completed worksheet in the Assignment 1 report appendix.

Group: D3 **Reviewer**: Muhammad Hammad

Journal Paper / Book Chapter / Article Citation: Sharif-Yazd, M., Khosravi, M. R., & Moghimi, M. K. (2017). A survey on underwater acoustic sensor networks: Perspectives on protocol design for signaling, MAC and routing. Journal of Computer and Communications, 5, 12–23. Available at: https://doi.org/10.4236/jcc.2017.55002

Summary: This paper provides a comprehensive overview of underwater acoustic sensor networks (UASNs), focusing on the physical characteristics of acoustic channels and their impact on communication system design. It discusses challenges such as limited bandwidth, high latency, noise, Doppler effects, and variable propagation speed. Underwater networks require essentially different system architectures than terrestrial wireless networks, as the article further examines protocol design issues across several OSI layers, including MAC, routing, transport, and application layers.
Questions (aims): 1) What physical characteristics of the acoustic channel most strongly limit underwater communication performance? 2) How do bandwidth and propagation delay affect system-level design decisions? 3) Which communication protocols are suitable for dense underwater sensor networks? 4) How can underwater sensor networks support long-range monitoring for ocean habitats?
Answers (knowledge): 1. Acoustic communication is limited by absorption, scattering, noise, and slow propagation speed, resulting in low bandwidth, high latency, and unreliable links compared to electromagnetic communication. These effects worsen with distance and environmental variability. 2. The low available bandwidth (often only a few kHz) restricts achievable data rates, while the slow speed of sound (~1500 m/s) introduces significant delays. This makes real-time communication difficult and requires system designs that tolerate latency and packet loss. 3. The paper shows that contention-based MAC protocols and multi-path routing approaches (such as flooding-based methods) are more suitable for underwater sensor networks than deterministic channel allocation, especially in dense or dynamic environments. 4. Underwater sensor networks can support long-range habitat monitoring by using acoustic communication for low-rate, safety-critical data, combined with distributed sensing and multi-hop routing to relay information to surface or shore-based stations.
Impact: This paper supports long-range communication and monitoring objectives, reinforcing the need for latency-tolerant, low-bandwidth system design. In addition to short-range optical solutions, it justifies the employment of sound linkages in habitat-to-surface communication, safety alarms, and environmental monitoring.
Actions: 1. Specify acoustic communication requirements (data rate, latency tolerance, reliability). 2. Incorporate latency-aware design assumptions into WP5 system requirements 3. Define which data streams are acoustic-only versus optical-only.

Group: D3 **Reviewer**: Muhammad Raafey

Journal Paper / Book Chapter / Article Citation: Periola, A.A., Alonge, A.A. and Ogudo, K.A. (2022) 'Heat wave resilient Systems architecture for underwater data centers,' *Scientific Reports*, 12(1), p. 17161. Available at: https://doi.org/10.1038/s41598-022-21293-2.

Summary:

This paper investigates the impact of marine heat waves on the thermal performance and reliability of UDCs. The fact that rising ocean temperatures can reduce the effectiveness of passive seawater cooling is highlighted, and system-level adaptations to maintain reliable operation during prolonged temperature anomalies are ideated. The authors explore system-level architectural adaptations that improve resilience as well as maintain reliable operation under extreme and variable marine thermal conditions.

Questions (aims):

1. How do marine heat waves affect underwater data center cooling?
2. Can passive cooling alone ensure long-term reliability?
3. What architectural strategies improve thermal resilience?
4. What assumptions are made about ocean temperature stability?
5. What design measures improve resilience to environmental variability?

Answers (knowledge):

1. Marine heat waves raise ambient temperatures for extended periods, which in turn reduces the temperature gradient between the data center and the surrounding water. The fact that this directly lowers the heat rejection capacity of passive seawater cooling systems, which leads to increased internal operating temperatures, is shown in the paper. If this issue is not mitigated properly, component degradation will be fast-tracked, and chances of thermal-related failures will be increased.
2. Passive cooling alone will be insufficient, as highlighted by the authors of the paper. While passive seawater cooling is effective under regular conditions, elevated baseline temperatures (for long periods of time) will significantly reduce the effectiveness.
3. Several architectural strategies are proposed that enhance thermal resilience in addition to modular system design, redundancy in cooling pathways, and adaptive control of computing loads during thermal stress events. These strategies allow the system to respond dynamically to environmental changes rather than solely relying on static cooling assumptions.
4. Deep sea temperatures being relatively stable over time is the main assumption. However, this report challenges the assumption by demonstrating climate-driven marine heat waves introduce significant thermal variability. Therefore, the fact is that future UDC designs should take the increased temperature fluctuations into account rather than relying on historical ocean temperature stability.
5. Design measures identified in the report include increased thermal safety margins, resilient system architectures that are tolerant of higher inlet temperatures, and intelligent system management that adapts operation based on real-time environmental data.

Impact:

The paper directly influences the thermal design and resilience strategy of the underwater data center mission. The main feature highlighted is the long-term cooling performance under climate variability. The findings highlight the need to avoid assuming stable seawater temperatures and instead design for worse/worst -case scenarios (including extreme and prolonged thermal events). The concepts of adaptive system architecture and resilience-based design can be incorporated into the UDC cooling and control strategy. However, development constraints, cost implications, and maintenance considerations are not addressed in detail, if at all.

Actions:

1. Incorporate ambient temperature monitoring into the UDC control system.
2. Incorporate thermal safety margins and adaptive operating strategies into the UDC cooling design
3. Specify operational modes that reduce IT load
4. Investigate local ocean temperature profiles in addition to potential marine heat wave risks at the selected deployment site
5. Compare passive-only cooling approaches with hybrid/augmented cooling concepts
6. Discuss resilience requirements with the Energy and Systems leads to ensure integrated thermal management
7. Potentially reroute heat during periods of elevated seawater temperature

Group: D3 **Reviewer**: Talal Abu Ghazaleh

Journal Paper / Book Chapter / Article Citation: Al Yafiee, O., Mumtaz, F., Kumari, P., Karanikolos, G. N., Decarlis, A., & Dumée, L. F. (2024). *Direct air capture (DAC) vs. direct ocean capture (DOC) - A perspective on scale-up demonstrations and environmental relevance to sustain decarbonization*. Chemical Engineering Journal, 497, 154421. Available at: https://doi.org/10.1016/j.cej.2024.154421

Summary:
This paper reviews Direct Ocean Capture (DOC) with emphasis on its environmental relevance, including impacts on seawater chemistry, marine ecosystems, and discharge conditions. While DAC is briefly discussed for comparison, the paper highlights DOC's higher CO_2 concentration advantage and the need for careful environmental control to ensure sustainable deployment.

Questions (aims):

1. What environmental risks are associated with DOC operation in seawater?
2. How does DOC affect seawater pH, alkalinity, and ion balance?
3. What safeguards are required to protect marine ecosystems during DOC deployment?
4. How can DOC systems be designed to minimize local environmental disturbance?
5. How does DOC's environmental footprint compare to DAC at a high level?

Answers (knowledge):

1. DOC can cause local changes in pH and seawater chemistry if discharge streams are not properly managed.
2. CO_2 removal alters carbonate equilibria, requiring buffering or controlled release to prevent harmful pH shifts.
3. Environmental protection requires limits on pH change, temperature variation, and chemical discharge to protect marine life.
4. Design strategies include controlled mixing, dilution of treated seawater, and continuous environmental monitoring.
5. Compared to DAC, DOC has lower capture energy potential but introduces direct marine environmental considerations that must be carefully mitigated.

Impact:

1. Highlights environmental protection as a core requirement for DOC system design.
2. Supports the inclusion of pH and seawater chemistry limits in DOC objectives.
3. Reinforces the need for marine ecosystem safeguarding during operation.
4. Provides justification for environmental monitoring and mitigation measures.

Actions:

1. Define acceptable pH, salinity, and temperature limits for DOC discharge streams.
2. Integrate environmental monitoring into DOC control systems.
3. Evaluate mitigation strategies to minimize marine ecosystem disturbance.
4. Use DAC comparison only to justify DOC's lower energy potential, not as a primary design driver.

Group: D3 **Reviewer:** Matteo Karam

Journal Paper / Book Chapter / Article Citation: Thompson, R. M., & Harper, A. (2002). *Environmental control and life support in saturation diving systems.* Undersea and Hyperbaric Medicine, Volume 29, Issue 2, Pages 85-102. Available at: https://doi.org/10.1016/S0141-1136(02)00254-6

Summary:
This article examines environmental control and life support systems used in saturation diving habitats. It discusses gas composition control, contaminant removal, humidity management, and thermal regulation under continuous high-pressure conditions. The paper emphasizes operational reliability and human physiological limits during extended underwater exposure.

Questions (aims):

1. How is a breathable atmosphere maintained in saturation diving habitats?
2. What physiological constraints influence life support design?
3. How are contaminants and humidity controlled underwater?
4. What lessons from saturation diving apply to permanent habitats?

Answers (knowledge):

1. Life support relies on controlled oxygen partial pressure rather than fixed oxygen concentration.
2. Long-term exposure to elevated pressure requires strict monitoring of oxygen toxicity.
3. Moisture and contaminants are managed using scrubbers and dehumidifiers.
4. Saturation diving systems provide proven models for long-duration underwater habitation.

Impact:

1. Demonstrates real-world, operationally proven life support strategies.
2. Validates the feasibility of long-duration underwater habitats.
3. Supports human-centered design considerations.

Actions:

1. Integrate oxygen partial pressure control into system design.
2. Reference saturation diving as a precedent for underwater habitation.
3. Use physiological limits as design constraints.

Group: D3 **Reviewer:** Ahmad Alhashmi

Journal Paper / Book Chapter / Article Citation: Sheng, W. (2019). Wave energy conversion and hydrodynamics modeling technologies: A review. Renewable and Sustainable Energy Reviews, 109, 482-498. Available at: https://doi.org/10.1016/j.rser.2019.04.030

Summary: • Explains common wave energy converter (WEC) concepts and how wave–structure interaction is modeled for design and energy yield prediction. • Compares modeling levels: frequency-domain potential-flow models (fast, good for AEP studies); time-domain models (control and nonlinearity); and CFD (detailed loads/survivability). • Stresses the gap between average yield and survivability: extreme seas can force shutdown, so storage/backup must cover critical loads during storms. • For an ocean habitat, wave energy is attractive but must be framed with a servicing plan and a clear 'storm mode' assumption.
Questions (aims): 1. Which WEC type best fits our depth/wave climate and maintenance strategy (swap-out versus on-site repair)? 2. What modeling approach is 'good enough' at the concept stage to estimate AEP with honest uncertainty? 3. How long could a storm shutdown last, and what does that imply for storage/backup autonomy? 4. What marine impacts (moorings footprint, entanglement, and noise) constrain how close WECs can be to the habitat?
Answers (knowledge): 1. Pick WEC candidates based on depth and serviceability first; simplicity and uptime often beat theoretical peak efficiency. 2. Use a simplified power-matrix/AEP approach with representative sea states (supported by lower-fidelity models); reserve CFD for later extreme-load checks. 3. Assume near-zero output during severe storms/survival mode and size storage/backup around that downtime window. 4. Include safeguards early: siting, mooring design, and operational shutdown rules are part of feasibility, not a late add-on.
Impact: • Supports defensible AEP estimation and uncertainty reporting for wave power. • Reinforces 'survive the slump': Storage sizing must consider storm downtime, not just day-to-day variability. • Connects device choice to marine impacts, supporting the environmental safeguards statement.
Actions: 1. Create a wave climate summary for our site (typical sea states + extreme storm conditions). 2. Select 1-2 plausible WEC concepts and justify them using depth + serviceability + survivability mode. 3. Estimate AEP using a simple power matrix and state an uncertainty range (best/typical/worst). 4. Add an explicit 'storm mode' assumption to the energy model and re-check storage/backup requirements.

Group: D3 **Reviewer:** Saba Falah

Journal Paper / Book Chapter / Article Citation: Hu, Q., Li, Z., Zhai, X., & Zheng, H. (2022). Development of Hydraulic Lifting System of Deep-Sea Mineral Resources. Minerals, 12(10), 1319. Available at: https://doi.org/10.3390/min12101319

Summary:

This article presents the design of a hydraulic lifting system intended for deep-sea mineral harvesting, with particular attention to challenges such as high hydrostatic pressure and extended riser pipe lengths. Key system components include pumps, riser pipes, and control mechanisms, and performance simulations validate the system's operational efficiency in deep-sea conditions. The proposed hydraulic system facilitates reliable and cost-effective extraction of polymetallic nodules while aiming to minimize environmental impact. Seawater and nodules are transported to the mining vessel via the riser system, as illustrated in Figure 26. In airlifting systems, parameters such as pipe diameter, depth of compressed air injection, and particle size significantly affect throughput and energy consumption.

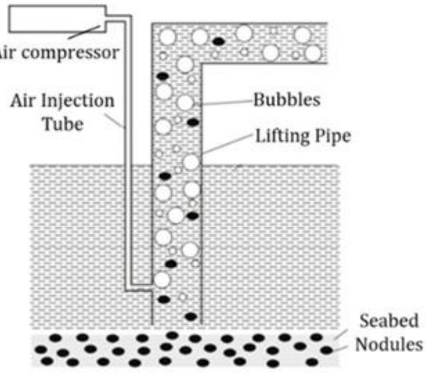


Figure 26. *Diagram of the air-lifting system with compartments.*

The hydraulic mining system primarily consists of three main components: the overwater support subsystem, the lifting subsystem, and the lifting subsystem. The overall configuration of the mining system is illustrated in Figure 27.

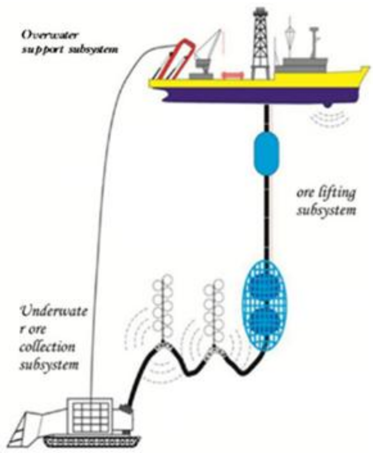


Figure 27. *The general schematic of a deep-sea hydraulic mining system.*

Although the pneumatic lifting method is characterized by structural simplicity, it tends to require excessively large overall dimensions when deployed in high-capacity commercial exploitation systems. Its lifting energy consumption is also substantially higher than that of hydraulic lifting, as illustrated in Figure 28, with the full comparative analysis provided in Table 38.

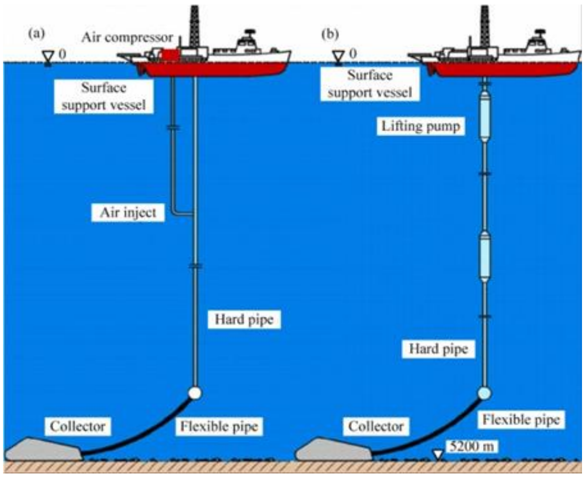


Figure 28. *The two underwater lifting systems: (a) pneumatic; (b) hydraulic.*

Table 38. *Comparing pneumatic lifting and hydraulic lifting systems.*

Performance	Pneumatic lifting	Hydraulic lifting
Structure reliability	Simple structure; low failure rate; easy maintenance	Complex structure and inconvenient maintenance
Pipe inner diameter	More than 1 m	About 0.4 m
Mineral particle size	No maximum particle size limitation	Exists a maximum particle size
Blocking risks	Low	Possibility of blockage
Collision damage in particle transportation	Not serious	Serious
Energy efficiency	Generally below 15 %	More than 40 %
Conveying concentration	3 %–4 %	About 15 %

Questions (aims):

1. What are the primary challenges in hydraulic lifting for deep-sea mining?
2. How does the proposed hydraulic system design overcome pressure and flow issues?
3. What simulation results validate the system's performance?
4. How can this technology support sustainable deep-sea resource extraction?

Answers (knowledge):

1. Deep-sea hydraulic lifting faces high hydrostatic pressures (up to 70 MPa at 7000m), long vertical distances causing pressure drops, and risks of pipe vibration or clogging from nodules.
2. The design uses a multi-stage centrifugal pump integrated with flexible riser pipes and dynamic pressure compensation, ensuring stable flow rates and reducing energy loss through optimized valve controls.
3. Simulations show lifting capacities of 100-300 tons/hour at depths beyond 5000m, with efficiency above 75% and minimal backpressure, confirming reliability under variable seabed conditions.
4. The system incorporates low-sediment disturbance features and modular scalability, allowing integration with ISA regulations for environmentally controlled mining operations.

Impact:

The hydraulic lifting system advances commercial deep-sea mining feasibility by improving extraction efficiency and reliability, potentially lowering costs for critical minerals like cobalt and nickel. It provides engineering benchmarks for scaling operations but requires field trials to assess real-world durability and ecological footprint in sensitive abyssal plains.

Actions:

1. Evaluate hydraulic system designs for integration into marine robotics projects.

2. Simulate similar pressure-flow dynamics using Python or Fusion 360 for mechanical engineering assignments.
3. Discuss applications in other sectors, drawing parallels to different fields’ mining hydraulics.
4. Incorporate findings into academic reports on sustainable resource extraction technologies.
5. Collaborate with engineering peers on prototype modeling for deep-sea analogs.
6. Include this completed worksheet in the Assignment 1 report appendix.

Group: D3 **Reviewer**: Muhammad Hammad

Journal Paper / Book Chapter / Article Citation: Silva-Campillo, A., Pérez-Arribas, F., Suárez-Bermejo, J. C. *Health-Monitoring Systems for Marine Structures: A Review*. Sensors, 2023, 23(4), 2099. Available at: https://doi.org/10.3390/s23042099

Summary:

This review paper presents a comprehensive overview of structural health monitoring (SHM) systems used in marine and offshore structures. Sensor technologies such as strain gauges, fiber optic sensors (FBG), accelerometers, pressure transducers, acoustic emission sensors, and fatigue damage sensors are covered. The study also discusses data processing techniques, sensor placement strategies, and alarm system and digital twin integration. Real-time monitoring is the main focus in order to enhance safety and avoid structural failure.

Questions (aims):

1) What sensor types are most suitable for continuous hull health monitoring?
2) How can leaks and structural damage be detected early?
3) What monitoring architecture supports real-time alarms and decision-making?
4) How should sensors be distributed across a marine structure?

Answers (knowledge):

1) Electrical strain gauges and fiber Bragg grating (FBG) sensors are widely used for hull stress and strain monitoring due to high accuracy and reliability.
2) Leak detection can be achieved using pressure transducers, acoustic emission sensors, and distributed fiber optic sensing.
3) A centralized monitoring system combining sensor data with data processing algorithms enables alarm triggering and condition assessment.
4) Optimal sensor placement is achieved using methods such as the inverse finite element method (iFEM) and modal analysis to maximize information capture.

Impact:

This paper supports structural safety and automation objectives by defining suitable sensor technologies and monitoring architectures for underwater structures It provides direct information for autonomous operation's alarm-driven decision support, leak detection, and hull integrity monitoring. Autonomous fault response and control logic still require further development.

Technology	Response	Pros	Cons
Capacitive Micro-Electro-Mechanical Systems (MEMS)	DC	Inexpensive, small size and easy to integrate into electrical systems	Poor signal-to-noise ratio, limited bandwidth
Piezoresistive (PR)	DC	Accurately calculating velocity or displacement	Low sensitivity, temperature compensation
Charge mode piezoelectric (PE)	AC	Good sensitivity, easy installation, durable in hostile environments	Need special cabling to shield from noise, requires a charge amplifier
Voltage mode Internal Electronic Piezoelectric (IEPE)	AC	Good sensitivity, easy installation, low noise levels, easily integrated with other systems	Ability to tolerate hostile environments when compared to PE accelerometers

Table 39. *Commonly available accelerator types.*

Actions:

1) Define a baseline SHM sensor suite mapped to failure modes.
2) Allocate sensors to structural zones based on risk and load paths.
3) Define alarm thresholds and data outputs for integration with the control system.

Group: D3 **Reviewer**: Muhammad Raafey

Journal Paper / Book Chapter / Article Citation: Tempfortransfer (2020) *Microsoft finds underwater datacenters are reliable, practical and use energy sustainably*. Available at: https://news.microsoft.com/source/features/sustainability/project-natick-underwater-datacenter/.

Summary:
The feasibility of deploying and operating UDCs for extended periods of time using seawater cooling is demonstrated by Project Natick in a real-world experimental scenario. It provides evidence on system reliability, failure rates, and energy sustainability, therefore supporting the practicality of sealed underwater data computing systems.

Questions (aims):

1. Are UDCs feasible in practice?
2. What failure rates were observed?
3. How reliable are sealed underwater systems?
4. How sustainable is the energy usage?
5. What are the operational benefits?

Answers (knowledge):

1. Microsoft's Project Natick demonstrates that UDCs are in fact feasible in practice with a high potential for success, as Microsoft successfully managed to deploy, operate, and test an underwater data center for many years without human intervention. The system remained functional throughout its entire deployment, thus validating the concept beyond theoretical modeling or lab testing.
2. Microsoft reported significantly lower server failure rates compared to conventional land-based data centers. The controlled, sealed environment reduces exposure to humidity and oxygen as well as disturbances induced through human activity/intervention, which are common causes of hardware failure in traditional data centers.
3. The sealed underwater system proved to be highly reliable. Through eliminating human access and maintaining a stable internal environment, the system experienced significantly reduced component failures. This suggests that sealed designs can indeed enhance reliability, provided all the components are designed for long-term autonomous operation.
4. The project demonstrated that UDCs can operate using renewable energy sources such as offshore wind or tidal power. Seawater cooling significantly reduced cooling energy demand (albeit not completely; if used with another passive cooling method, it should completely mitigate the energy demand for cooling), thus contributing to improved overall energy efficiency and supporting data center operation.
5. Operational benefits include but are not limited to reduced cooling energy requirements and lower failure rates, in addition to minimal maintenance requirements due to the sealed deployment. Additionally, proximity to coastal population centers can significantly reduce data latency, in turn offering performance benefits alongside energy and reliability requirements.

Impact:
The report provides strong real-world validation for the feasibility and reliability of underwater data centers, directly supporting the rationale for selecting a UDC. The observed reduction in failure rates and improved energy efficiency strengthen the case for sealed, passively cooled designs. These findings can be used to justify key design assumptions related to reliability and sustainability. However, the article does not provide detailed technical specifications or design calculations, which means that it must be supplemented through peer-reviewed engineering studies.

Actions:

1. Use Project Natick as a precedent to justify underwater deployment, passive cooling, and autonomous operation in the UDC design.
2. Combine observed failure-rate trends with academic models to inform reliability assumptions.
3. Investigate renewable energy integration options suitable for the proposed deployment location.
4. Identify design features that enable long-term sealed operation without maintenance access.

Group: D3 **Reviewer**: Talal Abu Ghazaleh

Journal Paper / Book Chapter / Article Citation: Karunarathne, S., Andrenacci, S., Carranza-Abaid, A., Jayarathna, C., Maelume, M., Skagestad, R., & Haugen, H. A.. (2025). Review on CO_2 removal from ocean with an emphasis on direct ocean capture (DOC) technologies. Separation and Purification Technology, 353, 128598. Available at: https://doi.org/10.1016/j.seppur.2024.128598

Summary:

This review covers ocean CO_2 removal methods with a focus on DOC, including electrochemical capture, mineralization, and ocean alkalinity enhancement. It discusses scale-up, industrial feasibility, and key performance metrics, especially capture efficiency and electricity/energy consumption, plus environmental and operational challenges.

Questions (aims):

1. What DOC approaches are most energy-efficient based on reported metrics?
2. What energy ranges (e.g., GJ per tonne CO_2) are reported for electrochemical DOC?
3. What factors drive energy use in electrodialysis/BPMED-type DOC systems?
4. What environmental/operational constraints must be controlled during deployment?
5. For your project, what design requirements should follow from the energy discussion?

Answers (knowledge):

1. The review highlights electrodialysis-type DOC as a strong candidate when optimized, with high reported capture efficiencies and low electricity consumption in the best cases.
2. Reported examples include 2.4-2.8 GJ/tonne CO_2 (high-efficiency cases) and 5.5 GJ/tonne CO_2 (prototype BPMED example).
3. Energy is driven mainly by separation work and electrochemical operation (cell design, membranes, pH control, and resistive losses).
4. Main constraints include managing seawater chemistry changes, by-products, discharge conditions, and practical scale-up issues (materials, fouling, and durability).
5. Your DOC concept should set clear targets for energy/tonne CO_2, capture efficiency, and seawater-chemistry limits, plus monitoring and safe discharge requirements.

Reference	Category	Max. capture efficiency	Min. or range in specific energy consumption ($GJ/tonne_{CO2}$)	What is included in specific energy consumption	TRL
Eisaman et al. (2012)	Electrochemical – acid	68 % at 6.5 $GJ/tonne_{CO2}$	5.5 $GJ/tonne_{CO2}$ with 59 % capture efficiency	Only electrical energy	3
Digdaya et al. (2020)	Electrochemical – acid + CO2RR	77 %	3.4 $GJ/tonne_{CO2}$ with 71 % capture efficiency	Only electrical energy	2–3
Willauer et al. (2014)	Electrochemical – acid + water electrolysis	92 %	2.1 $MJ/tonne_{H2}$	Only electrical energy	4
Yan et al. (2022)	Electrochemical – acid	91 %	2.4 $GJ/tonne_{CO2}$	Only electrical energy	2
Kim et al (2023)	Electrochemical – acid	87 %	2.8 $GJ/tonne_{CO2}$	Only electrical energy	2
Lannoy et al. (2017) and Eisaman et al. (2018) – acid process	Electrochemical – acid	N/A	8.95 (ED stack alone) or 11.3 $GJ/tonne_{CO2}$ (system*)	**ED stack: electrical energy. The system includes pumps, vacuum pumps and pretreatment	3
Lannoy et al. (2017) and Eisaman et al. (2018) – base process	Electrochemical – base + evt. chemical process	N/A	14.95 (ED stack alone) or 15.8 $GJ/tonne_{CO2}$ (system*)	**ED stack: electrical energy. The system includes pumps, vacuum pumps and pretreatment	3
La Plante et al. (2021)	Electrochemical – base	N/A	2.5 to 8.3 $GJ/tonne_{CO2}$	**System exclusive of water intake	1
Sharifian et al. (2022)	Electrochemical – base	60 (real SW)– 85 (synthetic SW)	7.23 $GJ/tonne_{CO2}$ (real SW)	Only electrical energy	3
Renforth et al. (2013)	Ocean Alkalinity enhancement	NE/A	0.1 to 1.3 $GJ/tonne_{CO2}$	Extraction, calcination, hydration and surface ocean dispersion of lime	1
Gauntlett R.B.	Air stripping	75 %	1.8	Only gas stripping, CO_2 separation not included	2

* The system considered has CO_2 extraction capacity of 7202 metric tons of CO_2 per year.

** By techno-economic analysis.

Table 40. *Summary of parameters to describe physicochemical ONETs.*

Impact:

1. Provides energy benchmarks for DOC (useful for your energy efficiency objective).
2. Summarizes which DOC routes look most scalable/feasible and why.
3. Highlights environmental/industrial constraints that should become requirements.
4. Helps justify objective-tree targets using literature-level performance metrics.

Actions:

1. Set an initial energy target (GJ/tonne CO_2) for your system and justify it from this review.

2. Decide which DOC route you're assuming (e.g., electrodialysis/BPMED-type) and list the key energy drivers to minimise.
3. Add seawater-chemistry + discharge constraints into requirements (pH/ion balance/by-products monitoring).
4. Find 1 more paper that reports pilot/scale-up energy data to validate the target range.

Group: D3 **Reviewer:** Matteo Karam

Journal Paper / Book Chapter / Article Citation: National Oceanic and Atmospheric Administration (NOAA). (n.d.). *Aquarius Underwater Laboratory.* NOAA Education. Available at: https://www.noaa.gov/education/resource-collections/ocean-coasts/aquarius-underwater-laboratory

Summary:
This article describes the life support and habitability systems of the Aquarius Underwater Laboratory, the world's only operational undersea research habitat. It outlines air supply, CO_2 scrubbing, temperature control, humidity management, and emergency systems. Aquarius serves as an analogue for both underwater and space habitats.

Questions (aims):

1. How does Aquarius maintain a safe living environment underwater?
2. What life-support technologies are used in a real underwater habitat?
3. How is crew safety ensured during long missions?
4. What analog value does Aquarius provide for space exploration?

Answers (knowledge):

1. Aquarius uses surface-supplied air with continuous CO_2 scrubbing.
2. Environmental conditions are monitored in real time for crew safety.
3. Emergency systems allow safe evacuation if life support fails.
4. Aquarius demonstrates parallels between underwater and space habitats.

Impact:

1. Provides a real-world case study of underwater life support.
2. Strengthens the credibility of underwater habitat concepts.
3. Demonstrates operational feasibility.

Actions:

1. Reference Aquarius as a benchmark system.
2. Adapt proven technologies to the conceptual habitat.
3. Use Aquarius to justify design assumptions.

Group: D3 **Reviewer:** Ahmad Alhashmi

Journal Paper / Book Chapter / Article Citation: Walker, S., & Thies, P. R. (2021). A review of component and system reliability in tidal turbine deployments. Renewable and Sustainable Energy Reviews, 151, 111495. Available at: https://doi.org/10.1016/j.rser.2021.111495

Summary: A deployment-based review (58 projects, 2003-Aug 2020) focusing on downtime drivers. • Shows that reliability and maintenance access dominate cost and viability in tidal energy; predictable tides do not automatically mean reliable power. • Recurring issues include marine degradation (corrosion, biofouling, sealing) and failures in drivetrain/electrical interfaces; access windows strongly affect repair time. • Highlights the value of modular retrieval strategies and condition monitoring to reduce downtime. • For our habitat, tidal is attractive for predictability, but only if hardware is designed for swap-out maintenance and fault isolation.
Questions (aims): 1. Which subsystems most often drive downtime, and how do we design redundancy/modularity around them? 2. What maintenance concept fits our self-sufficient habitat: in-situ repair, swap-out modules, or full retrieval? 3. How do we choose the number/size of turbines to tolerate one unit being offline while still meeting critical loads? 4. What safeguards/mitigations are needed for marine life interaction, seabed disturbance, and noise?
Answers (knowledge): 1. Assume a small set of components dominates downtime; design around them with monitoring, spares, and modular replacement. 2. A swap-out approach is most compatible with isolation: treat the turbine as a 'service cartridge' recoverable with minimal specialist tooling. 3. Use multiple turbines (N+1 philosophy) so a single failure does not cause loss of critical power; document the footprint/cabling trade-off. 4. Mitigate impacts via siting, guarding, and operational controls (e.g., shutdown conditions) and include them in the safeguards statement.
Impact: • Supports redundancy and maintenance assumptions used in the energy system design and CONOPS. • Strengthens corrosion/biofouling control as a core energy-reliability requirement. • Provides evidence for including condition monitoring and fault detection in the energy architecture.
Actions: 1. Draft a simple tidal-turbine FMEA-lite and identify design mitigations (seals, coatings, modularity, monitoring). 2. Define a maintenance and retrieval concept (tools, access, who does what) suitable for long-duration autonomy. 3. Add an availability assumption into the power model and run sensitivity (what if availability is lower?). 4. Decide an N+1 turbine count that keeps life-support loads powered during a turbine outage.

Group: D3 **Reviewer:** Saba Falah

Journal Paper / Book Chapter / Article Citation: Zhang, Q., Chen, X., Luan, L., Sha, F., & Liu, X. (2025). Technology and equipment of deep-sea mining: State of the art and perspectives. Earth Energy Science, 1(1), 65-84. Available at: https://doi.org/10.1016/j.ees.2024.08.002

Summary:
This review article delineates the technological evolution and contemporary status of deep-sea mining systems targeting polymetallic nodules, sulphides, and cobalt-rich crusts, propelled by escalating demand for critical high-tech minerals.

1. Key Components:
   - Mining vehicles incorporating collection mechanisms and ambulatory locomotion modes.
   - Subsea lifting systems for material transport.
   - Surface support vessels for operational oversight.
   - Design imperatives: minimized environmental impact, enhanced reliability, optimized energy efficiency, and scalability for commercial deployment.
2. Mineral Resources:
   Polymetallic nodules (PMN), polymetallic sulphides (PMS), and cobalt-rich ferromanganese crusts (CFC) exhibit substantial commercial potential, with abundant reserves of metals and rare earth elements, as depicted in Figure 29.
3. Challenges and Directions:
   Ongoing hurdles include achieving self-propelled crawler mobility, advancing hydraulic lifting innovations, and integrating intelligent automation to facilitate viable commercial exploitation.

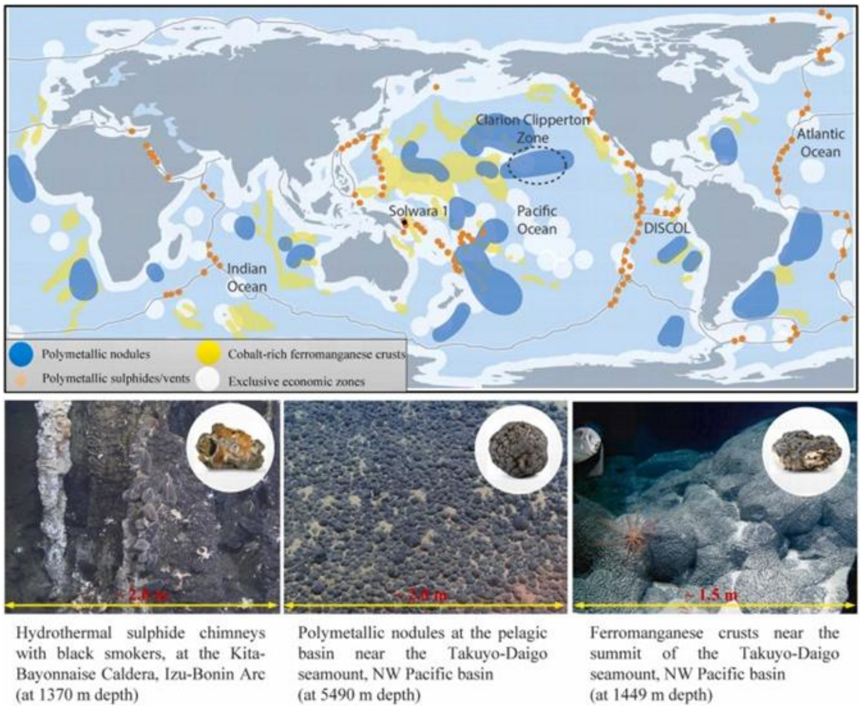


Figure 29. *Distribution of deep-sea mineral resources by type.*

Deep-sea mineral resources can be summarized as follows, representing prime targets for commercial exploitation due to their rich critical metal content (Ni, Co, Cu, Mn, REEs) vital for energy transition technologies.

1. **Polymetallic Nodules (PMN):** Layered iron-manganese oxide concretions, enriched with metals, manifest as 0.5-15 cm spherical or irregular nodules on deep-sea sediment surfaces at depths of 4,000-6,000 m.
2. **Cobalt-Rich Ferromanganese Crusts (CFC):** High-grade (>50-60%) sulphidic mineral deposits derived from hydrothermal activity, predominantly concentrated on seamount slopes and summits at depths of 800-2,500 m.
3. **Polymetallic Massive Sulphides (PMS):** Layered manganese-iron oxide deposits bearing metals, formed on hard rock substrates at tectonic plate boundaries, mid-ocean ridges, and volcanic arcs at depths of 1,000-4,000 m.

Seabed collection of these nodules entails "picking" them onto a mining vehicle (see Figure 30). Given sparse distributions (few to tens of kilograms per square meter), the vehicle must traverse soft seabed sediment continuously to maintain collection efficiency.

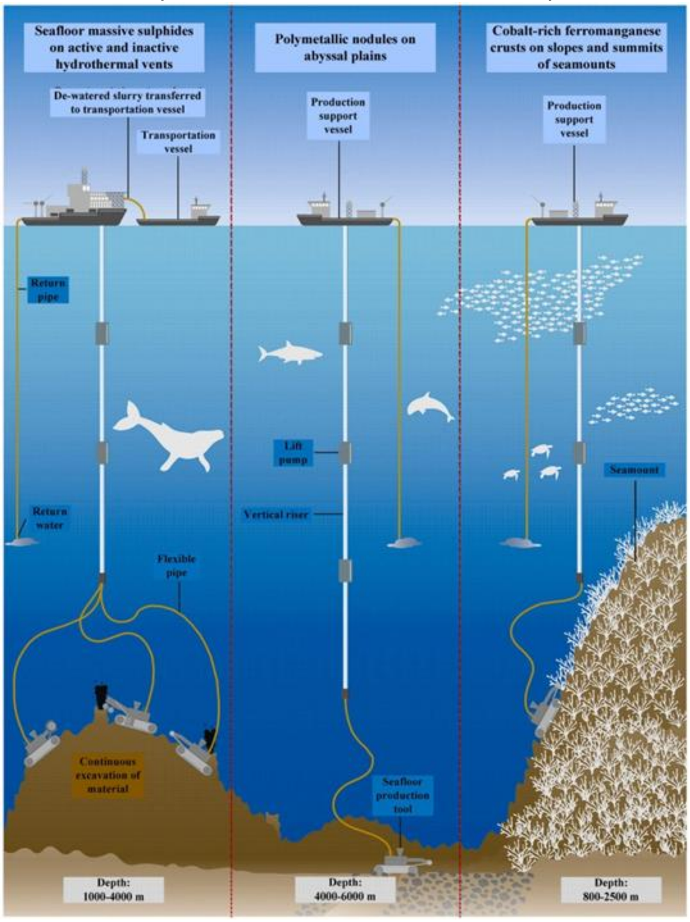


Figure 30. *Deep-sea mining processes for three mineral deposit types.*

Mineral particles are entrained within the water flow and conveyed through the transport pipeline. This methodology was pioneered by the Japanese Technology Research Association of Ocean Mineral Resources Mining System (TRAM) in 1981, with successful sea trials conducted in 1997 (Figure 31).

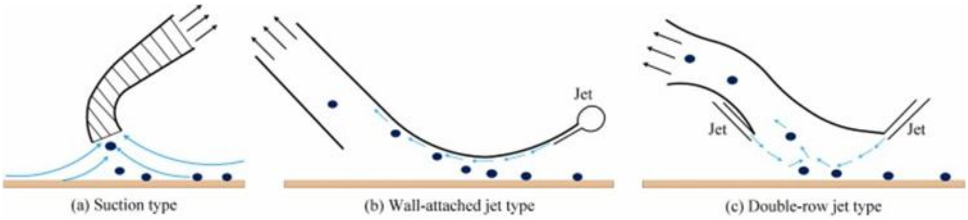


Figure 31. *Schematic of three hydraulic collection* models: (a) Suction; (b) Wall-attached; (c) Double-row.

Deep-sea mining vehicles play a crucial role in deep-sea mining operations. These vehicles consist of several integrated parts, such as the collecting system, propulsion system, buoyancy aids, control unit, crushing mechanism, and chassis structure (see Figure 32).

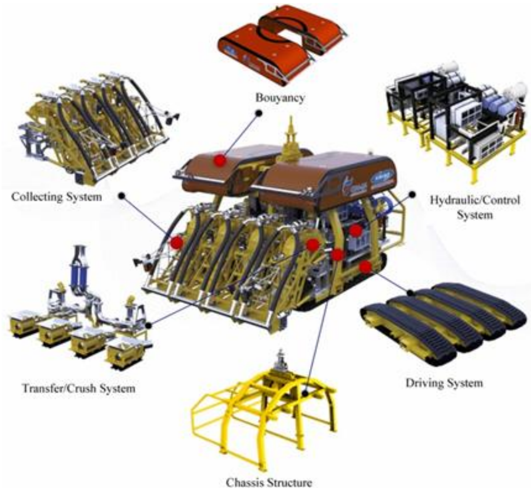


Figure 32. *Representation of the mining vehicle system.*

Questions (aims):

1. What are the primary technological requirements for commercial deep-sea mining?
2. How have deep-sea mining vehicles evolved in collection and mobility?
3. What are the key components and challenges of subsea lifting systems?
4. What future developments are needed for sustainable deep-sea mining?

Answers (knowledge):

1. Commercial systems must prioritize low environmental disturbance, high reliability under extreme pressures, energy-efficient operations, and high collection efficiency to compete with land-based mining.
2. Mining vehicles have progressed from towed collectors to autonomous crawlers with track-based mobility and innovative nodule pickup mechanisms like suction or mechanical grippers, improving adaptability to uneven seabeds.
3. Subsea lifting involves riser pipes, pumps, and dynamic control systems to transport minerals from 4000-7000 m depths; challenges include pressure differentials, clogging, and vertical stability in currents.
4. Prospects include eco-friendly self-propelled vehicles, advanced hydraulic pipelines, AI-driven intelligence, and integrated environmental monitoring compliant with ISA regulations.

Impact:

The article provides a comprehensive roadmap for advancing deep-sea mining from trials to commercial scale, informing equipment design for critical mineral supply chains vital to energy transitions. It highlights pathways to balance resource extraction with ocean protection but stresses the need for integrated field testing and regulatory alignment to overcome technical hurdles and public concerns.

Actions:

1. Analyze mining vehicle designs for mechanical engineering projects.
2. Model lifting system dynamics using Fusion 360 or Python simulations for academic reports.
3. Present key technologies in discussions, comparing them to other sectors' resource extraction.
4. Integrate findings into sustainability modules for engineering competitions.
5. Collaborate with peers on deep-sea mining technology prototypes.
6. Include this completed worksheet in the Assignment 1 report appendix.

Group: D3 **Reviewer**: Muhammad Hammad

Journal Paper / Book Chapter / Article Citation: Sharma, R., et al. Design and Development of Integrated Monitoring Systems for Marine Structures. International Journal of Naval Architecture and Ocean Engineering, 2014. Available at: https://iaeme.com/Home/article_id/IJNAOER_06_01_001

Summary:
This paper focuses on the design of integrated sensor networks for marine structures, emphasizing internal monitoring, fault detection, and alarm philosophy. It covers automated warning systems, real-time monitoring, sensor fusion, and system design. The study emphasizes how crucial data integration, redundancy, and dependability are for offshore and undersea systems.

Questions (aims):
1) How can multiple sensors be integrated into one monitoring system?
2) What alarm philosophy is suitable for unmanned marine structures?
3) How can internal control and monitoring be automated?
4) What role does redundancy play in safety-critical systems?

Answers (knowledge):

1) Sensors are integrated through a centralized control unit using wired or fiber-optic communication.

Sensor Type	Key Features	Advantages	Limitations
Piezoelectric	Vibration and wave sensing	High sensitivity, low cost	Susceptible to noise
Fiber-optic (FBG)	Strain and temperature sensing	High resolution, EMI immunity	Requires complex installation
Acoustic Emission	Passive crack monitoring	Early detection of crack growth	Limited by signal attenuation
Ultrasonic Sensors	Active defect imaging	Precise flaw localization	Needs coupling media

Table 41. *Different* types *of sensors that can be integrated together.*

2) A multi-level alarm philosophy is recommended (warning, critical, and emergency).
3) Automated monitoring uses threshold-based alarms combined with trend analysis.
4) Redundant sensors and communication paths improve reliability and fault tolerance.

Impact:
It facilitates safe operation without constant human supervision and autonomous operating needs. By addressing system-level automation, it enhances the sensor-focused article.

Actions:
1) Design a multi-level alarm philosophy for hull stress and leakage.
2) Create a monitoring and control system block diagram.
3) Define redundancy requirements for safety-critical sensors.

Group: D3 **Reviewer**: Muhammad Raafey

Journal Paper / Book Chapter / Article Citation: Liu, F. *et al.* (2021) 'Design approach for subsea data center based on thermodynamic theory under the premise of building energy conservation,' *Thermal Science*, 25(6 Part A), pp. 4209–4216. https://doi.org/10.2298/tsci2106209l.

Summary:
The paper proposes a thermodynamic-based design approach for marine data centres, focusing on passive cooling and energy conservation. Heat transfer and energy balance principles are applied to model how waste heat from computing equipment can be efficiently dissipated into the surrounding seawater. The study provides a theoretical framework for sizing and designing subsea data centres that maximise cooling efficiency while minimising energy consumption.

Questions (aims):

1. What design considerations improve passive cooling efficiency?
2. How is thermodynamic theory applied to UDC design?
3. How do convective heat transfer principles apply to UDCs?
4. What assumptions are used in heat transfer modeling?
5. How does energy conservation guide system scaling?
6. What materials and configurations maximize energy conservation?

Answers (knowledge):

1. The paper identifies effective heat exchange between the data center enclosure and surrounding seawater as a key design consideration. Key factors such as maximizing external surface area, ensuring good thermal contact with seawater, and minimizing internal thermal resistance all contribute to improved passive cooling efficiency.
2. Thermodynamic theory is applied through energy balance equations that relate heat generated by servers to heat rejected into the surrounding seawater. The system is modeled as a closed energy system, using steady-state assumptions to estimate temperature gradients and cooling capacity under defined operating conditions.
3. Convective heat transfer governs how heat is transferred from the data center enclosure to the surrounding seawater. The report uses convection coefficients to describe heat dissipation, highlighting that seawater flow and temperature differences significantly influence overall cooling performance.
4. Key assumptions include steady-state operation, uniform internal heat generation, and relatively stable seawater temperatures. Consistent material properties and neglected transient effects are also assumed, therefore, simplifying the model while acknowledging that real-world conditions may introduce variability.
5. Energy conservation principles are utilized to relate total computing load to required heat rejection capacity. As system scale increases, cooling surface area and thermal exchange capacity must scale accordingly to maintain acceptable operating temperatures, guiding safe and efficient system scaling as shown by the paper.
6. Materials with high thermal conductivity are favored to minimize internal thermal resistance. It is also recommended that compact yet thermally efficient configurations that promote uniform heat distribution, reducing energy losses and improving overall cooling effectiveness, should be implemented into the design.

Impact:
The paper provides a strong theoretical foundation for the thermal and energy design for the UDC. The thermodynamic framework can be directly used to justify cooling assumptions, system scalability, and energy efficiency claims. The emphasis on passive cooling and energy conservation aligns closely with the sustainability objectives of the project. However, the paper doesn't cover operational challenges, environmental variability, or long-term reliability, which must be supplemented by empirical and case-study sources.

Actions:

1. Apply the thermodynamic framework to estimate cooling capacity for the proposed UDC computing load.
2. Use energy conservation principles to justify system scaling decisions in the design report.
3. Compare theoretical assumptions with real-world data from Project Natick and other case studies.
4. Identify materials and enclosure configurations suitable for efficient heat transfer in the marine environment.

Group: D3 **Reviewer**: Talal Abu Ghazaleh

Journal Paper / Book Chapter / Article Citation: Michaelides, E. E. (2021). Thermodynamic analysis and power requirements of CO_2 capture, transportation, and storage in the ocean. Energy, 230, 120804. Available at: https://doi.org/10.1016/j.energy.2021.120804

Summary:
The paper explains the full CCS chain (Carbon Capture & Storage): separate CO_2 from flue gas, compress it, transport it (pipeline), store it deep in the ocean, and monitor it. It derives a benchmark "minimum work" (exergy) for separating CO_2 from a gas mixture using an ideal reversible membrane process and argues any real capture method must use ≥ that minimum. It then shows why CO_2 should be transported at supercritical pressures (> 8 MPa) to avoid two-phase flow/bubbles and reduce friction losses, and it runs a case study on a 1 GW coal plant showing CCS can consume a big fraction of generated power: 16% minimum (ideal) but ~58% with realistic efficiencies.

Questions (aims):

1. What are the main stages of CCS in this paper?
2. What is the minimum theoretical work required to separate CO_2 from flue gas?
3. Why must real capture processes use at least this minimum work?
4. Why is CO_2 transported at supercritical pressure (> 8 MPa)?
5. What depth/conditions are discussed for ocean storage, and what's the basic idea?

Answers (knowledge):

1. CCS is described as CO_2 separation from exhaust, then compression, transportation (pipeline), and long-term storage/monitoring.

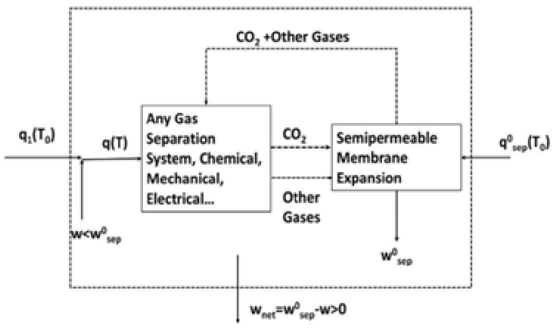
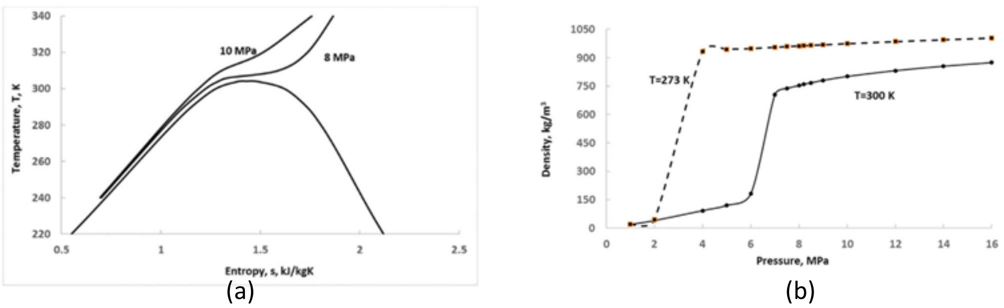


Figure 33. *Conceptual representation of a generic gas separation system coupled with a semipermeable membrane expansion stage.*

2. The paper gives a benchmark minimum separation work of 106.8 kJ/kg CO_2 (25°C) (reversible isothermal membrane separation).
3. If a process used less than the minimum, it would violate the second law, so real methods must consume ≥ this benchmark.
4. Supercritical transport avoids two-phase flow/bubbles and reduces friction losses; practical operation should be above the critical pressure and typically > 8 MPa.
5. Ocean storage is discussed at depths > 500 m, considering injection/droplet dissolution behavior and storage rationale.



(a) (b)

***Figure 34**. (a) Pressure*-density *diagram for carbon dioxide; (b) T-s diagram of carbon dioxide.*

Impact:

1. Gives a clear energy benchmark for CO_2 separation (useful for your DOC energy-efficiency targets).
2. Shows why energy penalty is unavoidable (2nd-law argument), supporting realistic requirements.
3. Provides guidance on CO_2 handling/transport conditions (supercritical pressure) relevant to CO_2 processing choices.
4. Quantifies how big the system-level power penalty can be, helping justify efficiency as a core objective.

Actions:

1. Use 106.8 kJ/kg CO_2 as a baseline benchmark when discussing “minimum energy” claims.
2. Set a DOC requirement/target for energy per ton of CO_2 and explain the gap between ideal and real performance.
3. Note the importance of supercritical CO_2 handling when proposing downstream CO_2 storage/transport assumptions.
4. Add a short “system efficiency penalty” discussion to your report to justify why DOC energy efficiency matters.

Group: D3 **Reviewer:** Matteo Karam

Journal Paper / Book Chapter / Article Citation: Martinez, S., Lee, H., & Patel, G. (2020). *Closed and semi-closed life support systems in extreme environments.* Sustainability, Volume 12, Article 1987. Available at: https://doi.org/10.3390/su12051987

Summary: This paper analyzes closed and semi-closed life support systems used in extreme environments, including underwater habitats, submarines, and space stations. It compares physicochemical and bioregenerative approaches to air and water recycling, highlighting trade-offs between complexity, reliability, and sustainability.
Questions (aims): 1. What level of closure is practical for underwater habitats? 2. How do closed-loop systems improve mission sustainability? 3. What technologies are most reliable underwater? 4. What trade-offs exist between openness and autonomy?
Answers (knowledge): 1. Fully closed-loop systems are complex but reduce resupply needs. 2. Semi-closed systems balance reliability and sustainability. 3. Physicochemical systems are currently more reliable underwater. 4. Biological systems may supplement life support but require backup.
Impact: 1. Supports the selection of semi-closed life support architecture. 2. Provides justification for hybrid system design. 3. Aligns underwater habitats with sustainable engineering principles.
Actions: 1. Adopt a semi-closed life support approach. 2. Use hybrid physicochemical systems with limited bioregeneration. 3. Reference this work for justifying system architecture choices.

Group: D3 **Reviewer:** Ahmad Alhashmi

Journal Paper / Book Chapter / Article Citation: Xu, L., Guerrero, J. M., Lashab, A., Wei, B., & Bazmohammadi, N. (2022). A Review of DC Shipboard Microgrids—Part I: Power Architectures, Energy Storage, and Power Converters. IEEE Transactions on Power Electronics, 37(5), 5155-5172. Available at: https://doi.org/10.1109/TPEL.2021.3128417

Summary:

A structured review of DC shipboard microgrid architectures and why topology matters for resilience.

• Reviews DC shipboard microgrids: architecture options (radial/ring/zonal), converter blocks, storage interfaces, and integration trade-offs.

• Highlights DC advantages for isolated systems: fewer conversion stages for storage/electronic loads, flexible integration of diverse sources, and compactness.

• Emphasizes topology for fault containment: segmented/zonal layouts can prevent a single fault from cascading across the facility.

• Directly relevant to our habitat: we need black-start capability, load shedding (life support first), and a clear protection philosophy.

Questions (aims):

1. Should we use AC, DC, or hybrid distribution, and what is the justification (efficiency, protection, or equipment compatibility)?
2. Which topology best matches our habitat layout and supports fault containment (zonal versus simple radial)?
3. What voltage levels are plausible for main distribution versus local conversion at the concept stage?
4. How do we integrate storage so it supports energy balancing and transient stability?

Answers (knowledge):

1. A hybrid system is realistic: a main DC bus for storage + electronic loads, with local AC where motors/legacy equipment require it.
2. A zonal/segmented architecture maps naturally to habitat zones and improves resilience by isolating faults.
3. Voltage selection is a trade-off between conductor mass and safety/protection complexity; propose a candidate range and note standards-driven refinement later.
4. Use bidirectional converters so storage can regulate bus voltage during transients and cover energy deficits during renewable lulls.

Impact:

• Provides authoritative framing for microgrid topology and converter/interface choices in our report.

• Supports load prioritization + shedding as an engineered control strategy, not an afterthought.

• Improves safety argument: segmentation and fault isolation reduce single-point failures.

Actions:

1. Draw the habitat single-line diagram (sources → buses → converters → critical/non-critical loads).
2. Select a candidate topology (e.g., zonal DC with tiebreakers) and justify it using fault containment and expansion logic.
3. Write a load-priority and shedding scheme (life support first) with simple trigger conditions.
4. Define a black-start concept: which source/storage re-energizes the bus and in what sequence.

Group: D3 **Reviewer:** Saba Falah

Journal Paper / Book Chapter / Article Citation: Deng, L., Lu, H., Yang, J., Guo, R., Zhang, B., & Sun, P. (2025). Vortex-Induced Vibration of Deep-Sea Mining Riser Under Different Currents and Tension Conditions Using Wake Oscillator Model. Journal of Marine Science and Engineering, 13(8), 1565. Available at: https://doi.org/10.3390/jmse13081565

Summary:

This study investigates vortex-induced vibrations (VIV) in deep-sea mining risers through a wake oscillator model, evaluating the influences of varying ocean currents and top tension on dynamic responses.

1) Modeling Approach:
Fluid-structure interactions are modeled via coupled Van der Pol wake oscillator and beam equations, simulating cross-flow (CF) and in-line (IL) vibrations under realistic ocean conditions.
2) Key Findings:
Results elucidate fatigue life, displacement amplitudes, and mitigation strategies to enhance riser stability during mineral lifting operations.
3) Riser Design Configuration:
To minimize drag forces from ocean currents at operational depths, buoyancy modules are omitted in the upper riser section. Wastewater discharge occurs at 600 m depth; the riser span from 100-600 m integrates buoyancy modules and wastewater pipes (Figure 35.a). The lower section (600-4,000 m) incorporates buoyancy modules as configured in Figure 35.b.

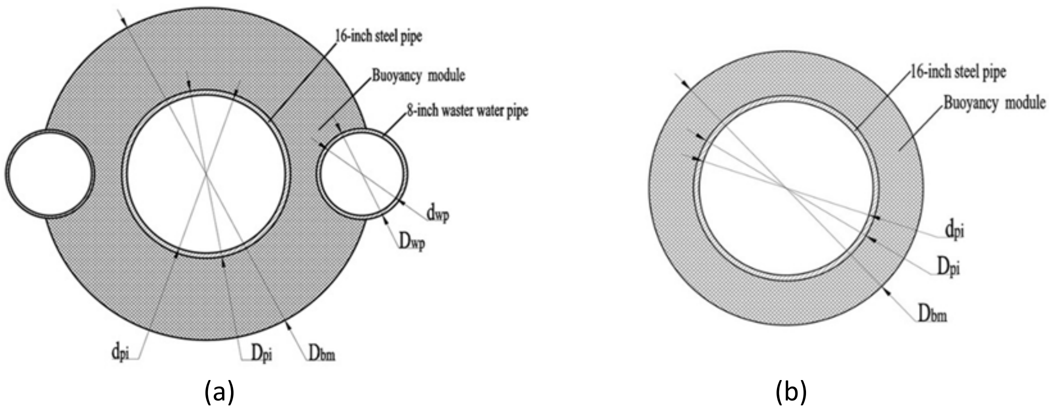


Figure 35. (a) Cross-sectional view of a riser featuring integrated return pipes and a buoyancy module; (b) Cross-sectional view of a riser equipped solely with a buoyancy module.

Questions (aims):

1. How does the wake oscillator model predict VIV in deep-sea mining risers?
2. What impact do current profiles and top tension have on riser vibration responses?
3. How do VIV modes (standing vs. traveling waves) affect riser fatigue life?
4. What parameters optimize riser design for deep-sea mining reliability?

Answers (knowledge):

1. The wake oscillator model couples nonlinear Van der Pol equations for fluid wake dynamics with structural beam equations, capturing lock-in phenomena and multi-modal VIV responses efficiently without full CFD computation.
2. Sheared currents increase dominant mode orders and peak displacements, while higher top tension reduces RMS amplitudes and shifts weak points, extending fatigue life; uniform flows show standing waves, and sheared flows add traveling components.

3. Multi-modal VIV combines standing and traveling waves, with CF vibrations dominating displacement but IL contributing equally to fatigue due to doubled frequency; buoyancy modules and fairings mitigate this by altering stress distribution. 4. Optimal designs balance tension (20-30% submerged weight), buoyancy placement, and suppression devices, targeting <0.2D normalized displacement and >10-year fatigue life under 1-2 knot currents at 4000-6000 m depths.
Impact: The model provides predictive tools for riser fatigue assessment, enabling safer deep-sea mining by quantifying VIV risks under operational variability. It guides design iterations to minimize downtime and environmental disturbance from vibrations but requires experimental validation in full-scale ISA test sites to confirm predictions against real seabed currents.
Actions: 1. Simulate riser VIV dynamics in Python or MATLAB for mechanical engineering vibration assignments. 2. Model tension-current interactions using Fusion 360 for optimization analogues. 3. Integrate wake oscillator methods into academic reports on marine/offshore engineering. 4. Prototype scaled riser tests with peers for the project. 5. Include this completed worksheet in the Assignment 1 report appendix.

Group: D3 **Reviewer**: Muhammad Hammad

Journal Paper / Book Chapter / Article Citation: Yazdani, A. M., Sammut, K., Yakimenko, O., & Lammas, A. (2019). A Survey of Underwater Docking Guidance Systems. Robotics and Autonomous Systems, 124, 103382. Available at: https://doi.org/10.1016/j.robot.2019.103382

Summary: This paper presents a comprehensive survey of underwater docking guidance systems for autonomous underwater vehicles (AUVs). It reviews the complete docking process, including homing, guidance, approach, capture, and retention. In addition to discussing environmental issues such as currents, visibility, and positional uncertainty, the paper compares acoustic, optical, electromagnetic, and hybrid guidance methods. The study emphasizes that long-duration autonomous missions require dependable docking that is resilient to misalignment and external disruptions.
Questions (aims): 1) What guidance methods are most suitable for reliable autonomous docking underwater? 2) What environmental factors most strongly affect docking success? 3) How should docking interfaces accommodate misalignment and uncertainty? 4) What functional requirements should a subsea docking system satisfy?
Answers (knowledge): 1) Acoustic guidance is suitable for long-range homing, while optical systems provide high-precision short-range alignment. Hybrid approaches offer the best overall reliability. 2) Ocean currents, poor visibility, sensor noise, and vehicle positioning errors significantly reduce docking accuracy and stability. 3) Docking interfaces must tolerate positional and angular misalignment through funnel-shaped structures, compliant mechanisms, or guided capture geometries. 4) A docking system must support autonomous homing, controlled approach, secure capture, and stable retention under environmental disturbances.
Impact: This paper directly informs the robotics objectives related to autonomous docking, alignment tolerance, and environmental robustness. It encourages the inclusion of objectives, including safe docking, regulated vehicle approach, and operation in unpredictable circumstances. The necessity of a docking interface that reduces reliance on exact human control is also justified in the study. Remaining gaps include detailed design choices for power and data transfer once docking is achieved.

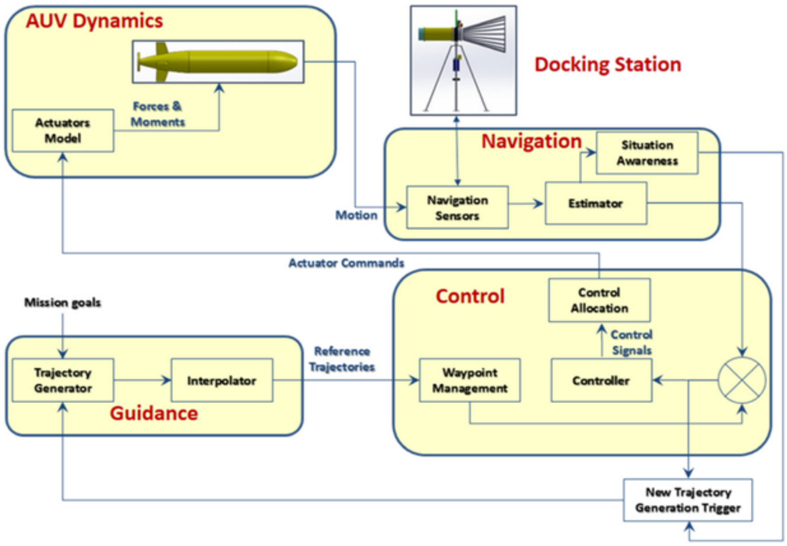


Figure 36. *Typical docking GNC architecture.*

Actions:

1. Develop a functional flow diagram for the autonomous docking sequence.
2. Define alignment and misalignment tolerance requirements.
3. Identify environmental design constraints affecting docking reliability.

Group: D3 **Reviewer**: Muhammad Raafey

Journal Paper / Book Chapter / Article Citation: Port_Admin (2025) *Marine data centers: floating, submerged, and more efficient*. Available at: https://piernext.portdebarcelona.cat/en/technology/marine-data-centers-floating-submerged-and-more-efficient/.

Summary: The article explores the emergence of marine-based data centers, both floating and submerged, as sustainable and more efficient alternatives to land-based facilities. It outlines operational advantages such as reduced energy consumption through efficient cooling with seawater, lower failure rates, and proximity to data demand centers. The paper also discusses engineering challenges and environmental considerations that need to be addressed for marine data centers to become a successful scalable solution for growing digital infrastructure needs.
Questions (aims): 1. What are the key operational benefits of marine-based data centers? 2. How do efficiency gains compare to land-based data centers? 3. What engineering challenges arise in marine environments? 4. What future trends are identified? 5. How do submerged UDCs compare with floating models?
Answers (knowledge): 1. Marine data centers can exploit the natural cooling capacity of seawater, which lessens the requirement for active mechanical cooling systems and significantly lowers energy consumption. The article highlights that submerged centers benefit from stable, cold-water conditions that minimize thermal variation and reduce hardware failure rates compared to terrestrial facilities. Furthermore, by placing data centers near coastal regions or ports, latency for end users can be reduced due to closer geographic proximity to populations and data demand centers. 2. The paper highlights reported gains such as up to 70% improved cooling efficiency in addition to 30% reduced energy consumption for some marine data centers versus land-based equivalents, largely because the surrounding marine environment absorbs waste heat passively and consistently. These figures suggest that marine-based cooling can deliver significantly enhanced energy performance compared to traditional air-cooled systems. 3. Operating in marine environments introduces several challenges. Corrosion and biological fouling require robust materials and coatings to protect infrastructure. Maintenance access is more complex, especially for submerged units, since repairs may necessitate extraction to the surface and thus would involve a change in environment (which could be harmful for components through means such as rust/corrosion). Marine ecosystems and environmental legislation also impose constraints: variations in seawater temperature, currents, and potential heat discharge effects must be mitigated. 4. The article discusses the proliferation of projects in different formats: submerged, floating, and hybrid models. in regions such as California, China, and Japan, with varying design aims. Floating models offer flexibility and mobility and can be relocated based on demand, while submerged projects prioritize stability and thermal consistency, which means that they require long-term/permanent solutions until their end of service. Integration with renewable energy sources (offshore wind and tidal) and selective siting near coastal hubs are identified as key trends for future marine data center deployment. 5. Submerged UDCs are described as less sensitive to environmental variation, with more consistent cooling performance due to deeper and more stable water temperatures.

	Floating models may face more variability from wave action, currents, and temperature changes, but they offer greater operational flexibility, such as relocation capacity, and so can still benefit from natural cooling. Each model thus presents distinct trade-offs between stability and adaptability.
Impact: This article provides practical context for different marine data center configurations, highlighting how environmental conditions and deployment strategies influence operational performance. For the design of the UDC, the comparison between submerged and floating models will help inform decisions about deployment depth, cooling strategies, and system flexibility. The documented efficiency gains strengthen the rationale for selecting marine cooling approaches. Furthermore, the discussion of engineering challenges such as corrosion, maintenance access, and environmental regulation points to important design constraints that must be incorporated into the technical analysis. However, the article does not quantify specific design parameters (e.g., thermal load limits or structural requirements); therefore, further detailed technical sources will still be needed.	
Actions: 1. Identify marine site characteristics (depth, temperature range, and currents) that will affect the choice between submerged versus floating UDC designs. 2. Integrate efficiency benchmark figures (e.g., energy reduction percentages) into the UDC energy model for comparative performance analysis. 3. Research corrosion-resistant materials and protective coatings suitable for long-term marine deployment. 4. Review environmental and permitting regulations relevant to submerged infrastructure deployment (especially if considering coastal ports) to inform feasibility analysis.	

Group: D3 **Reviewer**: Talal Abu Ghazaleh

Journal Paper / Book Chapter / Article Citation: Li, H., Tang, Z., Xing, X., Guo, D., Cui, L., Mao, X.-Z. (2018). Study of CO_2 capture by seawater and its reinforcement. Energy, 164, 1135–1144. Available at: https://doi.org/10.1016/j.energy.2018.09.066

Summary:

This paper tests seawater as a CO_2 absorbent and measures how CO_2 solubility changes with temperature, pressure, and salinity. It finds a trade-off: lower temperature + lower salinity help solubility (thermodynamics via Henry's constant), while higher temperature + lower salinity can increase the absorption rate (kinetics). To "reinforce" seawater capture and encourage carbonate fixation, the authors add CaO (like in industrial alkaline wastes) and report that 0.4% CaO increases CO_2 solubility by ~79.25%, making this a potentially useful coastal CO_2 capture route.

Questions (aims):

1. How do pressure and temperature affect CO_2 solubility in seawater (capture performance)?
2. How does salinity affect CO_2 solubility and capture behavior?
3. What's the key thermo explanation (Henry's constant), and what's the kinetic result (absorption rate)?
4. How does adding CaO reinforce seawater CO_2 capture, and how big is the effect?

Answers (knowledge):

1. CO_2 solubility increases with higher pressure and lower temperature (better capture capacity).

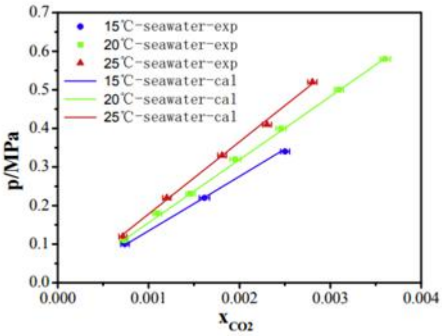


Figure 37. *CO_2 solubility in seawater at different temperatures.*

2. Salinity: Higher salinity reduces CO_2 solubility (worse capacity).

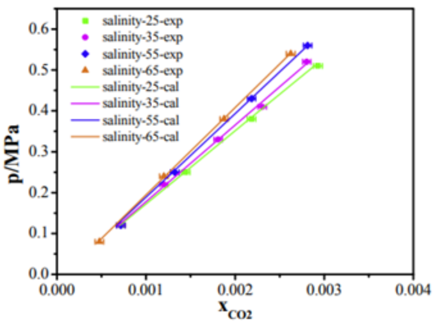


Figure 38. *CO_2 solubility in seawater at different salinities.*

3. Temperature/salinity effects are explained mainly through Henry's constant changes (capacity). Higher temperature can increase $K_{(Ga)}$ and speed up absorption rate, even though solubility drops.
4. Adding CaO boosts seawater absorption and supports carbonate fixation; reported 0.4% CaO → CO_2 solubility +79.25%

Impact:

1. Gives clear evidence for what improves CO_2 capture capacity in seawater (pressure, temperature). 2. Shows salinity effect (important for site choice / seawater conditions). 3. Provides a realistic "boost" method using CaO that can significantly improve capture performance.
Actions: 1. Turn the results into requirements: "capture improves with lower T / lower salinity / higher pressure" (define your realistic operating window). 2. Add a design note: if using alkaline reinforcement, start from the paper's example (0.4% CaO) as a reference case. 3. Add monitoring constraints to protect seawater chemistry if you propose CaO/precipitation (tie to your environmental objective).

Group: D3 **Reviewer:** Matteo Karam

Journal Paper / Book Chapter / Article Citation: Donovan, E., & Roberts, T. (2015). *Human factors and life support requirements for long-duration undersea habitats.* Human Factors and Ergonomics in Manufacturing & Service Industries, Volume 25, Issue 4, Pages 345-358.
Available at: https://doi.org/10.1177/0018720815598273

Summary: This article examines the human factors that influence life support requirements in long-duration undersea habitats. It examines air quality, thermal comfort, noise, humidity, and psychological well-being, emphasizing their integration into life support design.
Questions (aims): 1. How do human factors influence life support design? 2. What environmental parameters affect crew performance? 3. How can life support systems improve habitability? 4. What risks arise from poor environmental control?
Answers (knowledge): 1. Poor air quality and humidity reduce cognitive performance. 2. Thermal discomfort increases stress and fatigue. 3. Life support must balance engineering efficiency with human comfort. 4. Habitability is essential for mission success.
Impact: 1. Highlights the importance of crew-centered design. 2. Links life support performance to human productivity. 3. Supports inclusion of environmental comfort parameters.
Actions: 1. Include habitability metrics in life support requirements. 2. Design for long-term comfort, not just survival. 3. Reference human factors when justifying system parameters.

Group: D3 **Reviewer:** Ahmad Alhashmi

Journal Paper / Book Chapter / Article Citation: Li, S., & Tian, X. (2024). Progress of seawater batteries: From mechanisms, materials to applications. Journal of Power Sources, 617, 235161. Available at: https://doi.org/10.1016/j.jpowsour.2024.235161

Summary: A review of seawater battery mechanisms and materials challenges under real seawater conditions. • Seawater batteries use seawater as electrolyte (and in some designs as an active cathode medium), aiming for safer, marine-compatible storage. • Distinguishes primary 'reserve' seawater batteries (activate when needed) from rechargeable designs; maturity and performance vary widely. • Materials durability is the main constraint: chloride chemistry, corrosion, membrane stability, and biofouling can reduce lifetime and efficiency. • For our habitat, the strongest near-term role is emergency/backup storage, where non-flammable aqueous chemistry is valuable.
Questions (aims): 1. Is the best use case emergency reserve power or daily cycling, and which seawater battery type fits that role? 2. How do energy density, efficiency, and lifetime compare with lithium-ion and hydrogen for the same function? 3. What packaging and safety measures prevent corrosion/leakage in an enclosed habitat? 4. What environmental safeguards are needed if any electrolyte or products contact the ocean?
Answers (knowledge): 1. Primary reserve seawater batteries align well with emergency power; rechargeable concepts are promising but should be treated cautiously unless proven at scale. 2. Expect lower energy density and uncertain cycle life versus lithium-ion, so use seawater batteries as a secondary/backup layer in a hybrid storage plan. 3. Design for containment: corrosion-resistant housing, double barriers, leakage detection, and easy module replacement. 4. Safeguards should include monitored containment and an end-of-life recovery/disposal plan; avoid direct release to the environment.
Impact: • Enables a credible discussion of a marine-specific storage technology (and its limits). • Supports electrical/fire safety thinking: lower flammability can help, but corrosion/leak risks must be engineered out. • Reinforces lifecycle planning and environmental safeguards as part of storage selection.
Actions: 1. Include seawater batteries as an emergency-reserve option in the storage trade study versus lithium-ion and hydrogen. 2. Define the safety concept (containment + leak detection + isolation) for any battery technology used in the habitat. 3. Decide a clear role (e.g., emergency-only) and document the evidence threshold to expand its role later (cycle life, packaging demos). 4. Write a short environmental safeguards paragraph covering leakage containment and end-of-life handling.

Group: D3 **Reviewer:** Saba Falah

Journal Paper / Book Chapter / Article Citation: Xiao, J., Cheng, P., Cao, J., Lin, R., Luo, M., Yu, C., Yao, B., & Hu, Y. (2024). Dynamic analysis and simulations of a deep-sea floating mining vehicle multi-body system under real-world operating conditions. Ocean Engineering, 306, 117871. Available at: https://doi.org/10.1016/j.oceaneng.2024.117871

Summary: This study introduces an eco-friendly deep-sea floating mining vehicle multi-body system (DFMVMS), featuring a central floating mining vehicle (FMV) connected via a main pipe cable, an umbilical pipe cable (UPC), an ore transit station, and a surface mother ship. Using lump-mass-and-spring modeling, it analyzes coupled dynamics across production (collection) and transfer (ore movement) operations, revealing intercomponent motion interactions and critical UPC tension variations. The approach validates system stability while minimizing benthic ecosystem disturbance through floating design.
Questions (aims): 1. What is the architecture of the proposed DFMVMS for deep-sea mining? 2. How does the lump-mass-and-spring method model multibody dynamics? 3. What are the key motion interactions during production and transfer operations? 4. How does FMV motion influence overall system tension and stability?
Answers (knowledge): 1. The system comprises surface mother ship → main pipe cable → ore transit station → umbilical pipe cable (UPC) → floating mining vehicle (FMV), enabling nodule collection without seabed crawlers to reduce habitat damage. 2. Lump-mass-and-spring discretizes flexible components (cables/pipes) into rigid masses connected by axial/torsional springs and dampers, coupling 6-DOF rigid body motions via boundary conditions for time-domain simulations. 3. Production involves FMV heave/pitch dominating UPC surge tension spikes; transfer sees coupled roll/yaw amplifying pipe stress. Simulations show 20-50% tension variation from FMV positioning alone. 4. FMV's three core motions (heave, pitch, and roll) propagate upward, creating resonance risks; optimal transfer routes minimize UPC peak loads (target <1.5x static tension) through phase-aligned positioning.
Impact: The DFMVMS advances sustainable mining by replacing tracked vehicles with floating collectors, reducing sediment plumes while maintaining productivity. Dynamic insights guide route planning and controller design for commercial operations, though scale-model testing is needed to validate against real currents and validate eco-claims in ISA permit zones.
Actions: 1. Model multi-body cable dynamics in Fusion 360 for analysis. 2. Simulate lump-mass-spring systems using Python for mechanical engineering dynamics assignments. 3. Integrate DFMVMS findings into the offshore engineering project report. 4. Prototype scaled UPC tension tests with engineering peers for societal demonstrations. 5. Include this completed worksheet in the Assignment 1 report appendix.

Group: D3 **Reviewer**: Muhammad Hammad

Journal Paper / Book Chapter / Article Citation: Teeneti, C. R., Truscott, T. T., Beal, D. N., & Pantic, Z. (2021). Review of Wireless Charging Systems for Autonomous Underwater Vehicles. IEEE Journal of Oceanic Engineering, 46(1), 68–91. Available at: https://doi.org/10.1109/JOE.2019.2953015

Summary:
This paper reviews state-of-the-art underwater charging systems for AUVs, with a focus on inductive wireless power transfer (IWPT). It compares contact-based and wireless charging approaches and highlights the limitations of wet-mate connectors in subsea environments. The paper also discusses docking station designs, alignment challenges, retention mechanisms, and the integration of power and data transfer systems. Environmental effects such as seawater conductivity, pressure, temperature, and biofouling are examined in detail.

Questions (aims):
1) Why are contact-based charging systems problematic underwater?
2) What are the advantages of inductive wireless power transfer for AUVs?
3) How do docking stability and alignment affect charging efficiency?
4) How should power and data transfer be integrated into docking systems?

Answers (knowledge):

1. Contact-based systems require high-precision alignment and are vulnerable to corrosion, biofouling, and electrical safety risks.
2. IWPT offers safer, more reliable charging with greater tolerance to misalignment and reduced maintenance requirements.
3. Charging efficiency depends on coil alignment, retention stability, and docking station robustness against currents.
4. Power and data transfer should be integrated into a single docking operation to enable autonomous energy replenishment and mission data exchange.

Impact:
This paper strongly supports robotics and systems objectives related to autonomous charging, data transfer, and long-duration operation. It justifies objectives such as enabling safe power transfer, reducing human intervention, and supporting sustained robotic missions. The paper also links docking, charging, and data exchange into a unified subsystem requirement.
Remaining gaps include project-specific power levels and detailed interface geometry.

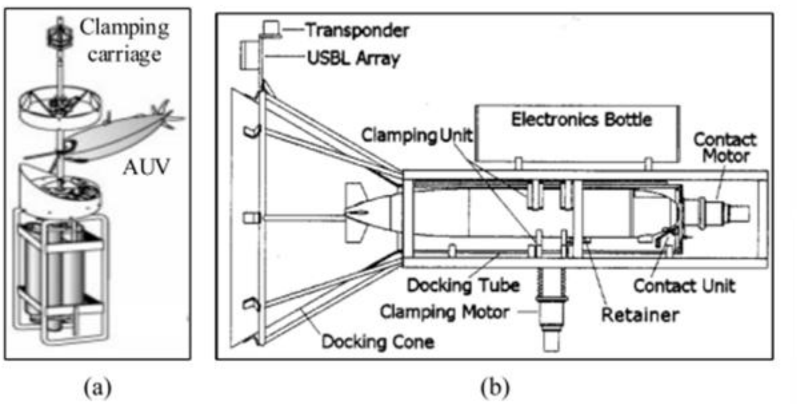


Figure 39. (a) A dock *mooring system; (b) REMUS docking station.*

Actions:
1. Define power transfer and data exchange requirements for docked vehicles.
2. Perform a trade-off study of underwater charging approaches.
3. Identify electrical safety and isolation constraints for subsea operations.

Group: D3 **Reviewer**: Muhammad Raafey

Journal Paper / Book Chapter / Article Citation: Sheldon, J. *et al.* (2024) 'AquaSonic: Acoustic Manipulation of Underwater Data Center Operations and Resource Management,' *AquaSonic: Acoustic Manipulation of Underwater Dazta Center Operations*, pp. 331–349. Available at: https://doi.org/10.1109/sp54263.2024.00201.

<table>
<tr><td>
Summary:
This paper investigates unique security vulnerabilities of UDC operations arising from underwater acoustic propagation. It is demonstrated that sound waves can be used to manipulate hardware and software behavior in submerged data centers, causing reduced performance, resource misallocation, and reliability issues. Through closed-water and open-water experiments, the work characterizes how acoustic signals at certain frequencies impact storage throughput, distributed file systems, and load-balancing mechanisms. The study also evaluates basic defense strategies and proposes a machine learning-based detection system to identify and mitigate these acoustic injection attacks.
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<tr><td>
Questions (aims):
1. Are there unique security vulnerabilities specific to UDC environments?
2. How do acoustic signals interact with data center hardware and software?
3. What are the risks or limitations?
4. What mitigation strategies can be implemented?
</td></tr>
<tr><td>
Answers (knowledge):
1. Underwater environments do support long-range sound propagation with slower attenuation than in air, meaning acoustic signals can travel further and with less energy loss. The article shows that this allows malicious acoustic “injection attacks” to interact with hardware and software in ways not seen in land-based data centers. These vulnerabilities affect fault-tolerant storage systems, distributed file systems, and resource managers that assume stable physical environments.
2. Acoustic signals couple into submerged data center structures and produce mechanical vibrations in server components such as hard disk drives. Because water transmits sound efficiently, resonant frequencies can induce physical oscillations inside enclosures, leading to throughput degradation and increased latency, in addition to automatic node removal in distributed systems. Software resource managers respond to these performance issues by reallocating workloads prematurely/incorrectly, therefore further disrupting performance.
3. The study identifies several risks:
• Performance degradation: Acoustic injection can reduce throughput by up to 100% under sustained attack.
• Distributed system disruption: Distributed file systems may remove nodes automatically due to perceived faults.
• Resource misallocation: Load-balancing mechanisms can incorrectly redistribute resources, creating overloads.
• Remote attack potential: Acoustic attacks could theoretically be launched several meters away using commercially available speakers.
• Limitations include the controlled nature of experiments and the fact that the work does not evaluate all possible environmental variables (e.g., varying depth, currents, or marine noise interference).
4. The authors investigate several mitigation methods and propose a machine learning-based detection system that identifies abnormal storage performance behavior with very low false positive rates. They also discuss structural defences like sound-absorbing enclosures, but note that these may increase thermal resistance and counteract marine
</td></tr>
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<tr><td></td><td>cooling benefits. Software-oriented solutions, detecting and responding to acoustic perturbations at the systems level, offer a promising balance between performance protection and thermal efficiency.</td></tr>
<tr><td colspan="2">Impact:
This paper highlights a category of operational risk that is unique to underwater deployment, acoustic security vulnerabilities, that isn’t yet addressed in conventional UDC cooling or reliability literature. For the UDC design, this introduces a critical consideration beyond thermal and energy performance: how environmental acoustic propagation could affect system integrity, performance, and reliability. The acoustic attack vectors identified show that hardware and software behavior may be influenced by the marine medium itself and not just due to heat or pressure. Incorporating resilient and detection-oriented software controls, together with considering physical enclosure design to attenuate sound, will therefore be crucial design factors. Nevertheless, the paper primarily focuses on the security threat itself rather than architectural solutions for robust acoustic hardening; this remains an area for further research.</td></tr>
<tr><td colspan="2">Actions:
1. Assess acoustic environmental profiles at intended deployment locations to understand background noise and possible interference sources.
2. Evaluate storage technology choices (e.g., SSD versus HDD) for susceptibility to acoustic vibration in UDC designs.
3. Integrate acoustic monitoring and anomaly detection into the UDC management stack, potentially adopting or adapting machine learning-based detection.
4. Explore structural design options (e.g., damping materials, acoustic shielding) that minimize mechanical coupling without significantly reducing thermal transfer efficiency.</td></tr>
</table>

Group: D3 **Reviewer**: Talal Abu Ghazaleh

Journal Paper / Book Chapter / Article Citation: El-Naas, M. H., Mohammad, A. F., Suleiman, M. I., Al Musharfy, M., & Al-Marzouqi, A. H. (2017). A new process for the capture of CO_2 and reduction of water salinity. Desalination, 411, 69–75. Available at: https://doi.org/10.1016/j.desal.2017.02.005

Summary:

This paper presents a CO_2 capture method using brine (salty water) where adding $CaO/Ca(OH)_2$ increases pH and drives CO_2 to convert into stable solid products (mainly $NaHCO_3$ precipitate), achieving very high capture efficiency (up to 99%) under best conditions. It also reports a secondary benefit: the same reactions reduce sodium in the water (partial salinity reduction), but the key takeaway is that CO_2 capture depends strongly on CaO dose, temperature, and gas flow (residence time), with performance generally better at lower temperatures and high flow rates.

Questions (aims):

1. What is the CO_2 capture mechanism and what form is CO_2 stored in?
2. What capture efficiency is achieved and under what conditions?
3. How does CaO concentration affect CO_2 capture efficiency?
4. How do temperature and gas flow rate affect CO_2 capture (and why)?
5. Why is this approach attractive for real deployment (energy/safety/integration)?

Answers (knowledge):

1. CaO becomes $Ca(OH)_2$, which raises pH and promotes CO_2 conversion into solid bicarbonate/carbonate products (e.g., $NaHCO_3$ precipitate), meaning CO_2 ends up stored in a solid phase.

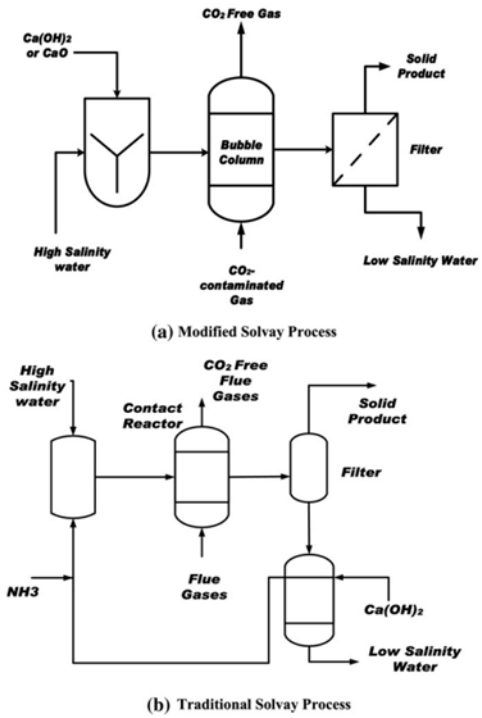

Figure 40. *Schematic diagrams of the main units of (a) modified; (b) traditional Solvay process.*

2. The paper reports very high CO_2 capture efficiency (up to 99%) at optimum conditions (notably at lower temperatures and appropriate CaO dosing / flow).
3. Increasing CaO generally supports strong capture by keeping pH high and maintaining the driving force for CO_2 conversion (they show capture remains very high near the tested optimum around their higher CaO case).

4. Capture decreases at higher temperatures because CO_2 dissolves less and product formation/precipitation conditions become less favorable; higher gas flow can reduce capture because residence time is shorter, so less CO_2 reacts before leaving.

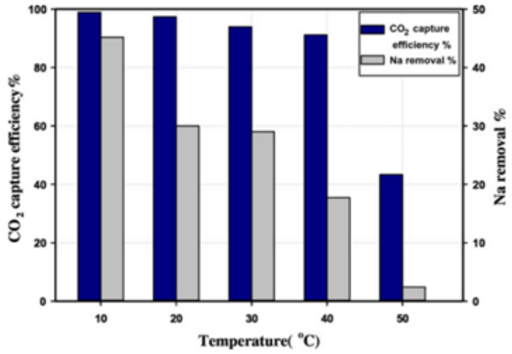

Figure 41. *Sodium removal and carbon dioxide capture efficiency versus temperature.*

5. Removing ammonia avoids a hazardous chemical and eliminates an energy-intensive
6. regeneration step, making the concept simpler and potentially lower-energy than conventional Solvay-style capture routes.

Impact:

1. Provides a CO_2 capture pathway with very high capture efficiency (up to ~99%), useful as a benchmark target.

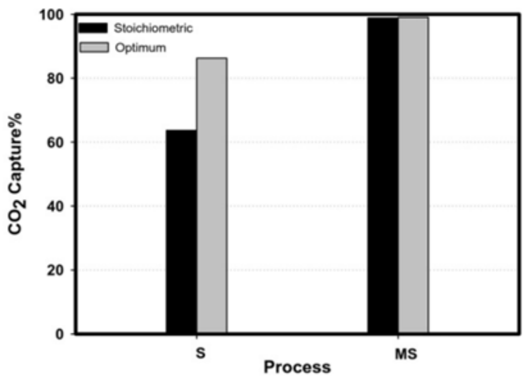

Figure 42.*C02 Capture for Stoichiometric and Optimum processes.*

2. Identifies key operating levers for capture: CaO dose, temperature, gas residence time.
3. Improves practical viability by avoiding ammonia (safety + fewer unit operations + lower energy burden).
4. Suggests an integration-friendly concept (bubble contactor + solids separation), but you must still consider discharge handling and solids management.

Actions:

1. Use the paper's reported high capture efficiency as a literature justification for an ambitious capture-performance target.
2. Define your operating window around the capture drivers: temperature range, CaO dosing, and gas flow/residence time.
3. Add a simple operational requirement: maintain high pH safely (chemical handling + monitoring).

4. Decide what you do with the captured form: solid collection/filtration + storage/use, and specify safe handling of the remaining brine.

Group: D3 **Reviewer:** Matteo Karam

Journal Paper / Book Chapter / Article Citation: Khan, A., & Li, Y. (2019). *Emergency life support and safety systems in subsea habitats.* Safety Science, Volume 118, Pages 25-36.
Available at: https://doi.org/10.1016/j.ssci.2019.01.019

Summary: This paper focuses on emergency life support and safety systems in subsea habitats. It discusses failure modes, redundancy, emergency breathing systems, compartment isolation, and evacuation planning under submerged conditions.
Questions (aims): 1. What are the main life support failure risks underwater? 2. How can emergency systems mitigate these risks? 3. What redundancy is required for crew survival? 4. How does underwater evacuation influence design?
Answers (knowledge): 1. Major risks include flooding, power loss, and gas imbalance. 2. Emergency oxygen and scrubbers provide short-term survival. 3. Compartment isolation limits damage from failures. 4. Evacuation constraints require conservative safety design.
Impact: 1. Strengthens safety and risk analysis sections. 2. Provides justification for redundancy and emergency systems. 3. Reinforces conservative engineering design philosophy.
Actions: 1. Integrate emergency life support subsystems. 2. Include failure-mode analysis in the report. 3. Use this paper to justify safety-driven design decisions.

Group: D3 **Reviewer:** Ahmad Alhashmi

Journal Paper / Book Chapter / Article Citation: Alkrush, A. A., Salem, M. S., Abdelrehim, O., & Hegazi, A. A. (2024). Data centers cooling: A critical review of techniques, challenges, and energy saving solutions. International Journal of Refrigeration, 160, 246-262. Available at: https://doi.org/10.1016/j.ijrefrig.2024.02.007

Summary:
A critical review of cooling methods and energy-saving strategies for high heat-density systems.
• Reviews air cooling, free cooling, liquid cooling, immersion, and thermal storage, showing how cooling design strongly affects total energy demand.
• Highlights that cooling choice depends on heat flux, reliability requirements, and the available heat sink; an ocean habitat has a major advantage: abundant cold seawater.
• Seawater enables efficient heat rejection (the 'free cooling' concept) but requires corrosion/biofouling control and careful separation from sensitive equipment.
• Waste heat can be recovered for domestic hot water, desalination preheating, or space heating, improving whole-system efficiency.

Questions (aims):
1. What cooling architecture best fits our modules (air vs liquid loop versus immersion), and what drives that choice?
2. How do we quantify the benefit of using seawater as a heat sink in our power budget?
3. Where can we reuse waste heat, and how do we show it in the system energy balance?
4. What redundancy and safety features are needed to prevent cooling failures from becoming mission failures?

Answers (knowledge):
1. A closed-loop liquid cooling system with a seawater heat exchanger is a strong default because it exploits seawater cooling while keeping seawater out of critical equipment.
2. Represent the seawater cooling benefit by using a lower 'cooling power per kW heat rejected' assumption than air cooling, and justify it with literature.
3. Waste heat recovery is most valuable with steady heat sources; show it as a heat-exchanger interface feeding hot water/desalination processes.
4. Design for resilience: N+1 pumps, isolation valves, leak detection, and a biofouling management plan; define a safe degraded mode.

Impact:
• Directly supports the thermal rejection & heat recovery branch of the energy objective tree.
• Improves energy sizing credibility: better cooling efficiency reduces required generation and storage capacity.
• Links energy work package to desalination/habitat services through heat recovery opportunities.

Actions:
1. Build a heat-load table (kW) for major systems and allocate a cooling approach to each module.
2. Sketch the thermal loop (equipment → coolant → seawater HX → ocean) with an optional heat-recovery branch to hot water/desalination.
3. Define redundancy and safety measures (N+1 pumps, isolation, leak detection, fouling control) for the cooling loop.
4. Add heat recovery and cooling assumptions into the overall energy balance to show system-level benefits.

APPENDIX D: DESIGN IDEAS

Supportive Images From CAD

Figure 43. *Bedroom and life supports layout.*

Figure 44. *Energy systems monitoring and storage.*

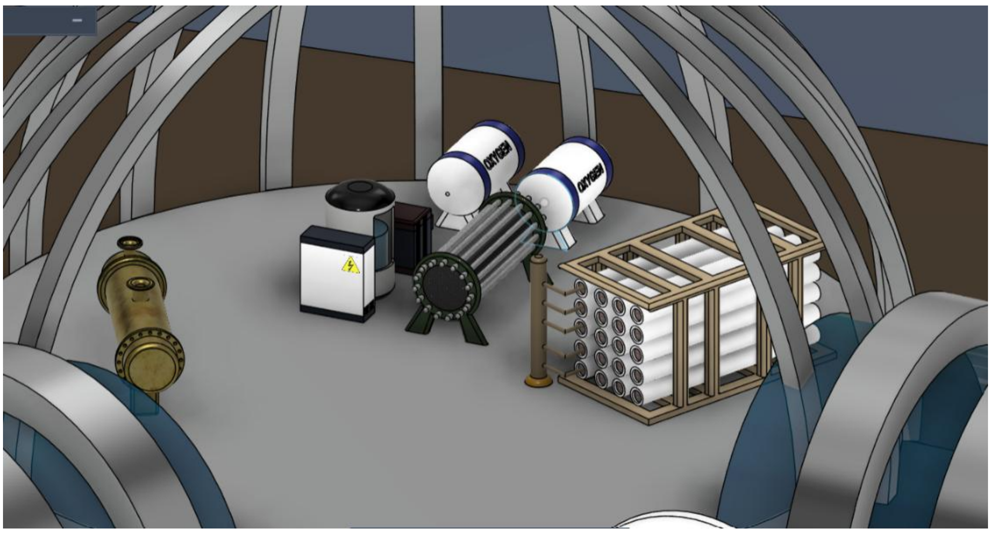

Figure 45. *Life Support Setup.*

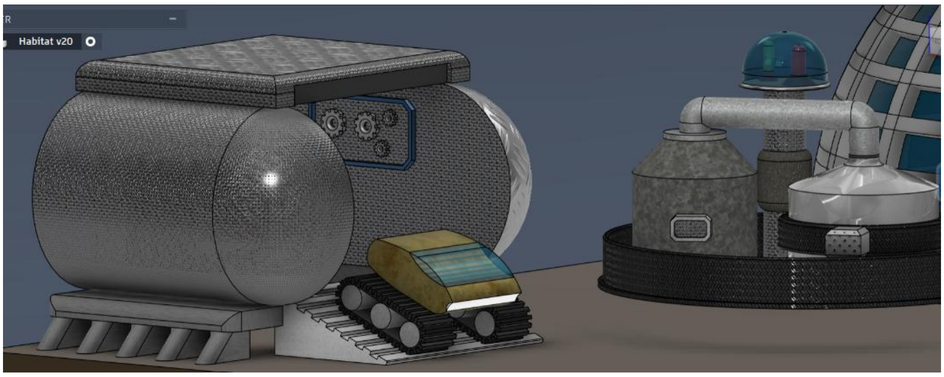

Figure 46. *Deep Sea Mining Setup.*

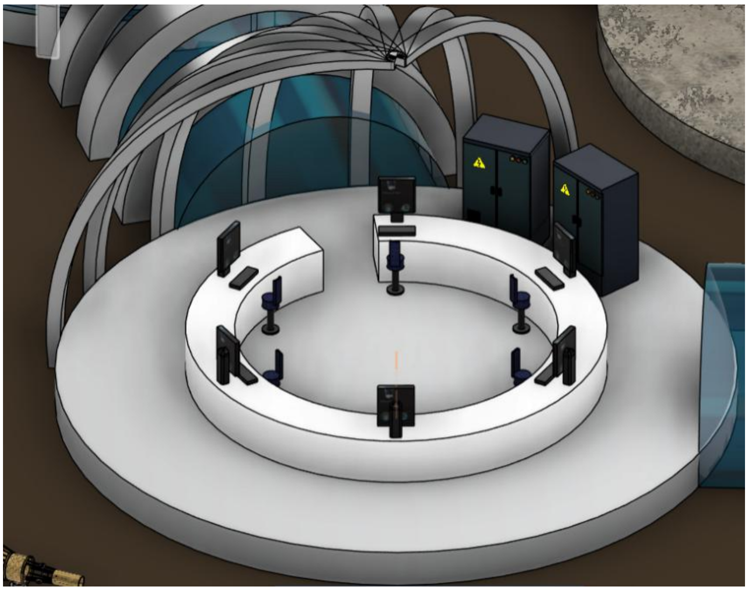

Figure 47. *Systems setup.*

Figure 48. *Pantry and agriculture setup.*

Rich Picture

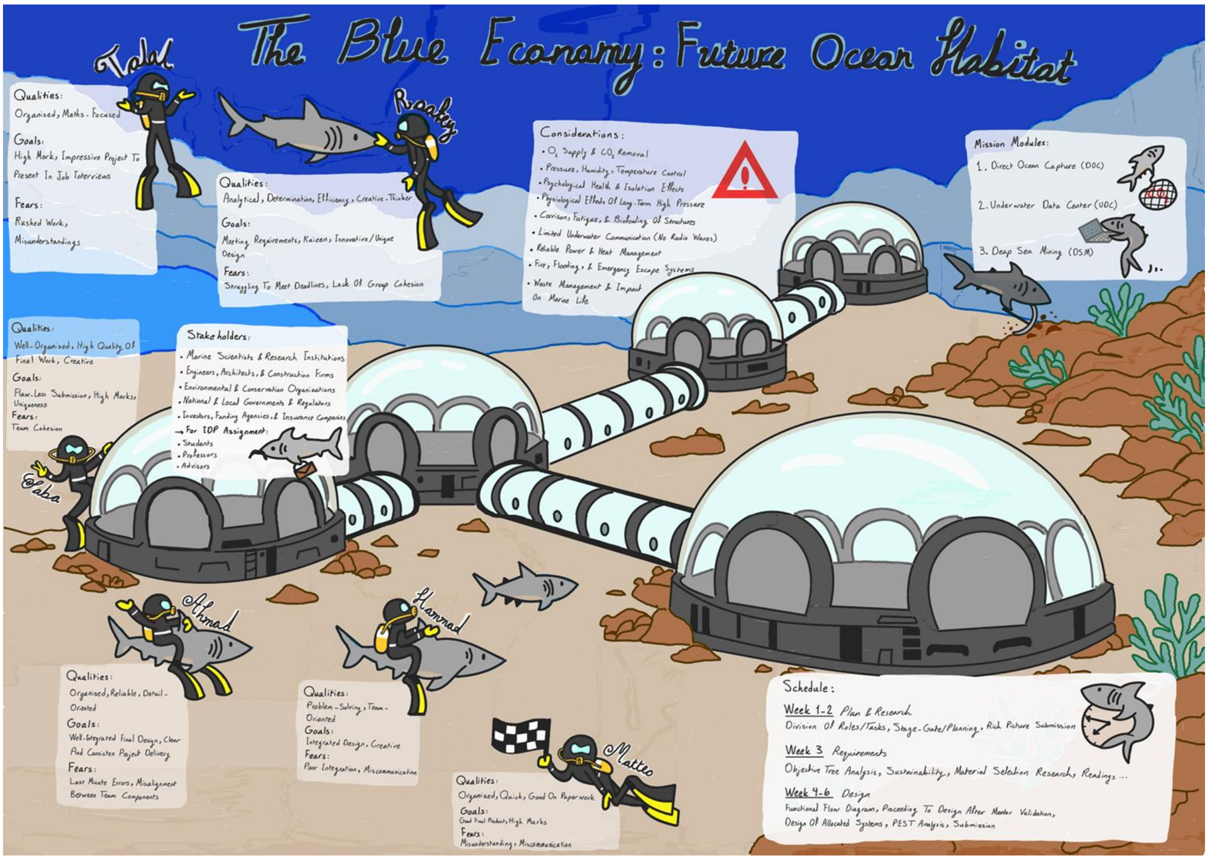

Figure 49. *Rich picture of "The Blue Economy: Future Ocean Habitat."*

Figure 49 presents the rich picture developed by the team, focusing on underwater habitat and the aim of producing an innovative design suitable for high academic performance and future professional presentation. Key stakeholders include marine scientists and research institutions, engineers and construction firms, environmental organizations, governmental regulators, and investors and funding bodies, as well as internal stakeholders such as students, professors, and academic advisors.

The diagram highlights major technical and environmental considerations, including life-support functions (oxygen supply and carbon dioxide removal), environmental control (pressure, humidity, and temperature), psychological and physiological effects of long-term high-pressure exposure, structural degradation, communication limitations, power and heat management, emergency systems, and waste management impacts on marine ecosystems. It also outlines three mission modules, Direct Ocean Capture, an Underwater Data Center, and Deep-Sea Mining, alongside each team member's qualities, goals, and fears and a phased schedule from initial planning and research to requirements definition and detailed design over weeks 1-6.

Design Thinking

To identify and classify critical tasks within the energy work package for the future ocean habitat, a prioritization grid (Figure 50) evaluates energy-system activities. Tasks map against user importance (crew safety, life-support continuity, and mission uptime) and group feasibility (scoping realism at the concept stage), grouping into four categories:

- Major Projects (high impact, complex execution)
- Quick Wins (high impact, low effort)
- Maybes (low impact, challenging)
- Fill-Ins (low impact, simple).

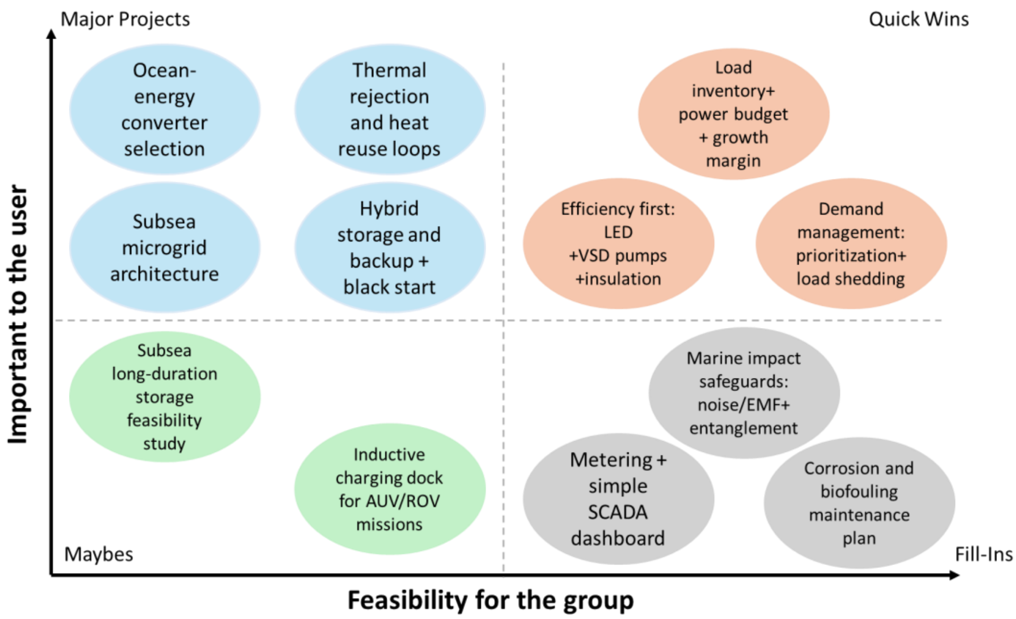

Figure 50. *Energy prioritization grid for the "Future Ocean Habitat."*

This ensures efficient resource allocation by prioritizing generation, storage, distribution, and thermal management.

1. Major Projects: Ocean-energy converters, hybrid storage/backup, subsea microgrid, thermal rejection.
2. Quick Wins: Load inventory, demand management, and efficiency upgrades.
3. Maybes: Long-duration storage study, inductive AUV charging.
4. Fill-Ins: Metering/SCADA, corrosion plans, marine safeguards.

Brainstorming – first concepts

Structure First Concept

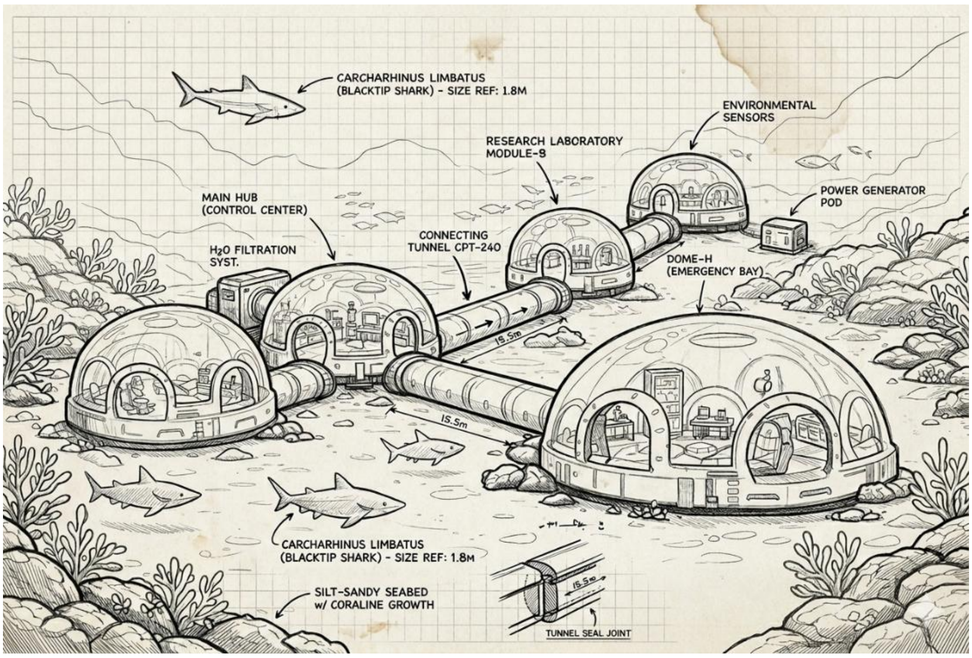

Figure 51. *Initial Structure Concept.*

The initial brainstorming phase for the underwater habitat concept focused on establishing a modular, functionally distributed structural layout capable of withstanding deep-sea hydrostatic pressures while supporting continuous scientific operations. Early engineering considerations prioritized the separation of critical systems into discrete, pressurized pods, including a central command hub, research laboratory, power generation unit, and dedicated emergency bay, interconnected via rigid, watertight tunnels. This modular approach offered significant advantages in terms of constructability, allowing for surface fabrication and staged seabed assembly, as well as operational resilience by isolating potential failures to individual modules without compromising the entire habitat. Structural geometry was influenced by pressure vessel theory, favoring spherical or cylindrical forms for the primary habitable modules to efficiently manage external loading.

Further development addressed the interface between the habitat and the seabed, considering foundation stability on a silt-sandy substrate. The integration of dedicated environmental sensor arrays and a standalone water filtration system reflected an early commitment to self-sufficiency and long-duration autonomy. The overall scale and proportions of structural elements, such as tunnel diameters, module spacing, and access hatch placements, were iteratively refined to accommodate internal systems integration, emergency egress requirements, and maintenance accessibility. This conceptual framework established the fundamental architectural and structural parameters that would guide subsequent detailed engineering analysis.

Energy First Concept

The first WP3 Energy concept for the Future Ocean Habitat uses a hybrid marine-renewable approach. Wave energy buoys at the surface and a seabed tidal turbine generate electricity that is exported via subsea cables to a wet-mate junction hub. Power is conditioned and regulated into a DC microgrid bus inside the habitat, where a Battery Energy Storage System (BESS) provides smoothing, ride-through and black-start capability. For longer low-resource periods and planned maintenance outages, an optional hydrogen reserve (electrolyser, storage and fuel cell) provides multi-day autonomy for Tier 1 loads. Waste heat from power electronics and storage is rejected using a seawater heat exchanger, leveraging the ocean as a cold sink.

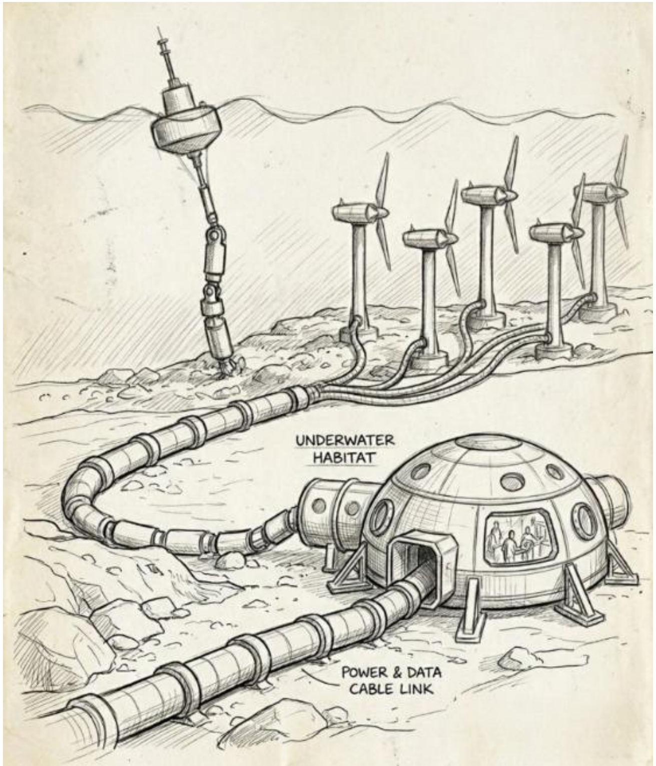

Figure 52. *Initial WP3 Energy Concept*

Main weaknesses at this stage are marine resource variability (calm seas and neap tides), storm survivability of surface devices, corrosion and biofouling of subsea hardware, connector/cable reliability, and safety management for energy storage (battery thermal runaway and hydrogen leak/venting). These issues guide the next design iteration (improved redundancy, clear isolation points, protective housings, condition monitoring and conservative safety zoning).

Systems First Concept

The first concept is described as an autonomous underwater habitat with the use of surface communication buoys for long-distance communication and acoustic transceivers that allow interaction with AUVs and seafloor assets. The central seabed systems control pod is depicted as the main communication, monitoring, and control point, reflecting the early recognition of the limitations of underwater communication and the need for a well-defined system boundary.

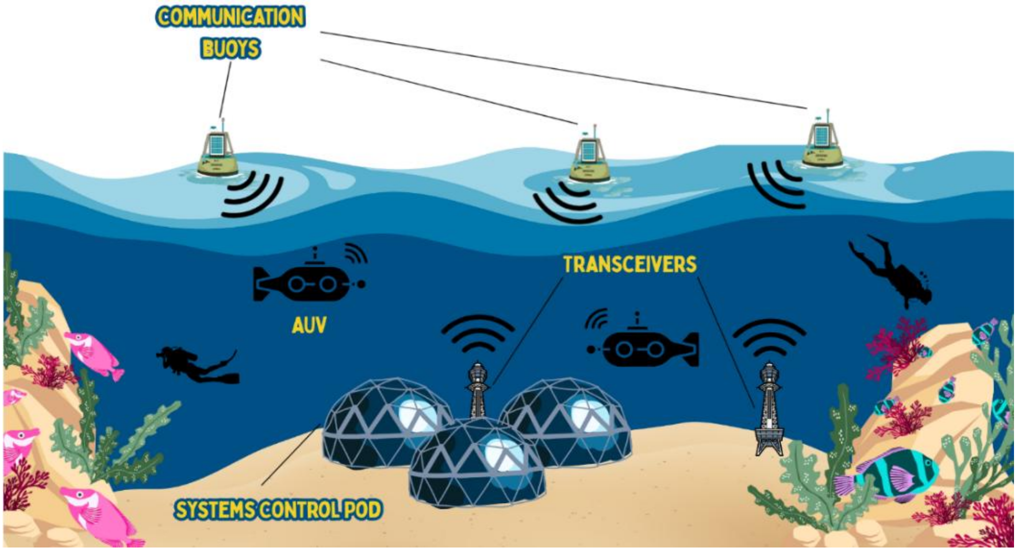

Figure 53. *Initial Systems concept.*

Critical analysis revealed several weaknesses, including the use of a single control pod, a lack of redundancy, and no defined response to loss of communication. The concept also failed to adequately model data prioritization, autonomy, and the integration of monitoring and safety systems.

The design was later improved to a more robust system with distributed control, hybrid acoustic and optical communication, and autonomous failback. Such improvements enhanced the fault tolerance and helped shape the functional flow diagram, requirements specification, and final system design.

Life Support System First Concept

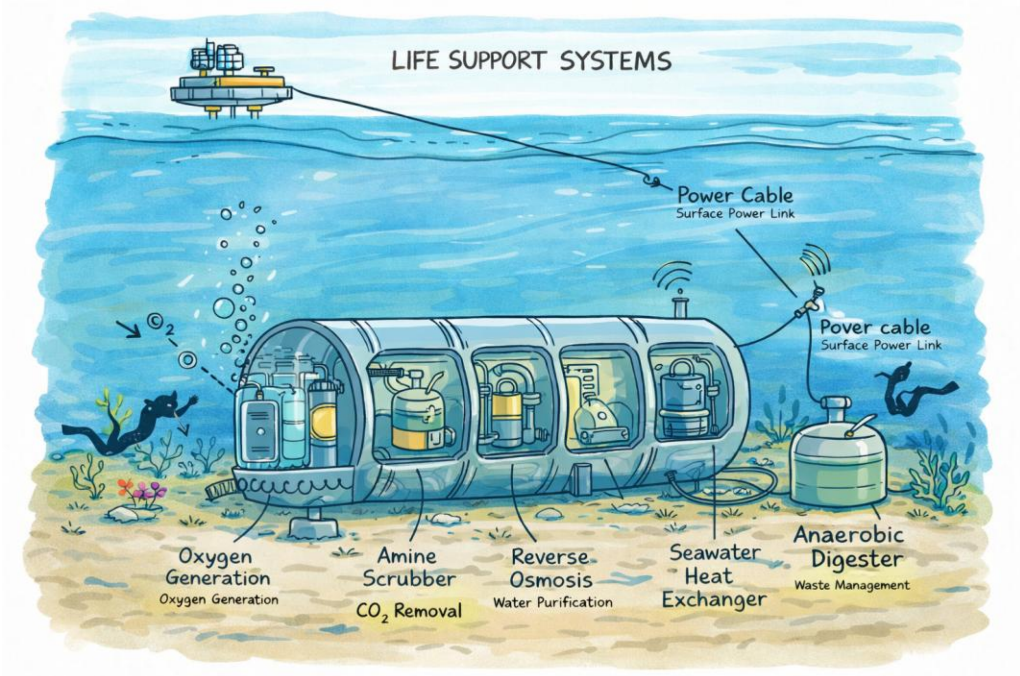

Figure 54. *Initial Life Support System concept.*

The first concept presents an integrated life support system designed to operate within an autonomous underwater habitat. The system combines oxygen generation through water electrolysis, carbon dioxide removal using regenerative scrubbers, and potable water production via reverse osmosis. Wastewater is treated and partially recycled to minimise environmental discharge, while excess heat is rejected through a seawater heat exchanger.

All subsystems are connected through a central control unit that continuously monitors oxygen levels, carbon dioxide concentration, pressure, temperature, and salinity. This early concept highlights the need for redundancy, reliable power supply, and clear system boundaries to ensure safe and sustained human habitation underwater.

Direct Ocean Capture First Concept

Our first DOC concept is a skid-mounted unit inside the underwater habitat. Seawater is filtered, pumped through an electrochemical stack to shift dissolved carbon into CO_2, then a degasser removes CO_2 gas. Treated seawater is returned to the ocean and CO_2 is sent to a small buffer/storage line. The DOC is powered by the habitat power system and monitored by the control system.

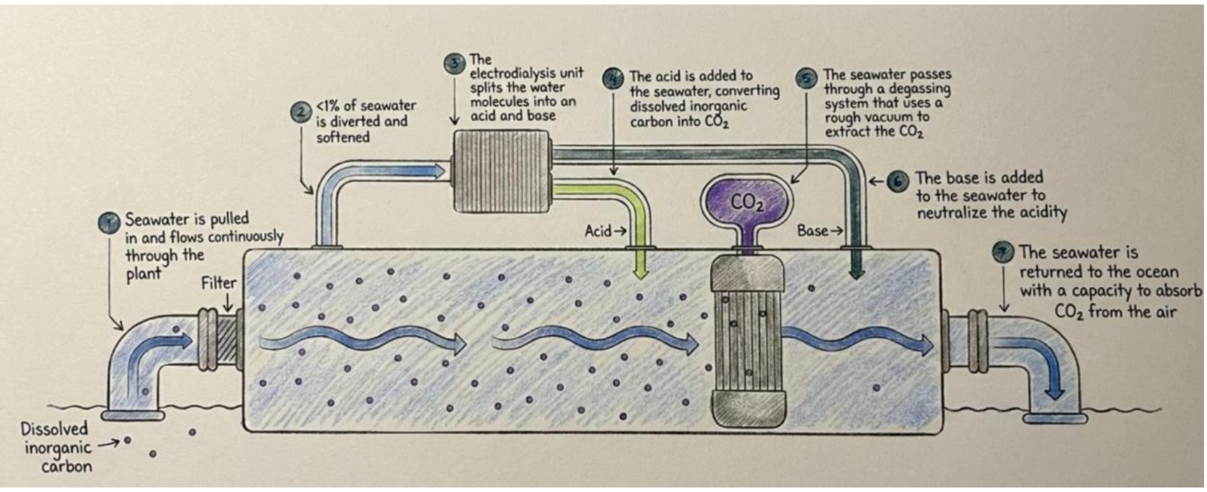

Figure 55. *Initial DOC concept*

Main weaknesses are power demand, membrane fouling/maintenance, pump redundancy, and safe handling/storage of the captured CO_2. These issues guide the next design iteration (added monitoring, redundancy, and clearer interfaces).

Underwater Data Center First Concept

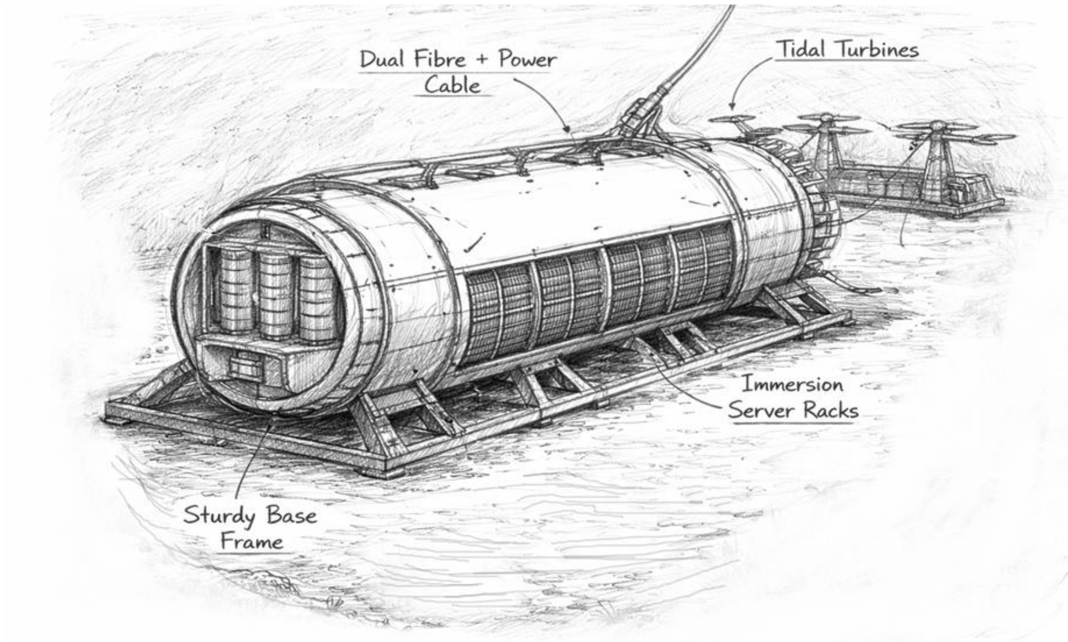

Figure 56. *Underwater Data Center Initial Concept.*

The first concept demonstrates the UDC designed to operate on the seabed (potentially near the rest of the habitat). The main features of the UDC system are the sturdy base frame, power distributor, immersion server racks, dual fibre + power cable, tidal turbines. The sturdy base frame is to protect the internal components from impact damage if any as well as pressure fluctuations and corrosion, the power distributor manages and distributes power throughout the entire system (is supposed to be covered but is shown for demonstration purposes), the immersion server racks cool the servers and have dual-loop cooling included inside as well as a heat exchanger with vents along the sides. The dual fibre + power cable supply the system with power and transmit data and the tidal turbines generate the power.

This early concept demonstrates the baseline design for the UDC and presents the full functionality of it. The concept also showcases the need for a robust design, good cooling (as data centres generally generate lots of heat), good reliable and robust power supply and management.

Deep Sea Mining First Concept

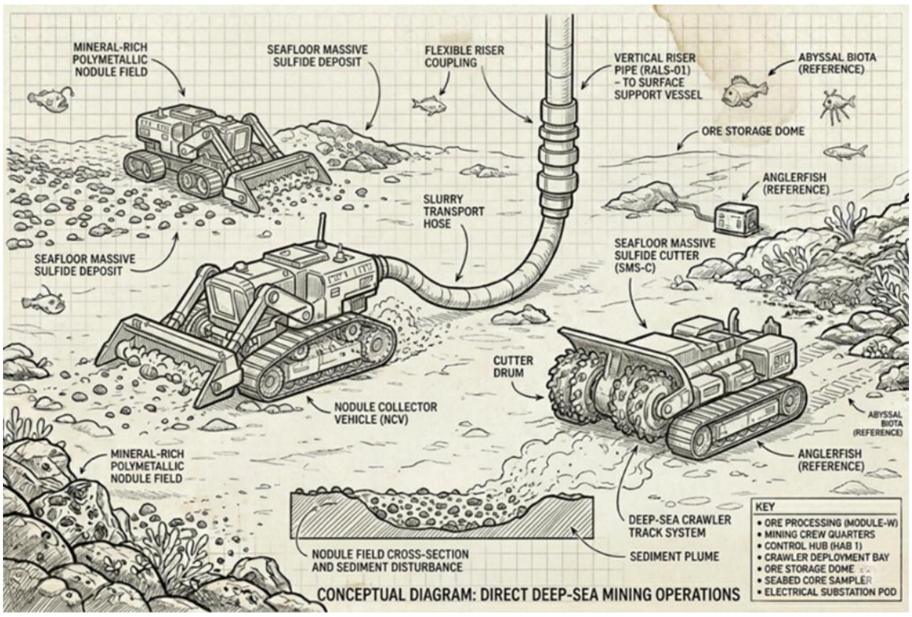

Figure 57. *Deep Sea Mining Initial Concept.*

The initial brainstorming phase for the deep sea mining system centered on developing a cohesive operational workflow between seafloor collection and surface support, with early engineering focus placed on the design of a remotely operated crawler and a dedicated offshore workshop. The crawler concept was conceived as a heavy-duty, tracked vehicle capable of traversing uneven sedimentary substrates while deploying hydraulic suction or mechanical cutting mechanisms to harvest polymetallic nodules. Structural considerations included ballasting for negative buoyancy, corrosion-resistant materials for prolonged seawater exposure, and redundant drive systems to mitigate the risk of immobilization on the seafloor. The workshop, envisioned as a modular processing and maintenance hub, was planned to house primary mineral dewatering equipment, teleoperation control stations, and repair facilities for crawler components, thereby reducing reliance on surface vessel support and enabling extended field operations.

Supporting infrastructure was concurrently developed to address power distribution, riser transfer, and environmental containment. The workshop was positioned conceptually as an intermediate node between the seafloor operations and the surface support vessel, receiving slurry via a flexible riser system and performing initial separation before transport. Power supply options considered included direct umbilical connection to a surface vessel, subsea cables from offshore renewable sources, and hybrid battery systems for the crawler to enhance maneuverability. Environmental mitigation strategies, such as shrouded collectors and plume diffusers, were integrated into the early crawler design to address sediment disturbance concerns. This iterative conceptual framework established the fundamental mechanical interfaces, operational hierarchies, and environmental constraints that would inform subsequent detailed engineering development.

Morphological Chart

Table 42. *Structure Subsystem Morphological Chart.*

Feature (Identified from requirements table)	Means			
	Option 1	Option 2	Option 3	Option 4
WP2. Structure				
Hull Material	Steel	Acrylic	Aluminum alloy	Composite
Structural Shape	Sphere	Cylinder	Torus	Multi-dome
Foundation Type	Seabed resting	Pile-driven anchors	Tethered buoyancy	Rock bolting
Deployment Method	Surface tow + ballast	Modular assembly	Dry dock tow	Submarine transport
Pressure Hull Design	Stiffened cylinder	Unstiffened sphere	Ring-stiffened cone	Multi-lobed
Access & Egress	Wet porch	Moon pool	Airlock trunk	Submarine lockout
Emergency Systems	Ballast blow	Escape pod	Personnel lockout	Rescue bell
Corrosion Protection	Sacrificial anodes	Impressed current	Coatings	Biocide treatment

Table 43. *Energy System Morphological Chart.*

Feature (Identified from requirements table)	Means			
	Option 1	Option 2	Option 3	Option 4
WP3. Energy System				
Primary generation	Wave WEC buoys	Tidal stream turbines	OTEC (site-dependent)	Hybrid wave + tidal
Wave device type	Point absorber buoy	Oscillating water column	Attenuator (hinged)	Submerged pressure device
Tidal deployment	Fixed monopile/gravity base	Floating tethered turbine	Lift-off seabed frame (ROV)	Ducted turbine
Microgrid architecture	AC microgrid (400 VAC)	DC microgrid (380 VDC)	Hybrid AC/DC	MVDC export + conversion
Subsea connections	Hardwired (no disconnect)	Wet-mate + junction box	Inductive coupler	Surface umbilical only

Conversion layout	Centralised converter	Distributed at loads	Modular converter racks (N+1)	Direct AC tie-in
Short-term storage	Li-ion BESS (modular)	Flywheel	Supercapacitors	Seawater battery (primary)
Long-duration reserve	Hydrogen (electrolyser + fuel cell)	Flow battery (VRFB)	Compressed air (CAES)	Diesel genset
Emergency reserve	UPS only	Seawater-activated reserve	Isolated BESS string (Tier 1)	UPS + seawater reserve (Tier 1)
Energy management	Manual priority	Scheduled load shifting	Auto tiered shedding	Predictive EMS (forecast + shedding)
Protection & isolation	Fuses only	Breakers + relays	Solid-state breakers	Zonal protection + isolation
Heat rejection	Air radiators	Seawater plate HX (closed loop)	Shell-and-tube HX	Direct seawater cooling
Heat recovery	None	DHW preheat	Desal/process preheat	DHW + desal preheat loop
Wetted materials	316L stainless	Duplex SS + coatings	Titanium (premium)	Mixed: Ti HX + duplex + HDPE
Maintenance strategy	Time-based servicing	Coatings + anodes	Active chlorination dosing	ROV cleaning + condition monitoring

Table 44. *Life Support System Morphological Chart.*

Feature (Identified from requirements table)	Means					
	Option 1	Option 2	Option 3	Option 4	Option 5	Option 6
WP4. Life Support System						
Oxygen Production	Water electrolysis	Algae bioreactor	Solid oxide electrolysis	Chemical oxygen generators	Peroxide-based O2	O2 storage tanks
CO2 Removal	Zeolite adsorption	Lithium hydroxide scrubbers	Amine-based scrubbers	Algae CO2 fixation	MOF adsorbents	Cryogenic capture

Water Recovery & Purification	Particulate filtration	Biofilm-resistant membranes	Thermal distillation	Catalytic oxidation	Chemical treatment	Reverse osmosis
Waste Management	Anaerobic digestion	Composting	Plasma gasification	Microbial fuel cells	Pyrolysis	Incineration
Thermal Control	Seawater heat exchanger	Active liquid cooling loop	Heat pump	Phase change materials	Passive hull cooling	Redundant cooling loop
Power for LSS	Surface cable power	Battery storage	Fuel cells	Diesel generator	Hybrid	Renewable surface

Table 45. *Systems Morphological Chart.*

Feature (Identified from requirements table)	Means		
	Option 1	Option 2	Option 3
WP5. Systems			
External communication and relay (Req. 7.1)	Acoustic modem to surface buoy relay	Hybrid acoustic + satellite buoy	Fibre-optic tether
Internal data exchange and resilience (Req. 7.1)	Wired Ethernet backbone	Short range optical (blue laser) links	Hybrid wired + optical with buffering
Autonomy during communication loss (Req. 7.1)	Local autonomous mission control	Multi level automatic control and safe state operation	Pre programmed safe-state logic
Structural and environmental monitoring (Req. 7.2)	Strain + pressure sensors	Fibre optic sensing (FBG)	Multi parameter sensor network
Leak detection and alarm philosophy (Req. 7.2)	Threshold based alarms	Automated fault detection and severity classification	Predictive anomaly detection
Control system architecture (Req. 7.2)	Centralised control unit	Distributed control nodes	Hybrid central distributed control
Vehicle docking and retention (Req. 7.3)	Passive funnel and mechanical capture	Acoustic homing + soft capture	Vision guided active docking
Power and data transfer to vehicles (Req. 7.3)	Wet mate electrical + data connectors	Inductive wireless charging	Combined power data interface

Fault containment & shutdown (Req. 7.4)	Compartmentalisation and isolation	Automated emergency shutdown	Graceful degradation modes
Redundancy and lifecycle support (Req. 7.4)	N+1 hardware redundancy	Modular replaceable units	Predictive maintenance strategy

Table 46. *Direct Ocean Capture Subsystem Morphological Chart.*

Feature (Identified from requirements table)	Means			
	Option 1	Option 2	Option 3	Option 4
6A. Direct Ocean Capture				
Seawater intake	Open intake + coarse screen	Intake + strainer basket	Screen + anti-fouling mesh	Dual intake (duty/standby)
Voltage Filters + housings	Cartridge filter	Bag filter	Media/sand filter	Self-cleaning filter
Pumps (seawater + recirc)	Centrifugal pump	Diaphragm pump	Progressive cavity pump	2× pumps (redundant)
Pipework + routing	Simple manifold	3-way valve routing	Parallel lines (2 trains)	Bypass loop + isolation
Electrochemical “stack”	Single electrodialysis stack	Bipolar membrane stack	2 stacks in parallel	Stack + polishing stage
Membrane choice/layout	AEM/CEM set	BPM focus (acid/base)	Mixed AEM/CEM/BPM	Modular cassette change-out
CO_2 release step	pH shift only (acidify)	pH shift + mixing	pH shift + hold tank	pH shift + recirc loop
CO_2 separation unit	Membrane degasser	Vacuum stripper column	Gas–liquid separator vessel	Multi-stage degassing
CO_2 handling / storage	Small buffer tank	CO_2 cylinder bank	Bladder/bag buffer	Buffer + compressor (small)
Neutralise + discharge	Base stream recombine	Static mixer before out	Hold tank neutralisation	Controlled dosing + feedback
Sensors pack	pH + flow	pH + flow + pressure	Add CO_2 flow meter	Add temp + conductivity
Control cabinet / logic	Simple controller + alarms	PLC + basic HMI	PLC + remote logging	Redundant PLC + safe shutdown

Materials (wet parts)	316 SS	Duplex SS	Polymer-lined sections	Mixed: SS + polymers
Maintenance strategy	Manual service intervals	Quick-change filters	Modular swap units	Condition-based maintenance
Fault management	Alarm + manual reset	Auto shutdown on limits	Bypass mode (limp home)	Full redundancy on critical

Table 47. *Underwater Data Centre Morphological Chart.*

Feature (Identified from requirements table)	Means			
	Option 1	Option 2	Option 3	Option 4
6B. Underwater Data Centre				
Power Supply Input	Direct Subsea Cable from shore	Offshore Platform feed	Renewable hybrid (tidal + subsea cable)	Surface Buoy power relay
Voltage Conversion	Centralised transformer module	Distributed rack-level converters	Solid-state modular converters	Redundant dual conversion system
Server Configuration	Standard 19” rack servers	Blade servers	Modular micro-data centre pods	High-density immersion-ready servers
Data Storage	HDD arrays	SSD arrays	Hybrid HDD/SSD arrays	Distributed Storage Cluster
Cooling Method (Internal)	Air cooling	Liquid cold-plate cooling	Immersion cooling	Two-phase dielectric cooling
Heat Rejection Method	Seawater heat exchanger	Direct seawater cooling loop	Heat pipe array	Phase change heat sink system
Thermal Circulation	Pump-driven loop	Natural convection loop	Redundant dual-loop system	Variable speed pump system
Structural Integration	Independent internal racks	Frame-mounted internal system	Shock-isolated mounting	Pressure-balanced internal modules
Monitoring System	Basic sensor network	Distributed IoT sensors	AI predictive monitoring	Redundant sensor network
Data Transmission	Fibre optic cable	Redundant dual fibre	Satellite relay backup	Subsea mesh network node

Fault Management	Manual remote reset	Automatic isolation of faulty racks	Hot-swappable modules	Full redundancy parallel architecture
Maintenance Strategy	Full retrieval servicing	ROV-assisted maintenance	Modular replacement units	Long-life sealed system

Table 48. *Deep Sea Mining Morphological Chart.*

Feature (Identified from requirements table)	Means			
	Option 1	Option 2	Option 3	Option 4
6C. Deep Sea Mining				
Collector Mechanism	Hydraulic suction	Mechanical cutting	Drum cutter	Continuous line bucket
Riser System	Flexible pipe + pump	Rigid pipe + airlift	Modular riser	Subsea vertical transport
Support Vessel	DP mining ship	Semi-submersible	Barge	Converted drillship
Mineral Processing	Onboard dewatering	Onshore processing	Subsea separation	Seafloor processing
Nodule Crusher	Jaw crusher	Cone crusher	High-pressure rolls	Impact mill
Tailings Discharge	Seafloor disposal	Mid-water discharge	Surface discharge	Dewatering + return
Power Source	Surface vessel supply	Subsea cable	Diesel-electric	Hybrid battery
Plume Mitigation	Shrouded collector	Diffuser system	Coagulant injection	Dynamic positioning

TRIZ Contradiction Matrix

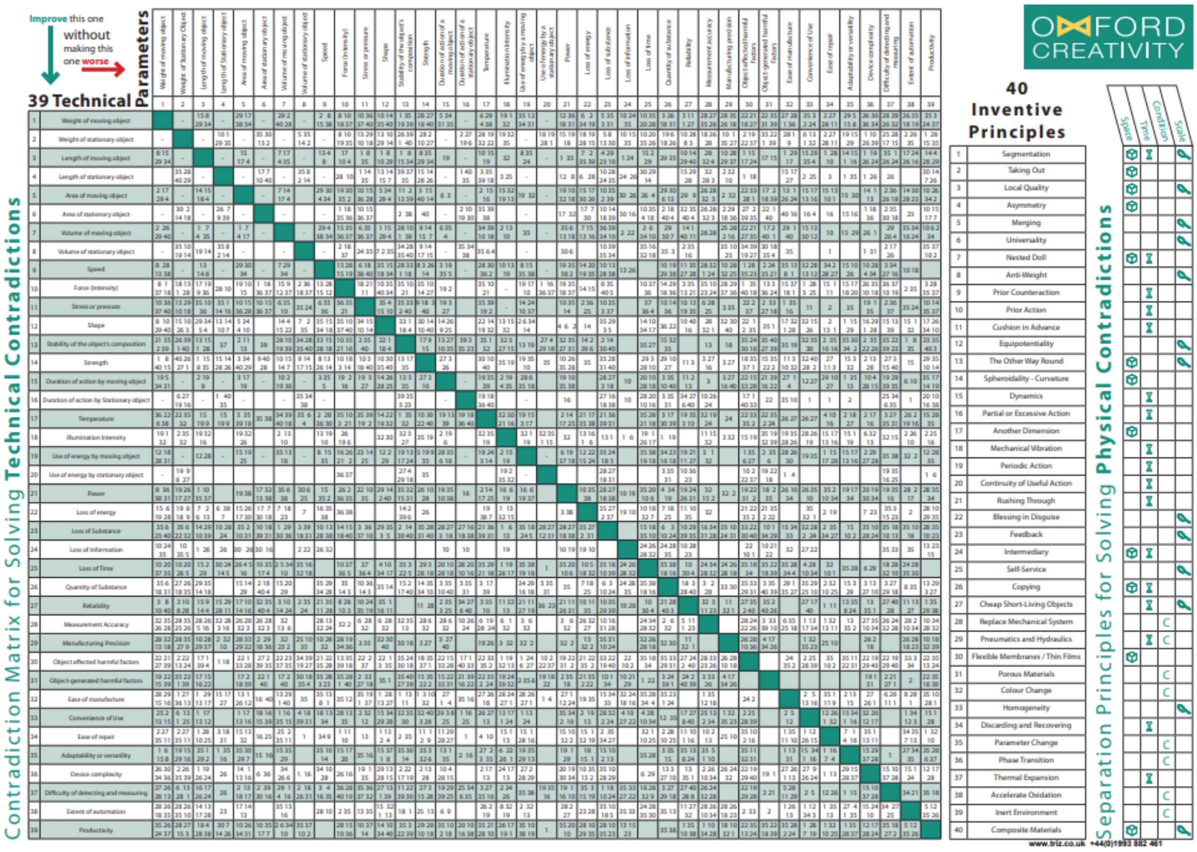

Figure 58. *Oxford Creativity TRIZ contradiction matrix (technical parameters and inventive principles). Source: Oxford Creativity Ltd (n.d.).*

Using the Oxford Creativity TRIZ matrix technical parameters and inventive principles (Figure 58), the following TRIZ contradiction table was created for the Underwater Habitat project (Table 43):

Table 49. *TRIZ Contradiction Matrix.*

Improve this one without making this one worse	Device complexity (36)	Quantity of substance / mass (26)	Area / space (6)	Loss of energy / inefficiency (22)	Temperature (17)
Reliability (27)	13, 35, 1	13, 35, 1	35, 1, 24	35, 19, 1	3, 35, 10
Productivity / throughput (39)	10, 35, 24	22, 30, 26	10, 15, 26	29, 34, 30	19, 35
Energy use (20)	-	35, 19	16, 35	10, 15, 19	35, 1, 24
Ease of repair (34)	35, 1, 13, 11	2, 28, 10, 25	16, 25	15, 13, 11	4, 10
Extent of automation (38)	15, 24, 10	35, 13	35, 13	26, 2, 19	35, 10, 18, 5
Harmful factors / environmental harm (31)	35, 19, 24	10, 35	35, 19	22, 35	35, 24

TRIZ Design Patterns Realised:

Designing a self-sustaining underwater habitat involves solving engineering contradictions, where improving one factor often makes another worse. Using the Oxford Creativity TRIZ contradiction matrix (Figure X), key system-level contradictions were selected for the whole habitat. The suggested inventive principles were then translated into practical design decisions that improve performance (reliability, throughput, energy use, maintainability) while reducing negatives (complexity, mass/space, inefficiency, environmental harm).

1) Improve Reliability (27) without increasing Device Complexity (36): Principles 13, 35, 1

The system must stay operational for long missions, but adding backup hardware can make it overly complex.

- (1) Segmentation: split the habitat into modular subsystems with standard interfaces (power/data/fluids). For example: a DOC skid, UDC rack/pod, DSM tool skid, comms unit, life support pack, and water/thermal packs. A failure in one module does not collapse the full habitat.
- (35) Parameter change: use adaptive operation instead of adding more mechanisms: variable pump speed (DOC/Water), adjustable cooling setpoints (Thermal), load-throttling (UDC computing), and controlled tool power (DSM) to keep stable operation under changing conditions.
- (13) The other way around: reduce mechanical “over-engineering” by shifting effort into monitoring + software logic (health checks, trend alarms, predictive maintenance) so reliability improves without adding many new moving parts.

2) Improve Ease of Repair (34) without increasing Quantity of Substance / Mass (26): Principles 2, 28, 10, 25

The habitat must be maintainable, but “easy access + extra tools + extra structure” increases mass/volume.

- (10) Preliminary action: design a service-first layout from day one (access panels, clear reach zones, lift points, quick-connect couplings). Prioritise high-maintenance items: DOC filters/pumps/valves, water filters, comms transceivers, UDC racks, DSM wear parts.
- (2) Taking out: put high-wear components into swap-out cartridges (filter cartridges, sensor pack modules, DSM tool heads, valve blocks). This reduces downtime without adding a lot of extra structure.
- (25) Self-service: build in self-diagnostics and guided checks (flow/pressure/leak tests for DOC and water; server health + fan status for UDC; link-quality alarms for comms; tool condition flags for DSM).
- (28) Mechanics substitution: replace manual inspection with sensors (vibration, pressure, conductivity, temperature, strain) + automatic alerts, so maintenance stays simple without extra physical inspection systems.

3) Reduce Harmful Factors / Environmental Harm (31) without increasing Loss of Energy / Inefficiency (22): Principles 22, 35

The habitat must avoid harming the marine environment, but “extra protection” can raise energy consumption.

- (35) Parameter change: introduce controlled operating modes: DOC discharge control (flow + chemistry limits), DSM low-disturbance mode, pump throttling, and “quiet” comms modes when needed—reducing impact without constant maximum power draw.

- (22) Blessing in disguise: recover and reuse “waste” energy: use UDC server heat + power electronics heat for habitat thermal needs (pre-heating, heat recovery loops) instead of dumping it into the sea, improving efficiency while reducing environmental stress.

4) Increase Productivity / Throughput (39) without increasing Device Complexity (36): Principles 10, 35, 24

You want higher mission output (DOC capture rate, UDC processing, DSM collection), but not a more complicated system.

- (10) Preliminary action: plan for throughput early with clear flow paths and interfaces (correct pipe sizing, standard manifolds, clean routing, scalable rack space), so you don’t add complex “patch fixes” later.
- (35) Parameter change: scale output by changing operating parameters (variable flow rate through DOC, adjustable vacuum/degassing duty, dynamic UDC processing schedules, DSM tool speed control) rather than adding more hardware stages.
- (24) Intermediary: use buffers and intermediate stages that simplify operation: small CO_2 buffer/storage, electrical bus regulation, thermal buffer tank/heat exchanger stage, and data buffering—so subsystems run smoothly without adding complex mechanisms.

APPENDIX E: Lab Exercises

Risk and Reliability Lab

Failure space-success space
Success Space:

- Doorbell rings only when button pressed
- Stops ringing when released
- Operates reliably for 10 years

Failure Space:

- No ring when pressed
- Rings without pressing
- Continuous ringing

Mechanisms and Failure Mechanisms

Mechanism	A	B	C
Wire	Open Circuit	Short Circuit	Intermittent Contact
Switch	Contact Corrosion	Stuck Closed	Mechanical Jam
Solenoid	Coil Burnout	Magnetic Short	Spring Failure

FMEA

Mechanism	Failure Mechanism	Failure Mode	Effect
Wire	Shock	Loose wire/lost connection.	Open circuit
Wire	Ground fault.	Loose wire	Short Circuit
Switch	Corrosion	High resistance	Open circuit
Solenoid	Coil Burnout	Overheating	Open Circuit
Spring	Fatigue	Won't return	Open Circuit

HAZOP

Guide word	Consequence	Causes
No	No ringing when pressed	Open circuit, dead battery, corroded switch
More/Less	Too loud or too quiet ringing	Voltage fluctuation, weak battery
As Well As/Part of	Ringing + electrical shock	Insulation failure
Reverse	Current flows incorrectly	Wiring reversed
Other than	Buzzing instead of ringing	Solenoid misalignment
Early/Late	Delay in ringing	Slow mechanical movement
Before/After	Rings before pressing or after release	Stuck switch

What value does HAZOP have over the information in FMEA?

HAZOP identifies deviations from design intent, considers unexpected system interactions, is more systematic with guidewords

How is HAZOP related to FMEA i.e. what information do they share and how can they be used together?

FMEA is Bottom-up (component-based), HAZOP is Top-down (system-based). Both identify failure modes and causes and can be combined for full coverage

RAG Risk Matrix

Risk	Likelihood	Impact	RAG	Mitigation
Electrolyser failure	Medium	High		Preventative maintenance, redundancy
Solar intermittency	High	Medium		Battery storage, grid backup
Water shortage	Low	High		Water recycling, desalination
Hydrogen leakage	Low	High		Leak detection systems
Market price drop	Medium	Medium		Long-term contracts
Minor equipment wear	High	Low		Scheduled servicing

Use of FS-SS, FMEA, FTA and Hazop

Failure Space / Success Space defines success as 200 kg/day production, net-zero emissions and failure as Underproduction, leaks, unsafe conditions

FMEA (Bottom-Up) analyses PEM membrane degradation, catalyst poisoning, pump failure and electrical inverter failure

FTA (Top-Down) with the top event: “Hydrogen production stops” is caused by power loss, electrolyser failure and/or water supply interruption

HAZOP:

- No flow = pump failure
- More pressure = explosion risk
- Reverse flow = system damage
- Early shutdown = grid instability

Applicability of reliability engineering techniques to your challenge:

Accelerated Life Testing: *An aircraft part is subjected to mechanical stresses such as compression, tension, shear, bending and torsion. Such forces are applied in an alternating fashion to simulate the fatigue the part may experience over time*	Antifragile: System improves via operational data feedback
Bulkhead: Separate hydrogen storage sections to prevent full failure	Change Control: Strict documentation for design modifications
Cold Standby: Backup electrolyser unit	Defensive Design: Overpressure valves and safety margins
Derating: Operate electrolyser below max capacity	Design Debt: Avoid shortcuts in early prototype
Design Life: 10-15 year operational target	Design Thinking: Human-centered plant operation
Durability: Corrosion-resistant materials	Edge Case: Extreme heat in Sustainable City
Entropy: Account for gradual material degradation	Error Tolerance: System continues despite minor faults
Fail Well: Safe shutdown during power loss	Fail-safe: Automatic hydrogen venting
Fault Tolerance: Parallel power supply systems	Graceful Degradation: Reduced output instead of shutdown
Human Error: Operator training	Human Factors: Clear control panel interface
Latent Human Error: Maintenance checklist errors	Maintainability: Easy membrane replacement
Mistake Proofing: Connector designs prevent wrong assembly	No Fault Found: Diagnostic logging system
Pokayoke: Incorrect valve connections physically impossible	Resilience: Ability to recover after grid disturbance
Root Cause: Structured incident analysis	Safety By Design: Hydrogen leak detection integrated

Self-Healing: Smart control software adjusting voltage	Service Life: Defined replacement cycle
Systems Thinking: Integration of renewables + storage + electrolysis	Testbed: Pilot-scale hydrogen plant
Wear And Tear: Monitor catalyst degradation	

Exercise A: Life-Cycle Thinking

1:

Product	Energy intensive phase
Coffee Maker	Use phase
Bicycle	Material production
Motorbike	Use phase
LPG fired patio heater	Use phase

2:

Product	Functional unit
Washing machines	E.g. Energy (kW.hr or MJ) or litres of water *per kg of clothes washed*
Refrigerators	kWh per litre cooled per year
Lighting	Lumens per kWh
Public transport	Passenger-km per MJ

3:

	Material	Manufacture	Transport	Use	Disposal
Materials resources (high use =0, none =4)	2	2	1	0	1
Energy Use (high use =0, none =4)	3	3	1	4	1
Global Warming (much CO_2 = 0 , no CO_2 = 4)	2	2	1	4	1
Human Health (Toxic emissions or waste?)	1	1	0	1	1
Column Totals	8	8	3	9	4

Total across columns= 32

Sustainable Development Goals Worksheet:

Positive and Negative Impacts of project

Environment Impacts	Impact	Societal Impacts	Impact
Land	Moderate land use required for 400 kW solar + 200 kW wind installation	Fertility	Indirect positive impact via improved air quality
Water Demand	Electrolysis requires purified water (moderate demand)	Population	Supports growing energy needs of urban population
Water Supply	Could strain supply unless desalination or recycling is used	Mortality	Reduced pollution lowers respiratory deaths
Emissions and Waste	Very low CO_2 emissions (green hydrogen), minor equipment waste	Health	Improved air quality therefore better public health
Soil	Minimal impact if foundations properly designed	Education	Can act as training/research facility
Bio Diversity	Wind turbines may affect birds; solar farms may disturb habitats	Vehicles	Enables hydrogen fuel cell vehicles
Material Consumption	Requires rare metals (platinum catalyst in PEM)	Infrastructure	Promotes hydrogen refuelling infrastructure
Electricity Generation	100% renewable – strong positive impact	Employment	Creates skilled green-energy jobs
Energy Supply	Improves clean energy security	Income Distribution	Expands green economy opportunities
Energy Consumption	High energy demand but renewable-powered	Poverty	Long-term sustainable economic growth

United Nations 2030 Sustainable Development Goals:

SDG Goals (including links)	The SDG target your project could progress (and how)
GOAL 1: No Poverty	Target 1.2: Job creation in the renewable sector
GOAL 2: Zero Hunger	Indirect via clean energy for agriculture
GOAL 3: Good Health and Well-being	Target 3.9: Reduce deaths from air pollution
GOAL 4: Quality Education	Target 4.4: Technical skills development
GOAL 5: Gender Equality	Equal employment opportunity policies
GOAL 6: Clean Water and Sanitation	Sustainable water management in electrolysis
GOAL 7: Affordable and Clean Energy	Target 7.2: Increase renewable energy share
GOAL 8: Decent Work and Economic Growth	Target 8.4: Improve resource efficiency
GOAL 9: Industry, Innovation and Infrastructure	Target 9.4: Upgrade infrastructure sustainably
GOAL 10: Reduced Inequality	Promote access to clean energy
GOAL 11: Sustainable Cities and Communities	Hydrogen transport systems
GOAL 12: Responsible Consumption and Production	Efficient hydrogen production

GOAL 13: Climate Action	Target 13.2: Integrate climate measures
GOAL 14: Life Below Water	Reduced ocean acidification
GOAL 15: Life on Land	Reduced ecosystem degradation
GOAL 16: Peace and Justice Strong Institutions	Strong governance frameworks
GOAL 17: Partnerships to achieve the Goal	Public-private renewable partnerships

APPENDIX F: Generative AI

In accordance with the university's guidance on the ethical use of generative AI, this statement confirms that generative AI was used as a support tool during the completion of this assignment. All final ideas, analyses, and conclusions presented in this report are the result of my own critical thinking and engineering judgment. The AI was used strictly as a thought partner and editorial assistant, as outlined below.

Declaration

Group D3 confirms that has used generative AI in the following ways:

- Scoping out a topic or developing initial ideas: Yes
- Summarising and understanding research: No
- To generate code which I have used in my assignment: No
- Editing and proofreading my work: Yes
- Other uses: No

A summary of the prompts used and their application is provided below:

1. Scoping Out Initial Ideas (Brainstorming)

At the early stage of the design process, AI was used to generate a broad range of possible engineering features for an underwater habitat.

Prompt Used: "Generate ideas for an underwater habitat structure. Suggest features like hull material, structural shape, foundation type, deployment method, pressure hull design, access & egress, emergency systems, and corrosion protection."

How the Output Was Used: The AI provided a list of plausible engineering options. Team reviewed these for technical feasibility, selected the most relevant ones, and refined them using team's knowledge of marine engineering and pressure vessel design.

2. Editing and Proofreading (Draft Refinement)

After drafting the technical description of the habitat, group used AI to suggest improvements to clarity and conciseness. The AI acted as a "fresh pair of eyes" to identify awkward phrasing.

Prompt Used: "Review the following paragraph for clarity and conciseness. Suggest improvements while maintaining a professional engineering tone."

How the Output Was Used: Team accepted some suggestions where they improved readability but rejected those that altered the technical meaning. The final text remains original work, with the AI acting only as an editor.